

F360TrackerLib Qualification Test Specification

Version 8.8.8

Generated by Doxygen 1.8.17

1 Covered Requirements	1
2 Module Documentation	17
2.1 F360_cluster_grouping_qualtest	17
2.1.1 Detailed Description	17
2.1.2 Function Documentation	17
2.1.2.1 f360_cluster_grouping_qualtest()	17
2.1.2.2 f360_cluster_grouping_qualtest_Test_Cluster_Grouping_Check_If_Dealiased_Flag_Is_Set_Correctly_1()	17
2.1.2.3 f360_cluster_grouping_qualtest_Test_Cluster_Grouping_Check_If_Dealiased_Flag_Is_Set_Correctly_2()	17
2.1.2.4 f360_cluster_grouping_qualtest_Test_Cluster_Grouping_Check_If_Range_Rates_Are_Dealiased_Correctly	17
2.2 F360_merge_two_clusters	20
2.2.1 Detailed Description	20
2.2.2 Function Documentation	20
2.2.2.1 f360_merge_two_clusters()	20
2.2.2.2 f360_merge_two_clusters_Test_Merge_Two_Clusters_Check_If_All_4_Dets_Are_Dealiased()	20
2.2.2.3 f360_merge_two_clusters_Test_Merge_Two_Clusters_Check_If_Dets_4_Are_Dealiased_1_Not_Dealiased	20
2.2.2.4 f360_merge_two_clusters_Test_Merge_Two_Clusters_Check_Range_Rate_Is_Dealiased_Correctly_1()	21
2.3 Initialize_clusters	22
2.3.1 Detailed Description	22
2.3.2 Function Documentation	22
2.3.2.1 initialize_qualtest_clusters()	22
2.3.2.2 initialize_qualtest_clusters_Test_Initialize_Clusters_Many_Detection_To_Cluster()	22
2.3.2.3 initialize_qualtest_clusters_Test_Initialize_Clusters_One_Detection_To_Cluster()	23
2.4 F360_allocate_id_for_initialized_object	24
2.4.1 Detailed Description	24
2.4.2 Function Documentation	24
2.4.2.1 f360_allocate_id_for_initialized_object()	24
2.4.2.2 f360_allocate_id_for_initialized_object_no_active_objects()	24
2.4.2.3 f360_allocate_id_for_initialized_object_objects_saturation()	25
2.4.2.4 f360_allocate_id_for_initialized_object_single_active_object()	25
2.5 F360_find_detection_inliers_single_object	26
2.6 F360_find_detection_inliers_find_valid_detections	27
2.6.1 Detailed Description	27
2.6.2 Function Documentation	27
2.6.2.1 f360_find_detection_inliers_find_valid_detections_Select_Detections_To_Mark_As_Inliers_Nearby_Wheel	27
2.6.2.2 f360_find_detection_inliers_find_valid_detections_Select_Detections_To_Mark_As_Inliers_Object_Wheel	27
2.6.2.3 f360_find_detection_inliers_find_valid_detections_Select_Detections_To_Mark_As_Inliers_Pair_Wheel_Spi	27
2.7 F360_Test_Determine_Association_Hypothesis_qualtest	29
2.7.1 Detailed Description	29
2.7.2 Function Documentation	29
2.7.2.1 f360_Test_Determine_Association_Hypothesis_qualtest()	29
2.7.2.2 f360_Test_Determine_Association_Hypothesis_qualtest_Test_Determine_Association_Hypothesis_A()	29

2.7.2.3 f360_Test_Determine_Association_Hypothesis_qualtest_Test_Determine_Association_Hypothesis_B()	30
2.8 F360_Test_Is_Det_Allowed_To_Associate_qualtest	31
2.8.1 Detailed Description	31
2.8.2 Function Documentation	31
2.8.2.1 f360_Test_Determine_Association_Hypothesis_qualtest_Test_Determine_Association_Hypothesis_C()	31
2.8.2.2 f360_Test_Determine_Association_Hypothesis_qualtest_Test_Determine_Association_Hypothesis_D()	32
2.8.2.3 f360_Test_Is_Det_Allowed_To_Associate_qualtest()	32
2.8.2.4 f360_Test_Is_Det_Allowed_To_Associate_qualtest_Test_Is_Det_Allowed_To_Associate_Det_Not_Ok_To_U	
2.8.2.5 f360_Test_Is_Det_Allowed_To_Associate_qualtest_Test_Is_Det_Allowed_To_Associate_Det_On_SEP_Obj	
2.8.2.6 f360_Test_Is_Det_Allowed_To_Associate_qualtest_Test_Is_Det_Allowed_To_Associate_Detection_Behind	
2.8.2.7 f360_Test_Is_Det_Allowed_To_Associate_qualtest_Test_Is_Det_Allowed_To_Associate_Mirror_Obj_Ambig	
2.8.2.8 f360_Test_Is_Det_Allowed_To_Associate_qualtest_Test_Is_Det_Allowed_To_Associate_Object_Behind_SE	
2.8.2.9 f360_Test_Is_Det_Allowed_To_Associate_qualtest_Test_Is_Det_Allowed_To_Associate_Object_Behind_SE	
2.9 F360_detect_wheelspin_object_qualtest	36
2.9.1 Detailed Description	36
2.9.2 Function Documentation	36
2.9.2.1 f360_detect_wheelspin_object_qualtest()	36
2.9.2.2 f360_detect_wheelspin_object_qualtest_f360_detect_wheelspin_object_qualtest_correctly_marked_as_ob	
2.9.2.3 f360_detect_wheelspin_object_qualtest_f360_detect_wheelspin_object_qualtest_correctly_not_marked_a	
2.10 F360_detect_wheelspin_nearby_qualtest	38
2.10.1 Detailed Description	38
2.10.2 Function Documentation	38
2.10.2.1 f360_detect_wheelspin_nearby_qualtest()	38
2.10.2.2 f360_detect_wheelspin_nearby_qualtest_f360_detect_wheelspin_nearby_qualtest_correctly_marked_as	
2.10.2.3 f360_detect_wheelspin_nearby_qualtest_f360_detect_wheelspin_nearby_qualtest_correctly_not_marked	
2.11 F360_fill_init_obj_track_props_qualtest	40
2.11.1 Detailed Description	40
2.11.2 Function Documentation	40
2.11.2.1 f360_fill_init_obj_track_props_qualtest()	40
2.11.2.2 f360_fill_init_obj_track_props_qualtest_TestCalcInitObjTrackMovingFlagMoving()	40
2.11.2.3 f360_fill_init_obj_track_props_qualtest_TestCalcInitObjTrackMovingFlagMovingValidForLiberalTracking()	4
2.11.2.4 f360_fill_init_obj_track_props_qualtest_TestCalcInitObjTrackMovingFlagNotMovingHighFracAz()	41
2.11.2.5 f360_fill_init_obj_track_props_qualtest_TestCalcInitObjTrackMovingFlagNotMovingLowSpeed()	42
2.11.2.6 f360_fill_init_obj_track_props_qualtest_TestCalcInitObjTrackMovingFlagNotMovingLowSpeedAndHighFrac	
2.11.2.7 f360_fill_init_obj_track_props_qualtest_TestCalcInitObjVCSHeadingMovingObj()	43
2.11.2.8 f360_fill_init_obj_track_props_qualtest_TestCalcInitObjVCSHeadingNotMovingObj()	43
2.11.2.9 f360_fill_init_obj_track_props_qualtest_TestFillInitObjTrackPropsNominal()	44
2.12 F360_init_obj_track_covariances	45
2.12.1 Detailed Description	45
2.12.2 Function Documentation	45
2.12.2.1 f360_init_obj_track_covariances()	45
2.12.2.2 f360_init_obj_track_covariances_TestCCANominalCase()	45

2.13 F360_fill_init_obj_track_props_determine_initial_size_qualtest	46
2.13.1 Detailed Description	46
2.13.2 Function Documentation	46
2.13.2.1 f360_fill_init_obj_track_props_determine_initial_size_qualtest()	46
2.13.2.2 f360_fill_init_obj_track_props_determine_initial_size_qualtest_Test_Initial_Size_Fast_Moving_Object() 46	
2.13.2.3 f360_fill_init_obj_track_props_determine_initial_size_qualtest_Test_Initial_Size_Slow_Moving_Object() 47	
2.13.2.4 f360_fill_init_obj_track_props_determine_initial_size_qualtest_Test_Initial_Size_Stationary_Object() 47	
2.14 F360_initialize_tracks	48
2.14.1 Detailed Description	48
2.14.2 Function Documentation	48
2.14.2.1 f360_initialize_tracked_objects()	48
2.14.2.2 f360_initialize_tracked_objects_Check_If_Objects_Are_Correctly_Sorted_By_Longitudinal_Position() 49	
2.14.2.3 f360_initialize_tracked_objects_Check_If_Objects_Are_Correctly_Sorted_By_Priority() 49	
2.14.2.4 f360_initialize_tracked_objects_Detections_Assignment_For_Single_Successfull_Object_Initialization() 50	
2.14.2.5 f360_initialize_tracked_objects_Detections_Assignment_For_Single_Unsuccessfull_Object_Initialization() 50	
2.14.2.6 f360_initialize_tracked_objects_Highest_Cluster_Priority_Initialization_Try_With_Obj_Saturation() 51	
2.14.2.7 f360_initialize_tracked_objects_Highest_Cluster_Priority_Reinitialization_With_Obj_Saturation() 51	
2.14.2.8 f360_initialize_tracked_objects_Lowest_Cluster_Priority_Initialization_Try_With_Obj_Saturation() 52	
2.14.2.9 f360_initialize_tracked_objects_Occluded_And_Non_Occluded_Cluster_Only_One_Successfull_Initialization 52	
2.14.2.10 f360_initialize_tracked_objects_Occluded_Cluster_With_Detections_Valid_For_Liberal_Initilization_Suc 52	
2.14.2.11 f360_initialize_tracked_objects_Single_Non_Occluded_Cluster__Successful_Initilization() 53	
2.14.2.12 f360_initialize_tracked_objects_Single_Occluded_Cluster__Unsuccessful_Initialization() 54	
2.14.2.13 f360_initialize_tracked_objects_Single_Successfull_Object_Initialization()	54
2.14.2.14 f360_initialize_tracked_objects_Single_Successfull_Object_Initialization_With_Init_Scheme_Check() 55	
2.14.2.15 f360_initialize_tracked_objects_Single_Unsuccessfull_Object_Initialization()	55
2.14.2.16 f360_initialize_tracked_objects_Two_Successfull_And_One_Unsuccessfull_Objects_Initialization() 56	
2.15 F360_try_to_init_obj_track_qualtest	57
2.15.1 Detailed Description	57
2.15.2 Function Documentation	57
2.15.2.1 f360_try_to_init_obj_track_qualtest()	57
2.15.2.2 f360_try_to_init_obj_track_qualtest_HostOverlapCheckPosLongNOTOverlapedPosLatNOTOverlappedtest 57	
2.15.2.3 f360_try_to_init_obj_track_qualtest_HostOverlapCheckPosLongNOTOverlapedPosLatOverlappedtest() 58	
2.15.2.4 f360_try_to_init_obj_track_qualtest_HostOverlapCheckPosLongOverlappedInRearAxleDistPosLatNOTOverlappedtest 58	
2.15.2.5 f360_try_to_init_obj_track_qualtest_HostOverlapCheckPosLongOverlappedInRearAxleDistPosLatOverlappedtest 58	
2.15.2.6 f360_try_to_init_obj_track_qualtest_LiberalInitializationTest()	59
2.16 Guardrail_detection	60
2.16.1 Detailed Description	60
2.16.2 Function Documentation	60
2.16.2.1 guardrail_detection()	60
2.16.2.2 guardrail_detection_capability()	60
2.16.2.3 guardrail_detection_no_guardrail_sanity_check()	61
2.17 F360_detect_wheelspin_pairs	62

2.17.1 Detailed Description	62
2.17.2 Function Documentation	62
2.17.2.1 f360_detect_wheelspin_pairs_qualtest_unittest()	62
2.17.2.2 f360_detect_wheelspin_pairs_qualtest_unittest_azimuth_spread_limits()	63
2.17.2.3 f360_detect_wheelspin_pairs_qualtest_unittest_doppler_spread_limits()	63
2.17.2.4 f360_detect_wheelspin_pairs_qualtest_unittest_lat_limits()	64
2.17.2.5 f360_detect_wheelspin_pairs_qualtest_unittest_lat_spread_limits()	64
2.17.2.6 f360_detect_wheelspin_pairs_qualtest_unittest_long_limits()	65
2.17.2.7 f360_detect_wheelspin_pairs_qualtest_unittest_long_spread_limits()	65
2.17.2.8 f360_detect_wheelspin_pairs_qualtest_unittest_max_wheelspin_pairs()	66
2.17.2.9 f360_detect_wheelspin_pairs_qualtest_unittest_multiple_candidates()	66
2.17.2.10 f360_detect_wheelspin_pairs_qualtest_unittest_proximity_detections()	67
2.17.2.11 f360_detect_wheelspin_pairs_qualtest_unittest_remarking()	67
2.18 F360_determine_dets_on_edge_of_fov	68
2.18.1 Detailed Description	68
2.18.2 Function Documentation	68
2.18.2.1 f360_determine_dets_on_edge_of_fov()	68
2.18.2.2 f360_determine_dets_on_edge_of_fov_lr_detections()	68
2.18.2.3 f360_determine_dets_on_edge_of_fov_mr_detections()	69
2.18.2.4 f360_determine_dets_on_edge_of_fov_range_limits()	70
2.19 F360_handle_host_vehicle_clutter_qualtest	71
2.19.1 Detailed Description	71
2.19.2 Function Documentation	71
2.19.2.1 f360_handle_host_vehicle_clutter_qualtest()	71
2.19.2.2 f360_handle_host_vehicle_clutter_qualtest_handle_host_vehicle_clutter__dets_marked_as_not_okay_to_use()	71
2.19.2.3 f360_handle_host_vehicle_clutter_qualtest_handle_host_vehicle_clutter__dets_marked_as_okay_to_use()	71
2.19.2.4 f360_handle_host_vehicle_clutter_qualtest_handle_host_vehicle_clutter__hvc_and_behind_guardrail_dets_marked_as_not_okay_to_use()	71
2.19.2.5 f360_handle_host_vehicle_clutter_qualtest_handle_host_vehicle_clutter__hvc_and_on_guardrail_dets_marked_as_not_okay_to_use()	71
2.20 F360_mark_water_spray_detections	74
2.20.1 Detailed Description	74
2.20.2 Function Documentation	74
2.20.2.1 f360_mark_water_spray_detections()	74
2.20.2.2 f360_mark_water_spray_detections_default_test()	74
2.20.2.3 f360_mark_water_spray_detections_det_high_rcs()	75
2.20.2.4 f360_mark_water_spray_detections_det_range_rate()	75
2.20.2.5 f360_mark_water_spray_detections_detections_outside_vcs_zone()	76
2.20.2.6 f360_mark_water_spray_detections_f_ok_to_use_false()	76
2.20.2.7 f360_mark_water_spray_detections_low_host_speed()	77
2.21 F360_msmt_update_obj_trks_cca_non_moveable_qualtest	78
2.21.1 Detailed Description	78
2.21.2 Function Documentation	78
2.21.2.1 f360_msmt_update_obj_trks_cca_non_moveable_qualtest()	78

2.21.2.2 f360_msmt_update_obj_trks_cca_non_moveable_qualtest_Msmt_Update_Obj_Trks_CCA_Non_Moveable	
2.22 F360_msmt_update_obj_trks_cca_moveable_qualtest	79
2.22.1 Detailed Description	79
2.22.2 Function Documentation	79
2.22.2.1 f360_msmt_update_obj_trks_cca_moveable_qualtest()	79
2.22.2.2 f360_msmt_update_obj_trks_cca_moveable_qualtest_Msmt_Update_Obj_Trks_CCA_Moveable_Check_H	
2.22.2.3 f360_msmt_update_obj_trks_cca_moveable_qualtest_Msmt_Update_Obj_Trks_CCA_Moveable_Check_U	
2.22.2.4 f360_msmt_update_obj_trks_cca_moveable_qualtest_Msmt_Update_Obj_Trks_CCA_Moveable_Check_U	
2.23 F360_msmt_update_obj_trks_ctca_qualtest	82
2.23.1 Detailed Description	82
2.23.2 Function Documentation	82
2.23.2.1 f360_msmt_update_obj_trks_ctca_qualtest()	82
2.23.2.2 f360_msmt_update_obj_trks_ctca_qualtest_Msmt_Update_Obj_Trks_CTCA_Check_Update_One_Det() 8	
2.24 F360_msmt_update_obj_trks_ctca_not_moving_qualtest	83
2.24.1 Detailed Description	83
2.24.2 Function Documentation	83
2.24.2.1 f360_msmt_update_obj_trks_ctca_not_moving_qualtest()	83
2.24.2.2 f360_msmt_update_obj_trks_ctca_not_moving_qualtest_Msmt_Update_ObjTrks_CTCA_curvature_head	
2.25 F360_msmt_update_object_timestamp_qualtest	84
2.25.1 Detailed Description	84
2.25.2 Function Documentation	84
2.25.2.1 f360_msmt_update_object_timestamp_qualtest()	84
2.25.2.2 f360_msmt_update_object_timestamp_qualtest_Msmt_Update_Object_Timestamp_One_Det() 84	
2.26 F360_msmt_update_supporting_functions_Select_Dets_For_RR_Update_qualtest	85
2.26.1 Detailed Description	85
2.26.2 Function Documentation	85
2.26.2.1 f360_msmt_update_supporting_functions_Select_Dets_For_RR_Update_qualtest() 85	
2.26.2.2 f360_msmt_update_supporting_functions_Select_Dets_For_RR_Update_qualtest_Test_Select_Dets_For	
2.27 F360_update_object_track_properties_Obj_Size	86
2.27.1 Detailed Description	86
2.27.2 Function Documentation	86
2.27.2.1 f360_update_object_track_properties_Calc_Obj_Size()	86
2.27.2.2 f360_update_object_track_properties_Calc_Obj_Size_Obj_Size_Is_Not_Updated_Obj_Coasting() 86	
2.27.2.3 f360_update_object_track_properties_Calc_Obj_Size_Obj_Size_Is_Not_Updated_Obj_One_Assoc_Det()	
2.27.2.4 f360_update_object_track_properties_Calc_Obj_Size_Obj_Size_Is_Updated()	87
2.28 F360_object_track_management_qualtest	88
2.28.1 Detailed Description	88
2.28.2 Function Documentation	88
2.28.2.1 f360_object_track_management_qualtest()	88
2.28.2.2 f360_object_track_management_qualtest_Obj_Trk_Status_Book_Keeping_NEW() 88	
2.28.2.3 f360_object_track_management_qualtest_Obj_Trk_Status_Book_Keeping_UPDATED() 89	
2.29 F360_adjust_fltr_type_dependet_params_qualtest	91

2.29.1 Detailed Description	91
2.29.2 Function Documentation	91
2.29.2.1 f360_adjust_filtr_type_dependet_params_qualtest()	91
2.29.2.2 f360_adjust_filtr_type_dependet_params_qualtest_Adjust_Filtr_Type_Dependent_Params_switch_to_CTC()	
2.30 F360_cancel_new_updated_trk_overlapping_confirmed_trks_qualtest	92
2.30.1 Detailed Description	92
2.30.2 Function Documentation	92
2.30.2.1 f360_cancel_new_updated_trk_overlapping_confirmed_trks_qualtest()	92
2.30.2.2 f360_cancel_new_updated_trk_overlapping_confirmed_trks_qualtest_Cancel_New_Updated_Trk_Overlapping_Confirmed_Trks()	
2.30.2.3 f360_cancel_new_updated_trk_overlapping_confirmed_trks_qualtest_Cancel_New_Updated_Trk_Overlapping_Confirmed_Trks_Switch_To_CTC()	
2.31 F360_update_object_track_properties_obj_height	94
2.31.1 Detailed Description	94
2.31.2 Function Documentation	94
2.31.2.1 f360_update_object_track_properties_calc_obj_height()	94
2.31.2.2 f360_update_object_track_properties_calc_obj_height_obj_height_decay_below_zero()	94
2.31.2.3 f360_update_object_track_properties_calc_obj_height_obj_height_decays_when_no_dets()	95
2.31.2.4 f360_update_object_track_properties_calc_obj_height_obj_height_is_not_calculated_when_stationary()	95
2.31.2.5 f360_update_object_track_properties_calc_obj_height_obj_height_is_updated_when_valid()	96
2.32 F360_object_based_water_spray_detector_qualtest	97
2.32.1 Detailed Description	97
2.32.2 Function Documentation	97
2.32.2.1 f360_object_based_water_spray_detector_qualtest()	97
2.32.2.2 f360_object_based_water_spray_detector_qualtest_Detect_Water_Spray_From_Objects()	97
2.33 F360_object_based_angle_jump_detector_Detect_Angle_Jumps_From_Objects_qualtest	98
2.33.1 Detailed Description	98
2.33.2 Function Documentation	98
2.33.2.1 f360_object_based_angle_jump_detector_Detect_Angle_Jumps_From_Objects_qualtest()	98
2.33.2.2 f360_object_based_angle_jump_detector_Detect_Angle_Jumps_From_Objects_qualtest_flagging_correct()	
2.34 F360_find_tracks_on_and_behind_guardrail_qualtest	99
2.34.1 Detailed Description	99
2.34.2 Function Documentation	99
2.34.2.1 f360_find_tracks_on_and_behind_guardrail_qualtest_Flag_Objects_On_SEP_ObjectOnGuardrailLeftSide()	
2.34.2.2 f360_find_tracks_on_and_behind_guardrail_qualtest_Flag_Objects_On_SEP_ObjectOnGuardrailRightSide()	
2.35 Detecting_multibounces_from_object	101
2.35.1 Detailed Description	101
2.35.2 Function Documentation	101
2.35.2.1 detecting_multibounces_from_object()	101
2.35.2.2 detecting_multibounces_from_object_det_marked_as_multibounce()	101
2.35.2.3 detecting_multibounces_from_object_det_NOT_marked_as_multibounce()	102
2.36 Object_wrt_to_sep_tests	103
2.36.1 Detailed Description	103
2.36.2 Function Documentation	103

2.36.2.1	object_wrt_to_sep_tests()	103
2.36.2.2	object_wrt_to_sep_tests_obejct_marked_as_on_a_sep()	103
2.36.2.3	object_wrt_to_sep_tests_obejct_NOT_marked_as_on_sep()	104
2.37	Sensor_Postprocessing	105
2.37.1	Detailed Description	105
2.37.2	Function Documentation	105
2.37.2.1	f360_Update_Detection_History_qualtest()	105
2.37.2.2	f360_Update_Detection_History_qualtest_Test_Update_Detection_History_Find_Next_Hist_Det_Iidx_false	
2.37.2.3	f360_Update_Detection_History_qualtest_Test_Update_Detection_History_n_occupied_higher_than_max	
2.37.2.4	f360_Update_Detection_History_qualtest_Test_Update_Detection_History_old_dets_greater_than_zero()	107
2.37.2.5	f360_Update_Detection_History_qualtest_Test_Update_Detection_History_update_one_detection()	107
2.38	F360_kill_coasted_tracks_qualtest	108
2.38.1	Detailed Description	108
2.38.2	Function Documentation	108
2.38.2.1	f360_kill_coasted_tracks_qualtest()	108
2.38.2.2	f360_kill_coasted_tracks_qualtest_Do_not_terminate_updated_tracks()	109
2.38.2.3	f360_kill_coasted_tracks_qualtest_Non_Movable_Coasting_Obj_Outside_Front_Only_FOV_Coasting_Tim	
2.38.2.4	f360_kill_coasted_tracks_qualtest_Non_Movable_Coasting_Obj_Outside_Front_Only_FOV_Config()	110
2.38.2.5	f360_kill_coasted_tracks_qualtest_Non_Movable_Coasting_Obj_Outside_Front_Only_FOV_Nonmoveable	
2.38.2.6	f360_kill_coasted_tracks_qualtest_Non_Movable_Coasting_Obj_Outside_Front_Only_FOV_Pos()	111
2.38.2.7	f360_kill_coasted_tracks_qualtest_Termination_of_coasting_confirmed_objects()	112
2.38.2.8	f360_kill_coasted_tracks_qualtest_Termination_of_coasting_mirror_objects_variation_of_mirror_probability	
2.38.2.9	f360_kill_coasted_tracks_qualtest_Termination_of_coasting_mirror_objects_variation_of_stage_time()	113
2.39	F360_object_occlusion_classification	115
2.39.1	Detailed Description	115
2.39.2	Function Documentation	115
2.39.2.1	f360_object_occlusion_classification()	115
2.39.2.2	f360_object_occlusion_classification_Verify_All_Active_Objects_Are_Considered()	116
2.39.2.3	f360_object_occlusion_classification_Verify_Single_Object_Status_Occluded()	116
2.39.2.4	f360_object_occlusion_classification_Verify_Single_Object_Status_On_Edge()	117
2.39.2.5	f360_object_occlusion_classification_Verify_Single_Object_Status_Undefined_Not_Moving()	118
2.39.2.6	f360_object_occlusion_classification_Verify_Single_Object_Status_Visible()	118
2.40	F360_object_underdrivability_classification_qt	120
2.40.1	Detailed Description	120
2.40.2	Function Documentation	120
2.40.2.1	f360_object_underdrivability_classification_qt()	120
2.40.2.2	f360_object_underdrivability_classification_qt_test_objs_invalid()	121
2.40.2.3	f360_object_underdrivability_classification_qt_test_valid_moving_objs_behind_host()	121
2.40.2.4	f360_object_underdrivability_classification_qt_test_valid_moving_objs_height_above_threshold()	122
2.40.2.5	f360_object_underdrivability_classification_qt_test_valid_moving_objs_height_below_threshold()	122
2.41	F360_assign_underdrivability_status_to_tracks_ocg_QT	123
2.41.1	Detailed Description	123

2.41.2 Function Documentation	123
2.41.2.1 f360_assign_underdrivability_status_to_tracks_ocg_QT()	123
2.41.2.2 f360_assign_underdrivability_status_to_tracks_ocg_QT_Assign_Underdrivability_Status_To_Tracks_Curr	
2.41.2.3 f360_assign_underdrivability_status_to_tracks_ocg_QT_Assign_Underdrivability_Status_To_Tracks_Curr	
2.41.2.4 f360_assign_underdrivability_status_to_tracks_ocg_QT_Assign_Underdrivability_Status_To_Tracks_Curr	
2.41.2.5 f360_assign_underdrivability_status_to_tracks_ocg_QT_Assign_Underdrivability_Status_To_Tracks_Curr	
2.41.2.6 f360_assign_underdrivability_status_to_tracks_ocg_QT_Assign_Underdrivability_Status_To_Tracks_Not	
2.41.2.7 f360_assign_underdrivability_status_to_tracks_ocg_QT_Assign_Underdrivability_Status_To_Tracks_Obj	
2.42 Object_motion_status_should_be_updated	127
2.42.1 Detailed Description	127
2.42.2 Function Documentation	127
2.42.2.1 object_motion_status_should_be_updated_qualtest()	127
2.42.2.2 object_motion_status_should_be_updated_qualtest_Object_Motion_Status_Should_Be_Updated_COAS	
2.42.2.3 object_motion_status_should_be_updated_qualtest_Object_Motion_Status_Should_Be_Updated_COAS	
2.42.2.4 object_motion_status_should_be_updated_qualtest_Object_Motion_Status_Should_Be_Updated_NEW	
2.42.2.5 object_motion_status_should_be_updated_qualtest_Object_Motion_Status_Should_Be_Updated_NEW	
2.42.2.6 object_motion_status_should_be_updated_qualtest_Object_Motion_Status_Should_Be_Updated_UPDA	
2.43 Update_object_properties	131
2.43.1 Detailed Description	131
2.43.2 Function Documentation	131
2.43.2.1 update_object_properties_qualtest()	131
2.43.2.2 update_object_properties_qualtest_update_object_properties__check_if_changes_moveable_flag()	131
2.43.2.3 update_object_properties_qualtest_update_object_properties__check_if_moveable_flag_is_not_reset()	132
2.44 Check_if_object_is_suspected_stationary	133
2.44.1 Detailed Description	133
2.44.2 Function Documentation	133
2.44.2.1 check_if_object_is_suspected_stationary_qualtest()	133
2.44.2.2 check_if_object_is_suspected_stationary_qualtest_check_if_object_is_suspected_stationary__cca_marke	
2.44.2.3 check_if_object_is_suspected_stationary_qualtest_check_if_object_is_suspected_stationary__cca_marke	
2.44.2.4 check_if_object_is_suspected_stationary_qualtest_check_if_object_is_suspected_stationary__cca_marke	
2.44.2.5 check_if_object_is_suspected_stationary_qualtest_check_if_object_is_suspected_stationary__cca_marke	
2.44.2.6 check_if_object_is_suspected_stationary_qualtest_check_if_object_is_suspected_stationary__cca_marke	
2.44.2.7 check_if_object_is_suspected_stationary_qualtest_check_if_object_is_suspected_stationary__cca_marke	
2.44.2.8 check_if_object_is_suspected_stationary_qualtest_check_if_object_is_suspected_stationary__ctca_marke	
2.44.2.9 check_if_object_is_suspected_stationary_qualtest_check_if_object_is_suspected_stationary__ctca_marke	
2.44.2.10 check_if_object_is_suspected_stationary_qualtest_check_if_object_is_suspected_stationary__ctca_mar	
2.44.2.11 check_if_object_is_suspected_stationary_qualtest_check_if_object_is_suspected_stationary__ctca_mar	
2.44.2.12 check_if_object_is_suspected_stationary_qualtest_check_if_object_is_suspected_stationary__ctca_mar	
2.45 CalcProbability	140
2.45.1 Detailed Description	140
2.45.2 Function Documentation	140
2.45.2.1 calcProbability_qualtest_calcProbability__check_if_sum_of_calculated_probabilities_is_equal_1()	140

2.46 Set_object_class	141
2.46.1 Detailed Description	141
2.46.2 Function Documentation	141
2.46.2.1 set_object_class_qualtest_set_object_class__check_if_classifies_as_undetermined()141	
2.46.2.2 set_object_class_qualtest_set_object_class__check_if_min_prob_value_to_classify_is_0_65()142	
2.46.2.3 set_object_class_qualtest_set_object_class__check_if_prioritizes_cars()	142
2.46.2.4 set_object_class_qualtest_set_object_class__check_if_prioritizes_motorcycles()	143
2.46.2.5 set_object_class_qualtest_set_object_class__check_if_prioritizes_pedestrian()	143
2.46.2.6 set_object_class_qualtest_set_object_class__check_if_prioritizes_trucks()	144
2.47 FilteringAndNormalizationOfProbabilities	145
2.47.1 Detailed Description	145
2.47.2 Function Documentation	145
2.47.2.1 f360_run_obj_class()	145
2.47.2.2 f360_run_obj_class_f360_run_obj_class_bicycle()	146
2.47.2.3 f360_run_obj_class_f360_run_obj_class_car()	146
2.47.2.4 f360_run_obj_class_f360_run_obj_class_motorcycle()	147
2.47.2.5 f360_run_obj_class_f360_run_obj_class_pedestrian()	147
2.47.2.6 f360_run_obj_class_f360_run_obj_class_truck()	148
2.47.2.7 f360_run_obj_class_f360_run_obj_class_undetermined()	148
2.47.2.8 filteringAndNormalizationOfProbabilities_qualtest_filteringAndNormalizationOfProbabilities__check_if_undetermined()	148
2.48 CalcUndetProb	150
2.48.1 Detailed Description	150
2.48.2 Function Documentation	150
2.48.2.1 calcUndetProb_qualtest()	150
2.48.2.2 calcUndetProb_qualtest_calcUndetProb__check_if_prob_unknown_is_ste_to_max_value_when_object_has_prob()	150
2.48.2.3 calcUndetProb_qualtest_calcUndetProb__check_if_prob_unknown_is_ste_to_max_value_when_object_has_prob()	150
2.48.2.4 calcUndetProb_qualtest_calcUndetProb__check_if_prob_unknown_is_ste_to_max_value_when_object_has_prob()	150
2.48.2.5 calcUndetProb_qualtest_calcUndetProb__check_if_prob_unknown_is_ste_to_max_value_when_object_has_prob()	150
2.49 Track_Downselection	153
2.49.1 Detailed Description	153
2.49.2 Function Documentation	153
2.49.2.1 f360_track_downselection_qualtest_f360_core_track_downselect__check_if_assigns_reduced_id_to_higher_priority()	153
2.49.2.2 f360_track_downselection_qualtest_f360_core_track_downselect__check_if_does_reduce_tracks_basing_on_priority()	153
2.49.2.3 f360_track_downselection_qualtest_f360_core_track_downselect__check_if_lowers_confidence_and_priority()	153
2.49.2.4 f360_track_downselection_qualtest_f360_core_track_downselect__check_if_lowers_confidence_of_tracks()	153
2.49.2.5 f360_track_downselection_qualtest_f360_core_track_downselect__check_if_lowers_confidence_of_tracks()	153
2.49.2.6 f360_track_downselection_qualtest_f360_core_track_downselect__check_if_resets_low_priority_target_reduced_id()	153
2.49.2.7 f360_track_downselection_qualtest_f360_core_track_downselect__check_if_tracks_are_downselected_as_higher_priority()	153
2.49.2.8 f360_track_downselection_qualtest_f360_core_track_downselect__check_if_tracks_with_higher_confidence_are_downselected()	153
2.49.2.9 f360_track_downselection_qualtest_f360_core_track_downselect__check_if_tracks_with_low_amount_of_tracks_are_downselected()	153
2.49.2.10 f360_track_downselection_qualtest_f360_core_track_downselect__check_if_tracks_with_lower_ttc_are_downselected()	153
2.49.2.11 f360_track_downselection_qualtest_f360_core_track_downselect__check_if_tracks_with_reduced_status_are_downselected()	153

2.49.2.12 f360_track_downselection_qualtest_f360_core_track_downselect__check_if_updated_tracks_are_prioriti	
2.49.2.13 f360_track_downselection_qualtest_f360_core_track_downselect__check_if_valid_trk_closer_to_host_is	
2.50 Track_Downselection__Calculate_Track_Priority_Non_Moveable_Objects	161
2.50.1 Detailed Description	161
2.50.2 Function Documentation	161
2.50.2.1 f360_track_downselection_calc_track_priority_qualtest_Host_Curvature_Is_Not_Close_To_Zero()	161
2.50.2.2 f360_track_downselection_calc_track_priority_qualtest_Host_Is_Moving()	162
2.50.2.3 f360_track_downselection_calc_track_priority_qualtest_Host_Is_Reversing()	162
2.50.2.4 f360_track_downselection_calc_track_priority_qualtest_Host_Is_Stationary()	163
2.50.2.5 f360_track_downselection_calc_track_priority_qualtest_Impact_oF_Host_Speeds_On_Object_Priorities()	163
2.51 F360_split_objects_in_orth_direction_qualtest	164
2.51.1 Detailed Description	164
2.51.2 Function Documentation	164
2.51.2.1 f360_split_objects_in_orth_direction_qualtest()	164
2.51.2.2 f360_split_objects_in_orth_direction_qualtest_Verify_Split_Logic()	164
2.52 F360_track_grouping	165
2.52.1 Detailed Description	165
2.52.2 Function Documentation	165
2.52.2.1 f360_track_grouping()	165
2.52.2.2 f360_track_grouping_Check_Consinsency_In_Obj_Id()	165
2.52.2.3 f360_track_grouping_Check_Updates_In_Long_Pos_Sort()	166
2.52.2.4 f360_track_grouping_Check_Updates_In_Priority_Sort()	166
2.52.2.5 f360_track_grouping_Test_Merged_Obj_Max_Size_Is_Not_excced()	167
2.52.2.6 f360_track_grouping_Test_Move_All_Dets_Assoc()	167
2.52.2.7 f360_track_grouping_Test_Move_Dets_Assoc()	168
2.52.2.8 f360_track_grouping_Test_Remain_Dets_Assoc()	168
2.52.2.9 f360_track_grouping_Test_Similar_Objects_Were_Merged()	169
2.53 F360_detect_multipath__Detect_And_Mark_Multipath_Objects_Qual	170
2.53.1 Detailed Description	170
2.53.2 Function Documentation	170
2.53.2.1 f360_detect_multipath__Detect_And_Mark_Multipath_Objects_Qual()	170
2.53.2.2 f360_detect_multipath__Detect_And_Mark_Multipath_Objects_Qual_default_scenario_where_host_speed	
2.53.2.3 f360_detect_multipath__Detect_And_Mark_Multipath_Objects_Qual_default_scenario_where_host_speed	
2.53.2.4 f360_detect_multipath__Detect_And_Mark_Multipath_Objects_Qual_scenario_where_reflector_objects	
2.53.2.5 f360_detect_multipath__Detect_And_Mark_Multipath_Objects_Qual_scenario_where_source_objects_d	
2.53.2.6 f360_detect_multipath__Detect_And_Mark_Multipath_Objects_Qual_scenario_where_source_objects_is	
2.53.2.7 f360_detect_multipath__Detect_And_Mark_Multipath_Objects_Qual_scenario_where_source_objects_is	
2.53.2.8 f360_detect_multipath__Detect_And_Mark_Multipath_Objects_Qual_scenario_where_source_objects_is	
2.53.2.9 f360_detect_multipath__Detect_And_Mark_Multipath_Objects_Qual_scenario_where_source_objects_u	
2.54 F360_determine_reflected_obj	175
2.54.1 Detailed Description	175
2.54.2 Function Documentation	175

2.54.2.1 f360_determine_reflected_obj_check_calibrations()	175
2.54.2.2 f360_determine_reflected_obj_Determine_Reflected_Obj__check_whether_mirror_track_has_its_mirror_p	
2.54.2.3 f360_determine_reflected_obj_Determine_Reflected_Obj__check_whether_mirror_track_has_its_mirror_p	
2.54.2.4 f360_determine_reflected_obj_Determine_Reflected_Obj__check_whether_track_behind_guardrail_has_i	
2.54.2.5 f360_determine_reflected_obj_Determine_Reflected_Obj__check_whether_track_that_is_not_a_mirror_tra	
2.54.2.6 f360_determine_reflected_obj_Determine_Reflected_Obj__check_whether_track_that_is_not_a_mirror_tra	
2.54.2.7 f360_determine_reflected_obj_Determine_Reflected_Obj__check_whether_track_that_is_not_a_mirror_tra	
2.54.2.8 f360_determine_reflected_obj_Determine_Reflected_Obj__check_whether_valid_track_behind_guardrail	
2.54.2.9 f360_determine_reflected_obj_Determine_Reflected_Obj__check_whether_valid_track_in_front_and_on_o	
2.55 F360_determine_reflected_obj_with_turned_hide_tracks_outside_guardrail	180
2.55.1 Detailed Description	180
2.55.2 Function Documentation	180
2.55.2.1 f360_determine_reflected_obj_with_turned_hide_tracks_outside_guardrail_prob_set_to_one_because_tra	
2.55.2.2 f360_determine_reflected_obj_with_turned_hide_tracks_outside_guardrail_prob_set_to_one_because_tra	
2.55.2.3 f360_determine_reflected_obj_with_turned_hide_tracks_outside_guardrail_prob_set_to_zero_because_tra	
2.55.2.4 f360_determine_reflected_obj_with_turned_hide_tracks_outside_guardrail_prob_set_to_zero_because_tra	
2.56 F360_update_exist_prob	183
2.56.1 Detailed Description	183
2.56.2 Function Documentation	183
2.56.2.1 f360_update_exist_prob_qualtest()	183
2.56.2.2 f360_update_exist_prob_qualtest_Update_Existence_Probability__check_whether_EP_depend_onobject	
2.56.2.3 f360_update_exist_prob_qualtest_Update_Existence_Probability__check_whether_maximum_EP_value_i	
2.56.2.4 f360_update_exist_prob_qualtest_Update_Existence_Probability__check_whether_minimum_EP_value_is	
2.57 F360_conf_track_confidence_level	186
2.57.1 Detailed Description	186
2.57.2 Function Documentation	186
2.57.2.1 f360_conf_track_confidence_level_qualtest()	186
2.57.2.2 f360_conf_track_confidence_level_qualtest_Conf_Track_Confidence_Level__if_confidence_is_calculated	
2.57.2.3 f360_conf_track_confidence_level_qualtest_Conf_Track_Confidence_Level__is_confidence_level_depend	
2.57.2.4 f360_conf_track_confidence_level_qualtest_Conf_Track_Confidence_Level__is_confidence_level_depend	
2.57.2.5 f360_conf_track_confidence_level_qualtest_Conf_Track_Confidence_Level__is_confidence_level_depend	
2.57.2.6 f360_conf_track_confidence_level_qualtest_Conf_Track_Confidence_Level__is_confidence_level_is_equa	
2.57.2.7 f360_conf_track_confidence_level_qualtest_Conf_Track_Confidence_Level__is_confidence_level_is_equa	
2.57.2.8 f360_conf_track_confidence_level_qualtest_Conf_Track_Confidence_Level__is_confidence_level_is_equa	
2.57.2.9 f360_conf_track_confidence_level_qualtest_Conf_Track_Confidence_Level__is_confidence_level_is_equa	
2.57.2.10 f360_conf_track_confidence_level_qualtest_Conf_Track_Confidence_Level__is_confidence_overlapping_o	
2.58 F360_input_diagnostics_qualtest	192
2.58.1 Detailed Description	193
2.58.2 Function Documentation	193
2.58.2.1 f360_input_diagnostics_qualtest_Fault_Should_Be_Returned_When_Core_Info_Cnt_Loops_Lower_Than	
2.58.2.2 f360_input_diagnostics_qualtest_Fault_Should_Be_Returned_When_Core_Info_Elapsed_Above_Upper_L	
2.58.2.3 f360_input_diagnostics_qualtest_Fault_Should_Be_Returned_When_Core_Info_Elapsed_Time_Below_Lo	

2.58.2.4	f360_input_diagnostics_qualtest_Fault_Should_Be_Returned_When_Core_Info_Time_us_Lower_Than_B	
2.58.2.5	f360_input_diagnostics_qualtest_Fault_Should_Be_Returned_When_Host_Info_Vehicle_index_Lower_Th	
2.58.2.6	f360_input_diagnostics_qualtest_No_Fault_Reported_For_Sensor_When_Calibration_Not_Valid()	196
2.58.2.7	f360_input_diagnostics_qualtest_No_Fault_Reported_For_Sensor_When_look_index_Overflow()	196
2.58.2.8	f360_input_diagnostics_qualtest_No_Increase_Fault_Should_Be_Returned_When_Look_Index_For_Sing	
2.58.2.9	f360_input_diagnostics_qualtest_No_Increase_Faults_Should_Be_Returned_When_Core_Signal_Receiv	
2.58.2.10	f360_input_diagnostics_qualtest_No_Increase_Faults_Should_Be_Returned_When_Sensors_Signal_Re	
2.58.2.11	f360_input_diagnostics_qualtest_Polarity_Fault_Should_Not_Be_Reported_When_Sensor_Polarity_Is_V	
2.58.2.12	f360_input_diagnostics_qualtest_Polarity_Fault_Should_Not_Be_Reported_When_Sensor_Polarity_Is_V	
2.58.2.13	f360_input_diagnostics_qualtest_Signal_Fault_Should_Be_Returned_When_Difference_Between_Senso	
2.58.2.14	f360_input_diagnostics_qualtest_Signal_Fault_Should_Be_Returned_When_Difference_Between_Senso	
2.58.2.15	f360_input_diagnostics_qualtest_Signal_Fault_Should_Not_Be_Set_When_No_Sensor_Data_In_Receiv	
2.58.2.16	f360_input_diagnostics_qualtest_Signal_Faults_Should_Be_Reported_When_Look_Index_Equal_To_Ze	
2.58.2.17	f360_input_diagnostics_qualtest_Signal_Faults_Should_Be_Reported_When_Sensor_Azimuth_Boresigh	
2.58.2.18	f360_input_diagnostics_qualtest_Signal_Faults_Should_Be_Reported_When_Sensor_Azimuth_Boresigh	
2.58.2.19	f360_input_diagnostics_qualtest_Signal_Faults_Should_Be_Reported_When_Sensor_Azimuth_Boresigh	
2.58.2.20	f360_input_diagnostics_qualtest_Signal_Faults_Should_Be_Reported_When_Sensor_Azimuth_Boresigh	
2.58.2.21	f360_input_diagnostics_qualtest_Signal_Faults_Should_Be_Reported_When_Sensor_Mounting_Position	
2.58.2.22	f360_input_diagnostics_qualtest_Signal_Faults_Should_Be_Reported_When_Sensor_Mounting_Position	
2.58.2.23	f360_input_diagnostics_qualtest_Signal_Faults_Should_Be_Reported_When_Sensor_Polarity_Is_Equal	
2.58.2.24	f360_input_diagnostics_qualtest_Signal_Faults_Should_Be_Reported_When_Sensor_Polarity_Is_Greate	
2.58.2.25	f360_input_diagnostics_qualtest_Signal_Faults_Should_Be_Reported_When_Sensor_Polarity_Is_Greate	
2.58.2.26	f360_input_diagnostics_qualtest_Signal_Faults_Should_Not_Be_Reported_When_Sensor_Mounting_Po	
2.58.2.27	f360_input_diagnostics_qualtest_Signal_Faults_Should_Not_Be_Reported_When_Sensor_Mounting_Po	
2.58.2.28	f360_input_diagnostics_qualtest_Signal_Faults_Should_Not_Be_Reported_When_Sensor_Mounting_Po	
2.58.2.29	f360_input_diagnostics_qualtest_Signal_Faults_Should_Not_Be_Reported_When_Sensor_Mounting_Po	
2.58.2.30	f360_input_diagnostics_qualtest_True_Should_Be_Returned_When_Big_Curvature_Rear_Or_VCS_Side	
2.58.2.31	f360_input_diagnostics_qualtest_True_Should_Be_Returned_When_Host_QF_Is_Accurate()	210
2.58.2.32	f360_input_diagnostics_qualtest_True_Should_Be_Returned_When_Host_QF_Is_NOT_Accurate()	211

3 File Documentation	213
3.1 Fusion360/cluster_grouping/unittest/f360_cluster_grouping_qualtest_unittest.cpp File Reference . . .	213
3.1.1 Detailed Description	213
3.2 Fusion360/cluster_grouping/unittest/f360_merge_two_clusters_qualtest_unittest.cpp File Reference .	213
3.2.1 Detailed Description	214
3.3 Fusion360/clustering/unittest/f360_initialize_clusters_qualtest_unittest.cpp File Reference	214
3.3.1 Detailed Description	214
3.4 Fusion360/common/unittest/f360_allocate_id_for_initialized_object_qualtest_unittest.cpp File Reference	214
3.4.1 Detailed Description	214
3.5 Fusion360/common/unittest/f360_find_detection_inliers_qualtest_unittest.cpp File Reference	215
3.5.1 Detailed Description	215
3.6 Fusion360/detection_to_track_association/unittest/f360_detection_to_track_association_qualtest_↔ unittest.cpp File Reference	215

3.6.1 Detailed Description	216
3.7 Fusion360/detection_to_track_association/unittest/f360_mark_detections_wheel_spin_from_↔ objects_qualtest_unittest.cpp File Reference	216
3.7.1 Detailed Description	216
3.8 Fusion360/detection_to_track_association/unittest/f360_nearby_wheel_spins_qualtest_unittest.cpp File Reference	216
3.8.1 Detailed Description	216
3.9 Fusion360/initialize_tracks/unittest/f360_fill_init_obj_track_props_qualtest_unittest.cpp File Reference	217
3.9.1 Detailed Description	217
3.10 Fusion360/initialize_tracks/unittest/f360_initialize_tracks_qualtest_unittest.cpp File Reference . . .	217
3.10.1 Detailed Description	218
3.11 Fusion360/initialize_tracks/unittest/f360_nees_cfmi_try_to_init_qualtest_unittest.cpp File Reference	218
3.11.1 Detailed Description	218
3.12 Fusion360/inputs_preprocessing/unittest/f360_inputs_preprocessing_qualtest_unittest.cpp File Ref- erence	218
3.12.1 Detailed Description	219
3.12.2 Tests	219
3.12.2.1 inputs_preprocessing_qualtest_Internal_Preprocessing_Input_Interface_Requirements()	219
3.13 Fusion360/internal_preprocessing/qualification_testing/qualification_testing/guardrail_detection_qualtest.cpp File Reference	219
3.13.1 Detailed Description	219
3.14 Fusion360/internal_preprocessing/unittest/f360_detect_wheelspin_pairs_qualtest_unittest.cpp File Reference	220
3.14.1 Detailed Description	220
3.15 Fusion360/internal_preprocessing/unittest/f360_determine_dets_on_edge_of_fov_qualtest_↔ unittest.cpp File Reference	220
3.15.1 Detailed Description	220
3.16 Fusion360/internal_preprocessing/unittest/f360_handle_host_vehicle_clutter_qualtest_unittest.cpp File Reference	221
3.16.1 Detailed Description	221
3.17 Fusion360/internal_preprocessing/unittest/f360_identify_and_flag_internal_reflections_qualtest_↔ unittest.cpp File Reference	221
3.17.1 Detailed Description	221
3.17.2 Tests	221
3.17.2.1 f360_identify_and_flag_internal_reflections__scenario_based_tests()	222
3.17.2.2 f360_identify_and_flag_internal_reflections__scenario_based_tests_flag_recurring_detections()	222
3.17.2.3 f360_identify_and_flag_internal_reflections__scenario_based_tests_reset_buffer_entries_with_low_occurences()	222
3.18 Fusion360/internal_preprocessing/unittest/f360_mark_host_water_spray_detections_qualtest_↔ unittest.cpp File Reference	224
3.18.1 Detailed Description	224
3.19 Fusion360/measurement_update_tracks/unittest/f360_measurement_update_tracks_qualtest_↔ unittest.cpp File Reference	224
3.19.1 Detailed Description	225
3.20 Fusion360/measurement_update_tracks/unittest/f360_msmt_update_support_functions_common_↔ _qualtest_unittest.cpp File Reference	225

3.20.1 Detailed Description	225
3.21 Fusion360/post_update_track_adjustments/unittest/f360_post_update_track_adjustments_↔ qualtest_unittest.cpp File Reference	225
3.21.1 Detailed Description	226
3.22 Fusion360/pre_association_track_management/unittest/f360_pre_association_track_management_↔ _qualtest_unittest.cpp File Reference	226
3.22.1 Detailed Description	226
3.23 Fusion360/sensor_postprocessing/unittest/f360_sensor_postprocessing_qualtest_unittest.cpp File Reference	226
3.23.1 Detailed Description	227
3.24 Fusion360/time_update_tracks/unittest/f360_time_update_tracks_qualtest_unittest.cpp File Reference	227
3.24.1 Detailed Description	227
3.25 Fusion360/track_classification/unittest/f360_object_occlusion_classification_qualtest_unittest.cpp File Reference	227
3.25.1 Detailed Description	228
3.26 Fusion360/track_classification/unittest/f360_object_underdrivability_classification_qualtest_↔ unittest.cpp File Reference	228
3.26.1 Detailed Description	228
3.27 Fusion360/track_classification/unittest/f360_track_classification_qualtest_unittest.cpp File Reference	228
3.27.1 Detailed Description	230
3.28 Fusion360/track_downselection/unittest/f360_track_downselection_qualtest_unittest.cpp File Refer- ence	230
3.28.1 Detailed Description	231
3.29 Fusion360/track_grouping/unittest/f360_split_objects_in_orth_direction_qualtest_unittest.cpp File Reference	231
3.29.1 Detailed Description	231
3.30 Fusion360/track_grouping/unittest/f360_track_grouping_qualtest_unittest.cpp File Reference	232
3.30.1 Detailed Description	232
3.31 Fusion360/track_validity/unittest/f360_detect_multipath_qualtest_unittest.cpp File Reference	232
3.31.1 Detailed Description	232
3.32 Fusion360/track_validity/unittest/f360_determine_reflected_obj_qualtest_unittest.cpp File Reference	233
3.32.1 Detailed Description	233
3.33 Fusion360/track_validity/unittest/f360_track_validity_qualtest_unittest.cpp File Reference	233
3.33.1 Detailed Description	234
3.34 StateManager/unittest/f360_input_diagnostics_qualtest_unittest.cpp File Reference	234
3.34.1 Detailed Description	235

Chapter 1

Covered Requirements

Member [calcProbability_qualtest_calcProbability__check_if_sum_of_calculated_probabilities_is_equal_1](#)
()
FTCP-11630

Member [calcUndetProb_qualtest_calcUndetProb__check_if_prob_unknown_is_ste_to_max_value_when_object_has_status](#)
()
FTCP-11454

Member [calcUndetProb_qualtest_calcUndetProb__check_if_prob_unknown_is_ste_to_max_value_when_object_has_status](#)
()
FTCP-11454

Member [calcUndetProb_qualtest_calcUndetProb__check_if_prob_unknown_is_ste_to_max_value_when_object_has_status](#)
()
FTCP-11454

Member [calcUndetProb_qualtest_calcUndetProb__check_if_prob_unknown_is_ste_to_max_value_when_object_has_status](#)
()
FTCP-11454

Member [check_if_object_is_suspected_stationary_qualtest_check_if_object_is_suspected_stationary__cca_marked_as_no](#)
()
FTCP-11471

Member [check_if_object_is_suspected_stationary_qualtest_check_if_object_is_suspected_stationary__cca_marked_as_no](#)
()
FTCP-11471

Member [check_if_object_is_suspected_stationary_qualtest_check_if_object_is_suspected_stationary__cca_marked_as_no](#)
()
FTCP-11471

Member [check_if_object_is_suspected_stationary_qualtest_check_if_object_is_suspected_stationary__cca_marked_as_no](#)
()
FTCP-11471

Member [check_if_object_is_suspected_stationary_qualtest_check_if_object_is_suspected_stationary__cca_marked_as_st](#)
()
FTCP-11471

Member [check_if_object_is_suspected_stationary_qualtest_check_if_object_is_suspected_stationary__cca_marked_as_st](#)
()
FTCP-11471

Member [check_if_object_is_suspected_stationary_qualtest_check_if_object_is_suspected_stationary__ctca_marked_as_n](#)
()
FTCP-11471

Member [check_if_object_is_suspected_stationary_qualtest_check_if_object_is_suspected_stationary_ctca_marked_as_n](#)
(
FTCP-11471

Member [check_if_object_is_suspected_stationary_qualtest_check_if_object_is_suspected_stationary_ctca_marked_as_n](#)
(
FTCP-11471

Member [check_if_object_is_suspected_stationary_qualtest_check_if_object_is_suspected_stationary_ctca_marked_as_n](#)
(
FTCP-11471

Member [check_if_object_is_suspected_stationary_qualtest_check_if_object_is_suspected_stationary_ctca_marked_as_s](#)
(
FTCP-11471

Member [detecting_multibounces_from_object_det_marked_as_multibounce](#) ()
FTCP-13286

Member [detecting_multibounces_from_object_det_NOT_marked_as_multibounce](#) ()
FTCP-13286

Member [f360_adjust_filtr_type_dependet_params_qualtest_Adjust_Filtr_Type_Dependent_Params_switch_to_CTCA](#)
(
FTCP-12604

Member [f360_allocate_id_for_initialized_object_no_active_objects](#) ()
FTCP-12399

Member [f360_allocate_id_for_initialized_object_objects_saturation](#) ()
FTCP-12399

Member [f360_allocate_id_for_initialized_object_single_active_object](#) ()
FTCP-12399

Member [f360_assign_underdrivability_status_to_tracks_ocg_QT_Assign_Underdrivability_Status_To_Tracks_Current_Sta](#)
(
FTCP-13916

Member [f360_assign_underdrivability_status_to_tracks_ocg_QT_Assign_Underdrivability_Status_To_Tracks_Current_Sta](#)
(
FTCP-13916

Member [f360_assign_underdrivability_status_to_tracks_ocg_QT_Assign_Underdrivability_Status_To_Tracks_Current_Sta](#)
(
FTCP-13916

Member [f360_assign_underdrivability_status_to_tracks_ocg_QT_Assign_Underdrivability_Status_To_Tracks_Current_Sta](#)
(
FTCP-13916

Member [f360_assign_underdrivability_status_to_tracks_ocg_QT_Assign_Underdrivability_Status_To_Tracks_Not_To_Con](#)
(
FTCP-13916

Member [f360_assign_underdrivability_status_to_tracks_ocg_QT_Assign_Underdrivability_Status_To_Tracks_Object_Not](#)
(
FTCP-13916

Member [f360_cancel_new_updated_trk_overlapping_confirmed_trks_qualtest_Cancel_New_Updated_Trk_Overlapping_Con](#)
(
FTCP-13165

Member [f360_cancel_new_updated_trk_overlapping_confirmed_trks_qualtest_Cancel_New_Updated_Trk_Overlapping_Con](#)
(
FTCP-13165

Member [f360_cluster_grouping_qualtest_Test_Cluster_Grouping_Check_If_Dealiased_Flag_Is_Set_Correctly_1](#)
 ()
 FTCP-8587

Member [f360_cluster_grouping_qualtest_Test_Cluster_Grouping_Check_If_Dealiased_Flag_Is_Set_Correctly_2](#)
 ()
 FTCP-8587

Member [f360_cluster_grouping_qualtest_Test_Cluster_Grouping_Check_If_Range_Rates_Are_Dealiased_Correctly](#)
 ()
 FTCP-8587

Member [f360_conf_track_confidence_level_qualtest_Conf_Track_Confidence_Level__if_confidence_is_calculated_for_upd](#)
 ()
 FTCP-12155

Member [f360_conf_track_confidence_level_qualtest_Conf_Track_Confidence_Level__is_confidence_level_depend_on_num](#)
 ()
 FTCP-12149

Member [f360_conf_track_confidence_level_qualtest_Conf_Track_Confidence_Level__is_confidence_level_depend_on_obje](#)
 ()
 FTCP-12149

Member [f360_conf_track_confidence_level_qualtest_Conf_Track_Confidence_Level__is_confidence_level_depend_on_time](#)
 ()
 FTCP-12149

Member [f360_conf_track_confidence_level_qualtest_Conf_Track_Confidence_Level__is_confidence_level_is_equal_or_high](#)
 ()
 FTCP-12232

Member [f360_conf_track_confidence_level_qualtest_Conf_Track_Confidence_Level__is_confidence_level_is_equal_or_high](#)
 ()
 FTCP-12232

Member [f360_conf_track_confidence_level_qualtest_Conf_Track_Confidence_Level__is_confidence_level_is_equal_or_low](#)
 ()
 FTCP-12232

Member [f360_conf_track_confidence_level_qualtest_Conf_Track_Confidence_Level__is_confidence_level_is_equal_or_low](#)
 ()
 FTCP-12232

Member [f360_conf_track_confidence_level_qualtest_Conf_Track_Confidence_Level__is_confidence_overlapping_object_is](#)
 ()
 FTCP-12230

Member [f360_detect_multipath__Detect_And_Mark_Multipath_Objects_Qual_default_scenario_where_host_speed_is_below](#)
 ()
 CPR-3768 CPR-3769 CPR-3771 CPR-3770

Member [f360_detect_multipath__Detect_And_Mark_Multipath_Objects_Qual_default_scenario_where_host_speed_is_too_h](#)
 ()
 CPR-3768

Member [f360_detect_multipath__Detect_And_Mark_Multipath_Objects_Qual_scenario_where__reflector_objects_does_not](#)
 ()
 CPR-3770

Member [f360_detect_multipath__Detect_And_Mark_Multipath_Objects_Qual_scenario_where__source_objects_does_not_e](#)
 ()
 CPR-3770

Member [f360_detect_multipath__Detect_And_Mark_Multipath_Objects_Qual_scenario_where__source_objects_is_beyond](#)
 ()
 CPR-3769

Member [f360_detect_multipath_Detect_And_Mark_Multipath_Objects_Qual_scenario_where__source_objects_is_inactive](#)
()
CPR-3769

Member [f360_detect_multipath_Detect_And_Mark_Multipath_Objects_Qual_scenario_where__source_objects_is_stationary](#)
()
CPR-3769

Member [f360_detect_multipath_Detect_And_Mark_Multipath_Objects_Qual_scenario_where__source_objects_use_CCV_f](#)
()
CPR-3769

Member [f360_detect_wheelspin_nearby_qualtest_f360_detect_wheelspin_nearby_qualtest__correctly_marked_as_nearby_v](#)
()
FTCP-8840

Member [f360_detect_wheelspin_nearby_qualtest_f360_detect_wheelspin_nearby_qualtest__correctly_not_marked_as_whe](#)
()
FTCP-8840

Member [f360_detect_wheelspin_object_qualtest_f360_detect_wheelspin_object_qualtest__correctly_marked_as_object_wh](#)
()
FTCP-8840

Member [f360_detect_wheelspin_object_qualtest_f360_detect_wheelspin_object_qualtest__correctly_not_marked_as_objec](#)
()
FTCP-8840

Member [f360_detect_wheelspin_pairs_qualtest_unittest_azimuth_spread_limits](#) ()
FTCP-8840

Member [f360_detect_wheelspin_pairs_qualtest_unittest_doppler_spread_limits](#) ()
FTCP-8840

Member [f360_detect_wheelspin_pairs_qualtest_unittest_lat_limits](#) ()
FTCP-8840

Member [f360_detect_wheelspin_pairs_qualtest_unittest_lat_spread_limits](#) ()
FTCP-8840

Member [f360_detect_wheelspin_pairs_qualtest_unittest_long_limits](#) ()
FTCP-8840

Member [f360_detect_wheelspin_pairs_qualtest_unittest_long_spread_limits](#) ()
FTCP-8840

Member [f360_detect_wheelspin_pairs_qualtest_unittest_max_wheelspin_pairs](#) ()
FTCP-8840

Member [f360_detect_wheelspin_pairs_qualtest_unittest_multiple_candidates](#) ()
FTCP-8840

Member [f360_detect_wheelspin_pairs_qualtest_unittest_proximity_detections](#) ()
FTCP-8840

Member [f360_detect_wheelspin_pairs_qualtest_unittest_remarking](#) ()
FTCP-8840

Member [f360_determine_dets_on_edge_of_fov_lr_detections](#) ()
FTCP-8838

Member [f360_determine_dets_on_edge_of_fov_mr_detections](#) ()
FTCP-8838

Member [f360_determine_dets_on_edge_of_fov_range_limits](#) ()
FTCP-8838

Member [f360_determine_reflected_obj_check_calibrations](#) ()

CPR-4874

Member [f360_determine_reflected_obj_Determine_Reflected_Obj__check_whether_mirror_track_has_its_mirror_prob_set](#) ()

CPR-4874 CPR-4875

Member [f360_determine_reflected_obj_Determine_Reflected_Obj__check_whether_mirror_track_has_its_mirror_prob_set](#) ()

CPR-4874 CPR-4875

Member [f360_determine_reflected_obj_Determine_Reflected_Obj__check_whether_track_behind_guardrail_has_its_mirror](#) ()

CPR-4874 CPR-4875

Member [f360_determine_reflected_obj_Determine_Reflected_Obj__check_whether_track_that_is_not_a_mirror_track_has](#) ()

CPR-4874 CPR-4875

Member [f360_determine_reflected_obj_Determine_Reflected_Obj__check_whether_track_that_is_not_a_mirror_track_has](#) ()

CPR-4874 CPR-4875

Member [f360_determine_reflected_obj_Determine_Reflected_Obj__check_whether_track_that_is_not_a_mirror_track_has](#) ()

CPR-4874 CPR-4875

Member [f360_determine_reflected_obj_Determine_Reflected_Obj__check_whether_valid_track_behind_guardrail_has_its_r](#) ()

CPR-4874 CPR-4875

Member [f360_determine_reflected_obj_Determine_Reflected_Obj__check_whether_valid_track_in_front_and_on_guardrail](#) ()

CPR-4874 CPR-4875

Member [f360_determine_reflected_obj_with_turned_hide_tracks_outside_guardrail_prob_set_to_one_because_track_is_no](#) ()

CPR-4874 CPR-4875

Member [f360_determine_reflected_obj_with_turned_hide_tracks_outside_guardrail_prob_set_to_one_because_track_is_no](#) ()

CPR-4874 CPR-4875

Member [f360_determine_reflected_obj_with_turned_hide_tracks_outside_guardrail_prob_set_to_zero_because_track_is_o](#) ()

CPR-4874 CPR-4875

Member [f360_determine_reflected_obj_with_turned_hide_tracks_outside_guardrail_prob_set_to_zero_because_track_is_o](#) ()

CPR-4874 CPR-4875

Member [f360_fill_init_obj_track_props_determine_initial_size_qualtest_Test_Initial_Size_Fast_Moving_Object](#) ()

FTCP-12402

Member [f360_fill_init_obj_track_props_determine_initial_size_qualtest_Test_Initial_Size_Slow_Moving_Object](#) ()

FTCP-12402

Member [f360_fill_init_obj_track_props_determine_initial_size_qualtest_Test_Initial_Size_Stationary_Object](#) ()

FTCP-12402

Member [f360_fill_init_obj_track_props_qualtest_TestCalcInitObjTrackMovingFlagMoving](#) ()

FTCP-12405

Member [f360_fill_init_obj_track_props_qualtest_TestCalcInitObjTrackMovingFlagMovingValidForLiberalTracking](#)
()
FTCP-12405

Member [f360_fill_init_obj_track_props_qualtest_TestCalcInitObjTrackMovingFlagNotMovingHighFracAz](#) ()
FTCP-12405

Member [f360_fill_init_obj_track_props_qualtest_TestCalcInitObjTrackMovingFlagNotMovingLowSpeed](#) ()
FTCP-12405

Member [f360_fill_init_obj_track_props_qualtest_TestCalcInitObjTrackMovingFlagNotMovingLowSpeedAndHighFracAz](#)
()
FTCP-12405

Member [f360_fill_init_obj_track_props_qualtest_TestCalcInitObjVCSHeadingMovingObj](#) ()
FTCP-12402

Member [f360_fill_init_obj_track_props_qualtest_TestCalcInitObjVCSHeadingNotMovingObj](#) ()
FTCP-12402

Member [f360_fill_init_obj_track_props_qualtest_TestFillInitObjTrackPropsNominal](#) ()
FTCP-12404, FTCP-12402 // TODO during industrialization: Need to check if these reqs must be updated

Member [f360_find_detection_inliers_find_valid_detections_Select_Detections_To_Mark_As_Inliers_Nearby_Wheel_Spin_De](#)
()
FTCP-13634

Member [f360_find_detection_inliers_find_valid_detections_Select_Detections_To_Mark_As_Inliers_Object_Wheel_Spin_De](#)
()
FTCP-13634

Member [f360_find_detection_inliers_find_valid_detections_Select_Detections_To_Mark_As_Inliers_Pair_Wheel_Spin_Dete](#)
()
FTCP-13634

Member [f360_find_detection_inliers_single_object_Find_Detection_Inliers_Close_Target_Detection](#) ()
FTCP-13634

Member [f360_find_tracks_on_and_behind_guardrail_qualtest_Flag_Objects_On_SEP_ObjectOnGuardrailLeftSide](#)
()
FTCP-12162

Member [f360_find_tracks_on_and_behind_guardrail_qualtest_Flag_Objects_On_SEP_ObjectOnGuardrailRightSide](#)
()
FTCP-12162

Member [f360_handle_host_vehicle_clutter_qualtest_handle_host_vehicle_clutter_dets_marked_as_not_okay_to_use](#)
()
FTCP-10043

Member [f360_handle_host_vehicle_clutter_qualtest_handle_host_vehicle_clutter_dets_marked_as_okay_to_use](#)
()
FTCP-10043

Member [f360_handle_host_vehicle_clutter_qualtest_handle_host_vehicle_clutter_hvc_and_behind_guardrail_dets_marke](#)
()
FTCP-10043

Member [f360_handle_host_vehicle_clutter_qualtest_handle_host_vehicle_clutter_hvc_and_on_guardrail_dets_marked_as](#)
()
FTCP-10043

Member [f360_identify_and_flag_internal_reflections_scenario_based_tests_flag_recurring_detections](#) ()
FTCP-13462

Member [f360_identify_and_flag_internal_reflections_scenario_based_tests_reset_buffer_entries_with_low_occurance](#)
()
FTCP-13462

Member [f360_init_obj_track_covariances_TestCCANominalCase](#) ()
FTCP-12406

Member [f360_initialize_tracked_objects_Check_If_Objects_Are_Correctly_Sorted_By_Longitudinal_Position](#)
()
FTCP-12401

Member [f360_initialize_tracked_objects_Check_If_Objects_Are_Correctly_Sorted_By_Priority](#) ()
FTCP-12400

Member [f360_initialize_tracked_objects_Detections_Assignment_For_Single_Successfull_Object_Initialization](#)
()
FTCP-12398

Member [f360_initialize_tracked_objects_Detections_Assignment_For_Single_Unsuccessfull_Object_Initialization](#)
()
FTCP-12398

Member [f360_initialize_tracked_objects_Highest_Cluster_Priority_Initialization_Try_With_Obj_Saturation](#)
()
FTCP-12408, FTCP-12399, FTCP-12402

Member [f360_initialize_tracked_objects_Highest_Cluster_Priority_Reinitialization_With_Obj_Saturation](#) ()
FTCP-12408, FTCP-12402

Member [f360_initialize_tracked_objects_Lowest_Cluster_Priority_Initialization_Try_With_Obj_Saturation](#) ()
FTCP-12397

Member [f360_initialize_tracked_objects_Occluded_And_Non_Occluded_Cluster__Only_One_Successfull_Initialization](#)
()
FTCP-13056

Member [f360_initialize_tracked_objects_Occluded_Cluster_With_Detections_Valid_For_Liberal_Initilization__Successfull_I](#)
()
FTCP-13056

Member [f360_initialize_tracked_objects_Single_Non_Occluded_Cluster__Successful_Initilization](#) ()
FTCP-13056

Member [f360_initialize_tracked_objects_Single_Occluded_Cluster__Unsuccessful_Initialization](#) ()
FTCP-13056

Member [f360_initialize_tracked_objects_Single_Successfull_Object_Initialization](#) ()
FTCP-12403

Member [f360_initialize_tracked_objects_Single_Successfull_Object_Initialization_With_Init_Scheme_Check](#)
()
FTCP-12404

Member [f360_initialize_tracked_objects_Single_Unsuccessfull_Object_Initialization](#) ()
FTCP-12403

Member [f360_initialize_tracked_objects_Two_Successfull_And_One_Unsuccessfull_Objects_Initialization](#)
()
FTCP-12403

Member [f360_input_diagnostics_qualtest_Fault_Should_Be_Returned_When_Core_Info_Cnt_Loops_Lower_Than_Before](#)
()
FTCP-8837

Member [f360_input_diagnostics_qualtest_Fault_Should_Be_Returned_When_Core_Info_Elapsed_Above_Upper_Limit](#)
()
FTCP-8604

- Member f360_input_diagnostics_qualtest_Fault_Should_Be_Returned_When_Core_Info_Elapsed_Time_Below_Lower_Limit**
 ()
 FTCP-8605
- Member f360_input_diagnostics_qualtest_Fault_Should_Be_Returned_When_Core_Info_Time_us_Lower_Than_Before**
 ()
 FTCP-8603
- Member f360_input_diagnostics_qualtest_Fault_Should_Be_Returned_When_Host_Info_Vehicle_index_Lower_Than_Before**
 ()
 FTCP-11448
- Member f360_input_diagnostics_qualtest_No_Fault_Reported_For_Sensor_When_Calibration_Not_Valid** ()
 FTCP-8592
- Member f360_input_diagnostics_qualtest_No_Fault_Reported_For_Sensor_When_look_index_Overflow** ()
 FTCP-8591
- Member f360_input_diagnostics_qualtest_No_Increase_Fault_Should_Be_Returned_When_Look_Index_For_Single_Sensor**
 ()
 FTCP-8591
- Member f360_input_diagnostics_qualtest_No_Increase_Faults_Should_Be_Returned_When_Core_Signal_Received_Twice**
 ()
 FTCP-8603, FTCP-8837, FTCP-11448,
- Member f360_input_diagnostics_qualtest_No_Increase_Faults_Should_Be_Returned_When_Sensors_Signal_Received_Twice**
 ()
 FTCP-8591
- Member f360_input_diagnostics_qualtest_Polarity_Fault_Should_Not_Be_Reported_When_Sensor_Polarity_Is_Valid_Negative**
 ()
 FTCP-8594
- Member f360_input_diagnostics_qualtest_Polarity_Fault_Should_Not_Be_Reported_When_Sensor_Polarity_Is_Valid_Positive**
 ()
 FTCP-8594
- Member f360_input_diagnostics_qualtest_Signal_Fault_Should_Be_Returned_When_Difference_Between_Sensor_And_Transducer**
 ()
 FTCP-8592
- Member f360_input_diagnostics_qualtest_Signal_Fault_Should_Be_Returned_When_Difference_Between_Sensor_And_Transducer**
 ()
 FTCP-8592
- Member f360_input_diagnostics_qualtest_Signal_Fault_Should_Not_Be_Set_When_No_Sensor_Data_In_Received**
 ()
 .NA
- Member f360_input_diagnostics_qualtest_Signal_Faults_Should_Be_Reported_When_Look_Index_Equal_To_Zero_Twice**
 ()
 FTCP-8591
- Member f360_input_diagnostics_qualtest_Signal_Faults_Should_Be_Reported_When_Sensor_Azimuth_Boresight_Angle_Is_Valid**
 ()
 FTCP-10137
- Member f360_input_diagnostics_qualtest_Signal_Faults_Should_Be_Reported_When_Sensor_Azimuth_Boresight_Angle_Is_Valid**
 ()
 FTCP-10137
- Member f360_input_diagnostics_qualtest_Signal_Faults_Should_Be_Reported_When_Sensor_Azimuth_Boresight_Angle_Is_Valid**
 ()
 FTCP-10137

-
- Member [f360_input_diagnostics_qualtest_Signal_Faults_Should_Be_Reported_When_Sensor_Azimuth_Boresight_Angle_Is_Within_Boresight_Angle](#)**
 ()
 FTCP-10137
- Member [f360_input_diagnostics_qualtest_Signal_Faults_Should_Be_Reported_When_Sensor_Mounting_Position_Is_Too_Far](#)**
 ()
 FTCP-8596
- Member [f360_input_diagnostics_qualtest_Signal_Faults_Should_Be_Reported_When_Sensor_Mounting_Position_Is_Too_Far_2](#)**
 ()
 FTCP-8596
- Member [f360_input_diagnostics_qualtest_Signal_Faults_Should_Be_Reported_When_Sensor_Polarity_Is_Equal_To_Zero](#)**
 ()
 FTCP-8594
- Member [f360_input_diagnostics_qualtest_Signal_Faults_Should_Be_Reported_When_Sensor_Polarity_Is_Greater_Than_Equal_To_Minimum](#)**
 ()
 FTCP-8594
- Member [f360_input_diagnostics_qualtest_Signal_Faults_Should_Be_Reported_When_Sensor_Polarity_Is_Greater_Than_Equal_To_Maximum](#)**
 ()
 FTCP-8594
- Member [f360_input_diagnostics_qualtest_Signal_Faults_Should_Not_Be_Reported_When_Sensor_Mounting_Position_Equals_Boresight_Angle](#)**
 ()
 FTCP-8596
- Member [f360_input_diagnostics_qualtest_Signal_Faults_Should_Not_Be_Reported_When_Sensor_Mounting_Position_Equals_Boresight_Angle_2](#)**
 ()
 FTCP-8596
- Member [f360_input_diagnostics_qualtest_Signal_Faults_Should_Not_Be_Reported_When_Sensor_Mounting_Position_Is_Within_Boresight_Angle](#)**
 ()
 FTCP-8596
- Member [f360_input_diagnostics_qualtest_Signal_Faults_Should_Not_Be_Reported_When_Sensor_Mounting_Position_Is_Within_Boresight_Angle_2](#)**
 ()
 FTCP-8596
- Member [f360_input_diagnostics_qualtest_True_Should_Be_Returned_When_Big_Curvature_Rear_Or_VCS_Sideslip](#)**
 ()
 FTCP-10857
- Member [f360_input_diagnostics_qualtest_True_Should_Be_Returned_When_Host_QF_Is_Accurate](#)** ()
 FTCP-8602, FTCP-10857, FTCP-13077, FTCP-13076
- Member [f360_input_diagnostics_qualtest_True_Should_Be_Returned_When_Host_QF_Is_NOT_Accurate](#)**
 ()
 FTCP-8602, FTCP-10857, FTCP-13077, FTCP-13076
- Member [f360_kill_coasted_tracks_qualtest_Do_not_terminate_updated_tracks](#)** ()
 FTCP-11715
- Member [f360_kill_coasted_tracks_qualtest_Non_Movable_Coasting_Obj_Outside_Front_Only_FOV_Coasting_Time](#)**
 ()
 CPR-4890
- Member [f360_kill_coasted_tracks_qualtest_Non_Movable_Coasting_Obj_Outside_Front_Only_FOV_Config](#)**
 ()
 CPR-4890
- Member [f360_kill_coasted_tracks_qualtest_Non_Movable_Coasting_Obj_Outside_Front_Only_FOV_Nonmoveable](#)**
 ()
 CPR-4890
-

Member [f360_kill_coasted_tracks_qualtest_Non_Movable_Coasting_Obj_Outside_Front_Only_FOV_Pos](#) ()
CPR-4890

Member [f360_kill_coasted_tracks_qualtest_Termination_of_coasting_confirmed_objects](#) ()
FTCP-11715

Member [f360_kill_coasted_tracks_qualtest_Termination_of_coasting_mirror_objects_variation_of_mirror_probability](#) ()
FTCP-11715

Member [f360_kill_coasted_tracks_qualtest_Termination_of_coasting_mirror_objects_variation_of_stage_time](#) ()
FTCP-11715

Member [f360_mark_water_spray_detections_default_test](#) ()
FTCP-8844

Member [f360_mark_water_spray_detections_det_high_rcs](#) ()
FTCP-8844

Member [f360_mark_water_spray_detections_det_range_rate](#) ()
FTCP-8844

Member [f360_mark_water_spray_detections_detections_outside_vcs_zone](#) ()
FTCP-8844

Member [f360_mark_water_spray_detections_f_ok_to_use_false](#) ()
FTCP-8844

Member [f360_mark_water_spray_detections_low_host_speed](#) ()
FTCP-8844

Member [f360_merge_two_clusters_Test_Merge_Two_Clusters_Check_If_All_4_Dets_Are_Dealiased](#) ()
FTCP-8587

Member [f360_merge_two_clusters_Test_Merge_Two_Clusters_Check_If_Dets_4_Are_Dealiased_1_Not_Dealiased](#) ()
FTCP-8587

Member [f360_merge_two_clusters_Test_Merge_Two_Clusters_Check_Range_Rate_Is_Dealiased_Correctly_1](#) ()
FTCP-8587

Member [f360_msmt_update_obj_trks_cca_moveable_qualtest_Msmt_Update_Obj_Trks_CCA_Moveable_Check_Head_Point](#) ()
FTCP-11426 FTCP-11024

Member [f360_msmt_update_obj_trks_cca_moveable_qualtest_Msmt_Update_Obj_Trks_CCA_Moveable_Check_Update_On](#) ()
FTCP-11024

Member [f360_msmt_update_obj_trks_cca_moveable_qualtest_Msmt_Update_Obj_Trks_CCA_Moveable_Check_Update_On](#) ()
FTCP-11024

Member [f360_msmt_update_obj_trks_cca_non_moveable_qualtest_Msmt_Update_Obj_Trks_CCA_Non_Moveable](#) ()
FTCP-11426 FTCP-11024

Member [f360_msmt_update_obj_trks_ctca_not_moving_qualtest_Msmt_Update_ObjTrks_CTCA__curvature_head_is_not_u](#) ()
FTCP-11424 FTCP-11024 FTCP-12609

Member [f360_msmt_update_obj_trks_ctca_qualtest_Msmt_Update_Obj_Trks_CTCA_Check_Update_One_Det](#) ()
FTCP-11024, FTCP-12609

Member [f360_msmt_update_object_timestamp_qualtest_Msmt_Update_Object_Timestamp_One_Det \(\)](#)
FTCP-11030

Member [f360_object_based_angle_jump_detector_Detect_Angle_Jumps_From_Objects_qualtest_flagging_correctness_of \(\)](#)
FTCP-12069 NA

Member [f360_object_based_water_spray_detector_qualtest_Detect_Water_Spray_From_Objects \(\)](#)
FTCP-12065 NA

Member [f360_object_occlusion_classification_Verify_All_Active_Objects_Are_Considered \(\)](#)
FTCP-13872, FTCP-13870

Member [f360_object_occlusion_classification_Verify_Single_Object_Status_Occluded \(\)](#)
FTCP-13872, FTCP-13870

Member [f360_object_occlusion_classification_Verify_Single_Object_Status_On_Edge \(\)](#)
FTCP-13872, FTCP-13870

Member [f360_object_occlusion_classification_Verify_Single_Object_Status_Undefined_Not_Moving \(\)](#)
FTCP-13872, FTCP-13870

Member [f360_object_occlusion_classification_Verify_Single_Object_Status_Visible \(\)](#)
FTCP-13872, FTCP-13870

Member [f360_object_track_management_qualtest_Obj_Trk_Status_Book_Keeping_NEW \(\)](#)
FTCP-12595

Member [f360_object_track_management_qualtest_Obj_Trk_Status_Book_Keeping_UPDATED \(\)](#)
FTCP-12603

Member [f360_object_underdrivability_classification_qt_test_objs_invalid \(\)](#)
FTCP-13970

Member [f360_object_underdrivability_classification_qt_test_valid_moving_objs_behind_host \(\)](#)
FTCP-13969 FTCP-13970

Member [f360_object_underdrivability_classification_qt_test_valid_moving_objs_height_above_threshold \(\)](#)
FTCP-13969

Member [f360_object_underdrivability_classification_qt_test_valid_moving_objs_height_below_threshold \(\)](#)
FTCP-13969

Member [f360_run_obj_class_f360_run_obj_class_bicycle \(\)](#)
FTCP-11450

Member [f360_run_obj_class_f360_run_obj_class_car \(\)](#)
FTCP-11450

Member [f360_run_obj_class_f360_run_obj_class_motorcycle \(\)](#)
FTCP-11450

Member [f360_run_obj_class_f360_run_obj_class_pedestrian \(\)](#)
FTCP-11450

Member [f360_run_obj_class_f360_run_obj_class_truck \(\)](#)
FTCP-11450

Member [f360_run_obj_class_f360_run_obj_class_undetermined \(\)](#)
FTCP-11450

Member [f360_split_objects_in_orth_direction_qualtest_Verify_Split_Logic \(\)](#)
FTCP-13121

Member f360_Test_Determine_Association_Hypothesis_qualtest_Test_Determine_Association_Hypothesis_A
 ()
 FTCP-10132

Member f360_Test_Determine_Association_Hypothesis_qualtest_Test_Determine_Association_Hypothesis_B
 ()
 FTCP-10132

Member f360_Test_Determine_Association_Hypothesis_qualtest_Test_Determine_Association_Hypothesis_C
 ()
 FTCP-12084

Member f360_Test_Determine_Association_Hypothesis_qualtest_Test_Determine_Association_Hypothesis_D
 ()
 FTCP-12084

Member f360_Test_Is_Det_Allowed_To_Associate_qualtest_Test_Is_Det_Allowed_To_Associate_Det_Not_Ok_To_Use
 ()
 FTCP-10125

Member f360_Test_Is_Det_Allowed_To_Associate_qualtest_Test_Is_Det_Allowed_To_Associate_Det_On_SEP_Obj_Moving
 ()
 FTCP-10041

Member f360_Test_Is_Det_Allowed_To_Associate_qualtest_Test_Is_Det_Allowed_To_Associate_Detection_Behind_SEP_Obj
 ()
 FTCP-10126

Member f360_Test_Is_Det_Allowed_To_Associate_qualtest_Test_Is_Det_Allowed_To_Associate_Mirror_Obj_Ambiguous_D
 ()
 FTCP-10127

Member f360_Test_Is_Det_Allowed_To_Associate_qualtest_Test_Is_Det_Allowed_To_Associate_Object_Behind_SEP_Obj
 ()
 FTCP-10126

Member f360_Test_Is_Det_Allowed_To_Associate_qualtest_Test_Is_Det_Allowed_To_Associate_Object_Behind_SEP_Obj
 ()
 FTCP-10126

Member f360_track_downselection_calc_track_priority_qualtest_Host_Curvature_Is_Not_Close_To_Zero ()
 FTCP-13920

Member f360_track_downselection_calc_track_priority_qualtest_Host_Is_Moving ()
 FTCP-13920

Member f360_track_downselection_calc_track_priority_qualtest_Host_Is_Reversing ()
 FTCP-13920

Member f360_track_downselection_calc_track_priority_qualtest_Host_Is_Stationary ()
 FTCP-13920

Member f360_track_downselection_calc_track_priority_qualtest_Impact_oF_Host_Speeds_On_Object_Priorities
 ()
 FTCP-13920

Member f360_track_downselection_qualtest_f360_core_track_downselect__check_if_assigns_reduced_id_to_higher_priori
 ()
 FTCP-10728

Member f360_track_downselection_qualtest_f360_core_track_downselect__check_if_does_reduce_tracks_basing_on_priori
 ()
 FTCP-10226

Member f360_track_downselection_qualtest_f360_core_track_downselect__check_if_lowers_confidence_and_priority_of_n
 ()
 FTCP-10524

Member [f360_track_downselection_qualtest_f360_core_track_downselect__check_if_lowers_confidence_of_tracks_behind](#)
()
FTCP-10524

Member [f360_track_downselection_qualtest_f360_core_track_downselect__check_if_lowers_confidence_of_tracks_that_ch](#)
()
FTCP-10524

Member [f360_track_downselection_qualtest_f360_core_track_downselect__check_if_resets_low_priority_target_reduced_i](#)
()
FTCP-10728

Member [f360_track_downselection_qualtest_f360_core_track_downselect__check_if_tracks_are_downselected_as_expect](#)
()
FTCP-10520

Member [f360_track_downselection_qualtest_f360_core_track_downselect__check_if_tracks_with_higher_confidence_level](#)
()
FTCP-10520

Member [f360_track_downselection_qualtest_f360_core_track_downselect__check_if_tracks_with_low_amount_of_average](#)
()
FTCP-10520

Member [f360_track_downselection_qualtest_f360_core_track_downselect__check_if_tracks_with_lower_ttc_are_prioritized](#)
()
FTCP-10520

Member [f360_track_downselection_qualtest_f360_core_track_downselect__check_if_tracks_with_reduced_status_set_as_i](#)
()
FTCP-10520

Member [f360_track_downselection_qualtest_f360_core_track_downselect__check_if_updated_tracks_are_prioritized_over](#)
()
FTCP-10520

Member [f360_track_downselection_qualtest_f360_core_track_downselect__check_if_valid_trk_closer_to_host_is_prioritize](#)
()
FTCP-10520

Member [f360_track_grouping_Check_Consinsency_In_Obj_Id](#) ()
FTCP-12944

Member [f360_track_grouping_Check_Updates_In_Long_Pos_Sort](#) ()
FTCP-13122

Member [f360_track_grouping_Check_Updates_In_Priority_Sort](#) ()
FTCP-12951

Member [f360_track_grouping_Test_Merged_Obj_Max_Size_Is_Not_excced](#) ()
FTCP-12947

Member [f360_track_grouping_Test_Move_All_Dets_Assoc](#) ()
FTCP-12952

Member [f360_track_grouping_Test_Move_Dets_Assoc](#) ()
FTCP-12952

Member [f360_track_grouping_Test_Remain_Dets_Assoc](#) ()
FTCP-12948

Member [f360_track_grouping_Test_Similar_Objects_Were_Merged](#) ()
FTCP-12943

Member [f360_try_to_init_obj_track_qualtest_HostOverlapCheckPosLongNOTOverlappedPosLatNOTOverlappedtest](#)
()
FTCP-12402

Member [f360_try_to_init_obj_track_qualtest_HostOverlapCheckPosLongNOTOverlapedPosLatOverlappedtest](#)
()
FTCP-12402

Member [f360_try_to_init_obj_track_qualtest_HostOverlapCheckPosLongOverlappedInRearAxleDistPosLatNOTOverlappedtest](#)
()
FTCP-12402

Member [f360_try_to_init_obj_track_qualtest_HostOverlapCheckPosLongOverlappedInRearAxleDistPosLatOverlappedtest](#)
()
FTCP-12402

Member [f360_try_to_init_obj_track_qualtest_LiberalInitializationTest](#) ()
FTCP-12402

Member [f360_Update_Detection_History_qualtest_Test_Update_Detection_History_Find_Next_Hist_Det_Idx_false](#)
()
FTCP-12845

Member [f360_Update_Detection_History_qualtest_Test_Update_Detection_History_n_occupied_higher_than_max](#)
()
FTCP-12845

Member [f360_Update_Detection_History_qualtest_Test_Update_Detection_History_old_dets_greater_than_zero](#)
()
FTCP-12845

Member [f360_Update_Detection_History_qualtest_Test_Update_Detection_History_update_one_detection](#)
()
FTCP-12845

Member [f360_update_exist_prob_qualtest_Update_Existence_Probability__check_whether_EP_depend_onobject_state_var](#)
()
FTCP-12167, FTCP-12387

Member [f360_update_exist_prob_qualtest_Update_Existence_Probability__check_whether_maximum_EP_value_is_1](#)
()
FTCP-12236, FTCP-12387

Member [f360_update_exist_prob_qualtest_Update_Existence_Probability__check_whether_minimum_EP_value_is_equal_t](#)
()
FTCP-12236, FTCP-12387

Member [f360_update_object_track_properties_calc_obj_height_obj_height_decay_below_zero](#) ()
FTCP-13974

Member [f360_update_object_track_properties_calc_obj_height_obj_height_decays_when_no_dets](#) ()
FTCP-13974

Member [f360_update_object_track_properties_calc_obj_height_obj_height_is_not_calculated_when_stationary](#)
()
FTCP-13974

Member [f360_update_object_track_properties_calc_obj_height_obj_height_is_updated_when_valid](#) ()
FTCP-13974

Member [f360_update_object_track_properties_Calc_Obj_Size_Obj_Size_Is_Not_Updated_Obj_Coasting](#) ()
FTCP-13529

Member [f360_update_object_track_properties_Calc_Obj_Size_Obj_Size_Is_Not_Updated_Obj_One_Assoc_Det](#)
()
FTCP-13529

Member [f360_update_object_track_properties_Calc_Obj_Size_Obj_Size_Is_Updated \(\)](#)

FTCP-13529

Member [filteringAndNormalizationOfProbabilities_qualtest_filteringAndNormalizationOfProbabilities__check_if_undetermined \(\)](#)

FTCP-11463

Member [guardrail_detection_capability \(\)](#)

FTCP-8839

Member [guardrail_detection_no_guardrail_sanity_check \(\)](#)

FTCP-8839

Member [initialize_qualtest_clusters_Test_Initialize_Clusters_Many_Detection_To_Cluster \(\)](#)

FTCP-8581

Member [initialize_qualtest_clusters_Test_Initialize_Clusters_One_Detection_To_Cluster \(\)](#)

FTCP-8581

Member [inputs_preprocessing_qualtest_Internal_Preprocessing_Input_Interface_Requirements \(\)](#)

FTCP-8256, FTCP-7799

Member [object_motion_status_should_be_updated_qualtest_Object_Motion_Status_Should_Be_Updated__COASTED__Timed \(\)](#)

FTCP-11692

Member [object_motion_status_should_be_updated_qualtest_Object_Motion_Status_Should_Be_Updated__COASTED__Timed \(\)](#)

FTCP-11692

Member [object_motion_status_should_be_updated_qualtest_Object_Motion_Status_Should_Be_Updated__NEW_COASTED__Timed \(\)](#)

FTCP-11692

Member [object_motion_status_should_be_updated_qualtest_Object_Motion_Status_Should_Be_Updated__NEW_UPDATED__Timed \(\)](#)

FTCP-11692

Member [object_motion_status_should_be_updated_qualtest_Object_Motion_Status_Should_Be_Updated__UPDATED__Timed \(\)](#)

FTCP-11692

Member [object_wrt_to_sep_tests_obejct_marked_as_on_a_sep \(\)](#)

FTCP-13327

Member [object_wrt_to_sep_tests_obejct_NOT_marked_as_on_sep \(\)](#)

FTCP-13327

Member [set_object_class_qualtest_set_object_class__check_if_classifies_as_undetermined \(\)](#)

FTCP-11631

Member [set_object_class_qualtest_set_object_class__check_if_min_prob_value_to_classify_is_0_65 \(\)](#)

FTCP-11536

Member [set_object_class_qualtest_set_object_class__check_if_prioritizes_cars \(\)](#)

FTCP-11457

Member [set_object_class_qualtest_set_object_class__check_if_prioritizes_motorcycles \(\)](#)

FTCP-11460

Member [set_object_class_qualtest_set_object_class__check_if_prioritizes_pedestrian \(\)](#)

FTCP-11459

Member [set_object_class_qualtest_set_object_class__check_if_prioritizes_trucks \(\)](#)

FTCP-11458

Member [update_object_properties_qualtest_update_object_properties__check_if_changes_moveable_flag](#)
()
FTCP-11469

Member [update_object_properties_qualtest_update_object_properties__check_if_moveable_flag_is_not_reset](#)
()
FTCP-11469

Chapter 2

Module Documentation

2.1 F360_cluster_grouping_qualtest

Functions

- test group [f360_cluster_grouping_qualtest](#) ()
Test group for f360_cluster_grouping_qualtest.
- [f360_cluster_grouping_qualtest_Test_Cluster_Grouping_Check_If_Range_Rates_Are_Dealiased_Correctly](#) ()
- [f360_cluster_grouping_qualtest_Test_Cluster_Grouping_Check_If_Dealiased_Flag_Is_Set_Correctly_1](#) ()
- [f360_cluster_grouping_qualtest_Test_Cluster_Grouping_Check_If_Dealiased_Flag_Is_Set_Correctly_2](#) ()

2.1.1 Detailed Description

2.1.2 Function Documentation

2.1.2.1 f360_cluster_grouping_qualtest()

```
test group f360_cluster_grouping_qualtest ( )
```

Test group for f360_cluster_grouping_qualtest.

Setup:\n

Setting up arguments for cluster_grouping function and assigning them with basic values

2.1.2.2 f360_cluster_grouping_qualtest_Test_Cluster_Grouping_Check_If_Dealiased_Flag_Is_Set_Correctly_1()

```
f360_cluster_grouping_qualtest_Test_Cluster_Grouping_Check_If_Dealiased_Flag_Is_Set_Correctly↔  
_1 ( )
```

Purpose:\n Test for fulfilling requirements of clustering for detections' range rate dealiasing flag for set of 6 valid detections

Requirements FTCP-8587

Precondition:\n

Configuring clusters and adding detections to them

Action:\n

Call Cluster_Grouping

Result:\n

Checking if dealiased flag is set correctly

2.1.2.3 f360_cluster_grouping_qualtest_Test_Cluster_Grouping_Check_If_Dealiased_Flag_Is_Set_Correctly_2()

```
f360_cluster_grouping_qualtest_Test_Cluster_Grouping_Check_If_Dealiased_Flag_Is_Set_Correctly↔  
_2 ( )
```

Purpose:\n Test for fulfilling requirements of clustering for detections' range rate dealiasing flag where two clusters shall not be merged/dealiased

Requirements FTCP-8587

Precondition:\n

Configuring clusters and adding detections to them

Action:\n

Call Cluster_Grouping

Result:\n

Checking if dealiased flag is set correctly

2.1.2.4 f360_cluster_grouping_qualtest_Test_Cluster_Grouping_Check_If_Range_Rates_Are_Dealiased_Correctly()

f360_cluster_grouping_qualtest_Test_Cluster_Grouping_Check_If_Range_Rates_Are_Dealiased_↔
Correctly ()

Purpose:\n Test for fulfilling requirements of clustering for detections' range rate dealiasing for set of 6 valid detections

Requirements FTCP-8587

Precondition:\n

Configuring clusters and adding detections to them

Action:\n

Call Cluster_Grouping

Result:\n

Checking if range rates are dealiased

2.2 F360_merge_two_clusters

Functions

- test group [f360_merge_two_clusters](#) ()
Test group for Merge_Two_Clusters function.
- [f360_merge_two_clusters_Test_Merge_Two_Clusters_Check_If_All_4_Dets_Are_Dealiased](#) ()
- [f360_merge_two_clusters_Test_Merge_Two_Clusters_Check_If_Dets_4_Are_Dealiased_1_Not_Dealiased](#) ()
- [f360_merge_two_clusters_Test_Merge_Two_Clusters_Check_Range_Rate_Is_Dealiased_Correctly_1](#) ()

2.2.1 Detailed Description

2.2.2 Function Documentation

2.2.2.1 f360_merge_two_clusters()

```
test group f360_merge_two_clusters ( )
```

Test group for Merge_Two_Clusters function.

Setup:\n

Nothing to setup in this test group

2.2.2.2 f360_merge_two_clusters_Test_Merge_Two_Clusters_Check_If_All_4_Dets_Are_Dealiased()

```
f360_merge_two_clusters_Test_Merge_Two_Clusters_Check_If_All_4_Dets_Are_Dealiased ( )
```

Purpose:\n Test to cover req if dealiased flag for detetions is properly set where all detection from two cluster are valid

Requirements FTCP-8587

Precondition:\n

Configuring clusters and adding detections to them

Action:\n

Call Merge_Two_Clusters

Result:\n

Checking if stationary cluster gets dealiased

2.2.2.3 f360_merge_two_clusters_Test_Merge_Two_Clusters_Check_If_Dets_4_Are_Dealiased_1_Not_Dealiased()

```
f360_merge_two_clusters_Test_Merge_Two_Clusters_Check_If_Dets_4_Are_Dealiased_1_Not_Dealiased  
( )
```

Purpose:\n Test to cover req if dealiased flag for detetions is properly set where one detection is valid and should be dealiased

Requirements FTCP-8587

Precondition:\n

Configuring clusters and adding detections to them

Action:\n

Call Merge_Two_Clusters

Result:\n

Checking if stationary cluster gets dealiased

2.2.2.4 f360_merge_two_clusters_Test_Merge_Two_Clusters_Check_Range_Rate_Is_Dealiased_Correctly_1()

```
f360_merge_two_clusters_Test_Merge_Two_Clusters_Check_Range_Rate_Is_Dealiased_Correctly_1 ( )
```

Purpose:\n Test to cover req if dealiased range rate for detetions is properly calculated

Requirements FTCP-8587

Precondition:\n

Configuring clusters and adding detections to them

Action:\n

Call Merge_Two_Clusters

Result:\n

Checking if stationary cluster gets dealiased

2.3 Initialize_clusters

Functions

- test group `initialize_qualtest_clusters` ()
Test group for initialize_clusters function.
- `initialize_qualtest_clusters_Test_Initialize_Clusters_One_Detection_To_Cluster` ()
- `initialize_qualtest_clusters_Test_Initialize_Clusters_Many_Detection_To_Cluster` ()

2.3.1 Detailed Description

2.3.2 Function Documentation

2.3.2.1 initialize_qualtest_clusters()

```
test group initialize_qualtest_clusters ( )
```

Test group for initialize_clusters function.

Setup:\n

Setting up arguments for initialize_clusters function and assigning them with basic values

2.3.2.2 initialize_qualtest_clusters_Test_Initialize_Clusters_Many_Detection_To_Cluster()

```
initialize_qualtest_clusters_Test_Initialize_Clusters_Many_Detection_To_Cluster ( )
```

Purpose:\n Test to cover req if cluster ids are properly set for 10 detetions from which one cluster shall be generated

Requirements FTCP-8581

Precondition:\n

Preparing initial clusters data and detections

Action:\n

Calling Initialize_Clusters

Result:\n

Checking if detections have correct set cluster ids

2.3.2.3 initialize_qualtest_clusters_Test_Initialize_Clusters_One_Detection_To_Cluster()

```
initialize_qualtest_clusters_Test_Initialize_Clusters_One_Detection_To_Cluster ( )
```

Purpose:\n Test to cover req if cluster id is properly set for one detetions from which cluster shall is generated

Requirements FTCP-8581

Precondition:\n

Preparing initial clusters data and detections

Action:\n

Calling Initialize_Clusters

Result:\n

Checking if one detections has correct set cluster id

2.4 F360_allocate_id_for_initialized_object

Functions

- test group [f360_allocate_id_for_initialized_object \(\)](#)
- [f360_allocate_id_for_initialized_object_no_active_objects \(\)](#)
- [f360_allocate_id_for_initialized_object_single_active_object \(\)](#)
- [f360_allocate_id_for_initialized_object_objects_saturation \(\)](#)

2.4.1 Detailed Description

2.4.2 Function Documentation

2.4.2.1 f360_allocate_id_for_initialized_object()

```
test_group f360_allocate_id_for_initialized_object ( )
```

Setup:\n

Initialize structures which are inputs of allocate_id_for_initialized_object

Teardown:\n

Nothing to teardown in this test group

2.4.2.2 f360_allocate_id_for_initialized_object_no_active_objects()

```
f360_allocate_id_for_initialized_object_no_active_objects ( )
```

Purpose:\n Test case where new initialized object is first ever active object, there is no other active tracks.

Requirements FTCP-12399

Precondition:\n

Tracker info array inactive_obj_ids contains all IDs.

Action:\n

Call function Allocate_Id_For_Initialized_Object

Result:\n

New ID should be equal to 1. Also check if now tracker_info.inactive_obj_ids first value is equal 2 (next free ID), and check if now tracker_info.active_obj_ids first value is equal 1.

2.4.2.3 f360_allocate_id_for_initialized_object_objects_saturation()

```
f360_allocate_id_for_initialized_object_objects_saturation ( )
```

Purpose:\n Test case where number of objects is staured.

Requirements FTCP-12399

Precondition:\n

Tracker info array active_obj_ids is full. Set tracker_info.num_active_objs to maximum value. Initialize lowest priority track with ID 10 and set tracker info pointer on this object.

Action:\n

Call function Allocate_Id_For_Initialized_Object

Result:\n

New ID should be equal to 2. Also check if now tracker_info.inactive_obj_ids first value is equal 3 (next free ID), and check if now tracker_info.active_obj_ids second value is equal 2.

2.4.2.4 f360_allocate_id_for_initialized_object_single_active_object()

```
f360_allocate_id_for_initialized_object_single_active_object ( )
```

Purpose:\n Test case where new initialized object is second ever active object, there is one other active tracks.

Requirements FTCP-12399

Precondition:\n

Tracker info array inactive_obj_ids contains all IDs except 1. Set tracker_info.active_obj_ids first value to be 1 and num_active_objs to 1.

Action:\n

Call function Allocate_Id_For_Initialized_Object

Result:\n

New ID should be equal to 2. Also check if now tracker_info.inactive_obj_ids first value is equal 3 (next free ID), and check if now tracker_info.active_obj_ids second value is equal 2.

2.5 F360_find_detection_inliers_single_object

Collaboration diagram for F360_find_detection_inliers_single_object:

2.6 F360_find_detection_inliers_find_valid_detections

Collaboration diagram for F360_find_detection_inliers_find_valid_detections:

Functions

- test group [f360_find_detection_inliers_find_valid_detections](#) ()
Testing of a function that finds detections that are valid candidates to be downselected to track.
- [f360_find_detection_inliers_find_valid_detections_Select_Detections_To_Mark_As_Inliers_Object_Wheel_Spin_Detection](#) ()
- [f360_find_detection_inliers_find_valid_detections_Select_Detections_To_Mark_As_Inliers_Pair_Wheel_Spin_Detection](#) ()
- [f360_find_detection_inliers_find_valid_detections_Select_Detections_To_Mark_As_Inliers_Nearby_Wheel_Spin_Detection](#) ()

2.6.1 Detailed Description

2.6.2 Function Documentation

2.6.2.1 f360_find_detection_inliers_find_valid_detections_Select_Detections_To_Mark_As_Inliers_Nearby_Wheel_Spin_De

```
f360_find_detection_inliers_find_valid_detections_Select_Detections_To_Mark_As_Inliers_↵
Nearby_Wheel_Spin_Detection ( )
```

Purpose:\n

Test that an associated detection with nearby wheel spin flag set is not considered to be a valid detection

Requirements FTCP-13634

Precondition:\n

Two detections, with no error flags set, have been associated to track in test group. Set nearby wheel spin flag for detection det_idx_2 to true Set expected number of valid detections to 1 Set expected valid detection indices to det_idx_1

Action:\n

Call function

Result:\n

Expect number of valid detection to be 1 Expect detection index to match the index of the first detection in test group

2.6.2.2 f360_find_detection_inliers_find_valid_detections_Select_Detections_To_Mark_As_Inliers_Object_Wheel_Spin_Detection

```
f360_find_detection_inliers_find_valid_detections_Select_Detections_To_Mark_As_Inliers_↵  
Object_Wheel_Spin_Detection ( )
```

Purpose:

Test that an associated detection with object wheel spin flag set is not considered to be a valid detection

Requirements FTCP-13634

Precondition:

Two detections, with no error flags set, have been associated to track in test group. Set object wheel spin flag for detection det_idx_2 to true Set expected number of valid detections to 1 Set expected valid detection indices to det_idx_1

Action:

Call function

Result:

Expect number of valid detection to be 1 Expect detection index to match the index of the first detection in test group

2.6.2.3 f360_find_detection_inliers_find_valid_detections_Select_Detections_To_Mark_As_Inliers_Pair_Wheel_Spin_Detection

```
f360_find_detection_inliers_find_valid_detections_Select_Detections_To_Mark_As_Inliers_Pair_↵  
Wheel_Spin_Detection ( )
```

Purpose:

Test that an associated detection with pair wheel spin flag set is not considered to be a valid detection

Requirements FTCP-13634

Precondition:

Two detections, with no error flags set, have been associated to track in test group. Set pair wheel spin flag for detection det_idx_2 to true Set expected number of valid detections to 1 Set expected valid detection indices to det_idx_1

Action:

Call function

Result:

Expect number of valid detection to be 1 Expect detection index to match the index of the first detection in test group

2.7 F360_Test_Determine_Association_Hypothesis_qualtest

Functions

- test group [f360_Test_Determine_Association_Hypothesis_qualtest](#) ()
Set up an object and place detections around it with various properties and check that expected detections are associated to expected parameters. More details can be found in DFD-175.
- [f360_Test_Determine_Association_Hypothesis_qualtest_Test_Determine_Association_Hypothesis_A](#) ()
- [f360_Test_Determine_Association_Hypothesis_qualtest_Test_Determine_Association_Hypothesis_B](#) ()

2.7.1 Detailed Description

2.7.2 Function Documentation

2.7.2.1 f360_Test_Determine_Association_Hypothesis_qualtest()

```
test_group f360_Test_Determine_Association_Hypothesis_qualtest ( )
```

Set up an object and place detections around it with various properties and check that expected detections are associated to expected parameters. More details can be found in DFD-175.

Setup:\n

Setup common data for all tests

Precondition:\n

Initialize detection score array

2.7.2.2 f360_Test_Determine_Association_Hypothesis_qualtest_Test_Determine_Association_Hypothesis_A()

```
f360_Test_Determine_Association_Hypothesis_qualtest_Test_Determine_Association_Hypothesis_A ( )
```

Purpose:\n Verify that all detections are associated to the first object in the object list

Requirements FTCP-10132

Action:\n

Call Determine_Association_Hypothesis

Result:\n

Check that correct parameters are returned

2.7.2.3 f360_Test_Determine_Association_Hypothesis_qualtest_Test_Determine_Association_Hypothesis_B()

```
f360_Test_Determine_Association_Hypothesis_qualtest_Test_Determine_Association_Hypothesis_B (
)
```

Purpose: Verify that all detections are associated as expected in scenario B described in document in DFD-175

Requirements FTCP-10132

Precondition:

Set up detections properties accordingly

Action:

Call function Determine_Association_Hypothesis twice for both test objects

Result:

Check that correct parameters are returned

2.8 F360_Test_Is_Det_Allowed_To_Associate_qualtest

Functions

- test group [f360_Test_Is_Det_Allowed_To_Associate_qualtest \(\)](#)

Included tests related to calculation of detection is allowed to associate based flags and properties.

- [f360_Test_Is_Det_Allowed_To_Associate_qualtest_Test_Is_Det_Allowed_To_Associate_Det_Not_Ok_To_Use \(\)](#)
- [f360_Test_Is_Det_Allowed_To_Associate_qualtest_Test_Is_Det_Allowed_To_Associate_Det_On_SEP_Obj_Moving_CTCA \(\)](#)
- [f360_Test_Is_Det_Allowed_To_Associate_qualtest_Test_Is_Det_Allowed_To_Associate_Mirror_Obj_Ambiguous_Detection \(\)](#)
- [f360_Test_Is_Det_Allowed_To_Associate_qualtest_Test_Is_Det_Allowed_To_Associate_Detection_Behind_SEP_Obj_Moving \(\)](#)
- [f360_Test_Is_Det_Allowed_To_Associate_qualtest_Test_Is_Det_Allowed_To_Associate_Object_Behind_SEP_Obj_Moving_C \(\)](#)
- [f360_Test_Is_Det_Allowed_To_Associate_qualtest_Test_Is_Det_Allowed_To_Associate_Object_Behind_SEP_Obj_Moving_C \(\)](#)
- [f360_Test_Determine_Association_Hypothesis_qualtest_Test_Determine_Association_Hypothesis_C \(\)](#)
- [f360_Test_Determine_Association_Hypothesis_qualtest_Test_Determine_Association_Hypothesis_D \(\)](#)

2.8.1 Detailed Description

2.8.2 Function Documentation

2.8.2.1 f360_Test_Determine_Association_Hypothesis_qualtest_Test_Determine_Association_Hypothesis_C()

```
f360_Test_Determine_Association_Hypothesis_qualtest_Test_Determine_Association_Hypothesis_C (
)
```

Purpose: Verify that for the detection associated to object which is not downselected, the Object ID is assigned.

Requirements FTCP-12084

Precondition:

Set up non downselected object

Action:

Call function Determine_Association_Hypothesis twice for both test objects

Result:

Check that object ID is assigned to the detection associated to object

2.8.2.2 f360_Test_Determine_Association_Hypothesis_qualtest_Test_Determine_Association_Hypothesis_D()

```
f360_Test_Determine_Association_Hypothesis_qualtest_Test_Determine_Association_Hypothesis_D (
)
```

Purpose: Verify that for the detection associated to object which is downselected, the Object ID is assigned.

Requirements FTCP-12084

Precondition:

Set up downselected object

Action:

Call function Determine_Association_Hypothesis twice for both test objects

Result:

Check that object ID is assigned to the detection associated to object

2.8.2.3 f360_Test_Is_Det_Allowed_To_Associate_qualtest()

```
test group f360_Test_Is_Det_Allowed_To_Associate_qualtest ( )
```

Included tests related to calculation of detection is allowed to associate based flags and properties.

Setup:

Setup test so that the highest complexity branch passes. Then tweak these parameters to reach full branch coverage in each individual test

2.8.2.4 f360_Test_Is_Det_Allowed_To_Associate_qualtest_Test_Is_Det_Allowed_To_Associate_Det_Not_Ok_To_Use()

```
f360_Test_Is_Det_Allowed_To_Associate_qualtest_Test_Is_Det_Allowed_To_Associate_Det_Not_Ok_↔
To_Use ( )
```

Purpose: This test will test that Is_Det_Allowed_To_Associate() returns false due to detection being flagged as not ok to use.

Requirements FTCP-10125

Precondition:

Set detections as not ok to use

Action:

Call Is_Det_Allowed_To_Associate

Result:

Check that detection flag f_allowed_to_associate is false

2.8.2.5 f360_Test_Is_Det_Allowed_To_Associate_qualtest_Test_Is_Det_Allowed_To_Associate_Det_On_SEP_Obj_Moving

f360_Test_Is_Det_Allowed_To_Associate_qualtest_Test_Is_Det_Allowed_To_Associate_Det_On_SEP_↔
Obj_Moving_CTCA ()

Purpose: This test will test that Is_Det_Allowed_To_Associate() returns false due to detection being flagged as
on some SEP and the object is a moving CTCA track.

Requirements FTCP-10041

Precondition:

Set object to moving Set detections to be on some SEP Set up an SEP association box, and set object position outside it

Action:

Call Is_Det_Allowed_To_Associate

Result:

Check that detection flag f_allowed_to_associate is false

2.8.2.6 f360_Test_Is_Det_Allowed_To_Associate_qualtest_Test_Is_Det_Allowed_To_Associate_Detection_Behind_SEP_Obj_Moving

f360_Test_Is_Det_Allowed_To_Associate_qualtest_Test_Is_Det_Allowed_To_Associate_Detection_↔
Behind_SEP_Obj_Moving ()

Purpose: This test will test that Is_Det_Allowed_To_Associate() returns false due to object and detection being on opposite sides of a SEP.
Detection is behind SEP and object is not.

Requirements FTCP-10126

Precondition:

Set detection behind SEP Object set to not behind SEP in TEST_SETUP Object set to moving in TEST_SETUP a SEP association zone is set up and object's position within it

Action:

Call Is_Det_Allowed_To_Associate

Result:

Check that detection flag f_allowed_to_associate is false

2.8.2.7 f360_Test_Is_Det_Allowed_To_Associate_qualtest_Test_Is_Det_Allowed_To_Associate_Mirror_Obj_Ambiguous_D

f360_Test_Is_Det_Allowed_To_Associate_qualtest_Test_Is_Det_Allowed_To_Associate_Mirror_Obj_↔
Ambiguous_Detection ()

Purpose: This test will test that Is_Det_Allowed_To_Associate() returns false due to object having a high mirror probability and detection motion status is ambiguous.

Requirements FTCP-10127

Precondition:

Set object mirror probability high Set detections motion status to ambiguous

Action:

Call Is_Det_Allowed_To_Associate

Result:

Check that detection flag f_allowed_to_associate is false

2.8.2.8 f360_Test_Is_Det_Allowed_To_Associate_qualtest_Test_Is_Det_Allowed_To_Associate_Object_Behind_SEP_Obj_I

f360_Test_Is_Det_Allowed_To_Associate_qualtest_Test_Is_Det_Allowed_To_Associate_Object_↔
Behind_SEP_Obj_Moving_CTCA ()

Purpose: This test will test that Is_Det_Allowed_To_Associate() returns false due to moving CTCA object and detection being on opposite sides of a SEP. The object is behind SEP while the detections is not.

Requirements FTCP-10126

Precondition:

Set object behind SEP Detection set to not behind SEP in TEST_SETUP Object set to moving in TEST_SETUP a SEP association zone is set up and object's position within it

Action:

Call Is_Det_Allowed_To_Associate

Result:

Check that detection flag f_allowed_to_associate is false

2.8.2.9 f360_Test_Is_Det_Allowed_To_Associate_qualtest_Test_Is_Det_Allowed_To_Associate_Object_Behind_SEP_Obj_

f360_Test_Is_Det_Allowed_To_Associate_qualtest_Test_Is_Det_Allowed_To_Associate_Object_↵
Behind_SEP_Obj_Moving_CTCA_Far_From_SEP ()

Purpose: This test will test that Is_Det_Allowed_To_Associate() returns true when a moving CTCA object and a detection are on opposite sides of a
the object is far from the SEP, i.e. outside of the SEP association box.

Requirements FTCP-10126

Precondition:

Set object behind the SEP Detection set to not behind SEP in TEST_SETUP Object set to moving in TEST↵
_SETUP A SEP association zone is set up Object's position is outside of the SEP association box

Action:

Call Is_Det_Allowed_To_Associate

Result:

Check that detection flag f_allowed_to_associate is true

2.9 F360_detect_wheelspin_object_qualtest

Functions

- test group [f360_detect_wheelspin_object_qualtest](#) ()
Group of tests checking if detections are correctly marked as object wheel spins.
- [f360_detect_wheelspin_object_qualtest_f360_detect_wheelspin_object_qualtest__correctly_marked_as_object_wheel_spin](#) ()
- [f360_detect_wheelspin_object_qualtest_f360_detect_wheelspin_object_qualtest__correctly_not_marked_as_object_wheel_spin](#) ()

2.9.1 Detailed Description

2.9.2 Function Documentation

2.9.2.1 f360_detect_wheelspin_object_qualtest()

```
test_group f360_detect_wheelspin_object_qualtest ( )
```

Group of tests checking if detections are correctly marked as object wheel spins.

Setup:\n

Init calibrations

2.9.2.2 f360_detect_wheelspin_object_qualtest_f360_detect_wheelspin_object_qualtest__correctly_marked_as_object_wheel_spin()

```
f360_detect_wheelspin_object_qualtest_f360_detect_wheelspin_object_qualtest__correctly_marked_as_object_wheel_spin ( )
```

Purpose:\n Checks if detections are correctly marked as OBJECT wheel spins

Requirements FTCP-8840

Precondition:\n

Setup object and two dets close to each other

Action:\n

Call Mark_Detections_Wheel_Spin_From_Objects()

Result:\n

Detections should be marked as OBJECT WHEEL SPIN

2.9.2.3 f360_detect_wheelspin_object_qualtest_f360_detect_wheelspin_object_qualtest__correctly_not_marked_as_object

f360_detect_wheelspin_object_qualtest_f360_detect_wheelspin_object_qualtest__correctly_not_marked_as_object_wheel_spin ()

Purpose: Checks if detections are correctly marked as OBJECT wheel spins

Requirements FTCP-8840

Precondition:

Setup object and two dets First det is not associated to object Second has small range rate

Action:

Call Mark_Detections_Wheel_Spin_From_Objects()

Result:

Detections should not be marked as OBJECT WHEEL SPIN

2.10 F360_detect_wheelspin_nearby_qualtest

Functions

- test group `f360_detect_wheelspin_nearby_qualtest` ()
Group of tests checking if detections are correctly marked as nearby wheel spins.
- `f360_detect_wheelspin_nearby_qualtest_f360_detect_wheelspin_nearby_qualtest__correctly_marked_as_nearby_wheel_spin` ()
- `f360_detect_wheelspin_nearby_qualtest_f360_detect_wheelspin_nearby_qualtest__correctly_not_marked_as_wheel_spin` ()

2.10.1 Detailed Description

2.10.2 Function Documentation

2.10.2.1 f360_detect_wheelspin_nearby_qualtest()

```
test_group f360_detect_wheelspin_nearby_qualtest ( )
```

Group of tests checking if detections are correctly marked as nearby wheel spins.

Setup:\n

Init calibrations

2.10.2.2 f360_detect_wheelspin_nearby_qualtest_f360_detect_wheelspin_nearby_qualtest__correctly_marked_as_nearby_wheel_spin()

```
f360_detect_wheelspin_nearby_qualtest_f360_detect_wheelspin_nearby_qualtest__correctly_↵  
marked_as_nearby_wheel_spin ( )
```

Purpose:\n Checks if detections are correctly marked as NEARBY wheel spins

Requirements FTCP-8840

Precondition:\n

Add det above wheel-spin and within limits and sort them

Action:\n

Call Detect_Nearby_Wheel_Spins().

Result:\n

Detection should be marked as NEARBY WHEEL SPIN

2.10.2.3 f360_detect_wheelspin_nearby_qualtest_f360_detect_wheelspin_nearby_qualtest__correctly_not_marked_as_wh

```
f360_detect_wheelspin_nearby_qualtest_f360_detect_wheelspin_nearby_qualtest__correctly_not_↵  
marked_as_wheel_spin ( )
```

Purpose: Checks if detections are correctly not marked as wheel spins

Requirements FTCP-8840

Precondition:

Add det which is out of the limits and sort them

Action:

Call Detect_Nearby_Wheel_Spins()

Result:

Detection should not be marked as WHEEL SPIN

2.11 F360_fill_init_obj_track_props_qualtest

Functions

- test group `f360_fill_init_obj_track_props_qualtest ()`
Test group for Fill_Init_Obj_Track_Props function.
- `f360_fill_init_obj_track_props_qualtest_TestCalcInitObjTrackMovingFlagMoving ()`
- `f360_fill_init_obj_track_props_qualtest_TestCalcInitObjTrackMovingFlagMovingValidForLiberalTracking ()`
- `f360_fill_init_obj_track_props_qualtest_TestCalcInitObjTrackMovingFlagNotMovingHighFracAz ()`
- `f360_fill_init_obj_track_props_qualtest_TestCalcInitObjTrackMovingFlagNotMovingLowSpeed ()`
- `f360_fill_init_obj_track_props_qualtest_TestCalcInitObjTrackMovingFlagNotMovingLowSpeedAndHighFracAz ()`
- `f360_fill_init_obj_track_props_qualtest_TestCalcInitObjVCSHeadingMovingObj ()`
- `f360_fill_init_obj_track_props_qualtest_TestCalcInitObjVCSHeadingNotMovingObj ()`
- `f360_fill_init_obj_track_props_qualtest_TestFillInitObjTrackPropsNominal ()`

2.11.1 Detailed Description

2.11.2 Function Documentation

2.11.2.1 f360_fill_init_obj_track_props_qualtest()

```
test group f360_fill_init_obj_track_props_qualtest ( )
```

Test group for Fill_Init_Obj_Track_Props function.

Setup:\n

Setting variables for fill init object track

2.11.2.2 f360_fill_init_obj_track_props_qualtest_TestCalcInitObjTrackMovingFlagMoving()

```
f360_fill_init_obj_track_props_qualtest_TestCalcInitObjTrackMovingFlagMoving ( )
```

Purpose:\n Checking object's f_moving flag assignment

Requirements FTCP-12405

Precondition:\n

Setting variables for object's speed check

Action:\n

Call Fill_Init_Obj_Track_Props

Result:\n

Checking if results match expectations

2.11.2.3 f360_fill_init_obj_track_props_qualtest_TestCalcInitObjTrackMovingFlagMovingValidForLiberalTracking()

```
f360_fill_init_obj_track_props_qualtest_TestCalcInitObjTrackMovingFlagMovingValidForLiberal↔  
Tracking ( )
```

Purpose:\n Checking object's f_moving flag assignment

Requirements FTCP-12405

Precondition:\n

Setting variables for object's speed check

Action:\n

Call Fill_Init_Obj_Track_Props

Result:\n

Checking if results match expectations

2.11.2.4 f360_fill_init_obj_track_props_qualtest_TestCalcInitObjTrackMovingFlagNotMovingHighFracAz()

```
f360_fill_init_obj_track_props_qualtest_TestCalcInitObjTrackMovingFlagNotMovingHighFracAz ( )
```

Purpose:\n Checking object's f_moving flag assignment

Requirements FTCP-12405

Precondition:\n

Setting variables for object's speed check

Action:\n

Call Fill_Init_Obj_Track_Props

Result:\n

Checking if results match expectations

2.11.2.5 f360_fill_init_obj_track_props_qualtest_TestCalcInitObjTrackMovingFlagNotMovingLowSpeed()

```
f360_fill_init_obj_track_props_qualtest_TestCalcInitObjTrackMovingFlagNotMovingLowSpeed ( )
```

Purpose: Checking object's f_moving flag assignment

Requirements FTCP-12405

Precondition:

Setting variables for object's speed check

Action:

Call Fill_Init_Obj_Track_Props

Result:

Checking if results match expectations

2.11.2.6 f360_fill_init_obj_track_props_qualtest_TestCalcInitObjTrackMovingFlagNotMovingLowSpeedAndHighFracAz()

```
f360_fill_init_obj_track_props_qualtest_TestCalcInitObjTrackMovingFlagNotMovingLowSpeedAnd↵  
HighFracAz ( )
```

Purpose: Checking object's f_moving flag assignment

Requirements FTCP-12405

Precondition:

Setting variables for object's speed check

Action:

Call Fill_Init_Obj_Track_Props

Result:

Checking if results match expectations

2.11.2.7 f360_fill_init_obj_track_props_qualtest_TestCalcInitObjVCSHeadingMovingObj()

f360_fill_init_obj_track_props_qualtest_TestCalcInitObjVCSHeadingMovingObj ()

Purpose:\n Checking object's VCS heading calculation

Requirements FTCP-12402

Precondition:\n

Setting variables for object's heading check

Action:\n

Call Fill_Init_Obj_Track_Props

Result:\n

Checking if results match expectations

2.11.2.8 f360_fill_init_obj_track_props_qualtest_TestCalcInitObjVCSHeadingNotMovingObj()

f360_fill_init_obj_track_props_qualtest_TestCalcInitObjVCSHeadingNotMovingObj ()

Purpose:\n Checking object's VCS heading calculation

Requirements FTCP-12402

Precondition:\n

Setting variables for object's heading check

Action:\n

Call Fill_Init_Obj_Track_Props

Result:\n

Checking if results match expectations

2.11.2.9 f360_fill_init_obj_track_props_qualtest_TestFillInitObjTrackPropsNominal()

```
f360_fill_init_obj_track_props_qualtest_TestFillInitObjTrackPropsNominal ( )
```

Purpose:\n Checking object's vehicular info update

Requirements FTCP-12404, FTCP-12402 // TODO during industrialization: Need to check if these reqs must be updated

Precondition:\n

Setting variables for object's vehicular info update

Action:\n

Call Fill_Init_Obj_Track_Props

Result:\n

Checking if results match expectations

2.12 F360_init_obj_track_covariances

Collaboration diagram for F360_init_obj_track_covariances:

Modules

- [F360_fill_init_obj_track_props_determine_initial_size_qualtest](#)

Functions

- test group [f360_init_obj_track_covariances](#) ()
Test group for Init_Obj_Track_Covariances function.
- [f360_init_obj_track_covariances_TestCCANominalCase](#) ()

2.12.1 Detailed Description

2.12.2 Function Documentation

2.12.2.1 f360_init_obj_track_covariances()

```
test group f360_init_obj_track_covariances ( )
```

Test group for Init_Obj_Track_Covariances function.

Setup:\n

Setting up calibrations, init info, obj track to init and test tolerance

2.12.2.2 f360_init_obj_track_covariances_TestCCANominalCase()

```
f360_init_obj_track_covariances_TestCCANominalCase ( )
```

Purpose:\n Checking initialization of track's covariances for nominal cases

Requirements FTCP-12406

Precondition:\n

Setting proper filter type

Action:\n

Call Init_Obj_Track_Covariances

Result:\n

Checking if results match expectations

2.13 F360_fill_init_obj_track_props_determine_initial_size_qualtest

Collaboration diagram for F360_fill_init_obj_track_props_determine_initial_size_qualtest:

Functions

- test group [f360_fill_init_obj_track_props_determine_initial_size_qualtest](#) ()
Test group for Determine_Initial_Size function.
- [f360_fill_init_obj_track_props_determine_initial_size_qualtest_Test_Initial_Size_Stationary_Object](#) ()
- [f360_fill_init_obj_track_props_determine_initial_size_qualtest_Test_Initial_Size_Slow_Moving_Object](#) ()
- [f360_fill_init_obj_track_props_determine_initial_size_qualtest_Test_Initial_Size_Fast_Moving_Object](#) ()

2.13.1 Detailed Description

2.13.2 Function Documentation

2.13.2.1 f360_fill_init_obj_track_props_determine_initial_size_qualtest()

```
test group f360_fill_init_obj_track_props_determine_initial_size_qualtest ( )
```

Test group for Determine_Initial_Size function.

Setup:\n

Setting variables for Determine_Initial_Size

2.13.2.2 f360_fill_init_obj_track_props_determine_initial_size_qualtest_Test_Initial_Size_Fast_Moving_Object()

```
f360_fill_init_obj_track_props_determine_initial_size_qualtest_Test_Initial_Size_Fast_Moving↔  
_Object ( )
```

Purpose:\n Test that initial object size is correctly determined for a fast moving object

Requirements FTCP-12402

Precondition:\n

Set speed above fast moving threshold. Expected length is 6 m and width 2 m.

Action:\n

Call Determine_Initial_Size

Result:\n

Check that length and width are determined correctly.

2.13.2.3 f360_fill_init_obj_track_props_determine_initial_size_qualtest_Test_Initial_Size_Slow_Moving_Object()

```
f360_fill_init_obj_track_props_determine_initial_size_qualtest_Test_Initial_Size_Slow_Moving↵  
_Object ( )
```

Purpose:\n Test that initial object size is correctly determined for a slow moving object

Requirements FTCP-12402

Precondition:\n

A slow moving target with speed above threshold for moving status but below threshold for fast moving is set up. Expected length is 3 m and width 1 m.

Action:\n

Call Determine_Initial_Size

Result:\n

Check that length and width are determined correctly.

2.13.2.4 f360_fill_init_obj_track_props_determine_initial_size_qualtest_Test_Initial_Size_Stationary_Object()

```
f360_fill_init_obj_track_props_determine_initial_size_qualtest_Test_Initial_Size_Stationary_↵  
Object ( )
```

Purpose:\n Test that initial object size is correctly determined for a stationary object

Requirements FTCP-12402

Precondition:\n

Set speed below threshold for moving status. Expected length and width are both 1 m.

Action:\n

Call Determine_Initial_Size

Result:\n

Check that length and width are determined correctly.

2.14 F360_initialize_tracks

Functions

- test group `f360_initialize_tracked_objects` ()
Test group for Initialize_Tracks function.
- `f360_initialize_tracked_objects_Lowest_Cluster_Priority_Initialization_Try_With_Obj_Saturation` ()
- `f360_initialize_tracked_objects_Highest_Cluster_Priority_Initialization_Try_With_Obj_Saturation` ()
- `f360_initialize_tracked_objects_Single_Successfull_Object_Initalization` ()
- `f360_initialize_tracked_objects_Single_Successfull_Object_Initalization_With_Init_Scheme_Check` ()
- `f360_initialize_tracked_objects_Detections_Assignment_For_Single_Successfull_Object_Initalization` ()
- `f360_initialize_tracked_objects_Detections_Assignment_For_Single_Unsuccessfull_Object_Initalization` ()
- `f360_initialize_tracked_objects_Single_Unsuccessfull_Object_Initalization` ()
- `f360_initialize_tracked_objects_Two_Successfull_And_One_Unsuccessfull_Objects_Initialization` ()
- `f360_initialize_tracked_objects_Highest_Cluster_Priority_Reinitialization_With_Obj_Saturation` ()
- `f360_initialize_tracked_objects_Check_If_Objects_Are_Correctly_Sorted_By_Priority` ()
- `f360_initialize_tracked_objects_Check_If_Objects_Are_Correctly_Sorted_By_Longitudinal_Position` ()
- `f360_initialize_tracked_objects_Single_Non_Occluded_Cluster__Successful_Initalization` ()
- `f360_initialize_tracked_objects_Single_Occluded_Cluster__Unsuccessful_Initialization` ()
- `f360_initialize_tracked_objects_Occluded_And_Non_Occluded_Cluster__Only_One_Successfull_Initialization` ()
- `f360_initialize_tracked_objects_Occluded_Cluster_With_Detections_Valid_For_Liberal_Initalization__Successfull_Initialization` ()

2.14.1 Detailed Description

2.14.2 Function Documentation

2.14.2.1 f360_initialize_tracked_objects()

```
test group f360_initialize_tracked_objects ( )
```

Test group for Initialize_Tracks function.

Setup:\n

Setting up calibrations, tracker info inactive ids and objects inactive ids to default values

2.14.2.2 f360_initialize_tracked_objects_Check_If_Objects_Are_Correctly_Sorted_By_Longitudinal_Position()

```
f360_initialize_tracked_objects_Check_If_Objects_Are_Correctly_Sorted_By_Longitudinal_Position  
( )
```

Purpose: High level initialization tracks testing. Checking if object after initialization are correctly sorted by longitudinal position

Requirements FTCP-12401

Precondition:

Setting up four clusters for initialization with different priority and position Initializing empty occlusion object - no occluding objects present

Action:

Calling Initialize_Tracks function

Result:

Four objects shall be sorted by longitudinal position

2.14.2.3 f360_initialize_tracked_objects_Check_If_Objects_Are_Correctly_Sorted_By_Priority()

```
f360_initialize_tracked_objects_Check_If_Objects_Are_Correctly_Sorted_By_Priority ( )
```

Purpose: High level initialization tracks testing. Checking if object after initialization are correctly sorted by priority

Requirements FTCP-12400

Precondition:

Setting up four clusters for initialization with different priority and position Initializing empty occlusion object - no occluding objects present

Action:

Calling Initialize_Tracks function

Result:

Priority in this case shall be based on distance (the higher distance from 0.0, 0.0 the lower priority)

2.14.2.4 f360_initialize_tracked_objects_Detections_Assignment_For_Single_Successfull_Object_Initialization()

```
f360_initialize_tracked_objects_Detections_Assignment_For_Single_Successfull_Object_Initialization  
( )
```

Purpose:\n High level initialization tracks testing. Checking if all detections from cluster that were used for init,
 \will be assigned to new object

Requirements FTCP-12398

Precondition:\n

Setting cluster with 2 hist dets and 2 dets from current time instance. Initializing empty occlusion object - no occluding objects present

Action:\n

Calling Initialize_Tracks function

Result:\n

Checking if 2 detections from current time instance were associated to new object

2.14.2.5 f360_initialize_tracked_objects_Detections_Assignment_For_Single_Unsuccessfull_Object_Initialization()

```
f360_initialize_tracked_objects_Detections_Assignment_For_Single_Unsuccessfull_Object_Initialization  
( )
```

Purpose:\n High level initialization tracks testing. Checking if all detections from cluster that were used for init,
 \will not be assigned any object due lack of initialization

Requirements FTCP-12398

Precondition:\n

Setting cluster with 2 hist dets and 2 dets from current time instance for unsuccessfull initialization Initializing empty occlusion object - no occluding objects present

Action:\n

Calling Initialize_Tracks function

Result:\n

Checking if no detections are associated to any object

2.14.2.6 f360_initialize_tracked_objects_Highest_Cluster_Priority_Initialization_Try_With_Obj_Saturation()

```
f360_initialize_tracked_objects_Highest_Cluster_Priority_Initialization_Try_With_Obj_Saturation  
( )
```

Purpose: High level initialization tracks testing

Requirements FTCP-12408, FTCP-12399, FTCP-12402

Precondition:

Setting up 1001 cluster for objects initialization, from first 1001 objects shall be initialized and the last one shall overwrite object with lowest priority

Action:

Calling Initialize_Tracks function

Result:

Checking if 1000 have been initialized where one is overwritten

2.14.2.7 f360_initialize_tracked_objects_Highest_Cluster_Priority_Reinitialization_With_Obj_Saturation()

```
f360_initialize_tracked_objects_Highest_Cluster_Priority_Reinitialization_With_Obj_Saturation  
( )
```

Purpose: High level initialization tracks testing

Requirements FTCP-12408, FTCP-12402

Precondition:

Setting up cluster with high enough priority for successful object initialization

Action:

Calling Initialize_Tracks function

Result:

Checking if new object had been initialized

2.14.2.8 f360_initialize_tracked_objects_Lowest_Cluster_Priority_Initialization_Try_With_Obj_Saturation()

```
f360_initialize_tracked_objects_Lowest_Cluster_Priority_Initialization_Try_With_Obj_Saturation
( )
```

Purpose: High level initialization tracks testing

Requirements FTCP-12397

Precondition:

Setting up 251 cluster for objects initialization, from first 250 objects shall be initialized but the last one shall not

Action:

Calling Initialize_Tracks function

Result:

Checking if 250 object will be initialized from first 250 clusters

2.14.2.9 f360_initialize_tracked_objects_Occluded_And_Non_Occluded_Cluster__Only_One_Successfull_Initialization()

```
f360_initialize_tracked_objects_Occluded_And_Non_Occluded_Cluster__Only_One_Successfull_↔
Initialization ( )
```

Purpose: Check if only one new object will be initialized when two clusters valid for initialization exist, but one of them is occluded

Requirements FTCP-13056

Precondition:

Setting up three cluster. Two valid for initialization, all with 2 dets form current time instant and 2 hist dets
Setting up moving high confidence object occluding first cluster and triggering occlusion map creation

Action:

Calling Initialize_Tracks function

Result:

Checking if only single additional object had been initialized

2.14.2.10 f360_initialize_tracked_objects_Occluded_Cluster_With_Detections_Valid_For_Liberal_Initilization__Successful

```
f360_initialize_tracked_objects_Occluded_Cluster_With_Detections_Valid_For_Liberal_Initilization↔
__Successful Initialization ( )
```

Purpose: Check if new object will be initialized when cluster is occluded by existing object, but contains one detection valid for liberal initialization

Requirements FTCP-13056

Precondition:

Setting up cluster with 2 dets from current time instance and 2 hist dets - valid for initalization Setting f_valid↔
_for_liberal_initialization flag for one detection in initialized cluster Set host speed to activate liberal tracking
Setting up moving high confidence object occluding created cluster and triggering occlusion map creation

Action:

Calling Initialize_Tracks function

Result:

Checking if only single additional object had been initialized

2.14.2.11 f360_initialize_tracked_objects_Single_Non_Occluded_Cluster__Successful_Initilization()

```
f360_initialize_tracked_objects_Single_Non_Occluded_Cluster__Successful_Initilization ( )
```

Purpose: Check if new object will be initilized from cluster even though other moving non-occluding object exists

Requirements FTCP-13056

Precondition:

Setting up cluster with 2 dets from current time instance and 2 hist dets - valid for initalization Setting up
moving high confidence object which does not occlude created cluster and triggering occlusion map creation

Action:

Calling Initialize_Tracks function

Result:

Checking if new object had been initialized correctly with proper status One active object already exists, so
creation of second object is expected

2.14.2.12 f360_initialize_tracked_objects_Single_Occluded_Cluster__Unsuccessful_Initialization()

```
f360_initialize_tracked_objects_Single_Occluded_Cluster__Unsuccessful_Initialization ( )
```

Purpose:\n Check if new object initialization will be stopped from cluster which is occluded by existing object

Requirements FTCP-13056

Precondition:\n

Setting up cluster with 2 dets from current time instance and 2 hist dets - valid for initialization Setting up moving high confidence object occluding created cluster and triggering occlusion map creation

Action:\n

Calling Initialize_Tracks function

Result:\n

Checking if no additional object has been initialized (one already exists as set in test preconditions)

2.14.2.13 f360_initialize_tracked_objects_Single_Successfull_Object_Initialization()

```
f360_initialize_tracked_objects_Single_Successfull_Object_Initialization ( )
```

Purpose:\n High level initialization tracks testing

Requirements FTCP-12403

Precondition:\n

Setting up cluster with 2 dets from current time instance and 2 hist dets for success initialization Initializing empty occlusion object - no occluding objects present

Action:\n

Calling Initialize_Tracks function

Result:\n

Checking if object had been initialized correctly with proper status

2.14.2.14 f360_initialize_tracked_objects_Single_Successfull_Object_Initialization_With_Init_Scheme_Check()

```
f360_initialize_tracked_objects_Single_Successfull_Object_Initialization_With_Init_Scheme_Check  
( )
```

Purpose:\n High level initialization tracks testing. Checking initialiation scheme

Requirements FTCP-12404

Precondition:\n

Setting up cluster with 2 dets from current time instance and 2 hist dets for success initialization Initializing empty occlusion object - no occluding objects present

Action:\n

Calling Initialize_Tracks function

Result:\n

Checking if object had been initialized correctly with proper initialization scheme

2.14.2.15 f360_initialize_tracked_objects_Single_Unsuccessfull_Object_Initialization()

```
f360_initialize_tracked_objects_Single_Unsuccessfull_Object_Initialization ( )
```

Purpose:\n High level initialization tracks testing

Requirements FTCP-12403

Precondition:\n

Setting cluster with 2 hist dets and 2 dets from current time instance for unsuccessfull initialization Initializing empty occlusion object - no occluding objects present

Action:\n

Calling Initialize_Tracks function

Result:\n

Checking if no object had been initialized

2.14.2.16 f360_initialize_tracked_objects_Two_Successfull_And_One_Unsuccessfull_Objects_Initialization()

```
f360_initialize_tracked_objects_Two_Successfull_And_One_Unsuccessfull_Objects_Initialization (
)
```

Purpose:\n High level initialization tracks testing

Requirements FTCP-12403

Precondition:\n

Setting up three cluster. Two for initialization, all with 2 dets form current time instant and 2 hist dets Initializing empty occlusion object - no occluding objects present

Action:\n

Calling Initialize_Tracks function

Result:\n

Checking if two objects have been initialized with proper status

2.15 F360_try_to_init_obj_track_qualtest

Functions

- test group [f360_try_to_init_obj_track_qualtest](#) ()
Test group for Try_To_Init_Obj_Track.
- [f360_try_to_init_obj_track_qualtest_HostOverlapCheckPosLongOverlappedInRearAxleDistPosLatNOTOverlappedtest](#) ()
- [f360_try_to_init_obj_track_qualtest_HostOverlapCheckPosLongOverlappedInRearAxleDistPosLatOverlappedtest](#) ()
- [f360_try_to_init_obj_track_qualtest_HostOverlapCheckPosLongNOTOverlapedPosLatNOTOverlappedtest](#) ()
- [f360_try_to_init_obj_track_qualtest_HostOverlapCheckPosLongNOTOverlapedPosLatOverlappedtest](#) ()
- [f360_try_to_init_obj_track_qualtest_LiberalInitializationTest](#) ()

2.15.1 Detailed Description

2.15.2 Function Documentation

2.15.2.1 f360_try_to_init_obj_track_qualtest()

```
test group f360_try_to_init_obj_track_qualtest ( )
```

Test group for Try_To_Init_Obj_Track.

Setup:\n

Setting up need parameters for Try_To_Init_Obj_Track

2.15.2.2 f360_try_to_init_obj_track_qualtest_HostOverlapCheckPosLongNOTOverlapedPosLatNOTOverlappedtest()

```
f360_try_to_init_obj_track_qualtest_HostOverlapCheckPosLongNOTOverlapedPosLatNOTOverlappedtest  
( )
```

Purpose:\n Test for checking proper objects initialization

Requirements FTCP-12402

Precondition:\n

Setting up cluster stationary cluster with certain position

Action:\n

Run Try_To_Init_Obj_Track function

Result:\n

Checking if results matching the expectations

2.15.2.3 f360_try_to_init_obj_track_qualtest_HostOverlapCheckPosLongNOTOverlapedPosLatOverlappedtest()

```
f360_try_to_init_obj_track_qualtest_HostOverlapCheckPosLongNOTOverlapedPosLatOverlappedtest (
)
```

Purpose:\n Test for checking proper objects initialization

Requirements FTCP-12402

Precondition:\n

Setting up cluster stationary cluster with certain position

Action:\n

Run Try_To_Init_Obj_Track function

Result:\n

Checking if results matching the expectations

2.15.2.4 f360_try_to_init_obj_track_qualtest_HostOverlapCheckPosLongOverlappedInRearAxleDistPosLatNOTOverlappedtest()

```
f360_try_to_init_obj_track_qualtest_HostOverlapCheckPosLongOverlappedInRearAxleDistPosLatNOT↔
Overlappedtest ( )
```

Purpose:\n Test for checking proper objects initialization

Requirements FTCP-12402

Precondition:\n

Setting up cluster stationary cluster with certain position

Action:\n

Run Try_To_Init_Obj_Track function

Result:\n

Checking if results matching the expectations

2.15.2.5 f360_try_to_init_obj_track_qualtest_HostOverlapCheckPosLongOverlappedInRearAxleDistPosLatOverlappedtest

```
f360_try_to_init_obj_track_qualtest_HostOverlapCheckPosLongOverlappedInRearAxleDistPosLat↔  
Overlappedtest ( )
```

Purpose: Test for checking proper objects initialization

Requirements FTCP-12402

Precondition:

Setting up cluster stationary cluster with certain position

Action:

Run Try_To_Init_Obj_Track function

Result:

Checking if results matching the expectations

2.15.2.6 f360_try_to_init_obj_track_qualtest_LiberalInitializationTest()

```
f360_try_to_init_obj_track_qualtest_LiberalInitializationTest ( )
```

Purpose: Test for checking proper objects initialization

Requirements FTCP-12402

Precondition:

Setting up host properties and creating single cluster with 4 detections.

Action:

Run Try_To_Init_Obj_Track function

Result:

Checking if results matching the expectations

2.16 Guardrail_detection

Functions

- test group [guardrail_detection](#) ()
- [guardrail_detection_capability](#) ()
- [guardrail_detection_no_guardrail_sanity_check](#) ()

2.16.1 Detailed Description

2.16.2 Function Documentation

2.16.2.1 [guardrail_detection](#)()

```
test_group guardrail_detection ( )
```

Setup:\n

Initialize calibrations Set up a sensor with id 1 with mounting location forward right Set up a static environment polynomial (sep) that represents the guardrail, spanning from -20m to 20m longitudinally with a lateral position of 10m.

2.16.2.2 [guardrail_detection_capability](#)()

```
guardrail_detection_capability ( )
```

Purpose:\n Verify that the tracker can find and flag detections that are likely to originate from a guardrail.

Requirements FTCP-8839

Precondition:\n

In test group setup:

- calibrations have been initialized.
- A sensor has been set up with id 1 with mounting location forward right
- A Static Environment Polynomial (SEP) that represents the guardrail has been set up. The sep spans from -20m to 20m longitudinally with a lateral position of 10m. Create a detection that is located on the SEP.

Action:\n

Call [Internal_Preprocessing](#)()

Result:\n

Detection is likely to originate from guardrail, check that it is flagged as on SEP.

2.16.2.3 guardrail_detection_no_guardrail_sanity_check()

```
guardrail_detection_no_guardrail_sanity_check ( )
```

Purpose:\n Verify that not all detections are marked as on guardrail

Requirements FTCP-8839

Action:\n

Call Internal_Preprocessing()

Result:\n

Detection should not be marked as guardrail.

2.17 F360_detect_wheelspin_pairs

Functions

- test group [f360_detect_wheelspin_pairs_qualtest_unittest](#) ()

Test group including all input to Detect_Wheel_Spin_Pairs and a helper file to add detections used for testing. It also includes some constants located in Detect_Wheel_Spin_Pairs that could be used to quickly adapt tests if the internal constants change.

- [f360_detect_wheelspin_pairs_qualtest_unittest_doppler_spread_limits](#) ()
- [f360_detect_wheelspin_pairs_qualtest_unittest_azimuth_spread_limits](#) ()
- [f360_detect_wheelspin_pairs_qualtest_unittest_long_spread_limits](#) ()
- [f360_detect_wheelspin_pairs_qualtest_unittest_lat_spread_limits](#) ()
- [f360_detect_wheelspin_pairs_qualtest_unittest_long_limits](#) ()
- [f360_detect_wheelspin_pairs_qualtest_unittest_lat_limits](#) ()
- [f360_detect_wheelspin_pairs_qualtest_unittest_remarking](#) ()
- [f360_detect_wheelspin_pairs_qualtest_unittest_multiple_candidates](#) ()
- [f360_detect_wheelspin_pairs_qualtest_unittest_proximity_detections](#) ()
- [f360_detect_wheelspin_pairs_qualtest_unittest_max_wheelspin_pairs](#) ()

2.17.1 Detailed Description

2.17.2 Function Documentation

2.17.2.1 f360_detect_wheelspin_pairs_qualtest_unittest()

```
test_group f360_detect_wheelspin_pairs_qualtest_unittest ( )
```

Test group including all input to Detect_Wheel_Spin_Pairs and a helper file to add detections used for testing. It also includes some constants located in Detect_Wheel_Spin_Pairs that could be used to quickly adapt tests if the internal constants change.

Setup:\n

Nothing to setup in this test group

Teardown:\n

Nothing to teardown in this test group

2.17.2.2 f360_detect_wheelspin_pairs_qualtest_unittest_azimuth_spread_limits()

```
f360_detect_wheelspin_pairs_qualtest_unittest_azimuth_spread_limits ( )
```

Purpose: Test that check that the algorithm evaluated the azimuth difference between pairs properly.

Requirements FTCP-8840

Precondition:

Setup three pairs of detection separated in space. The pairs should have azimuth difference under, on and above the limit for marking wheelspin respectively.

Action:

Run the wheelspin pair detector as defined by the test group

Result:

Check that the pairs with azimuth spread under the limit get marked as wheelspin but the pairs with spread and on and above the limit does not.

2.17.2.3 f360_detect_wheelspin_pairs_qualtest_unittest_doppler_spread_limits()

```
f360_detect_wheelspin_pairs_qualtest_unittest_doppler_spread_limits ( )
```

Purpose: Test that check that the algorithm evaluated the spread in range rate of a detection pair properly.

Requirements FTCP-8840

Precondition:

Setup three pairs of detection separated in space. The pairs should have range rate spread under, on and above the limit for marking wheelspin respectively.

Action:

Run the wheelspin pair detector as defined by the test group

Result:

Check that the pair with range rate spread under the limit is not marked as wheelspin but the other two pairs are.

2.17.2.4 f360_detect_wheelspin_pairs_qualtest_unittest_lat_limits()

```
f360_detect_wheelspin_pairs_qualtest_unittest_lat_limits ( )
```

Purpose: Test that check that the algorithm evaluated the absolute lat position of detections properly.

Requirements FTCP-8840

Precondition:

Setup three pairs with high enough doppler spread to be wheelspin pair but have lat pos of the detections under, on and above the limit for when to consider wheelspin.

Action:

Run the wheelspin pair detector as defined by the test group

Result:

Verify that the lat pos limit is inclusive so that the detections under and on the limit get marked as wheelspin but those above don't.

2.17.2.5 f360_detect_wheelspin_pairs_qualtest_unittest_lat_spread_limits()

```
f360_detect_wheelspin_pairs_qualtest_unittest_lat_spread_limits ( )
```

Purpose: Test that check that the algorithm evaluated the spread in lat position of a detection pair properly.

Requirements FTCP-8840

Precondition:

Setup two pairs with high enough doppler spread to be wheelspin pair but have lat pos spread under and above the limit.

Action:

Run the wheelspin pair detector as defined by the test group

Result:

Verify that the pair with lat spread below the limit get marked but the one with spread above don't.

2.17.2.6 f360_detect_wheelspin_pairs_qualtest_unittest_long_limits()

```
f360_detect_wheelspin_pairs_qualtest_unittest_long_limits ( )
```

Purpose: Test that check that the algorithm evaluated the absolute long position of detections properly.

Requirements FTCP-8840

Precondition:

Setup three pairs with high enough doppler spread to be wheelspin pair but have long pos of the detections under, on and above the limit for when to consider wheelspin.

Action:

Run the wheelspin pair detector as defined by the test group

Result:

Verify that the long pos limit is inclusive so that the detections under and on the limit get marked as wheelspin but those above don't.

2.17.2.7 f360_detect_wheelspin_pairs_qualtest_unittest_long_spread_limits()

```
f360_detect_wheelspin_pairs_qualtest_unittest_long_spread_limits ( )
```

Purpose: Test that check that the algorithm evaluated the spread in long position of a detection pair properly.

Requirements FTCP-8840

Precondition:

Setup two pairs with high enough doppler spread to be wheelspin pair but have long pos spread under and above the limit.

Action:

Run the wheelspin pair detector as defined by the test group

Result:

Verify that the pair with long spread below the limit get marked but the one with spread above don't.

2.17.2.8 f360_detect_wheelspin_pairs_qualtest_unittest_max_wheelspin_pairs()

```
f360_detect_wheelspin_pairs_qualtest_unittest_max_wheelspin_pairs ( )
```

Purpose:\n Test that no more than the maximum specified number of wheelspin pairs can be marked.

Requirements FTCP-8840

Precondition:\n

Setup possible more possible wheelspin pairs than the internal structures of the algorithm allows and verify that no more than the maximum allowed wheelspin pairs gets marked in the end.

Action:\n

Run the wheelspin pair detector as defined by the test group

Result:\n

Verify that no more than the maximum allowed pairs get marked.

2.17.2.9 f360_detect_wheelspin_pairs_qualtest_unittest_multiple_candidates()

```
f360_detect_wheelspin_pairs_qualtest_unittest_multiple_candidates ( )
```

Purpose:\n Test that verifies the behavior of the algorithm when we have multiple possible candidates for wheelspin pairs.

Requirements FTCP-8840

Precondition:\n

Setup three detections that could all form a detection pair with each other. Also place detections between them so that they could not be marked as wheelspin by being close to another formed pair.

Action:\n

Run the wheelspin pair detector as defined by the test group

Result:\n

Verify that the detections in the pair with the highest doppler spread is marked as wheelspin while the other is not marked at all.

2.17.2.10 f360_detect_wheelspin_pairs_qualtest_unittest_proximity_detections()

```
f360_detect_wheelspin_pairs_qualtest_unittest_proximity_detections ( )
```

Purpose: Test the marking of detections close to detection pairs.

Requirements FTCP-8840

Precondition:

Setup two detections that could form a wheelspin pair but add other detections that could not form pairs between them in the sorted detection props list. Some fill the requirements to be considered close to the wheelspin pair some don't.

Action:

Run the wheelspin pair detector as defined by the test group

Result:

Verify that all detections that are either in the wheelspin pair or considered close to a pair are marked as wheelspin but that all others don't.

2.17.2.11 f360_detect_wheelspin_pairs_qualtest_unittest_remarking()

```
f360_detect_wheelspin_pairs_qualtest_unittest_remarking ( )
```

Purpose: Test that checks that only detection that have not already been marked as wheelspin is considered

Purpose: in the algorithm.

Requirements FTCP-8840

Precondition:

Setup three detection pairs. The first two should have high enough doppler spread to be marked but have one of the two detections in the pair already marked as wheelspin. The third pair have too low doppler spread to be marked but both detections have already been marked.

Action:

Run the wheelspin pair detector as defined by the test group

Result:

Verify that no detection status is changed for any of the detections in the pairs

2.18 F360_determine_dets_on_edge_of_fov

Functions

- test group [f360_determine_dets_on_edge_of_fov](#) ()

Test group that sets up two sensors with located in origo. One aligned with the x-axis and one aligned with the y-axis. Their MR FOV is 90 degree so they overlap with 45 degrees while their LR FOV is 45 degrees so they align with their respective borders. The test group also contains helper functions to add detections to the respective sensors.

- [f360_determine_dets_on_edge_of_fov_mr_detections](#) ()
- [f360_determine_dets_on_edge_of_fov_lr_detections](#) ()
- [f360_determine_dets_on_edge_of_fov_range_limits](#) ()

2.18.1 Detailed Description

2.18.2 Function Documentation

2.18.2.1 f360_determine_dets_on_edge_of_fov()

```
test group f360_determine_dets_on_edge_of_fov ( )
```

Test group that sets up two sensors with located in origo. One aligned with the x-axis and one aligned with the y-axis. Their MR FOV is 90 degree so they overlap with 45 degrees while their LR FOV is 45 degrees so they align with their respective borders. The test group also contains helper functions to add detections to the respective sensors.

Setup:

Creates two sensors in origo. One aligned with the x-axis and the other one with the y-axis. Both have 90 and 45 degree FOV for the MR and LR looks respectively.

2.18.2.2 f360_determine_dets_on_edge_of_fov_lr_detections()

```
f360_determine_dets_on_edge_of_fov_lr_detections ( )
```

Purpose: Test different detection position inside/outside and border of FOV for the two sensors if the sensor

Purpose: with detections have look id corresponding to LR look.

Requirements FTCP-8838

Precondition:

Set the look id to LR on the sensor that should measure all detections together with a bunch of detections that will test inside, outside and border conditions.

Action:

Run the action defined by the test group.

Result:

Check that the flag of all detections have correct value. The detection outside owner FOV and inside the LR or MR FOV of the other sensor should be true.

2.18.2.3 f360_determine_dets_on_edge_of_fov_mr_detections()

```
f360_determine_dets_on_edge_of_fov_mr_detections ( )
```

Purpose: Test different detection position inside/outside and border of FOV for the two sensors if the sensor

Purpose: with detections have look id corresponding to MR look.

Requirements FTCP-8838**Precondition:**

Set the look id to MR on the sensor that should measure all detections together with a bunch of detections that will test inside, outside and border conditions.

Action:

Run the action defined by the test group.

Result:

Check that the flag of all detetions have correct value. Only the detection outside the owner sensors FOV but inside the other sensors MR FOV should be true. All other includeing border cases should be false.

2.18.2.4 f360_determine_dets_on_edge_of_fov_range_limits()

f360_determine_dets_on_edge_of_fov_range_limits ()

Purpose: Test that the range limits of the look type its used to exclude matches correctly.

Requirements FTCP-8838

Precondition:

Set the look id of the sensor to LR so that only the more narrow FOV is valid. Define two angles of detections that will be outside the ownser FOV but inside either both look types or the MR look type of the other sensor. Then setup detections with these two angles that also test inside, outside and border of that look types range limit.

Action:

Run the action defined by the test group.

Result:

Check that the correct flag have been set. Only reanges inside or on the border of the range limits should be true.

2.19 F360_handle_host_vehicle_clutter_qualtest

Functions

- test group `f360_handle_host_vehicle_clutter_qualtest` ()
Test group of HVC detections qualification tests. Tests verify whether requirements are fulfilled.
- `f360_handle_host_vehicle_clutter_qualtest_handle_host_vehicle_clutter__dets_marked_as_not_okay_to_use` ()
- `f360_handle_host_vehicle_clutter_qualtest_handle_host_vehicle_clutter__dets_marked_as_okay_to_use` ()
- `f360_handle_host_vehicle_clutter_qualtest_handle_host_vehicle_clutter__hvc_and_behind_guardrail_dets_marked_as_not_o` ()
- `f360_handle_host_vehicle_clutter_qualtest_handle_host_vehicle_clutter__hvc_and_on_guardrail_dets_marked_as_not_okay` ()

2.19.1 Detailed Description

2.19.2 Function Documentation

2.19.2.1 f360_handle_host_vehicle_clutter_qualtest()

```
test_group f360_handle_host_vehicle_clutter_qualtest ( )
```

Test group of HVC detections qualification tests. Tests verify whether requirements are fulfilled.

Setup:\n

Initialize tracker calibrations Set selected detection parameters Set sensors parameters

2.19.2.2 f360_handle_host_vehicle_clutter_qualtest_handle_host_vehicle_clutter__dets_marked_as_not_okay_to_use()

```
f360_handle_host_vehicle_clutter_qualtest_handle_host_vehicle_clutter__dets_marked_as_not_okay_to_use ( )
```

Purpose:\n

Purpose of this test is to verify whether detections inside HVC TP area are marked as not okay to use

Requirements FTCP-10043

Precondition:\n

Set detection position to be placed inside HVC TP area

Action:\n

Call tested function

Result:\n

Check whether detection `f_ok_to_use` flag was set to false

2.19.2.3 f360_handle_host_vehicle_clutter_qualtest_handle_host_vehicle_clutter__dets_marked_as_okay_to_use()

```
f360_handle_host_vehicle_clutter_qualtest_handle_host_vehicle_clutter__dets_marked_as_okay_to_use ↔  
to_use ( )
```

Purpose:\n

Purpose of this test is to verify whether detections outside HVC TP zone, not on and not behind guardrail do not have their f_ok_to_use flag changed

Requirements FTCP-10043

Precondition:\n

Set detection position to be placed outside HVC TP area Set detection f_ok_to_use flag as true

Action:\n

Call tested function

Result:\n

Check whether detection f_ok_to_use flag was not changed

2.19.2.4 f360_handle_host_vehicle_clutter_qualtest_handle_host_vehicle_clutter__hvc_and_behind_guardrail_dets_marked_as_not_okay_to_use()

```
f360_handle_host_vehicle_clutter_qualtest_handle_host_vehicle_clutter__hvc_and_behind_guardrail ↔  
_dets_marked_as_not_okay_to_use ( )
```

Purpose:\n

Purpose of this test is to verify whether detections that are behind guardrail and HVC are marked as not okay to use

Requirements FTCP-10043

Precondition:\n

Set detection position to be placed inside HVC TP area

Action:\n

Call tested function

Result:\n

Check whether detection f_ok_to_use flag was set to false

2.19.2.5 f360_handle_host_vehicle_clutter_qualtest_handle_host_vehicle_clutter__hvc_and_on_guardrail_dets_marked_a

```
f360_handle_host_vehicle_clutter_qualtest_handle_host_vehicle_clutter__hvc_and_on_guardrail_↵  
dets_marked_as_not_okay_to_use ( )
```

Purpose:\n

Purpose of this test is to verify whether detections that are on guardrail and HVC are marked as not okay to use

Requirements FTCP-10043**Precondition:**\n

Set detection position to be placed inside HVC TP area

Action:\n

Call tested function

Result:\n

Check whether detection f_ok_to_use flag was set to false

2.20 F360_mark_water_spray_detections

Functions

- test group [f360_mark_water_spray_detections](#) ()
This test group contains tests for Mark_Water_Spray_Detections function.
- [f360_mark_water_spray_detections_default_test](#) ()
- [f360_mark_water_spray_detections_detections_outside_vcs_zone](#) ()
- [f360_mark_water_spray_detections_low_host_speed](#) ()
- [f360_mark_water_spray_detections_f_ok_to_use_false](#) ()
- [f360_mark_water_spray_detections_det_high_rcs](#) ()
- [f360_mark_water_spray_detections_det_range_rate](#) ()

2.20.1 Detailed Description

2.20.2 Function Documentation

2.20.2.1 f360_mark_water_spray_detections()

```
test group f360_mark_water_spray_detections ( )
```

This test group contains tests for Mark_Water_Spray_Detections function.

Setup:\n

Default setup will change f_water_spray for all detections to true after function call

2.20.2.2 f360_mark_water_spray_detections_default_test()

```
f360_mark_water_spray_detections_default_test ( )
```

Purpose:\n

Verify that detections have been flagged as water spray correctly with default preconditions

Requirements FTCP-8844

Precondition:\n

Use default settings

Action:\n

Execute the function

Result:\n

Check that f_water_spray is set to true for all detections

2.20.2.3 f360_mark_water_spray_detections_det_high_rcs()

```
f360_mark_water_spray_detections_det_high_rcs ( )
```

Purpose:\n Test behaviour of host water spray detector for detection with high rcs

Requirements FTCP-8844

Precondition:\n

Set rcs above limit for det 1

Action:\n

Execute the function

Result:\n

Check that f_water_spray is set to false only for det 1

2.20.2.4 f360_mark_water_spray_detections_det_range_rate()

```
f360_mark_water_spray_detections_det_range_rate ( )
```

Purpose:\n Test behaviour of host water spray detector with det range rate outside of limits

Requirements FTCP-8844

Precondition:\n

Set range rate below limit for det 2 and above limit for det 3

Action:\n

Execute the function

Result:\n

Check that f_water_spray is set to false only for det 2 and det 3

2.20.2.5 f360_mark_water_spray_detections_detections_outside_vcs_zone()

```
f360_mark_water_spray_detections_detections_outside_vcs_zone ( )
```

Purpose:\n

Test that detections that are outside host water spray zone are correctly flagged as not being water spray detections

Requirements FTCP-8844

Precondition:\n

- Detection 0 is placed in front of host
- Detection 1 is placed to the to the left of host water spray zone
- Detection 2 is placed to the to the right of host water spray zone
- Detection 3 is placed behind of host water spray zone

Action:\n

Execute the function

Result:\n

Check that f_water_spray is set to false for the first 4 detections

2.20.2.6 f360_mark_water_spray_detections_f_ok_to_use_false()

```
f360_mark_water_spray_detections_f_ok_to_use_false ( )
```

Purpose:\n Test behavior of host water spray detector when f_ok_to_use is set to false for one detection

Requirements FTCP-8844

Precondition:\n

Set f_ok_to_use to false for det 0

Action:\n

Execute the function

Result:\n

Check that f_water_spray is set to false only for det 0

2.20.2.7 f360_mark_water_spray_detections_low_host_speed()

f360_mark_water_spray_detections_low_host_speed ()

Purpose:

Verify that detections have been correctly flagged as not water spray detections when host speed is below limit

Requirements FTCP-8844**Precondition:**

Set host speed below speed limit

Action:

Execute the function

Result:

Check that f_water_spray is false for all detections

2.21 F360_msmt_update_obj_trks_cca_non_moveable_qualtest

Functions

- test group `f360_msmt_update_obj_trks_cca_non_moveable_qualtest` ()
Setup general values on object states and detection properties that is needed to perform a measurement update.
- `f360_msmt_update_obj_trks_cca_non_moveable_qualtest_Msmt_Update_Obj_Trks_CCA_Non_Moveable` ()

2.21.1 Detailed Description

2.21.2 Function Documentation

2.21.2.1 f360_msmt_update_obj_trks_cca_non_moveable_qualtest()

```
test_group f360_msmt_update_obj_trks_cca_non_moveable_qualtest ( )
```

Setup general values on object states and detection properties that is needed to perform a measurement update.

Setup:\n

Setup general values for object and detection states.

2.21.2.2 f360_msmt_update_obj_trks_cca_non_moveable_qualtest_Msmt_Update_Obj_Trks_CCA_Non_Moveable()

```
f360_msmt_update_obj_trks_cca_non_moveable_qualtest_Msmt_Update_Obj_Trks_CCA_Non_Moveable ( )
```

Purpose:\n

Verify that object states are updated when 1 eligible detection is associated

Requirements FTCP-11426 FTCP-11024

Precondition:\n

Set number of selected detections for msmt update to 1. Set object to not moving Copy states of object before function is called.

Action:\n

Call function

Result:\n

Verify that object heading and pointing haven't changed Verify other states have changed. Verify that object vcs_position and bbox center are still equal.

Result:\n

Verify that object states have changed

2.22 F360_msmt_update_obj_trks_cca_moveable_qualtest

Functions

- test group `f360_msmt_update_obj_trks_cca_moveable_qualtest` ()
Testing that a call to `Msmt_Update_Obj_Trks_CCA()` with one eligible detection is updating objects states.
- `f360_msmt_update_obj_trks_cca_moveable_qualtest_Msmt_Update_Obj_Trks_CCA_Moveable_Check_Update_One_Det_Obj` ()
- `f360_msmt_update_obj_trks_cca_moveable_qualtest_Msmt_Update_Obj_Trks_CCA_Moveable_Check_Update_One_Det_Obj` ()
- `f360_msmt_update_obj_trks_cca_moveable_qualtest_Msmt_Update_Obj_Trks_CCA_Moveable_Check_Head_Point_Not_Updated` ()

2.22.1 Detailed Description

2.22.2 Function Documentation

2.22.2.1 f360_msmt_update_obj_trks_cca_moveable_qualtest()

```
test_group f360_msmt_update_obj_trks_cca_moveable_qualtest ( )
```

Testing that a call to `Msmt_Update_Obj_Trks_CCA()` with one eligible detection is updating objects states.

Setup:\n

Setup general values for object and detection states. Object:

- filter type: CCA
- vcs position: [10, 5] [m]
- vcs velocity: [5, 0.1] [m/s]
- vcs acceleration: [1.7, -0.6] [m/s^2]
- state error covariance: random covariance matrix, [6.516902025806293, 0.194512490773417, -0.038455610730064, -0.385033075453222, 0.128525832942101 0.093852564482656; 0.194512490773417, 6.663822593242870, 0.104355463057540, 0.166129709039925, 0.427907717373890, -0.184096722846899; -0.038455610730064, 0.104355463057540, 6.571751616753093, -0.046312466759188, -0.047463652995767, 0.405748620499433; -0.385033075453222, 0.166129709039925, -0.046312466759188, 6.948385410472763, -0.072459134186868, -0.409329684004222; 0.128525832942101, 0.427907717373890, -0.047463652995767, -0.072459134186868, 6.449236517036669, -0.276834993063314; 0.093852564482656, -0.184096722846899, 0.405748620499433, -0.409329684004222, -0.276834993063314, 6.957360618835150] is used in this test
- pseudo vcs position: `vcs_position + [0.1, -0.1] [m]`
- meascov: random covariance matrix, [1, 0.5; 0.5, 1] is used in the test
- number of associated detections: 1
- detection id of associated detection: 23
- `f_moving`: true
- object box orientation: 10deg
- object heading rate: -0.1rad/s

- object `cca_pnt_filter_cov`: random covariance matrix, $[1, 0.5; 0.5, 2]$ is used in the test. Setup selected detections for measurement update
- the single detection with id 23 which is associated to the object Setup one detection:
- id 23
- vcs azimuth: 5deg
- cos vcs az: $\cos(5\text{deg})$
- sin vcs az: $\sin(5\text{deg})$
- vcs position: $[10, 0.5]$ [m]
- compensated range rate: 5.1m/s

2.22.2.2 f360_msmt_update_obj_trks_cca_moveable_qualtest_Msmt_Update_Obj_Trks_CCA_Moveable_Check_Head_Poin

```
f360_msmt_update_obj_trks_cca_moveable_qualtest_Msmt_Update_Obj_Trks_CCA_Moveable_Check_Head↔
_Point_Not_Updated_Due_To_Not_Moving ( )
```

Purpose:\n

Verify that object states are updated when 1 eligible detection is associated to a CCA object and object is not moving (i.e object bbox orientation should not also be updated)

Requirements FTCP-11426 FTCP-11024

Precondition:\n

Use default test data from test group except for:

- set `f_moving` to false

Action:\n

Copy the object before function call so that we can later make comparisons Call function

Result:\n

Verify that object heading and pointing haven't changed and that heading rate has been set to 0 Verify other states have changed

2.22.2.3 f360_msmt_update_obj_trks_cca_moveable_qualtest_Msmt_Update_Obj_Trks_CCA_Moveable_Check_Update_O

```
f360_msmt_update_obj_trks_cca_moveable_qualtest_Msmt_Update_Obj_Trks_CCA_Moveable_Check_↔
Update_One_Det_Obj_Moving ( )
```

Purpose:\n

Verify that object states are updated when 1 eligible detection is associated to a CCA object and object is moving (i.e object bbox orientation should also be updated)

Requirements FTCP-11024

Precondition:\n

Use default test data from test group

Action:\n

Copy the object before function call so that we can later make comparisons Call function

Result:\n

Verify that object states have changed

2.22.2.4 f360_msmt_update_obj_trks_cca_moveable_qualtest_Msmt_Update_Obj_Trks_CCA_Moveable_Check_Update_O

```
f360_msmt_update_obj_trks_cca_moveable_qualtest_Msmt_Update_Obj_Trks_CCA_Moveable_Check_↔
Update_One_Det_Obj_Moving_SpeedLow ( )
```

Purpose:\n

Verify that object heading_rate and curvature is set to 0 when 1 eligible detection is associated to a CCA object and object is moving slow.

Requirements FTCP-11024

Precondition:\n

calib.k_cca_min_speed_to_update_pnt = 10.0F is used this test

Action:\n

Copy the object before function call so that we can later make comparisons Call function

Result:\n

Verify that object heading and pointing haven't changed and that heading rate has been set to 0 Verify other states have changed

2.23 F360_msmt_update_obj_trks_ctca_qualtest

Functions

- test group [f360_msmt_update_obj_trks_ctca_qualtest](#) ()
Setup general values on object states and detection properties that is needed to perform a measurement update.
- [f360_msmt_update_obj_trks_ctca_qualtest_Msmt_Update_Obj_Trks_CTCA_Check_Update_One_Det](#) ()

2.23.1 Detailed Description

2.23.2 Function Documentation

2.23.2.1 f360_msmt_update_obj_trks_ctca_qualtest()

```
test group f360_msmt_update_obj_trks_ctca_qualtest ( )
```

Setup general values on object states and detection properties that is needed to perform a measurement update.

Setup:\n

Setup general values for object and detection states.

2.23.2.2 f360_msmt_update_obj_trks_ctca_qualtest_Msmt_Update_Obj_Trks_CTCA_Check_Update_One_Det()

```
f360_msmt_update_obj_trks_ctca_qualtest_Msmt_Update_Obj_Trks_CTCA_Check_Update_One_Det ( )
```

Purpose:\n

Verify object that states are updated when 1 eligible detection is associated

Requirements FTCP-11024, FTCP-12609

Precondition:\n

Set one detection as eligible for measurement update Create copy of object before call to measurement update

Action:\n

Call function

Result:\n

Verify that object was measurement updated, i.e. that object states and covariance matrix differ after update

2.24 F360_msmt_update_obj_trks_ctca_not_moving_qualtest

Functions

- test group [f360_msmt_update_obj_trks_ctca_not_moving_qualtest](#) ()
Test group for tests related to freezing of heading and curvature for CTCA tracks.
- [f360_msmt_update_obj_trks_ctca_not_moving_qualtest_Msmt_Update_ObjTrks_CTCA__curvature_head_is_not_updated_if](#) ()

2.24.1 Detailed Description

2.24.2 Function Documentation

2.24.2.1 f360_msmt_update_obj_trks_ctca_not_moving_qualtest()

```
test group f360_msmt_update_obj_trks_ctca_not_moving_qualtest ( )
```

Test group for tests related to freezing of heading and curvature for CTCA tracks.

Setup:\n

Initialize tracker calibrations Set up track parameters to be placed on left side of host, close to y vcs axis. Set up parameters of two detections to make valid measurment

2.24.2.2 f360_msmt_update_obj_trks_ctca_not_moving_qualtest_Msmt_Update_ObjTrks_CTCA__curvature_head_is_not

```
f360_msmt_update_obj_trks_ctca_not_moving_qualtest_Msmt_Update_ObjTrks_CTCA__curvature_head_↔  
is_not_updated_if_movable_and_speed_too_low ( )
```

Purpose:\n

Purpose of this test is to verify that when object is moveable but its speed is too low its curvature and heading is not updated

Requirements FTCP-11424 FTCP-11024 FTCP-12609

Precondition:\n

Set track as movable Set its speed below threshold Initialize track to make measurement update step return valid numbers

Action:\n

Call tested function.

Result:\n

Check that curvature, heading and pointing of track was not changed. Check that all other states were updated

2.25 F360_msmt_update_object_timestamp_qualtest

Functions

- test group `f360_msmt_update_object_timestamp_qualtest` ()
Create needed default variables needed for update of object measurement timestamp.
- `f360_msmt_update_object_timestamp_qualtest_Msmt_Update_Object_Timestamp_One_Det` ()

2.25.1 Detailed Description

2.25.2 Function Documentation

2.25.2.1 f360_msmt_update_object_timestamp_qualtest()

```
test_group f360_msmt_update_object_timestamp_qualtest ( )
```

Create needed default variables needed for update of object measurement timestamp.

Setup:\n

Setup default values for easy testing between multiple test cases

2.25.2.2 f360_msmt_update_object_timestamp_qualtest_Msmt_Update_Object_Timestamp_One_Det()

```
f360_msmt_update_object_timestamp_qualtest_Msmt_Update_Object_Timestamp_One_Det ( )
```

Purpose:\n

Verify that object measurement timestamp is equal to detection timestamp when only one detection is used for update.

Requirements FTCP-11030

Action:\n

Call function

Result:\n

Check that object timestamp is equal to the only detection used in measurement update

2.26 F360_msmt_update_supporting_functions_Select_Dets_For_RR_Update_qualtest

Functions

- test group [f360_msmt_update_supporting_functions_Select_Dets_For_RR_Update_qualtest](#) ()
Purpose is to very that function Select_Dets_For_RR_Update works as intended. Initialize an object and assign some detections. Assign detection properties to cover all possible branches inside the function.
- [f360_msmt_update_supporting_functions_Select_Dets_For_RR_Update_qualtest_Test_Select_Dets_For_RR_Update](#) ()

2.26.1 Detailed Description

2.26.2 Function Documentation

2.26.2.1 f360_msmt_update_supporting_functions_Select_Dets_For_RR_Update_qualtest()

```
test_group f360_msmt_update_supporting_functions_Select_Dets_For_RR_Update_qualtest ( )
```

Purpose is to very that function Select_Dets_For_RR_Update works as intended. Initialize an object and assign some detections. Assign detection properties to cover all possible branches inside the function.

Setup:\n

Create an object, detections and return parameters along with expected data.

Setup:\n

Create detections to cover all possible branches

2.26.2.2 f360_msmt_update_supporting_functions_Select_Dets_For_RR_Update_qualtest_Test_Select_Dets_For_RR_Update

```
f360_msmt_update_supporting_functions_Select_Dets_For_RR_Update_qualtest_Test_Select_Dets_For_RR_Update ( )
```

Action:\n

Call the function Select_Dets_For_RR_Update

Result:\n

Check that function have selected correct detections

2.27 F360_update_object_track_properties_Obj_Size

Functions

- test group [f360_update_object_track_properties_Calc_Obj_Size](#) ()
Group of tests to check that object length and width are updated under correct circumstances.
- [f360_update_object_track_properties_Calc_Obj_Size_Obj_Size_Is_Updated](#) ()
- [f360_update_object_track_properties_Calc_Obj_Size_Obj_Size_Is_Not_Updated_Obj_Coasting](#) ()
- [f360_update_object_track_properties_Calc_Obj_Size_Obj_Size_Is_Not_Updated_Obj_One_Assoc_Det](#) ()

2.27.1 Detailed Description

2.27.2 Function Documentation

2.27.2.1 f360_update_object_track_properties_Calc_Obj_Size()

```
test_group f360_update_object_track_properties_Calc_Obj_Size ( )
```

Group of tests to check that object length and width are updated under correct circumstances.

Setup:\n

Set up an object with some detections

Setup:\n

Set up a general scenario with calibrations, one object with one detection on each side of the object and relevant tracker info parameters.

2.27.2.2 f360_update_object_track_properties_Calc_Obj_Size_Obj_Size_Is_Not_Updated_Obj_Coasting()

```
f360_update_object_track_properties_Calc_Obj_Size_Obj_Size_Is_Not_Updated_Obj_Coasting ( )
```

Purpose:\n

Test that object size is not updated when object has status coasted and 2 associated detections

Requirements FTCP-13529

Precondition:\n

An object and two detections have been set up Assign objects initial length and width to temporary variables

Action:\n

Call Update_Object_Track_Properties()

Result:\n

Check that object dimensions have not been updated

2.27.2.3 f360_update_object_track_properties_Calc_Obj_Size_Obj_Size_Is_Not_Updated_Obj_One_Assoc_Det()

```
f360_update_object_track_properties_Calc_Obj_Size_Obj_Size_Is_Not_Updated_Obj_One_Assoc_Det (
)
```

Purpose:\n

Test that object size is not updated when object has status updated but only 1 associated detection

Requirements FTCP-13529

Precondition:\n

An object and two detections have been set up Assign objects initial length and width to temporary variables

Action:\n

Call Update_Object_Track_Properties()

Result:\n

Check that object dimensions have been updated

2.27.2.4 f360_update_object_track_properties_Calc_Obj_Size_Obj_Size_Is_Updated()

```
f360_update_object_track_properties_Calc_Obj_Size_Obj_Size_Is_Updated ( )
```

Purpose:\n

Test that object size is updated when object has status updated and 2 associated detections

Requirements FTCP-13529

Precondition:\n

An object and two detections have been set up Assign objects initial length and width to temporary variables

Action:\n

Call Update_Object_Track_Properties()

Result:\n

Check that object dimensions have been updated

2.28 F360_object_track_management_qualtest

Functions

- test group [f360_object_track_management_qualtest](#) ()
Includes tests that will test the behavior of functions implemented in f360_object_track_management.cpp.
- [f360_object_track_management_qualtest_Obj_Trk_Status_Book_Keeping_NEW](#) ()
- [f360_object_track_management_qualtest_Obj_Trk_Status_Book_Keeping_UPDATED](#) ()

2.28.1 Detailed Description

2.28.2 Function Documentation

2.28.2.1 f360_object_track_management_qualtest()

```
test_group f360_object_track_management_qualtest ( )
```

Includes tests that will test the behavior of functions implemented in f360_object_track_management.cpp.

Setup:\n

Setting up common variables for test group f360_object_track_management.

2.28.2.2 f360_object_track_management_qualtest_Obj_Trk_Status_Book_Keeping_NEW()

```
f360_object_track_management_qualtest_Obj_Trk_Status_Book_Keeping_NEW ( )
```

Purpose:\n This test will test Obj_Trk_Status_Book_Keeping() when the status is NEW in some way.

Requirements FTCP-12595

Step 1:\n

Check that a NEW_UPDATED object which is set to COASTED also gets f_ghost_NU_2_C set to true.

Precondition:\n

Create initialized input. Set parameters so we will be able to enter wanted cases.

Action:\n

Call Obj_Trk_Status_Book_Keeping() which affects time_since_stage_start, time_since_track_updated, status and f_ghost_NU_2_C.

Result:\n

The status is expected to be COASTED, time_since_stage_start = 0, time_since_track_updated to be constant and f_ghost_NU_2_C to be true.

Step 2:\n

Check that a NEW_UPDATED object which receives valid detections is set to UPDATED.

Precondition:\n

Create initialized input. Set parameters so we will be able to enter wanted cases.

Action:\n

Call Obj_Trk_Status_Book_Keeping() which affects time_since_stage_start, time_since_track_updated, status and f_ghost_NU_2_C.

Result:\n

The status is expected to be UPDATED, time_since_stage_start to be constant, time_since_track_updated = 0 and f_ghost_NU_2_C to be false.

2.28.2.3 f360_object_track_management_qualtest_Obj_Trk_Status_Book_Keeping_UPDATED()

f360_object_track_management_qualtest_Obj_Trk_Status_Book_Keeping_UPDATED ()

Purpose:\n This test will test Obj_Trk_Status_Book_Keeping() when the status is F360_OBJECT_STATUS_UPDATED.

Requirements FTCP-12603

Step 1:\n

Check that an UPDATED object stays UPDATED when enough approved detections have been associated, i.e. when there are enough num_rr_inlier_dets (> 0 and > #wheelspin detections).

Precondition:\n

Create initialized input. Set parameters so we will be able to enter wanted cases.

Action:\n

Call `Obj_Trk_Status_Book_Keeping()` which affects `time_since_stage_start`, `time_since_track_updated`, `status` and `f_ghost_NU_2_C`.

Result:\n

The status is expected to stay as `UPDATED`, `time_since_stage_start` to be constant, `time_since_track_updated` = 0 and `f_ghost_NU_2_C` to be false.

Step 2:\n

Check that an `UPDATED` object is set to `COASTED` when there are no satisfying detections associated.

Precondition:\n

Create initialized input. Set parameters so we will be able to enter wanted cases.

Action:\n

Call `Obj_Trk_Status_Book_Keeping()` which affects `time_since_stage_start`, `time_since_track_updated`, `status` and `f_ghost_NU_2_C`.

Result:\n

The status is expected to be set as `COASTED`, `time_since_track_updated` to be constant, `time_since_stage_start` = 0 and `f_ghost_NU_2_C` to be false.

2.29 F360_adjust_filtr_type_dependet_params_qualtest

Functions

- test group [f360_adjust_filtr_type_dependet_params_qualtest](#) ()
- [f360_adjust_filtr_type_dependet_params_qualtest_Adjust_Filtr_Type_Dependent_Params__switch_to_CTCA](#) ()

2.29.1 Detailed Description

2.29.2 Function Documentation

2.29.2.1 f360_adjust_filtr_type_dependet_params_qualtest()

```
test_group f360_adjust_filtr_type_dependet_params_qualtest ( )
```

Setup:\n

2.29.2.2 f360_adjust_filtr_type_dependet_params_qualtest_Adjust_Filtr_Type_Dependent_Params__switch_to_CTCA()

```
f360_adjust_filtr_type_dependet_params_qualtest_Adjust_Filtr_Type_Dependent_Params__switch_to_↵  
CTCA ( )
```

Purpose:\nCorrect switching to CTCA (CCV before)

Requirements FTCP-12604

Precondition:\n

Action:\n

Result:\n

2.30 F360_cancel_new_updated_trk_overlapping_confirmed_trks_↔ qualtest

Functions

- test group `f360_cancel_new_updated_trk_overlapping_confirmed_trks_qualtest ()`
Tests that an objects is terminated correctly when overlapping with another.
- `f360_cancel_new_updated_trk_overlapping_confirmed_trks_qualtest_Cancel_New_Updated_Trk_Overlapping_Confirmed_Trk_Overlapping_Confirmed_Trks_Second_Obj_Coasted ()`
- `f360_cancel_new_updated_trk_overlapping_confirmed_trks_qualtest_Cancel_New_Updated_Trk_Overlapping_Confirmed_Trk_Overlapping_Confirmed_Trks_Second_Obj_Coasted ()`

2.30.1 Detailed Description

2.30.2 Function Documentation

2.30.2.1 f360_cancel_new_updated_trk_overlapping_confirmed_trks_qualtest()

```
test_group f360_cancel_new_updated_trk_overlapping_confirmed_trks_qualtest ( )
```

Tests that an objects is terminated correctly when overlapping with another.

Setup:\n

This tests sets up a scenario with two objects whose areas are overlapping.

2.30.2.2 f360_cancel_new_updated_trk_overlapping_confirmed_trks_qualtest_Cancel_New_Updated_Trk_Overlapping_Confirmed_Trks_Second_Obj_Coasted()

```
f360_cancel_new_updated_trk_overlapping_confirmed_trks_qualtest_Cancel_New_Updated_Trk_Overlapping_Confirmed_Trks_Second_Obj_Coasted ( )
```

Purpose:\n

Purpose of this test is to check that when two moving objects, one with status F360_OBJECT_STATUS_NEW_UPDATED and the other with F360_OBJECT_STATUS_COASTED, have bounding boxes with enough overlap, the object with status F360_OBJECT_STATUS_NEW_UPDATED is terminated, given that the object with status F360_OBJECT_STATUS_COASTED has a confidence level above 0.5 and time_since_cluster_created above 0.4 seconds.

Requirements FTCP-13165

Precondition:

- Two objects have been set up close to each other such that their bounding boxes overlap is > 75%.
- The first object's is moving and its status is set to F360_OBJECT_STATUS_NEW_UPDATED.
- The second object's is moving and its status is set to F360_OBJECT_STATUS_COASTED, it's confidence level and time_since_cluster_created above their respective thresholds.
- detection_posn_hist has been set up with some arbitrary properties for a slot reserved for the first object.

Action:

Call function

Result:

Verify that all object 1's properties that were assigned in TEST_SETUP are reset to default values and that the slot reserved for object 1 in detection_posn_hist has been reset.

2.30.2.3 f360_cancel_new_updated_trk_overlapping_confirmed_trks_qualtest_Cancel_New_Updated_Trk_Overlapping_Co

```
f360_cancel_new_updated_trk_overlapping_confirmed_trks_qualtest_Cancel_New_Updated_Trk_↵
Overlapping_Confirmed_Trks_Second_Obj_Updated ( )
```

Purpose:

Purpose of this test is to check that when two moving objects, one with status F360_OBJECT_STATUS_NEW_UPDATED and the other with F360_OBJECT_STATUS_UPDATED, have bounding boxes with enough overlap, the object with status F360_OBJECT_STATUS_NEW_UPDATED is terminated, given that the object with status F360_OBJECT_STATUS_UPDATED has a confidence level above 0.5 and time_since_cluster_created above 0.4 seconds.

Requirements FTCP-13165**Precondition:**

- Two objects have been set up close to each other such that their bounding boxes overlap is > 75%.
- The first object's is moving and its status is set to F360_OBJECT_STATUS_NEW_UPDATED.
- The second object's is moving and its status is set to F360_OBJECT_STATUS_UPDATED, it's confidence level and time_since_cluster_created above their respective thresholds.
- detection_posn_hist has been set up with some arbitrary properties for a slot reserved for the first object.

Action:

Call function

Result:

Verify that all object 1's properties that were assigned in TEST_SETUP are reset to default values and that the slot reserved for object 1 in detection_posn_hist has been reset.

2.31 F360_update_object_track_properties_obj_height

Functions

- test group [f360_update_object_track_properties_calc_obj_height](#) ()
Group of tests to check that objects over the ground height is updated under correct circumstances.
- [f360_update_object_track_properties_calc_obj_height_obj_height_is_updated_when_valid](#) ()
- [f360_update_object_track_properties_calc_obj_height_obj_height_is_not_calculated_when_stationary](#) ()
- [f360_update_object_track_properties_calc_obj_height_obj_height_decays_when_no_dets](#) ()
- [f360_update_object_track_properties_calc_obj_height_obj_height_decay_below_zero](#) ()

2.31.1 Detailed Description

2.31.2 Function Documentation

2.31.2.1 f360_update_object_track_properties_calc_obj_height()

```
test_group f360_update_object_track_properties_calc_obj_height ( )
```

Group of tests to check that objects over the ground height is updated under correct circumstances.

Setup:\n

Set up common variables used by Update_Object_Track_Properties

Setup:\n

Initialize tracker calibrations Set up a valid moving object in front of the host Set up initial over the ground height to 0.0F Set up initial historic ndets count to 10.0F Assign valid detections to the object with non-zero vcs_position_z

2.31.2.2 f360_update_object_track_properties_calc_obj_height_obj_height_decay_below_zero()

```
f360_update_object_track_properties_calc_obj_height_obj_height_decay_below_zero ( )
```

Purpose:\n

Test that object over the ground height does not reach negative values when decaying effect could bring the value below zero

Requirements FTCP-13974

Precondition:\n

Object has no detections assigned

Action:\n

Call Update_Object_Track_Properties()

Result:\n

Check if object height did not reach negative values

2.31.2.3 f360_update_object_track_properties_calc_obj_height_obj_height_decays_when_no_dets()

```
f360_update_object_track_properties_calc_obj_height_obj_height_decays_when_no_dets ( )
```

Purpose:\n

Test that object over the ground height is properly decayed when object is valid but has no detections associated

Requirements FTCP-13974

Precondition:\n

Object has no detections assigned Object over the ground height is above zero

Action:\n

Call Update_Object_Track_Properties()

Result:\n

Check if object height has decayed properly

2.31.2.4 f360_update_object_track_properties_calc_obj_height_obj_height_is_not_calculated_when_stationary()

```
f360_update_object_track_properties_calc_obj_height_obj_height_is_not_calculated_when_stationary  
( )
```

Purpose:\n

Test that object over the ground height is not updated when object is stationary

Requirements FTCP-13974

Precondition:\n

Object f_moving flag set to false. Object over the ground state before update is stored as a variable

Action:\n

Call Update_Object_Track_Properties()

Result:\n

Check if object height has not been updated

2.31.2.5 f360_update_object_track_properties_calc_obj_height_obj_height_is_updated_when_valid()

```
f360_update_object_track_properties_calc_obj_height_obj_height_is_updated_when_valid ( )
```

Purpose:

Test that object over the ground height is updated when object is valid and has valid associated detections

Requirements FTCP-13974

Precondition:

Default test setup from TEST GROUP is used.

Action:

Call Update_Object_Track_Properties()

Result:

Check if object height has been updated

2.32 F360_object_based_water_spray_detector_qualtest

Functions

- test group [f360_object_based_water_spray_detector_qualtest](#) ()
Test group contains variables and test cases related to flagging of detections as water spray from objects. Three objects and five detections are defined in the test group, please see DFD-390 for a detailed description of the setup.
- [f360_object_based_water_spray_detector_qualtest_Detect_Water_Spray_From_Objects](#) ()

2.32.1 Detailed Description

2.32.2 Function Documentation

2.32.2.1 f360_object_based_water_spray_detector_qualtest()

```
test_group f360_object_based_water_spray_detector_qualtest ( )
```

Test group contains variables and test cases related to flagging of detections as water spray from objects. Three objects and five detections are defined in the test group, please see DFD-390 for a detailed description of the setup.

Setup:\n

Setup an object so that it is a valid object to check for water spray detections

2.32.2.2 f360_object_based_water_spray_detector_qualtest_Detect_Water_Spray_From_Objects()

```
f360_object_based_water_spray_detector_qualtest_Detect_Water_Spray_From_Objects ( )
```

Purpose:\n

Verify that detections have been flagged as object water spray correctly in scenario defined in test group

Requirements FTCP-12065 NA

Action:\n

Call function

Result:\n

Compare against expected data

2.33 F360_object_based_angle_jump_detector_Detect_Angle_Jumps_From_Objects_qualtest ↩

Functions

- test group [f360_object_based_angle_jump_detector_Detect_Angle_Jumps_From_Objects_qualtest](#) ()
Test group for testing Detect_Angle_Jumps_From_Objects()
- [f360_object_based_angle_jump_detector_Detect_Angle_Jumps_From_Objects_qualtest_flagging_correctness_of_f_ok_to_use](#) ()

2.33.1 Detailed Description

2.33.2 Function Documentation

2.33.2.1 f360_object_based_angle_jump_detector_Detect_Angle_Jumps_From_Objects_qualtest()

```
test_group f360_object_based_angle_jump_detector_Detect_Angle_Jumps_From_Objects_qualtest ( )
```

Test group for testing Detect_Angle_Jumps_From_Objects()

Setup:\n

2.33.2.2 f360_object_based_angle_jump_detector_Detect_Angle_Jumps_From_Objects_qualtest_flagging_correctness_of_f_ok_to_use()

```
f360_object_based_angle_jump_detector_Detect_Angle_Jumps_From_Objects_qualtest_flagging_↩  
correctness_of_f_ok_to_use ( )
```

Purpose:\n

Detect_Angle_Jumps_From_Objects should set correctly f_ok_to_use

Requirements FTCP-12069 NA

Precondition:\n

Set input in order to set only one detection

Action:\n

Call Detect_Angle_Jumps_From_Objects()

Result:\n

Det[1] is not ok to use

2.34 F360_find_tracks_on_and_behind_guardrail_qualtest

Functions

- test group [f360_find_tracks_on_and_behind_guardrail_qualtest](#) ()
Test group of Find_Tracks_On_And_Behind_Guardrail.
- [f360_find_tracks_on_and_behind_guardrail_qualtest_Flag_Objects_On_SEP_ObjectOnGuardrailLeftSide](#) ()
- [f360_find_tracks_on_and_behind_guardrail_qualtest_Flag_Objects_On_SEP_ObjectOnGuardrailRightSide](#) ()

2.34.1 Detailed Description

2.34.2 Function Documentation

2.34.2.1 f360_find_tracks_on_and_behind_guardrail_qualtest_Flag_Objects_On_SEP_ObjectOnGuardrailLeftSide()

`f360_find_tracks_on_and_behind_guardrail_qualtest_Flag_Objects_On_SEP_ObjectOnGuardrailLeftSide ( )`

Purpose: Test that object is correctly marked as on guardrail when object is on SEP on left side

Requirements FTCP-12162

Precondition:

In test group setup:

- A SEP has been set up on the left side of host.
- An object has been placed on the SEP.
- The object has been placed inside tracker info struct.

Action:

Call `Flag_Objects_On_SEP`

Result:

Make sure that object is flagged as on SEP with id 1

2.34.2.2 f360_find_tracks_on_and_behind_guardrail_qualtest_Flag_Objects_On_SEP_ObjectOnGuardrailRightSide()

f360_find_tracks_on_and_behind_guardrail_qualtest_Flag_Objects_On_SEP_ObjectOnGuardrailRightSide ()

Purpose: Test that object is correctly marked as on guardrail when object is on SEP on right side

Requirements FTCP-12162

Precondition:

- A SEP has been set up on the right side of host.
- An object has been placed inside tracker info struct. Set object position to be on the right SEP.

Action:

Call Flag_Objects_On_SEP

Result:

Make sure that object is flagged as on SEP with id 2

2.35 Detecting_multibounces_from_object

Functions

- test group [detecting_multibounces_from_object](#) ()
Test requirements w.r.t. multibounce detections.
- [detecting_multibounces_from_object_det_marked_as_multibounce](#) ()
- [detecting_multibounces_from_object_det_NOT_marked_as_multibounce](#) ()

2.35.1 Detailed Description

2.35.2 Function Documentation

2.35.2.1 detecting_multibounces_from_object()

test group [detecting_multibounces_from_object](#) ()

Test requirements w.r.t. multibounce detections.

Setup:\n

Setup all needed data like radars, detections and object

2.35.2.2 detecting_multibounces_from_object_det_marked_as_multibounce()

[detecting_multibounces_from_object_det_marked_as_multibounce](#) ()

Purpose:\n

Check if detection is marked as multibounces if all conditions are met.

Requirements FTCP-13286

Precondition:\n

Same as setup

Action:\n

Call [Pre_Association_Track_Management](#)().

Result:\n

Detection is marked as multi bounce and should be not ok to use

2.35.2.3 detecting_multibounces_from_object_det_NOT_marked_as_multibounce()

detecting_multibounces_from_object_det_NOT_marked_as_multibounce ()

Purpose:\n

Check if detection is NOT marked as multibounces if one condition is not met.

Requirements FTCP-13286

Precondition:\n

Same as setup with below changes. Set detection range to high value

Action:\n

Call Pre_Association_Track_Management().

Result:\n

Detection is not marked as multi bounce and should ok to use

2.36 Object_wrt_to_sep_tests

Functions

- test group [object_wrt_to_sep_tests](#) ()
Test requirements w.r.t. object relation to a SEP.
- [object_wrt_to_sep_tests_obejct_marked_as_on_a_sep](#) ()
- [object_wrt_to_sep_tests_obejct_NOT_marked_as_on_sep](#) ()

2.36.1 Detailed Description

2.36.2 Function Documentation

2.36.2.1 object_wrt_to_sep_tests()

```
test group object_wrt_to_sep_tests ( )
```

Test requirements w.r.t. object relation to a SEP.

Setup:\n

Set up required variables. Scenario: One object that lies on a SEP

2.36.2.2 object_wrt_to_sep_tests_obejct_marked_as_on_a_sep()

```
object_wrt_to_sep_tests_obejct_marked_as_on_a_sep ( )
```

Purpose:\n Check if object is correctly marked on a SEP

Requirements FTCP-13327

Precondition:\n

Same as setup

Action:\n

Call Pre_Association_Track_Management().

Result:\n

Object is on SEP (sep id = 1)

2.36.2.3 object_wrt_to_sep_tests_obejct_NOT_marked_as_on_sep()

```
object_wrt_to_sep_tests_obejct_NOT_marked_as_on_sep ( )
```

Purpose:\n Check if object is correctly NOT marked on a SEP

Requirements FTCP-13327

Precondition:\n

Same as setup with below changes move sep behind the object laterally

Action:\n

Call Pre_Association_Track_Management().

Result:\n

Object is on SEP (sep id = F360_INVALID_UNSIGNED_ID)

2.37 Sensor_Postprocessing

Functions

- test group `f360_Update_Detection_History_qualtest ()`
This test suit aims to test Update_Detection_History function.
- `f360_Update_Detection_History_qualtest_Test_Update_Detection_History_update_one_detection ()`
- `f360_Update_Detection_History_qualtest_Test_Update_Detection_History_old_dets_greater_than_zero ()`
- `f360_Update_Detection_History_qualtest_Test_Update_Detection_History_n_occupied_higher_than_max ()`
- `f360_Update_Detection_History_qualtest_Test_Update_Detection_History_Find_Next_Hist_Det_Idx_false ()`

2.37.1 Detailed Description

2.37.2 Function Documentation

2.37.2.1 f360_Update_Detection_History_qualtest()

```
test group f360_Update_Detection_History_qualtest ( )
```

This test suit aims to test Update_Detection_History function.

Setup:\n

Set common variables used in all tests. Setup one detection and its properties as follows: range_rate_↔
dealiased to 0.5F vcs_az to 0.2F time_since_measurement to 0.01F

Teardown:\n

Nothing to teardown in this test group

2.37.2.2 f360_Update_Detection_History_qualtest_Test_Update_Detection_History_Find_Next_Hist_Det_Idx_false()

```
f360_Update_Detection_History_qualtest_Test_Update_Detection_History_Find_Next_Hist_Det_Idx_↔  
false ( )
```

Purpose:\n Verify that no new detection is stored in detection history structure when Find_Next_Hist_Det_Idx fails to find the next detection history in

Requirements FTCP-12845

Precondition:\n

- Set the cluster num_old_dets to MAX_DETS_IN_OBJ
- Set the cluster old_det_idx to MAX_NUMBER_OF_HISTORIC_DETECTIONS

Action:\n

- Call Sensor_Postprocessing module function

Result:\n

- Find_Next_Hist_Det_Idx function fails to find new index and then Detections history isn't updated

2.37.2.3 f360_Update_Detection_History_qualtest_Test_Update_Detection_History_n_occupied_higher_than_max()

```
f360_Update_Detection_History_qualtest_Test_Update_Detection_History_n_occupied_higher_than_↵  
max ( )
```

Purpose: Verify that no new detections is stored in the detections history when it reached max occupation and there is no old detections in the cluster

Requirements FTCP-12845

Precondition:

- Set detection history occupation to MAX_NUMBER_OF_HISTORIC_DETECTIONS
- Set cluster num_old_dets to zero

Action:

- Call Sensor_Postprocessing module function

Result:

- Detections history isn't updated and remains at the initial value

2.37.2.4 f360_Update_Detection_History_qualtest_Test_Update_Detection_History_old_dets_greater_than_zero()

```
f360_Update_Detection_History_qualtest_Test_Update_Detection_History_old_dets_greater_than_↵  
zero ( )
```

Purpose: Verify that the oldest historical detection associated to the cluster is removed and replaced with the new one

Requirements FTCP-12845

Precondition:

- Set detection history occupation to MAX_NUMBER_OF_HISTORIC_DETECTIONS
- Set the cluster num_old_dets to 1 (higher than zero)
- Set old_det_idx to zero to set detection history index to zero(the index that will be replaced)

Action:

- Call Sensor_Postprocessing module function

Result:

- The oldest historical detection is replaced with the new one

2.37.2.5 f360_Update_Detection_History_qualtest_Test_Update_Detection_History_update_one_detection()

f360_Update_Detection_History_qualtest_Test_Update_Detection_History_update_one_detection ()

Purpose:\n verify that The tracker shall maintain a history of the detections that were used to form clusters

Requirements FTCP-12845

Precondition:\n

- prepare one detection to be stored in detection history.

Action:\n

- Call Sensor_Postprocessing module function

Result:\n

- Detections history is updated with the expected values from test setup

2.38 F360_kill_coasted_tracks_qualtest

Functions

- test group `f360_kill_coasted_tracks_qualtest` ()

This group provides tests ensuring specified behavior for termination of coasting objects.

- `f360_kill_coasted_tracks_qualtest_Do_not_terminate_updated_tracks` ()
- `f360_kill_coasted_tracks_qualtest_Termination_of_coasting_confirmed_objects` ()
- `f360_kill_coasted_tracks_qualtest_Termination_of_coasting_mirror_objects_variation_of_stage_time` ()
- `f360_kill_coasted_tracks_qualtest_Termination_of_coasting_mirror_objects_variation_of_mirror_probability` ()
- `f360_kill_coasted_tracks_qualtest_Non_Movable_Coasting_Obj_Outside_Front_Only_FOV_Coasting_Time` ()
- `f360_kill_coasted_tracks_qualtest_Non_Movable_Coasting_Obj_Outside_Front_Only_FOV_Pos` ()
- `f360_kill_coasted_tracks_qualtest_Non_Movable_Coasting_Obj_Outside_Front_Only_FOV_Config` ()
- `f360_kill_coasted_tracks_qualtest_Non_Movable_Coasting_Obj_Outside_Front_Only_FOV_Nonmoveable` ()

2.38.1 Detailed Description

2.38.2 Function Documentation

2.38.2.1 f360_kill_coasted_tracks_qualtest()

```
test_group f360_kill_coasted_tracks_qualtest ( )
```

This group provides tests ensuring specified behavior for termination of coasting objects.

Setup:\n

Define calibration thresholds needed for function-under-test

- set threshold for mirror probability for termination of coasting objects to 0.3
- set threshold for stage age of confirmed coasting tracks for termination to 0.55 s
- set threshold for stage age of mirrored coasting tracks for termination to 0.25 s

2.38.2.2 f360_kill_coasted_tracks_qualtest_Do_not_terminate_updated_tracks()

```
f360_kill_coasted_tracks_qualtest_Do_not_terminate_updated_tracks ( )
```

Purpose:

This test ensures that updated objects are not terminated, even if their remaining attributes fulfill conditions for being terminated.

Requirements FTCP-11715

Precondition:

Create an object

- set status to UPDATED
- set time since stage start to a value exceeding the threshold for termination of coasting objects
- set mirror probability to a value exceeding the threshold for termination of coasting objects Set number number of active objects and values for active object id array in tracker information structure accordingly

Action:

Call function-under-test

Result:

Object is not terminated

2.38.2.3 f360_kill_coasted_tracks_qualtest_Non_Movable_Coasting_Obj_Outside_Front_Only_FOV_Coasting_Time()

```
f360_kill_coasted_tracks_qualtest_Non_Movable_Coasting_Obj_Outside_Front_Only_FOV_Coasting_↵  
Time ( )
```

Purpose:

This test ensures that non-moveable coasting objects, outside FOV given front radar only config, are terminated, if their time since stage start exceeds a given threshold.

Requirements CPR-4890

Precondition:\n

Create three objects

- set status of all objects to COASTED
 - set mirror probability to 0.1 to prevent mirror probability kill branch from being triggered
 - set f_moveable to be FALSE
 - set object position outside sensor FOV
 - 1. object: set time since stage start to a value below the threshold for termination of coasting objects
 - 2. object: set time since stage start to a value that equals the threshold for termination of coasting objects
 - 3. object: set time since stage start to a value above the threshold for termination of coasting objects
- With the object alive time near the threshold of k_max_coast_time_outside_fov, the normal coasting kill branch is not triggered. Set number number of active objects and values for active object id array in tracker information structure accordingly Set f_single_front_center_radar_only to TRUE

Action:\n

Call function-under-test

Result:\n

Coasting object with time since stage start at a value above the threshold is expected to be terminated, the other two objects keep their status

2.38.2.4 f360_kill_coasted_tracks_qualtest_Non_Movable_Coasting_Obj_Outside_Front_Only_FOV_Config()

```
f360_kill_coasted_tracks_qualtest_Non_Movable_Coasting_Obj_Outside_Front_Only_FOV_Config ( )
```

Purpose:\n

This test ensures that non-moveable long coasting objects given outside sensor FOV, are not terminated unless the config is front radar only

Requirements CPR-4890

Precondition:\n

Create one objects

- set status to COASTED
- set mirror probability to 0.1 to prevent mirror probability kill branch from being triggered
- set f_moveable to be FALSE
- set time since stage start to a value above the threshold for termination of coasting objects
- set object position outside FOV With the object alive time near the threshold of k_max_coast_time_outside_fov, the normal coasting kill branch is not triggered. Set number number of active objects and values for active object id array in tracker information structure accordingly Set f_single_front_center_radar_only to FALSE

Action:\n

Call function-under-test

Result:\n

Coasting object with position outside FOV is NOT expected to be terminated, if NOT in a front sensor only config

2.38.2.5 f360_kill_coasted_tracks_qualtest_Non_Movable_Coasting_Obj_Outside_Front_Only_FOV_Nonmoveable()

```
f360_kill_coasted_tracks_qualtest_Non_Movable_Coasting_Obj_Outside_Front_Only_FOV_Nonmoveable  
( )
```

Purpose:

This test ensures that coasting objects given outside sensor FOV and front radar only config, are not terminated, unless the object is non-moveable

Requirements CPR-4890**Precondition:**

Create two objects

- set status to COASTED
- set mirror probability to 0.1 to prevent mirror probability kill branch from being triggered
- set time since stage start to a value above the threshold for termination of coasting objects
- set object position outside FOV
- 1. object: set f_moveable to be TRUE
- 2. object: set f_moveable to be FALSE With the object alive time near the threshold of k_max_coast ↔ time_outside_fov, the normal coasting kill branch is not triggered. Set number number of active objects and values for active object id array in tracker information structure accordingly Set f_single_front ↔ center_radar_only to TRUE

Action:

Call function-under-test

Result:

Coasting objects with position outside FOV and in radar only config are NOT expected to be terminated, unless it is non-moveable

2.38.2.6 f360_kill_coasted_tracks_qualtest_Non_Movable_Coasting_Obj_Outside_Front_Only_FOV_Pos()

```
f360_kill_coasted_tracks_qualtest_Non_Movable_Coasting_Obj_Outside_Front_Only_FOV_Pos ( )
```

Purpose:

This test ensures that non-moveable long coasting objects given front radar only config, are terminated, if their position is outside sensor FOV.

Requirements CPR-4890

Precondition:\n

Create two objects

- set status of all objects to COASTED
- set mirror probability to 0.1 to prevent mirror probability kill branch from being triggered
- set f_moveable to be FALSE
- set time since stage start to a value above the threshold for termination of coasting objects
- 1. object: set object position within FOV
- 2. object: set object position outside FOV With the object alive time near the threshold of k_max_↵ coast_time_outside_fov, the normal coasting kill branch is not triggered. Set number number of active objects and values for active object id array in tracker information structure accordingly Set f_single_↵ front_center_radar_only to TRUE

Action:\n

Call function-under-test

Result:\n

Coasting object with position outside FOV is expected to be terminated, the other object keeps its status

2.38.2.7 f360_kill_coasted_tracks_qualtest_Termination_of_coasting_confirmed_objects()

f360_kill_coasted_tracks_qualtest_Termination_of_coasting_confirmed_objects ()

Purpose:\n

This test ensures that confirmed coasting objects are terminated, if their time since stage start exceeds a given threshold.

Requirements FTCP-11715**Precondition:**\n

Create three objects

- set status of all objects to COASTED
- 1. object: set time since stage start to a value below the threshold for termination of coasting objects
- 2. object: set time since stage start to a value that equals the threshold for termination of coasting objects
- 3. object: set time since stage start to a value above the threshold for termination of coasting objects Set number number of active objects and values for active object id array in tracker information structure accordingly

Action:\n

Call function-under-test

Result:\n

Coasting object with time since stage start at a value above the threshold is expected to be terminated, the other two objects keep their status

2.38.2.8 f360_kill_coasted_tracks_qualtest_Termination_of_coasting_mirror_objects_variation_of_mirror_probability()

```
f360_kill_coasted_tracks_qualtest_Termination_of_coasting_mirror_objects_variation_of_mirror↵
_probability ( )
```

Purpose:

This test ensures that mirrored coasting objects are terminated, if their time since stage start and their mirror probabilities exceed given thresholds.

Requirements FTCP-11715**Precondition:**

Create three objects

- set status of all objects to COASTED
 - set time since stage start for all objects to a value above the threshold for termination of coasting mirror objects
 - 1. object: set mirror probability to a value below the threshold for termination of coasting mirror objects
 - 2. object: set mirror probability to a value that equals the threshold for termination of coasting mirror objects
 - 3. object: set mirror probability to a value above the threshold for termination of coasting mirror objects
- Set number number of active objects and values for active object id array in tracker information structure accordingly

Action:

Call function-under-test

Result:

Coasting object with mirror probability at a value above the threshold is expected to be terminated, the other two objects keep their status

2.38.2.9 f360_kill_coasted_tracks_qualtest_Termination_of_coasting_mirror_objects_variation_of_stage_time()

```
f360_kill_coasted_tracks_qualtest_Termination_of_coasting_mirror_objects_variation_of_stage_↵
time ( )
```

Purpose:

This test ensures that mirrored coasting objects are terminated, if their time since stage start exceeds a given threshold.

Requirements FTCP-11715

Precondition:\n

Create three objects

- set status of all objects to COASTED
- set mirror probability for all objects to a value above threshold for termination of coasting mirror objects
- 1. object: set time since stage start to a value below the threshold for termination of coasting mirror objects
- 2. object: set time since stage start to a value that equals the threshold for termination of coasting mirror objects
- 3. object: set time since stage start to a value above the threshold for termination of coasting mirror objects Set number number of active objects and values for active object id array in tracker information structure accordingly

Action:\n

Call function-under-test

Result:\n

Coasting object with time since stage start at a value above the threshold is expected to be terminated, the other two objects keep their status

2.39 F360_object_occlusion_classification

Functions

- test group `f360_object_occlusion_classification` ()
Test group dedicated to verify the functionality of the function `Object_Occlusion_Classification()`. It is done by setting up a scenario with a single forward looking sensor and 2 objects. 1 object is intended to be tested and while the other object is placed between the test object and the sensor, occluding the test object. This scenario is varied slightly to achieve different levels of occlusion.
- `f360_object_occlusion_classification_Verify_Single_Object_Status_Occluded` ()
- `f360_object_occlusion_classification_Verify_Single_Object_Status_On_Edge` ()
- `f360_object_occlusion_classification_Verify_Single_Object_Status_Visible` ()
- `f360_object_occlusion_classification_Verify_Single_Object_Status_Undefined_Not_Moving` ()
- `f360_object_occlusion_classification_Verify_All_Active_Objects_Are_Considered` ()

2.39.1 Detailed Description

2.39.2 Function Documentation

2.39.2.1 f360_object_occlusion_classification()

```
test group f360_object_occlusion_classification ( )
```

Test group dedicated to verify the functionality of the function `Object_Occlusion_Classification()`. It is done by setting up a scenario with a single forward looking sensor and 2 objects. 1 object is intended to be tested and while the other object is placed between the test object and the sensor, occluding the test object. This scenario is varied slightly to achieve different levels of occlusion.

Setup:\n

Initialize tracker calibrations. Setup a single forward looking sensor such that

- its mounting position is in vcs origo.
- its boresight is vcs forward
- is valid
- has look id F360_DET_LOOK_ID_0
- for that look id, has a fov_min_az_rad of -1.5F
- for that look id, has a fov_max_az_rad of 1.5F
- for that look id, has a range limit of 150.0F Setup tracker_info to contain 2 active objects and object ids
- Setup the occluding object to
 - have vcs position (25,0)
 - be f_moving
 - be f_moveable
 - have status updated
 - have confidence_level 1
 - have reference point rear
 - have heading and pointing 0
 - have a length of 2 and width of 10
 - update bbox center to correctly set all private parameters in the bbox class

2.39.2.2 f360_object_occlusion_classification_Verify_All_Active_Objects_Are_Considered()

```
f360_object_occlusion_classification_Verify_All_Active_Objects_Are_Considered ( )
```

Purpose:\n

Verify that all active moving objects have their occlusion status evaluated

Requirements FTCP-13872, FTCP-13870

Precondition:\n

Setup the test object to be positioned at (28,6), i.e. there is a free line of sight between the sensor and the test object. Set the test object to also

- be moving
- be moveable
- status updated
- confidence_level 1
- reference point rear
- heading and pointing 0
- a length of 2 and width of 10
- update bbox center to correctly set all private parameters in the bbox class Initialize occlusion

Action:\n

Call Object_Occlusion_Classification().

Result:\n

Verify the test object occlusion_status.at_vcs_position have status VISIBLE

2.39.2.3 f360_object_occlusion_classification_Verify_Single_Object_Status_Occluded()

```
f360_object_occlusion_classification_Verify_Single_Object_Status_Occluded ( )
```

Purpose:\n

Verify that an object whose reference point is positioned behind another object is marked as occluded.

Requirements FTCP-13872, FTCP-13870

Precondition:\n

Setup the test object to be positioned at (28,0), i.e. behind the occluding object Set the test object to also

- be f_moving
- be moveable
- have status updated
- have confidence_level 1
- have reference point rear
- have heading and pointing 0
- have a length of 2 and width of 10
- update bbox center to correctly set all private parameters in the bbox class Initialize occlusion

Action:\n

Call Object_Occlusion_Classification().

Result:\n

Verify the test object occlusion_status.at_vcs_position have status OCCLUDED

2.39.2.4 f360_object_occlusion_classification_Verify_Single_Object_Status_On_Edge()

f360_object_occlusion_classification_Verify_Single_Object_Status_On_Edge ()

Purpose:\n

Verify that an object whose reference point is positioned behind another object corner is marked as on edge.

Requirements FTCP-13872, FTCP-13870

Precondition:\n

Setup the test object to be positioned at (28,5), i.e. behind the occluding object close to the right corner of the occluding object. Set the test object to also

- be f_moving
- be moveable
- status updated
- confidence_level 1
- reference point rear
- heading and pointing 0
- a length of 2 and width of 10
- update bbox center to correctly set all private parameters in the bbox class Initialize occlusion

Action:\n

Call Object_Occlusion_Classification().

Result:\n

Verify the test object occlusion_status.at_vcs_position have status ON_EDGE

2.39.2.5 f360_object_occlusion_classification_Verify_Single_Object_Status_Undefined_Not_Moving()

```
f360_object_occlusion_classification_Verify_Single_Object_Status_Undefined_Not_Moving ( )
```

Purpose:\n

Verify that an object's occlusion status is marked as UNDEFINED if the object is not moving

Requirements FTCP-13872, FTCP-13870

Precondition:\n

Setup the test object to be positioned at (28,0), i.e. behind the occluding object Set the test object to also

- be moveable
- status updated
- confidence_level 1
- reference point rear
- heading and pointing 0
- a length of 2 and width of 10
- update bbox center to correctly set all private parameters in the bbox class Initialize occlusion

Action:\n

Call Object_Occlusion_Classification().

Result:\n

Verify the test object occlusion_status.at_vcs_position have status UNDEFINED

2.39.2.6 f360_object_occlusion_classification_Verify_Single_Object_Status_Visible()

```
f360_object_occlusion_classification_Verify_Single_Object_Status_Visible ( )
```

Purpose:\n

Verify that an object whose reference point is not positioned behind another object is marked as visible.

Requirements FTCP-13872, FTCP-13870

Precondition:\n

Setup the test object to be positioned at (28,6), i.e. there is a free line of sight between the sensor and the test object. Set the test object to also

- be f_moving
- be moveable
- status updated
- confidence_level 1
- reference point rear
- heading and pointing 0
- a length of 2 and width of 10
- bbox center to correctly set all private parameters in the bbox class Initialize occlusion

Action:\n

Call Object_Occlusion_Classification().

Result:\n

Verify the test object occlusion_status.at_vcs_position have status VISIBLE

2.40 F360_object_underdrivability_classification_qt

Functions

- test group `f360_object_underdrivability_classification_qt()`

This test group will test the intended functionality of `Object_Underdrivability_Classification()`. Tests are designed to check.

- `f360_object_underdrivability_classification_qt_test_valid_moving_objs_height_above_threshold()`
- `f360_object_underdrivability_classification_qt_test_valid_moving_objs_height_below_threshold()`
- `f360_object_underdrivability_classification_qt_test_valid_moving_objs_behind_host()`
- `f360_object_underdrivability_classification_qt_test_objs_invalid()`

2.40.1 Detailed Description

2.40.2 Function Documentation

2.40.2.1 f360_object_underdrivability_classification_qt()

```
test_group f360_object_underdrivability_classification_qt ( )
```

This test group will test the intended functionality of `Object_Underdrivability_Classification()`. Tests are designed to check.

- That loop over objects are functioning correctly such that all active objects get their status updated but not inactive objects
- That correct classification method is used for moving and stationary objects respectively
- That stationary objects are not classified when occupancy grid is invalid

Setup:\n

Setup 10 objects that are valid and moving with UD status `NOT_TO_CONSIDER` and that has a very large estimated height such that they would be classified `CAN_PASS_UNDER` by `Assign_Underdrivability_Status↵_To_Moving_Object()` Setup 10 objects that are invalid and moving with UD status `NOT_TO_CONSIDER`. Since they are invalid their underdrivable status should be unchanged Remaining objects should be invalid with UD status `NOT_TO_CONSIDER`. Since they are invalid their underdrivable status should be unchanged Setup tracker timestamp to an arbitrary positive value Setup a valid occupancy grid with all zones to `CAN_NOT_↵PASS_UNDER`.

2.40.2.2 f360_object_underdrivability_classification_qt_test_objs_invalid()

```
f360_object_underdrivability_classification_qt_test_objs_invalid ( )
```

Purpose:

The purpose of this test is to check if objects are set as UNDERDRIVABLE_STATUS_NOT_TO_CONSIDER when their underdrivability status is not updated by either Assign_Underdrivability_Status_To_Moving_Object or Assign_Underdrivability_Status_To_Moving_Object functions.

Requirements FTCP-13970

Precondition:

Occupancy grid not valid, otherwise default test setup from TEST GROUP is used.

Action:

Call Object_Underdrivability_Classification()

Result:

Check that the underdrivability status for invalid objects were not updated, i.e. that their underdrivability status is NOT_TO_CONSIDER

2.40.2.3 f360_object_underdrivability_classification_qt_test_valid_moving_objs_behind_host()

```
f360_object_underdrivability_classification_qt_test_valid_moving_objs_behind_host ( )
```

Purpose:

The purpose of this test is to check such that valid (historical height mean is above the threshold for required number of consecutive scans) moving objects do not get their underdrivability status updated by Assign_Underdrivability_Status_To_Moving_Object() when they are behind the host vehicle.

Requirements FTCP-13969 FTCP-13970

Precondition:

Default test setup from TEST GROUP is used.

Action:

Call Object_Underdrivability_Classification()

Result:

Check that the underdrivability status for moving objects outside of classification area were properly classified by Assign_Underdrivability_Status_To_Stationary_Object(), i.e. that their underdrivability status is NOT_TO_CONSIDER

2.40.2.4 f360_object_underdrivability_classification_qt_test_valid_moving_objs_height_above_threshold()

```
f360_object_underdrivability_classification_qt_test_valid_moving_objs_height_above_threshold (
)
```

Purpose:\n

The purpose of this test is to check such that valid moving objects with height above the threshold get their underdrivability status updated by Assign_Underdrivability_Status_To_Moving_Object()

Requirements FTCP-13969

Precondition:\n

Default test setup from TEST GROUP is used.

Action:\n

Call Object_Underdrivability_Classification()

Result:\n

Check that the underdrivability status for valid moving objects were updated by Assign_Underdrivability_↔ Status_To_Moving_Object(), i.e. that their underdrivability status is CAN_PASS_UNDER

2.40.2.5 f360_object_underdrivability_classification_qt_test_valid_moving_objs_height_below_threshold()

```
f360_object_underdrivability_classification_qt_test_valid_moving_objs_height_below_threshold (
)
```

Purpose:\n

The purpose of this test is to check such that valid moving objects with height below the threshold get their underdrivability status updated by Assign_Underdrivability_Status_To_Moving_Object()

Requirements FTCP-13969

Precondition:\n

Default test setup from TEST GROUP is used.

Action:\n

Call Object_Underdrivability_Classification()

Result:\n

Check that the underdrivability status for valid moving objects were updated by Assign_Underdrivability_↔ Status_To_Moving_Object(), i.e. that their underdrivability status is CAN_NOT_PASS_UNDER

2.41 F360_assign_underdrivability_status_to_tracks_ocg_QT

Collaboration diagram for F360_assign_underdrivability_status_to_tracks_ocg_QT:

Modules

- [Object_motion_status_should_be_updated](#)

Functions

- test group [f360_assign_underdrivability_status_to_tracks_ocg_QT](#) ()
Test group of Assign_Underdrivability_Status_To_Tracks. Tests verify whether tracks are properly checked whether they can be marked as underdrivable, whether they do not have their underdrivability status modified when are not out of zones FOV and whether they are properly assigned new underdrivability status.
- [f360_assign_underdrivability_status_to_tracks_ocg_QT_Assign_Underdrivability_Status_To_Tracks__Object_Not_In_FOV](#) ()
- [f360_assign_underdrivability_status_to_tracks_ocg_QT_Assign_Underdrivability_Status_To_Tracks__Current_Status_Can_Pa](#) ()
- [f360_assign_underdrivability_status_to_tracks_ocg_QT_Assign_Underdrivability_Status_To_Tracks__Current_Status_Can_Pa](#) ()
- [f360_assign_underdrivability_status_to_tracks_ocg_QT_Assign_Underdrivability_Status_To_Tracks__Not_To_Consider_After](#) ()
- [f360_assign_underdrivability_status_to_tracks_ocg_QT_Assign_Underdrivability_Status_To_Tracks__Current_Status_Can_N](#) ()
- [f360_assign_underdrivability_status_to_tracks_ocg_QT_Assign_Underdrivability_Status_To_Tracks__Current_Status_Can_N](#) ()

2.41.1 Detailed Description

2.41.2 Function Documentation

2.41.2.1 f360_assign_underdrivability_status_to_tracks_ocg_QT()

```
test_group f360_assign_underdrivability_status_to_tracks_ocg_QT ( )
```

Test group of Assign_Underdrivability_Status_To_Tracks. Tests verify whether tracks are properly checked whether they can be marked as underdrivable, whether they do not have their underdrivability status modified when are not out of zones FOV and whether they are properly assigned new underdrivability status.

2.41.2.2 f360_assign_underdrivability_status_to_tracks_ocg_QT_Assign_Underdrivability_Status_To_Tracks__Current_St

```
f360_assign_underdrivability_status_to_tracks_ocg_QT_Assign_Underdrivability_Status_To_↵
Tracks__Current_Status_Can_Not_Pass_New_Can_Pass ( )
```

Purpose:\n

Purpose if this test is to verify whether when current object status is UNDERDRIVABLE_STATUS_CAN_NOT_PASS_UNDER and new one is UNDERDRIVABLE_STATUS_CAN_PASS_UNDER - status is increased. it also check if the object's probability_underdrivable is correct

Requirements FTCP-13916

Precondition:\n

Pick zone where track should be placed. Verify whether zone is not outside limits. Set up 2 object parameters to make it valid. Set up its status to UNDERDRIVABLE_STATUS_CAN_NOT_PASS_UNDER. Set status of zone where object is located to UNDERDRIVABLE_STATUS_CAN_PASS_UNDER. Set probabilities associated with the zones 0 to 3 to 0.2F, 0.1F, 0.5F, 0.3F respectively Set expected_probability_underdrivable to 0.3 since this is the probability that should be calculated for the target object

Action:\n

Call tested function.

Result:\n

Check whether object status was changed to UNDERDRIVABLE_STATUS_CAN_PASS_UNDER. Check whether object probability underdrivable was changed to the expected value

2.41.2.3 f360_assign_underdrivability_status_to_tracks_ocg_QT_Assign_Underdrivability_Status_To_Tracks__Current_St

```
f360_assign_underdrivability_status_to_tracks_ocg_QT_Assign_Underdrivability_Status_To_↵
Tracks__Current_Status_Can_Not_Pass_New_Is_Likely_To_Pass ( )
```

Purpose:\n

Purpose if this test is to verify whether when current object status is UNDERDRIVABLE_STATUS_CAN_NOT_PASS_UNDER and new one is UNDERDRIVABLE_STATUS_IS_LIKELY_TO_PASS_UNDER - status is increased. it also check if the object's probability_underdrivable is correct

Requirements FTCP-13916

Precondition:\n

Pick zone where track should be placed. Verify whether zone is not outside limits. Set up object parameters to make it valid. Set up its status to UNDERDRIVABLE_STATUS_CAN_NOT_PASS_UNDER. Set status of zone where object is located to UNDERDRIVABLE_STATUS_IS_LIKELY_TO_PASS_UNDER. Set probabilities associated with the zones 0 to 3 to 0.1F, 0.4F, 0.25F, 0.15F respectively Set expected_probability_underdrivable to 0.65 since this is the probability that should be calculated for the target object

Action:\n

Call tested function.

Result:\n

Check whether object status was changed to UNDERDRIVABLE_STATUS_IS_LIKELY_TO_PASS_UNDER. Check whether object probability underdrivable was changed to the expected value

2.41.2.4 f360_assign_underdrivability_status_to_tracks_ocg_QT_Assign_Underdrivability_Status_To_Tracks__Current_St

```
f360_assign_underdrivability_status_to_tracks_ocg_QT_Assign_Underdrivability_Status_To_↵
Tracks__Current_Status_Can_Pass_Under_New_One_Can_Not_Pass ( )
```

Purpose:\n

Purpose if this test is to verify whether when current object status is UNDERDRIVABLE_STATUS_CAN_P↵
ASS_UNDER and new one is UNDERDRIVABLE_STATUS_CAN_NOT_PASS_UNDER - status is lowered.
it also check if the object's probability_underdrivable is correct

Requirements FTCP-13916**Precondition:\n**

Pick zone where track should be placed. Verify whether zone is not outside limits. Set up object parameters to make it valid. Set up its status to UNDERDRIVABLE_STATUS_CAN_PASS_UNDER. Set status of zone where object is located to UNDERDRIVABLE_STATUS_CAN_NOT_PASS_UNDER. Set probabilities associated with the zones 0 to 3 to 0.5F, 0.2F, 0.1F, 0.3F respectively Set expected_probability_underdrivable to 0.3 since this is the probability that should be calculated for the target object The object is located in the cell with longitudinal index 15. The status and probabilities should be takes from the cell where the object is and not an neighbouring cell (second picked).

To insure that final test result neighbouring cells are set to invalid statuses.

Action:\n

Call tested function.

Result:\n

Check whether object status was changed to UNDERDRIVABLE_STATUS_CAN_NOT_PASS_UNDER.
Check whether object probability underdrivable was changed to to the expected value

2.41.2.5 f360_assign_underdrivability_status_to_tracks_ocg_QT_Assign_Underdrivability_Status_To_Tracks__Current_St

```
f360_assign_underdrivability_status_to_tracks_ocg_QT_Assign_Underdrivability_Status_To_↵
Tracks__Current_Status_Can_Pass_Under_New_One_Is_Likely_To_Pass ( )
```

Purpose:\n

Purpose if this test is to verify whether when current object status is UNDERDRIVABLE_STATUS_CAN_↵
PASS_UNDER but new one is UNDERDRIVABLE_STATUS_IS_LIKELY_TO_PASS_UNDER - status is not
lowered. it also check if the object's probability_underdrivable is correct

Requirements FTCP-13916**Precondition:\n**

Pick zone where track should be placed. Verify whether zone is not outside limits. Set up object parameters to make it valid. Set up its status to UNDERDRIVABLE_STATUS_CAN_PASS_UNDER. Set status of zone where object is located to UNDERDRIVABLE_STATUS_IS_LIKELY_TO_PASS_UNDER.

Action:\n

Call tested function.

Result:\n

Check whether object status was not changed. Check if probability underdrivable is correct

2.41.2.6 f360_assign_underdrivability_status_to_tracks_ocg_QT_Assign_Underdrivability_Status_To_Tracks__Not_To_Co

```
f360_assign_underdrivability_status_to_tracks_ocg_QT_Assign_Underdrivability_Status_To_↵
Tracks__Not_To_Consider_After_Status_Determined ( )
```

Purpose:\n

Purpose of this test is to verify whether when current object status is UNDERDRIVABLE_STATUS_NOT_↵
CAN_PASS_UNDER and the object is placed laterally outside occupancy grid, object's status is changed to
UNDERDRIVABLE_STATUS_NOT_TO_CONSIDER it also check if the object's probability_underdrivable is
correct

Requirements FTCP-13916

Precondition:\n

Pick zone where track should be placed. Verify whether zone is outside limits. Set up object parameters to
make it valid. Set up its status to UNDERDRIVABLE_STATUS_NOT_CAN_PASS_UNDER. Set expected_↵
probability_underdrivable to 0 since this is the probability that should be calculated for the target object

The object is located in the cell with longitudinal index 14. The status and probabilities are not relevant in this case
since the object should not be considered for underdrivability check.

Action:\n

Call tested function.

Result:\n

Check whether object status was changed to UNDERDRIVABLE_STATUS_NOT_TO_CONSIDER. Check
whether object probability underdrivable was changed to to the expected value

2.41.2.7 f360_assign_underdrivability_status_to_tracks_ocg_QT_Assign_Underdrivability_Status_To_Tracks__Object_Not

```
f360_assign_underdrivability_status_to_tracks_ocg_QT_Assign_Underdrivability_Status_To_↵
Tracks__Object_Not_In_FOV ( )
```

Purpose:\n

Purpose if this test is to verify whether object should be considered but its position is outside zones FOV,
its underdrivability status is set to UNDERDRIVABLE_STATUS_NOT_TO_CONSIDER it also check if the
object's probability_underdrivable is correct

Requirements FTCP-13916

Precondition:\n

Set up object parameters to make it valid. Set up its status to UNDERDRIVABLE_STATUS_CAN_PASS_↵
_UNDER Set object VCS position to be outside zones FOV. Set the expected_probability_underdrivable to
comapre with results of the tested method

Action:\n

Call tested function.

Result:\n

Check whether object status was not changed. Check if probability underdrivable is correct

2.42 Object_motion_status_should_be_updated

Collaboration diagram for Object_motion_status_should_be_updated:

Modules

- [Update_object_properties](#)

Functions

- test group [object_motion_status_should_be_updated_qualtest](#) ()
Test group of Object_Motion_Status_Should_Be_Updated function. Tests verify whether function properly verifies whether object motion status should be updated.
- [object_motion_status_should_be_updated_qualtest_Object_Motion_Status_Should_Be_Updated__NEW_UPDATED](#) ()
- [object_motion_status_should_be_updated_qualtest_Object_Motion_Status_Should_Be_Updated__UPDATED](#) ()
- [object_motion_status_should_be_updated_qualtest_Object_Motion_Status_Should_Be_Updated__COASTED__Time_Below](#) ()
- [object_motion_status_should_be_updated_qualtest_Object_Motion_Status_Should_Be_Updated__COASTED__Time_Above](#) ()
- [object_motion_status_should_be_updated_qualtest_Object_Motion_Status_Should_Be_Updated__NEW_COASTED__Time](#) ()

2.42.1 Detailed Description

2.42.2 Function Documentation

2.42.2.1 object_motion_status_should_be_updated_qualtest()

```
test group object_motion_status_should_be_updated_qualtest ( )
```

Test group of Object_Motion_Status_Should_Be_Updated function. Tests verify whether function properly verifies whether object motion status should be updated.

Setup:\n

Initialize calibrations structure

2.42.2.2 object_motion_status_should_be_updated_qualtest_Object_Motion_Status_Should_Be_Updated__COASTED__T

```
object_motion_status_should_be_updated_qualtest_Object_Motion_Status_Should_Be_Updated__COASTED__T  
TED__Time_Above_Threshold ( )
```

Purpose:

Purpose of this test is to verify whether function returns false if object status is COASTED and time since update is above threshold

Requirements FTCP-11692

Precondition:

Set status of object to COASTED

Action:

Call tested function

Result:

Check whether returned value is false

2.42.2.3 object_motion_status_should_be_updated_qualtest_Object_Motion_Status_Should_Be_Updated__COASTED__T

```
object_motion_status_should_be_updated_qualtest_Object_Motion_Status_Should_Be_Updated__COASTED__T  
TED__Time_Below_Threshold ( )
```

Purpose:

Purpose of this test is to verify whether function returns true if object status is COASTED and time since update is below threshold

Requirements FTCP-11692

Precondition:

Set status of object to COASTED

Action:

Call tested function

Result:

Check whether returned value is true

2.42.2.4 object_motion_status_should_be_updated_qualtest_Object_Motion_Status_Should_Be_Updated__NEW_COASTED__Time_Below_Threshold ()

```
object_motion_status_should_be_updated_qualtest_Object_Motion_Status_Should_Be_Updated__NEW_COASTED__Time_Below_Threshold ( )
```

Purpose:

Purpose of this test is to verify whether function returns false if object status is NEW_COASTED and time since update is below threshold

Requirements FTCP-11692

Precondition:

Set status of object to NEW_COASTED

Action:

Call tested function

Result:

Check whether returned value is false

2.42.2.5 object_motion_status_should_be_updated_qualtest_Object_Motion_Status_Should_Be_Updated__NEW_UPDATED ()

```
object_motion_status_should_be_updated_qualtest_Object_Motion_Status_Should_Be_Updated__NEW_UPDATED ( )
```

Purpose:

Purpose of this test is to verify whether function returns true if object status is NEW_UPDATED

Requirements FTCP-11692

Precondition:

Set status of object to NEW_UPDATED

Action:

Call tested function

Result:

Check whether returned value is true

2.42.2.6 object_motion_status_should_be_updated_qualtest_Object_Motion_Status_Should_Be_Updated__UPDATED()

object_motion_status_should_be_updated_qualtest_Object_Motion_Status_Should_Be_Updated__UPDA←
TED ()

Purpose:\n

Purpose of this test is to verify whether function returns true if object status is UPDATED

Requirements FTCP-11692

Precondition:\n

Set status of object to UPDATED

Action:\n

Call tested function

Result:\n

Check whether returned value is true

2.43 Update_object_properties

Collaboration diagram for Update_object_properties:

Modules

- [Check_if_object_is_suspected_stationary](#)

Functions

- test group [update_object_properties_qualtest](#) ()
Test group of Update_Object_Properties() function. Tests verify whether object motion properties are properly updated.
- [update_object_properties_qualtest_update_object_properties__check_if_changes_moveable_flag](#) ()
- [update_object_properties_qualtest_update_object_properties__check_if_moveable_flag_is_not_reset](#) ()

2.43.1 Detailed Description

2.43.2 Function Documentation

2.43.2.1 update_object_properties_qualtest()

```
test group update_object_properties_qualtest ( )
```

Test group of Update_Object_Properties() function. Tests verify whether object motion properties are properly updated.

Setup:\n

Initialize globals structure

2.43.2.2 update_object_properties_qualtest_update_object_properties__check_if_changes_moveable_flag()

```
update_object_properties_qualtest_update_object_properties__check_if_changes_moveable_flag ( )
```

Purpose:\n

Purpose of this test is to verify whether movable flag is set when object is marked as moving

Requirements FTCP-11469

Precondition:\n

Set status f_moving flag to true Set status f_moveable flag to false

Action:\n

Call tested function

Result:\n

Check whether object f_moveable flag was set

2.43.2.3 update_object_properties_qualtest_update_object_properties__check_if_moveable_flag_is_not_reset()

```
update_object_properties_qualtest_update_object_properties__check_if_moveable_flag_is_not_reset ( )
```

Purpose:

Purpose of this test is to verify whether movable flag is not reset if object is not moving

Requirements FTCP-11469

Precondition:

Set status f_moving flag to false Set status f_moveable flag to true

Action:

Call tested function

Result:

Check whether object f_moveable flag was not reset

2.44 Check_if_object_is_suspected_stationary

Collaboration diagram for Check_if_object_is_suspected_stationary:

Modules

- [CalcProbability](#)

Functions

- test group [check_if_object_is_suspected_stationary_qualtest](#) ()
Test group of Check_If_Object_Is_Suspected_Stationary() function. Test verify whether stationary suspected flag is properly set for both CTCA and CCA tracks.
- [check_if_object_is_suspected_stationary_qualtest_check_if_object_is_suspected_stationary__ctca_marked_as_stationary_if_](#)
()
- [check_if_object_is_suspected_stationary_qualtest_check_if_object_is_suspected_stationary__ctca_marked_as_not_stationary](#)
()
- [check_if_object_is_suspected_stationary_qualtest_check_if_object_is_suspected_stationary__ctca_marked_as_not_stationary](#)
()
- [check_if_object_is_suspected_stationary_qualtest_check_if_object_is_suspected_stationary__ctca_marked_as_not_stationary](#)
()
- [check_if_object_is_suspected_stationary_qualtest_check_if_object_is_suspected_stationary__ctca_marked_as_not_stationary](#)
()
- [check_if_object_is_suspected_stationary_qualtest_check_if_object_is_suspected_stationary__cca_marked_as_stationary_if_](#)
()
- [check_if_object_is_suspected_stationary_qualtest_check_if_object_is_suspected_stationary__cca_marked_as_stationary_if_](#)
()
- [check_if_object_is_suspected_stationary_qualtest_check_if_object_is_suspected_stationary__cca_marked_as_not_stationary](#)
()
- [check_if_object_is_suspected_stationary_qualtest_check_if_object_is_suspected_stationary__cca_marked_as_not_stationary](#)
()
- [check_if_object_is_suspected_stationary_qualtest_check_if_object_is_suspected_stationary__cca_marked_as_not_stationary](#)
()
- [check_if_object_is_suspected_stationary_qualtest_check_if_object_is_suspected_stationary__cca_marked_as_not_stationary](#)
()

2.44.1 Detailed Description

2.44.2 Function Documentation

2.44.2.1 check_if_object_is_suspected_stationary_qualtest()

```
test group check_if_object_is_suspected_stationary_qualtest ( )
```

Test group of Check_If_Object_Is_Suspected_Stationary() function. Test verify whether stationary suspected flag is properly set for both CTCA and CCA tracks.

Setup:\n

Initialize tracker calibrations

2.44.2.2 check_if_object_is_suspected_stationary_qualtest_check_if_object_is_suspected_stationary__cca_marked_as_r

```
check_if_object_is_suspected_stationary_qualtest_check_if_object_is_suspected_stationary__↵
cca_marked_as_not_stationary_if_has_enough_moving_detections ( )
```

Purpose:\n

Purpose of this test is to verify whether CCA object is not marked as stationary if has enough moving detections

Requirements FTCP-11471

Precondition:\n

Create occlusion object Set occlusion status to not occluded, so, that moving_speed_threshold = calib.k↵
 k_object_motion_min_speed Set object speed to be greater than calib.k_object_motion_min_speed, so that
 CCA conditions are reached Call Set_Object_Num_Moving_Dets(f_desired_moving = true), so that the detec-
 tions suggest that the object is not stationary Call Set_Object_Nees_Params(f_desired_moving = false), so
 that speed variance suggests that the object is stationary Call Set_Object_Parallel_Moving(f_desired_moving
 = false), so that the object is suspected to be stationary

Action:\n

Call Is_Object_Suspected_Stationary

Result:\n

Check if the object is suspected to be not stationary

2.44.2.3 check_if_object_is_suspected_stationary_qualtest_check_if_object_is_suspected_stationary__cca_marked_as_r

```
check_if_object_is_suspected_stationary_qualtest_check_if_object_is_suspected_stationary__↵
cca_marked_as_not_stationary_if_is_cross_moving ( )
```

Purpose:\n

Purpose of this test is to verify whether CCA object is not marked as stationary if is cross moving

Requirements FTCP-11471

Precondition:\n

Create occlusion object Set occlusion status to not occluded, so, that moving_speed_threshold = calib.k↵
 _object_motion_min_speed Set object speed to be greater than calib.k_object_motion_min_speed, so that
 CCA conditions are reached Call Set_Object_Num_Moving_Dets(f_desired_moving = false), so that the de-
 tectons suggest that the object is stationary Call Set_Object_Nees_Params(f_desired_moving = false), so that
 speed variance suggests that the object is not stationary Call Set_Object_Parallel_Moving(f_desired_moving
 = false), so that the object is suspected to be stationary Call Set_Object_Cross_Moving(tf_desired_moving =
 true), so that the object is suspected to be not stationary

Action:\n

Call Is_Object_Suspected_Stationary

Result:\n

Check if the object is suspected to be not stationary

2.44.2.4 check_if_object_is_suspected_stationary_qualtest_check_if_object_is_suspected_stationary__cca_marked_as_r

```
check_if_object_is_suspected_stationary_qualtest_check_if_object_is_suspected_stationary__↵
cca_marked_as_not_stationary_if_is_parallel_moving ( )
```

Purpose:\n

Purpose of this test is to verify whether CCA object is not marked as stationary if is parallel moving

Requirements FTCP-11471

Precondition:\n

Create occlusion object Set occlusion status to not occluded, so, that moving_speed_threshold = calib.k↵
_object_motion_min_speed Set object speed to be greater than calib.k_object_motion_min_speed, so that
CCA conditions are reached Call Set_Object_Num_Moving_Dets(f_desired_moving = false), so that the de-
tections suggest that the object is stationary Call Set_Object_Nees_Params(f_desired_moving = false), so
that speed variance suggests that the object is stationary Call Set_Object_Parallel_Moving(f_desired_moving
= true), so that the object is suspected to be not stationary

Action:\n

Call Is_Object_Suspected_Stationary

Result:\n

Check if the object is suspected to be not stationary

2.44.2.5 check_if_object_is_suspected_stationary_qualtest_check_if_object_is_suspected_stationary__cca_marked_as_r

```
check_if_object_is_suspected_stationary_qualtest_check_if_object_is_suspected_stationary__↵
cca_marked_as_not_stationary_if_meets_nees_condition ( )
```

Purpose:\n

Purpose of this test is to verify whether CTCA object is not marked as stationary if meets nees condition

Requirements FTCP-11471

Precondition:\n

Create occlusion object Set occlusion status to not occluded, so, that moving_speed_threshold = calib.k↵
_object_motion_min_speed Set object speed to be greater than calib.k_object_motion_min_speed, so that
CCA conditions are reached Call Set_Object_Num_Moving_Dets(f_desired_moving = false), so that the de-
tections suggest that the object is stationary Call Set_Object_Nees_Params(f_desired_moving = true), so that
speed variance suggests that the object is not stationary Call Set_Object_Parallel_Moving(f_desired_moving
= false), so that the object is suspected to be stationary

Action:\n

Call Is_Object_Suspected_Stationary

Result:\n

Check if the object is suspected to be not stationary

2.44.2.6 check_if_object_is_suspected_stationary_qualtest_check_if_object_is_suspected_stationary__cca_marked_as_s

```
check_if_object_is_suspected_stationary_qualtest_check_if_object_is_suspected_stationary__↔
cca_marked_as_stationary_if_does_not_meet_any_condition ( )
```

Purpose:\n

Purpose of this test is to verify whether CCA object is marked as stationary if does not meet any conditions to be moving.

Requirements FTCP-11471

Precondition:\n

Create occlusion object Set occlusion status to not occluded, so, that moving_speed_threshold = calib.k↔_object_motion_min_speed Set object speed to be greater than calib.k_object_motion_min_speed, so that CCA conditions are reached Call Set_Object_Num_Moving_Dets(f_desired_moving = false), so that the detections suggest that the object is stationary Set_Object_Nees_Params(f_desired_moving = false), so that the object is stationary Call Set_Object_Parallel_Moving(f_desired_moving = false), so that the object is suspected to be stationary

Action:\n

Call Is_Object_Suspected_Stationary

Result:\n

Check if the object is suspected to be stationary

2.44.2.7 check_if_object_is_suspected_stationary_qualtest_check_if_object_is_suspected_stationary__cca_marked_as_s

```
check_if_object_is_suspected_stationary_qualtest_check_if_object_is_suspected_stationary__↔
cca_marked_as_stationary_if_moveable_and_does_not_meet_any_moving_condition ( )
```

Purpose:\n

Purpose of this test is to verify whether CCA object is marked as stationary if is moveable and does not meet any other conditions to be moving.

Requirements FTCP-11471

Precondition:\n

Create occlusion object Set occlusion status to not occluded, so, that moving_speed_threshold = calib.k↵_object_motion_min_speed Set object speed to be greater than calib.k_object_motion_min_speed, so that CCA conditions are reached Call Set_Object_Num_Moving_Dets(f_desired_moving = false), so that the detections suggest that the object is stationary Call Set_Object_Nees_Params(f_desired_moving = false), so that speed variance suggests that the object is stationary Call Set_Object_Parallel_Moving(f_desired_moving = false), so that the object is suspected to be stationary Call Set_Object_Cross_Moving(f_desired_moving = false), so that the object is suspected to be stationary

Action:\n

Call Is_Object_Suspected_Stationary

Result:\n

Check if the object is suspected to be stationary

2.44.2.8 check_if_object_is_suspected_stationary_qualtest_check_if_object_is_suspected_stationary__ctca_marked_as_

```
check_if_object_is_suspected_stationary_qualtest_check_if_object_is_suspected_stationary__↵
ctca_marked_as_not_stationary_if_has_enough_moving_detections ( )
```

Purpose:\n

Purpose of this test is to verify whether CTCA object is not marked as stationary if has enough moving detections

Requirements FTCP-11471**Precondition:**\n

Set all but one object parameters to make it stationary

Action:\n

Call tested function

Result:\n

Check if f_stationary_suspected is false

2.44.2.9 `check_if_object_is_suspected_stationary_qualtest_check_if_object_is_suspected_stationary__ctca_marked_as_`

```
check_if_object_is_suspected_stationary_qualtest_check_if_object_is_suspected_stationary__↔  
ctca_marked_as_not_stationary_if_is_cross_moving ( )
```

Purpose:\n

Purpose of this test is to verify whether CTCA object is not marked as stationary if is cross moving

Requirements FTCP-11471

Precondition:\n

Set all but one object parameters to make it stationary

Action:\n

Call tested function

Result:\n

Check if `f_stationary_suspected` is false

2.44.2.10 `check_if_object_is_suspected_stationary_qualtest_check_if_object_is_suspected_stationary__ctca_marked_as_`

```
check_if_object_is_suspected_stationary_qualtest_check_if_object_is_suspected_stationary__↔  
ctca_marked_as_not_stationary_if_is_parallel_moving ( )
```

Purpose:\n

Purpose of this test is to verify whether CTCA object is not marked as stationary if is parallel moving

Requirements FTCP-11471

Precondition:\n

Set all but one object parameters to make it stationary

Action:\n

Call tested function

Result:\n

Check if `f_stationary_suspected` is false

2.44.2.11 check_if_object_is_suspected_stationary_qualtest_check_if_object_is_suspected_stationary__ctca_marked_as

```
check_if_object_is_suspected_stationary_qualtest_check_if_object_is_suspected_stationary__↔  
ctca_marked_as_not_stationary_if_meets_sigma_condition ( )
```

Purpose:\n

Purpose of this test is to verify whether CTCA object is not marked as stationary if meets sigma condition

Requirements FTCP-11471

Precondition:\n

Set all but one object parameters to make it stationary

Action:\n

Call tested function

Result:\n

Check if f_stationary_suspected is false

2.44.2.12 check_if_object_is_suspected_stationary_qualtest_check_if_object_is_suspected_stationary__ctca_marked_as

```
check_if_object_is_suspected_stationary_qualtest_check_if_object_is_suspected_stationary__↔  
ctca_marked_as_stationary_if_does_not_meet_any_condition ( )
```

Purpose:\n

Purpose of this test is to verify whether CTCA object is marked as stationary if does not meet any conditions to be moving.

Requirements FTCP-11471

Precondition:\n

Set all object parameters to make it stationary

Action:\n

Call tested function

Result:\n

Check if f_stationary_suspected is true

2.45 CalcProbability

Collaboration diagram for CalcProbability:

Modules

- [Set_object_class](#)

Functions

- test group [calcProbability_qualtest](#) ()
Test group of calcProbabilityfunction.
- [calcProbability_qualtest_calcProbability__check_if_sum_of_calculated_probabilities_is_equal_1](#) ()

2.45.1 Detailed Description

2.45.2 Function Documentation

2.45.2.1 [calcProbability_qualtest_calcProbability__check_if_sum_of_calculated_probabilities_is_equal_1\(\)](#)

```
calcProbability_qualtest_calcProbability__check_if_sum_of_calculated_probabilities_is_equal_1  
( )
```

Purpose:\n

Purpose of this test is to verify whether sum of calculated probabilities is equal to maximum defined value (1.0).

Requirements FTCP-11630

Precondition:\n

Setting required parameters

Action:\n

Call tested function.

Result:\n

Check whether sum of calculated probabilities is equal to `sum_of_probabilities_expected`.

2.46 Set_object_class

Collaboration diagram for Set_object_class:

Modules

- [FilteringAndNormalizationOfProbabilities](#)

Functions

- test group [set_object_class_qualtest \(\)](#)
Test group of setObjectClass function. Tests verify whether object class is properly assigned.
- [set_object_class_qualtest_set_object_class__check_if_prioritizes_pedestrian \(\)](#)
- [set_object_class_qualtest_set_object_class__check_if_prioritizes_motorcycles \(\)](#)
- [set_object_class_qualtest_set_object_class__check_if_prioritizes_cars \(\)](#)
- [set_object_class_qualtest_set_object_class__check_if_prioritizes_trucks \(\)](#)
- [set_object_class_qualtest_set_object_class__check_if_classifies_as_undetermined \(\)](#)
- [set_object_class_qualtest_set_object_class__check_if_min_prob_value_to_classify_is_0_65 \(\)](#)

2.46.1 Detailed Description

2.46.2 Function Documentation

2.46.2.1 set_object_class_qualtest_set_object_class__check_if_classifies_as_undetermined()

```
set_object_class_qualtest_set_object_class__check_if_classifies_as_undetermined ( )
```

Purpose:\n

Purpose of this test is to verify whether if object is not classified either as pedestrian, motorcycle car or truck, object is classified as undetermined.

Requirements FTCP-11631

Precondition:\n

Set pedestrian probability to be below threshold Set motorcycle probability to be below threshold Set car probability to be below threshold Set truck probability to be below threshold

Action:\n

Call tested function.

Result:\n

Check whether object was classified as truck

2.46.2.2 set_object_class_qualtest_set_object_class__check_if_min_prob_value_to_classify_is_0_65()

```
set_object_class_qualtest_set_object_class__check_if_min_prob_value_to_classify_is_0_65 ( )
```

Purpose:

Purpose of this test is to verify whether minimum probability value to classify track as specific class is equal to 0.65

Requirements FTCP-11536

Precondition:

Set pedestrian probability to be below threshold Set motorcycle probability to be below threshold Set car probability to be below threshold Set truck probability to be equal minimum probability

Action:

Call tested function.

Result:

Check whether object was classified as truck

2.46.2.3 set_object_class_qualtest_set_object_class__check_if_prioritizes_cars()

```
set_object_class_qualtest_set_object_class__check_if_prioritizes_cars ( )
```

Purpose:

Purpose of this test is to verify if object is not classified either as pedestrian or motorcycle, car class is prioritized

Requirements FTCP-11457

Precondition:

Set pedestrian probability to be below threshold Set motorcycle probability to be below threshold Set car probability to be above threshold Set truck probability to be above threshold

Action:

Call tested function.

Result:

Check whether object was classified as car

2.46.2.4 set_object_class_qualtest_set_object_class__check_if_prioritizes_motorcycles()

```
set_object_class_qualtest_set_object_class__check_if_prioritizes_motorcycles ( )
```

Purpose:\n

Purpose of this test is to verify if object is not classified as bicycle, motorcycle class is prioritized

Requirements FTCP-11460

Precondition:\n

Set pedestrian probability to be below threshold Set bicycle probability to be below threshold Set motorcycle probability to be above threshold Set car probability to be above threshold Set truck probability to be above threshold

Action:\n

Call tested function.

Result:\n

Check whether object was classified as motorcycle

2.46.2.5 set_object_class_qualtest_set_object_class__check_if_prioritizes_pedestrian()

```
set_object_class_qualtest_set_object_class__check_if_prioritizes_pedestrian ( )
```

Purpose:\n

Purpose of this test is to verify whether pedestrian class is prioritized over other classes.

Requirements FTCP-11459

Precondition:\n

Set pedestrian probability to be above threshold Set bicycle probability to be above threshold Set motorcycle probability to be above threshold Set car probability to be above threshold Set truck probability to be above threshold

Action:\n

Call tested function.

Result:\n

Check whether object was classified as pedestrian

2.46.2.6 `set_object_class_qualtest_set_object_class__check_if_prioritizes_trucks()`

```
set_object_class_qualtest_set_object_class__check_if_prioritizes_trucks ( )
```

Purpose:

Purpose of this test is to verify if object is not classified either as pedestrian, motorcycle or car, truck class is prioritized

Requirements FTCP-11458

Precondition:

Set pedestrian probability to be below threshold Set motorcycle probability to be below threshold Set car probability to be below threshold Set truck probability to be above threshold

Action:

Call tested function.

Result:

Check whether object was classified as truck

2.47 FilteringAndNormalizationOfProbabilities

Collaboration diagram for FilteringAndNormalizationOfProbabilities:

Modules

- [CalcUndetProb](#)

Functions

- test group [filteringAndNormalizationOfProbabilities_qualtest](#) ()
Test group of filteringAndNormalizationOfProbabilities function. Tests verify whether probability being undetermined is used to lower probabilities of object being specific class.
- [filteringAndNormalizationOfProbabilities_qualtest_filteringAndNormalizationOfProbabilities__check_if_undeterm_prob_value_i](#) ()
- test group [f360_run_obj_class](#) ()
Test group for run_obj_class. Purpose is to test that all object classes are possible object classifications, corresponding to the requirement FTCP-11450. The classes are.
 - [f360_run_obj_class_f360_run_obj_class_bicycle](#) ()
 - [f360_run_obj_class_f360_run_obj_class_pedestrian](#) ()
 - [f360_run_obj_class_f360_run_obj_class_motorcycle](#) ()
 - [f360_run_obj_class_f360_run_obj_class_car](#) ()
 - [f360_run_obj_class_f360_run_obj_class_truck](#) ()
 - [f360_run_obj_class_f360_run_obj_class_undetermined](#) ()

2.47.1 Detailed Description

2.47.2 Function Documentation

2.47.2.1 f360_run_obj_class()

```
test group f360_run_obj_class ( )
```

Test group for run_obj_class. Purpose is to test that all object classes are possible object classifications, corresponding to the requirement FTCP-11450. The classes are.

- pedestrian
- motorcycle
- bicycle
- car
- truck
- undetermined

Setup:\n

- Initialize calibrations
- Set host speed to 2 m/s
- Set up an object with
 - Length 2 m
 - Width 1.15 m
 - Speed 5.29 m/s
 - and status UPDATED

2.47.2.2 f360_run_obj_class_f360_run_obj_class_bicycle()

```
f360_run_obj_class_f360_run_obj_class_bicycle ( )
```

Purpose:\n

Purpose of this test is to verify that an object can be classified as a bicycle.

Requirements FTCP-11450

Precondition:\n

An object has been set up in the TEST_GROUP. Set expected object class to bicycle

Action:\n

Call run_obj_class.

Result:\n

Check that the object is classified as bicycle.

2.47.2.3 f360_run_obj_class_f360_run_obj_class_car()

```
f360_run_obj_class_f360_run_obj_class_car ( )
```

Purpose:\n

Purpose of this test is to verify that an object can be classified as a car.

Requirements FTCP-11450

Precondition:\n

Set expected object class to car. An object has been set up in the TEST_GROUP.

- Change speed to 20 m/s
- and length and width to 5 m and 2 m respectively.

Action:\n

Call run_obj_class.

Result:\n

Check that the object is classified as car.

2.47.2.4 f360_run_obj_class_f360_run_obj_class_motorcycle()

```
f360_run_obj_class_f360_run_obj_class_motorcycle ( )
```

Purpose:\n

Purpose of this test is to verify that an object can be classified as a motorcycle.

Requirements FTCP-11450

Precondition:\n

Set expected object class to motorcycle. An object has been set up in the TEST_GROUP.

- Change speed to 20 m/s
- and length to 2.5 m.

Action:\n

Call run_obj_class.

Result:\n

Check that the object is classified as motorcycle.

2.47.2.5 f360_run_obj_class_f360_run_obj_class_pedestrian()

```
f360_run_obj_class_f360_run_obj_class_pedestrian ( )
```

Purpose:\n

Purpose of this test is to verify that an object can be classified as a pedestrian.

Requirements FTCP-11450

Precondition:\n

Set expected object class to pedestrian. An object has been set up in the TEST_GROUP.

- Change speed to 1 m/s
- and length to 1 m.

Action:\n

Call run_obj_class.

Result:\n

Check that the object is classified as pedestrian.

2.47.2.6 f360_run_obj_class_f360_run_obj_class_truck()

```
f360_run_obj_class_f360_run_obj_class_truck ( )
```

Purpose:\n

Purpose of this test is to verify that an object can be classified as a truck.

Requirements FTCP-11450

Precondition:\n

Set expected object class to truck. An object has been set up in the TEST_GROUP.

- Change speed to 20 m/s
- Set length and width to 16 m and 2.5m respectively

Action:\n

Call run_obj_class.

Result:\n

Check that the object is classified as truck.

2.47.2.7 f360_run_obj_class_f360_run_obj_class_undetermined()

```
f360_run_obj_class_f360_run_obj_class_undetermined ( )
```

Purpose:\n

Purpose of this test is to verify that an object can be classified as a undetermined.

Requirements FTCP-11450

Precondition:\n

Set expected object class to undetermined. An object has been set up in the TEST_GROUP.

- Change speed to 0.002 m/s
- Set both length and width to 1 m
- Set status to NEW_UPDATED

Action:\n

Call run_obj_class.

Result:\n

Check that the object is classified as undetermined.

2.47.2.8 filteringAndNormalizationOfProbabilities_qualtest_filteringAndNormalizationOfProbabilities__check_if_undeterm

```
filteringAndNormalizationOfProbabilities_qualtest_filteringAndNormalizationOfProbabilities__↵  
check_if_undeterm_prob_value_is_used_to_lower_prob_of_other_class ( )
```

Purpose:

Purpose of this test is to verify whether probability of being undetermined is used to lower probabilities of object being specific class

Requirements FTCP-11463**Precondition:**

Setting required parameters

Action:

Call tested function.

Result:

Check whether object was classified as truck

2.48 CalcUndetProb

Collaboration diagram for CalcUndetProb:

Functions

- test group `calcUndetProb_qualtest` ()

Purpose of this test group is to verify whether probability unknown is set to max value when object has status: F360_OBJECT_STATUS_NEW or F360_OBJECT_STATUS_INVALID or F360_OBJECT_STATUS_NEW_UPDATED or F360_OBJECT_STATUS_NEW_COASTED.

- `calcUndetProb_qualtest_calcUndetProb__check_if_prob_unknown_is_set_to_max_value_when_object_has_status_F360_OBJECT_STATUS_NEW` ()
- `calcUndetProb_qualtest_calcUndetProb__check_if_prob_unknown_is_set_to_max_value_when_object_has_status_F360_OBJECT_STATUS_INVALID` ()
- `calcUndetProb_qualtest_calcUndetProb__check_if_prob_unknown_is_set_to_max_value_when_object_has_status_F360_OBJECT_STATUS_NEW_UPDATED` ()
- `calcUndetProb_qualtest_calcUndetProb__check_if_prob_unknown_is_set_to_max_value_when_object_has_status_F360_OBJECT_STATUS_NEW_COASTED` ()

2.48.1 Detailed Description

2.48.2 Function Documentation

2.48.2.1 `calcUndetProb_qualtest()`

```
test_group calcUndetProb_qualtest ( )
```

Purpose of this test group is to verify whether probability unknown is set to max value when object has status: F360_OBJECT_STATUS_NEW or F360_OBJECT_STATUS_INVALID or F360_OBJECT_STATUS_NEW_UPDATED or F360_OBJECT_STATUS_NEW_COASTED.

Setup:\n

2.48.2.2 calcUndetProb_qualtest_calcUndetProb__check_if_prob_unknown_is_ste_to_max_value_when_object_has_statu

```
calcUndetProb_qualtest_calcUndetProb__check_if_prob_unknown_is_ste_to_max_value_when_object_↵  
has_status_F360_OBJECT_STATUS_INVALID ( )
```

Purpose:

Purpose of this test is to verify whether probability unknown is set to max value when object has status:
F360_OBJECT_STATUS_INVALID

Requirements FTCP-11454

Precondition:

Setting required parameters

Action:

Call tested function.

Result:

Check whether object has undetermined probability equal 1.

2.48.2.3 calcUndetProb_qualtest_calcUndetProb__check_if_prob_unknown_is_ste_to_max_value_when_object_has_statu

```
calcUndetProb_qualtest_calcUndetProb__check_if_prob_unknown_is_ste_to_max_value_when_object_↵  
has_status_F360_OBJECT_STATUS_NEW ( )
```

Purpose:

Purpose of this test is to verify whether probability unknown is set to max value when object has status:
F360_OBJECT_STATUS_NEW

Requirements FTCP-11454

Precondition:

Setting required parameters

Action:

Call tested function.

Result:

Check whether object has undetermined probability equal 1.

2.48.2.4 calcUndetProb_qualtest_calcUndetProb__check_if_prob_unknown_is_ste_to_max_value_when_object_has_statu

```
calcUndetProb_qualtest_calcUndetProb__check_if_prob_unknown_is_ste_to_max_value_when_object_↵  
has_status_F360_OBJECT_STATUS_NEW_COASTED ( )
```

Purpose:\n

Purpose of this test is to verify whether probability unknown is set to max value when object has status:
F360_OBJECT_STATUS_NEW_COASTED

Requirements FTCP-11454

Precondition:\n

Setting required parameters

Action:\n

Call tested function.

Result:\n

Check whether object has undetermined probability equal 1.

2.48.2.5 calcUndetProb_qualtest_calcUndetProb__check_if_prob_unknown_is_ste_to_max_value_when_object_has_statu

```
calcUndetProb_qualtest_calcUndetProb__check_if_prob_unknown_is_ste_to_max_value_when_object_↵  
has_status_F360_OBJECT_STATUS_NEW_UPDATED ( )
```

Purpose:\n

Purpose of this test is to verify whether probability unknown is set to max value when object has status:
F360_OBJECT_STATUS_NEW_UPDATED

Requirements FTCP-11454

Precondition:\n

Setting required parameters

Action:\n

Call tested function.

Result:\n

Check whether object has undetermined probability equal 1.

2.49 Track_Downselection

Functions

- test group [f360_track_downselection_qualtest](#) ()

This test group is used for creatie data structure needed in further tests.

- [f360_track_downselection_qualtest_f360_core_track_downselect__check_if_does_reduce_tracks_basing_on_priority](#) ()
- [f360_track_downselection_qualtest_f360_core_track_downselect__check_if_updated_tracks_are_prioritized_over_new_tracks](#) ()
- [f360_track_downselection_qualtest_f360_core_track_downselect__check_if_tracks_with_reduced_status_set_as_invalid_are](#) ()
- [f360_track_downselection_qualtest_f360_core_track_downselect__check_if_tracks_with_higher_confidence_level_are_prioriti](#) ()
- [f360_track_downselection_qualtest_f360_core_track_downselect__check_if_valid_trk_closer_to_host_is_prioritized](#) ()
- [f360_track_downselection_qualtest_f360_core_track_downselect__check_if_tracks_with_lower_ttc_are_prioritized](#) ()
- [f360_track_downselection_qualtest_f360_core_track_downselect__check_if_tracks_with_low_amount_of_average_dets_are_c](#) ()
- [f360_track_downselection_qualtest_f360_core_track_downselect__check_if_tracks_are_downselected_as_expected_based_o](#) ()
- [f360_track_downselection_qualtest_f360_core_track_downselect__check_if_resets_low_priority_target_reduced_id](#) ()
- [f360_track_downselection_qualtest_f360_core_track_downselect__check_if_assigns_reduced_id_to_higher_priority_targets_a](#) ()
- [f360_track_downselection_qualtest_f360_core_track_downselect__check_if_lowers_confidence_of_tracks_that_changed_their](#) ()
- [f360_track_downselection_qualtest_f360_core_track_downselect__check_if_lowers_confidence_and_priority_of_mirror_detec](#) ()
- [f360_track_downselection_qualtest_f360_core_track_downselect__check_if_lowers_confidence_of_tracks_behind_guardrail](#) ()

2.49.1 Detailed Description

2.49.2 Function Documentation

2.49.2.1 [f360_track_downselection_qualtest_f360_core_track_downselect__check_if_assigns_reduced_id_to_higher_prio](#)

```
f360_track_downselection_qualtest_f360_core_track_downselect__check_if_assigns_reduced_id_to←
_higher_priority_targets_and_resets_reduced_id_of_lower_priority_targets ( )
```

Purpose:\n Purpose of this test is to verify whether tracks with high priority are assigned reduced ID. Low priority targets should not be assigned reduced id.

Requirements FTCP-10728

Precondition:

Create more tracks than NUMBER_OF_REDUCED_OBJECT_TRACKS. Assign reduced_id to tracks in range <1 - NUMBER_OF_REDUCED_TRACKS>. Set up parameters of selected tracks in range <1 - NUMBER_OF_REDUCED_TRACKS> so they will be treated as low priority. Set up parameters of all other tracks so they will be treated as high priority.

Action:

call Track_Downselection

Result:

Verify whether reduced id was taken from tracks that were qualified as low priority. Verify whether new tracks were assigned reduced_id.

2.49.2.2 f360_track_downselection_qualtest_f360_core_track_downselect__check_if_does_reduce_tracks_basing_on_prio

```
f360_track_downselection_qualtest_f360_core_track_downselect__check_if_does_reduce_tracks_basing_on_priority ( )
```

Purpose: Purpose of this test is to verify whether tracker does downselect tracks basing on their calculated priority.

Requirements FTCP-10226

Precondition:

Set up more tracks than NUMBER_OF_REDUCED_OBJECT_TRACKS Assigning them proper values so priority will be calculated for them Do not assign them reduced_id Increase lateral position of selected tracks to decrease their priority

Action:

call Track_Downselection

Result:

Verify whether specified tracks were not assigned reduced_id

2.49.2.3 f360_track_downselection_qualtest_f360_core_track_downselect__check_if_lowers_confidence_and_priority_of_

```
f360_track_downselection_qualtest_f360_core_track_downselect__check_if_lowers_confidence_and←  
_priority_of_mirror_detections ( )
```

Purpose:\n Purpose of this test is to verify whether tracks with high mirror probability have their confidence level and priority decreased

Requirements FTCP-10524

Precondition:\n

Set up tracks that fill requirements for having low priority

Action:\n

call Track_Downselection

Result:\n

Verify whether confidence level of first track was reduced and confidence of second track was not changed.
Verify whether both tracks were not assigned reduced_id

2.49.2.4 f360_track_downselection_qualtest_f360_core_track_downselect__check_if_lowers_confidence_of_tracks_behind

```
f360_track_downselection_qualtest_f360_core_track_downselect__check_if_lowers_confidence_of_←  
tracks_behind_guardrail ( )
```

Purpose:\n Purpose of this test is to verify whether tracks that are behind guardrail have their confidence level decreased.

Requirements FTCP-10524

Precondition:\n

Set up tracks that fill requirements for having low priority

Action:\n

call Track_Downselection

Result:\n

Verify whether confidence level of first track was reduced and confidence of second track was not changed.

2.49.2.5 f360_track_downselection_qualtest_f360_core_track_downselect__check_if_lowers_confidence_of_tracks_that_c

```
f360_track_downselection_qualtest_f360_core_track_downselect__check_if_lowers_confidence_of_↵  
tracks_that_changed_their_status_from_new_to_coasted ( )
```

Purpose:\n Purpose of this test is to verify whether tracks that changed their status
from new to coasted have their confidence level decreased.

Requirements FTCP-10524

Precondition:\n

Set up tracks to be marked as changing status from updated to coasted.

Action:\n

call Track_Downselection

Result:\n

Verify whether confidence evel of first track is higher than the second one Confidence level of third track should not change

2.49.2.6 f360_track_downselection_qualtest_f360_core_track_downselect__check_if_resets_low_priority_target_reduced

```
f360_track_downselection_qualtest_f360_core_track_downselect__check_if_resets_low_priority_↵  
target_reduced_id ( )
```

Purpose:\n Purpose of this test is to verify whether low priority objects have their reduced id taken

Requirements FTCP-10728

Precondition:\n

Set up tracks that fill requirements for having low priority

Action:\n

call Track_Downselection

Result:\n

Verify whether their reduced id was reseted.

2.49.2.7 f360_track_downselection_qualtest_f360_core_track_downselect__check_if_tracks_are_downselected_as_expected

```
f360_track_downselection_qualtest_f360_core_track_downselect__check_if_tracks_are_downselected↵  
_as_expected_based_on_overall_confidence ( )
```

Purpose:\n Purpose of this test is to verify that tracks are downselected correctly based on their overall confidence

Requirements FTCP-10520

Precondition:\n

Set up tracks to have same params and have different overall confidence

Action:\n

call Track_Downselection

Result:\n

Verify whether first track was not assigned reduced id Verify whether second track was assigned reduced id

2.49.2.8 f360_track_downselection_qualtest_f360_core_track_downselect__check_if_tracks_with_higher_confidence_level

```
f360_track_downselection_qualtest_f360_core_track_downselect__check_if_tracks_with_higher↵  
confidence_level_are_prioritized ( )
```

Purpose:\n Purpose of this test is to verify whether tracks with higher confidence level are prioritized

Requirements FTCP-10520

Precondition:\n

Set up tracks to have same params and different confidence level Tracks should not be qualified as near host

Action:\n

call Track_Downselection

Result:\n

Verify whether first track was not assigned reduced id Verify whether second track was assigned reduced id

2.49.2.9 f360_track_downselection_qualtest_f360_core_track_downselect__check_if_tracks_with_low_amount_of_average_dets_are_downprioritized ()

```
f360_track_downselection_qualtest_f360_core_track_downselect__check_if_tracks_with_low_amount_of_average_dets_are_downprioritized ( )
```

Purpose: Purpose of this test is to verify whether tracks with low number of average dets are downprioritized

Requirements FTCP-10520

Precondition:

Set up tracks that fill requirements for having high and low number of average dets

Action:

call Track_Downselection

Result:

Verify whether track with high number of total_reduced_dets was assigned reduced id and the other was not

2.49.2.10 f360_track_downselection_qualtest_f360_core_track_downselect__check_if_tracks_with_lower_ttc_are_prioritized ()

```
f360_track_downselection_qualtest_f360_core_track_downselect__check_if_tracks_with_lower_ttc_are_prioritized ( )
```

Purpose: Purpose of this test is to verify whether tracks with lower time to collision are prioritized

Requirements FTCP-10520

Precondition:

Set up tracks that fill requirements for having: high ttc, low ttc

Action:

call Track_Downselection

Result:

Verify whether track with lower ttc was prioritized

2.49.2.11 f360_track_downselection_qualtest_f360_core_track_downselect__check_if_tracks_with_reduced_status_set_as

```
f360_track_downselection_qualtest_f360_core_track_downselect__check_if_tracks_with_reduced_↵  
status_set_as_invalid_are_not_downselected ( )
```

Purpose:\n Purpose of this test is to verify whether tracks which are already reduced will not have their reduced id overwritten

Requirements FTCP-10520

Precondition:\n

Set up tracks to have same params and have different reduced status Tracks should not be qualified as near host

Action:\n

call Track_Downselection

Result:\n

Verify whether first track was not assigned reduced id Verify whether second track was assigned reduced id

2.49.2.12 f360_track_downselection_qualtest_f360_core_track_downselect__check_if_updated_tracks_are_prioritized_ov

```
f360_track_downselection_qualtest_f360_core_track_downselect__check_if_updated_tracks_are_↵  
prioritized_over_new_tracks ( )
```

Purpose:\n Purpose of this test is to verify whether tracks updated are prioritized over new tracks

Requirements FTCP-10520

Precondition:\n

Set up tracks to have same params and have different status Tracks should not be qualified as near host

Action:\n

call Track_Downselection

Result:\n

Verify whether first track was not assigned reduced id Verify whether second track was assigned reduced id

2.49.2.13 f360_track_downselection_qualtest_f360_core_track_downselect__check_if_valid_trk_closer_to_host_is_prioritized

```
f360_track_downselection_qualtest_f360_core_track_downselect__check_if_valid_trk_closer_to_↵  
host_is_prioritized ( )
```

Purpose:\n Purpose of this test is to verify whether tracks closer to host are prioritized

Requirements FTCP-10520

Precondition:\n

Set up tracks that fill requirements for being qualified as near host

Action:\n

call Track_Downselection

Result:\n

Verify whether tracks were prioritized (lower reduced id -> higher priority)

2.50 Track_Downselection__Calculate_Track_Priority_Non_Moveable_Objects

Functions

- test group [f360_track_downselection_calc_track_priority_qualtest \(\)](#)

This test group tests priority assignment strategy for non_moveable objects. It tests the following cases: If host speed should impact importance (for downselection), such that the objects in the direction of host motion, should have higher importance. For forward driving host, objects in front have higher importance. On the flip side, for reverse driving host, objects in the back of host have higher importance. The importance of the non_movable objects is decided based on their vcs longitudinal position and curvi lateral position (based on host trajectory). Objects longitudinally further away from the host are given lower importance for downselection, as compared to longitudinally closer objects. Objects laterally further away from the host trajectory are given lower importance for downselection, as compared to laterally closer objects.

- [f360_track_downselection_calc_track_priority_qualtest_Host_Is_Stationary \(\)](#)
- [f360_track_downselection_calc_track_priority_qualtest_Host_Is_Moving \(\)](#)
- [f360_track_downselection_calc_track_priority_qualtest_Host_Is_Reversing \(\)](#)
- [f360_track_downselection_calc_track_priority_qualtest_Impact_oF_Host_Speeds_On_Object_Priorities \(\)](#)
- [f360_track_downselection_calc_track_priority_qualtest_Host_Curvature_Is_Not_Close_To_Zero \(\)](#)

2.50.1 Detailed Description

2.50.2 Function Documentation

2.50.2.1 f360_track_downselection_calc_track_priority_qualtest_Host_Curvature_Is_Not_Close_To_Zero()

```
f360_track_downselection_calc_track_priority_qualtest_Host_Curvature_Is_Not_Close_To_Zero ( )
```

Purpose: Purpose of this test is to verify that priority of objects is decided correctly, when the host has non close to zero curvature

The test also checks if the host trajectory is taken into account, when deciding the priorities

Requirements FTCP-13920

Precondition:

Call the Calc_Track_Priority with zero host curvature and slow host speed (5m/s) After that, set the host curvature to be relatively higher (0.005 m^{-1}) The idea here is to test the impact of higher host curvature (0.005) on object priorities

Action:

Call Calc_Track_Priority

Result:

High priority value means less importance for downselection

2.50.2.2 f360_track_downselection_calc_track_priority_qualtest_Host_Is_Moving()

```
f360_track_downselection_calc_track_priority_qualtest_Host_Is_Moving ( )
```

Purpose:\n Purpose of this test is to verify that priority of objects is decided correctly, when host is moving forward

Requirements FTCP-13920

Precondition:\n

Set host speed to be positive (5.0 m/s) Host curvature is close to zero (0.0001)

Action:\n

Call Calc_Track_Priority

Result:\n

High priority value means less importance for downselection

2.50.2.3 f360_track_downselection_calc_track_priority_qualtest_Host_Is_Reversing()

```
f360_track_downselection_calc_track_priority_qualtest_Host_Is_Reversing ( )
```

Purpose:\n Purpose of this test is to verify that priority of objects is decided correctly, when host is reversing

The idea here is to check if the back objects are given more priority for downselection than front objects

Requirements FTCP-13920

Precondition:\n

Set host speed to be -3.0 m/s i.e the host is reversing Host curvature is close to zero (0.0001)

Action:\n

Call Calc_Track_Priority

Result:\n

High priority value means less importance for downselection Some checks are flipped as compared to when host is moving forward (see test: "Host_Is_Moving_Slow")

2.50.2.4 f360_track_downselection_calc_track_priority_qualtest_Host_Is_Stationary()

```
f360_track_downselection_calc_track_priority_qualtest_Host_Is_Stationary ( )
```

Purpose:\n Purpose of this test is to verify that priority of objects is decided correctly, when host is stationary

Requirements FTCP-13920

Precondition:\n

Same as the test setup i.e host speed and curvature is zero

Action:\n

Call Calc_Track_Priority

Result:\n

High priority value means less importance for downselection

2.50.2.5 f360_track_downselection_calc_track_priority_qualtest_Impact_oF_Host_Speeds_On_Object_Priorities()

```
f360_track_downselection_calc_track_priority_qualtest_Impact_oF_Host_Speeds_On_Object_Priorities  
( )
```

Purpose:\n Purpose of this test is to verify that priority of objects is decided correctly, when host is moving fast and forward

The test also checks the impact of host speed on object priority calculation

Requirements FTCP-13920

Precondition:\n

Call the Calc_Track_Priority with slow host speed (5m/s) After that, set the host speed to be high (30m/s) The Host curvature is close to zero (0.0001) in both cases The idea here is to test the impact of host speed on object priorities

Action:\n

Call Calc_Track_Priority

Result:\n

High priority value means less importance for downselection

2.51 F360_split_objects_in_orth_direction_qualtest

Functions

- test group [f360_split_objects_in_orth_direction_qualtest](#) ()
This purpose of this test group is for testing the functionality of the main function in f360_split_objects_in_orth_↔direction.cpp file.
- [f360_split_objects_in_orth_direction_qualtest_Verify_Split_Logic](#) ()

2.51.1 Detailed Description

2.51.2 Function Documentation

2.51.2.1 f360_split_objects_in_orth_direction_qualtest()

```
test_group f360_split_objects_in_orth_direction_qualtest ( )
```

This purpose of this test group is for testing the functionality of the main function in f360_split_objects_in_orth_↔direction.cpp file.

Setup:\n

Initialize common variable used in tests. Set tracker info to have 1 active object Initialize an object that should trigger a split.

2.51.2.2 f360_split_objects_in_orth_direction_qualtest_Verify_Split_Logic()

```
f360_split_objects_in_orth_direction_qualtest_Verify_Split_Logic ( )
```

Purpose:\n

Test checks that the function split an object in its orthogonal direction when the object have aggregated a large gap between its associated detections in the objects orthogonal direction.

Requirements FTCP-13121

Precondition:\n

Action:\n

call main function in f360_split_objects_in_orth_direction.cpp; Split_Objects_In_Orth_Direction().

Result:\n

Verify that split happened by checking that the number of active objects have increased by 1

2.52 F360_track_grouping

Functions

- test group [f360_track_grouping](#) ()
This purpose of this test group is for testing the functionality of the Track_Grouping() function.
- [f360_track_grouping_Test_Similar_Objects_Were_Merged](#) ()
- [f360_track_grouping_Test_Merged_Obj_Max_Size_Is_Not_exceed](#) ()
- [f360_track_grouping_Test_Remain_Dets_Assoc](#) ()
- [f360_track_grouping_Test_Move_All_Dets_Assoc](#) ()
- [f360_track_grouping_Test_Move_Dets_Assoc](#) ()
- [f360_track_grouping_Check_Consistency_In_Obj_Id](#) ()
- [f360_track_grouping_Check_Updates_In_Long_Pos_Sort](#) ()
- [f360_track_grouping_Check_Updates_In_Priority_Sort](#) ()

2.52.1 Detailed Description

2.52.2 Function Documentation

2.52.2.1 f360_track_grouping()

test group `f360_track_grouping` ()

This purpose of this test group is for testing the functionality of the Track_Grouping() function.

Setup:\n

Initialization of used variables.

2.52.2.2 f360_track_grouping_Check_Consistency_In_Obj_Id()

`f360_track_grouping_Check_Consistency_In_Obj_Id` ()

Purpose:\n The purpose of this test is to check,

if newly merged object has id of kept object.

Requirements FTCP-12944

Precondition:\n

Define order of sorted tracks.

Action:\n

Call the function Track_Grouping()

Result:\n

Merged object id should be equal to expected value.

2.52.2.3 f360_track_grouping_Check_Updates_In_Long_Pos_Sort()

```
f360_track_grouping_Check_Updates_In_Long_Pos_Sort ( )
```

Purpose:\n The purpose of this test is to check,
if sort list of sorted object by vcs long position was updated properly.

Requirements FTCP-13122

Precondition:\n

Fill relevant properties of object with same values, Define different IDs for this object. Define order of sorted tracks.

Action:\n

Call the function Track_Grouping()

Result:\n

Check if vcslong_sorted list changed as expected. Order of object idx by vcs long should have been changed from 2,4,3,0,1 to 2,4,3,0

2.52.2.4 f360_track_grouping_Check_Updates_In_Priority_Sort()

```
f360_track_grouping_Check_Updates_In_Priority_Sort ( )
```

Purpose:\n The purpose of this test is to check,
if object priority list was updated properly.

Requirements FTCP-12951

Precondition:\n

Fill relevant properties of object with same values, Define different IDs for this object. Define order of sorted tracks.

Action:\n

Call the function Track_Grouping()

Result:\n

Check if track priority list changed as expected. Order of object idx by priority should be 4,0,2,3

2.52.2.5 f360_track_grouping_Test_Merged_Obj_Max_Size_Is_Not_excced()

f360_track_grouping_Test_Merged_Obj_Max_Size_Is_Not_excced ()

Purpose:\n The purpose of this test is to check,

if size of merged object doesn't exceed maximum parameters.

Requirements FTCP-12947

Precondition:\n

Define order of sorted tracks. Define objects width which is greater that maximum width value.

Action:\n

Call the function Track_Grouping()

Result:\n

Merged object width should be less or equal than maximum object width

2.52.2.6 f360_track_grouping_Test_Move_All_Dets_Assoc()

f360_track_grouping_Test_Move_All_Dets_Assoc ()

Purpose:\n The purpose of this test is to check,

if detections previously associated to killed object are associated properly to newly merged one.

Requirements FTCP-12952

Precondition:\n

Define order of sorted tracks. Define detections associated to killed Object.

Action:\n

Call the function Track_Grouping()

Result:\n

Number of detections in merged object should be equal to expected value. Killed object should not contain any associated detections.

2.52.2.7 f360_track_grouping_Test_Move_Dets_Assoc()

```
f360_track_grouping_Test_Move_Dets_Assoc ( )
```

Purpose: The purpose of this test is to check,

if detections previously associated to killed object are associated properly to newly merged one, when sum of detections from both objects is greater than max number of detections associated to object.

Requirements FTCP-12952

Precondition:

Define order of sorted tracks. Define detections association to both objects.

Action:

Call the function Track_Grouping()

Result:

Number of detections in merged object should be equal to expected value. Killed object should not contain any associated detections.

2.52.2.8 f360_track_grouping_Test_Remain_Dets_Assoc()

```
f360_track_grouping_Test_Remain_Dets_Assoc ( )
```

Purpose: The purpose of this test is to check,

if all detections previously associated to original object are still associated to newly merged one.

Requirements FTCP-12948

Precondition:

Define order of sorted tracks. Define detections associated to object.

Action:

Call the function Track_Grouping()

Result:

Number of detections in merged object should be equal to expected value. Killed object should not contain any associated detections.

2.52.2.9 f360_track_grouping_Test_Similar_Objects_Were_Merged()

f360_track_grouping_Test_Similar_Objects_Were_Merged ()

Purpose:\n The purpose of this test is to check,

if two objects with same parameters of position, velocity and heading have been merged.

Requirements FTCP-12943

Precondition:\n

Define order of sorted tracks and number of active objects.

Action:\n

Call the function Track_Grouping()

Result:\n

One of defined object should be invalidated

2.53 F360_detect_multipath__Detect_And_Mark_Multipath_Objects_Qual

Functions

- test group `f360_detect_multipath__Detect_And_Mark_Multipath_Objects_Qual()`
Test group of Detect_And_Mark_Multipath_Objects from f360_detect_multipath module.
- `f360_detect_multipath__Detect_And_Mark_Multipath_Objects_Qual_default_scenario_where_host_speed_is_too_high()`
- `f360_detect_multipath__Detect_And_Mark_Multipath_Objects_Qual_default_scenario_where_host_speed_is_below_threshold()`
- `f360_detect_multipath__Detect_And_Mark_Multipath_Objects_Qual_scenario_where__source_objects_is_stationary()`
- `f360_detect_multipath__Detect_And_Mark_Multipath_Objects_Qual_scenario_where__source_objects_is_inactive()`
- `f360_detect_multipath__Detect_And_Mark_Multipath_Objects_Qual_scenario_where__source_objects_is_beyond_sensor_fov()`
- `f360_detect_multipath__Detect_And_Mark_Multipath_Objects_Qual_scenario_where__source_objects_use_CCV_filter()`
- `f360_detect_multipath__Detect_And_Mark_Multipath_Objects_Qual_scenario_where__reflector_objects_does_not_exist()`
- `f360_detect_multipath__Detect_And_Mark_Multipath_Objects_Qual_scenario_where__source_objects_does_not_exist()`

2.53.1 Detailed Description

2.53.2 Function Documentation

2.53.2.1 f360_detect_multipath__Detect_And_Mark_Multipath_Objects_Qual()

```
test_group f360_detect_multipath__Detect_And_Mark_Multipath_Objects_Qual ( )
```

Test group of Detect_And_Mark_Multipath_Objects from f360_detect_multipath module.

Setup:\n

Two rear sensors are set up. Three objects inside radar FOV are created: source, reflector and ghost. Properties of those objects are set according multipath theory. Tracker info is also fill properly for those three active objects.

2.53.2.2 f360_detect_multipath__Detect_And_Mark_Multipath_Objects_Qual_default_scenario_where_host_speed_is_belo

```
f360_detect_multipath__Detect_And_Mark_Multipath_Objects_Qual_default_scenario_where_host_↵  
speed_is_below_threshold ( )
```

Purpose:\n Check if Detect_And_Mark_Multipath_Objects() change ghost object mirror probability to default mirror probability when host speed is below threshold and all other conditions have been met.

Requirements CPR-3768 CPR-3769 CPR-3771 CPR-3770

Precondition:\n

Host speed is lower than max host speed for enabling multipath detection.

Action:\n

Call tested function

Result:\n

Ghost object mirror probability should be set to default mirror probability. The rest objects should have 0.0F

2.53.2.3 f360_detect_multipath__Detect_And_Mark_Multipath_Objects_Qual_default_scenario_where_host_speed_is_too

```
f360_detect_multipath__Detect_And_Mark_Multipath_Objects_Qual_default_scenario_where_host_↵  
speed_is_too_high ( )
```

Purpose:\n Check if Detect_And_Mark_Multipath_Objects() is not activated properly if host speed is above threshold.

Requirements CPR-3768

Precondition:\n

Host speed is minimum host speed for disabling multipath detection.

Action:\n

Call Detect_And_Mark_Multipath_Objects()

Result:\n

All objects mirror probability should be set to 0.0F

2.53.2.4 f360_detect_multipath__Detect_And_Mark_Multipath_Objects_Qual_scenario_where__reflector_objects_does_not_exist

```
f360_detect_multipath__Detect_And_Mark_Multipath_Objects_Qual_scenario_where__reflector_↵  
objects_does_not_exist ( )
```

Purpose:\n Check if Detect_And_Mark_Multipath_Objects() doesn't change ghost object mirror probability to default value when reflector does not exists.

Requirements CPR-3770

Precondition:\n

Remove reflector object.

Action:\n

Call tested function

Result:\n

Ghost object mirror probability should be set to 0.0F. Other object also should have 0.0F.

2.53.2.5 f360_detect_multipath__Detect_And_Mark_Multipath_Objects_Qual_scenario_where__source_objects_does_not_exist

```
f360_detect_multipath__Detect_And_Mark_Multipath_Objects_Qual_scenario_where__source_objects_↵  
_does_not_exist ( )
```

Purpose:\n Check if Detect_And_Mark_Multipath_Objects() doesn't change ghost object mirror probability to default value when source does not exists.

Requirements CPR-3770

Precondition:\n

Remove source object.

Action:\n

Call tested function

Result:\n

Ghost object mirror probability should be set to 0.0F. Other object also should have 0.0F.

2.53.2.6 f360_detect_multipath__Detect_And_Mark_Multipath_Objects_Qual_scenario_where__source_objects_is_beyond

```
f360_detect_multipath__Detect_And_Mark_Multipath_Objects_Qual_scenario_where__source_objects←  
_is_beyond_sensor_fov ( )
```

Purpose:\n Check if Detect_And_Mark_Multipath_Objects() doesn't change ghost object mirror probability to default value when source object meet all conditions except object is not in sensor FOV.

Requirements CPR-3769

Precondition:\n

Reduce sensors FOV.

Action:\n

Call tested function

Result:\n

Ghost object mirror probability should be set to 0.0F. Other objects also should have 0.0F

2.53.2.7 f360_detect_multipath__Detect_And_Mark_Multipath_Objects_Qual_scenario_where__source_objects_is_inactive

```
f360_detect_multipath__Detect_And_Mark_Multipath_Objects_Qual_scenario_where__source_objects←  
_is_inactive ( )
```

Purpose:\n Check if Detect_And_Mark_Multipath_Objects() doesn't change ghost object mirror probability to default value when source object meet all conditions except object is inactive.

Requirements CPR-3769

Precondition:\n

Deactivate source object

Action:\n

Call tested function

Result:\n

Ghost object mirror probability should be set to 0.0F. Other objects also should have 0.0F

2.53.2.8 f360_detect_multipath__Detect_And_Mark_Multipath_Objects_Qual_scenario_where__source_objects_is_stationary

```
f360_detect_multipath__Detect_And_Mark_Multipath_Objects_Qual_scenario_where__source_objects←  
_is_stationary ( )
```

Purpose: Check if Detect_And_Mark_Multipath_Objects() doesn't change ghost object mirror probability to default value when source object meet all conditions except motion status.

Requirements CPR-3769

Precondition:

Set source object moving flag to false

Action:

Call tested function

Result:

Ghost object mirror probability should be set to 0.0F. Other objects also should have 0.0F

2.53.2.9 f360_detect_multipath__Detect_And_Mark_Multipath_Objects_Qual_scenario_where__source_objects_use_CCV_filter

```
f360_detect_multipath__Detect_And_Mark_Multipath_Objects_Qual_scenario_where__source_objects←  
_use_CCV_filter ( )
```

Purpose: Check if Detect_And_Mark_Multipath_Objects() doesn't change ghost object mirror probability to default value when source object meet all conditions except object doesn't use CTCA motion model

Requirements CPR-3769

Precondition:

Set source object filter type for CCV.

Action:

Call tested function

Result:

Ghost object mirror probability should be set to 0.0F. Other objects also should have 0.0F

2.54 F360_determine_reflected_obj

Functions

- test group [f360_determine_reflected_obj](#) ()
Test group of f360_determine_reflected_obj.
- [f360_determine_reflected_obj_Determine_Reflected_Obj__check_whether_track_behind_guardrail_has_its_mirror_prob_set](#) ()
- [f360_determine_reflected_obj_Determine_Reflected_Obj__check_whether_track_that_is_not_a_mirror_track_has_its_mirror](#) ()
- [f360_determine_reflected_obj_Determine_Reflected_Obj__check_whether_track_that_is_not_a_mirror_track_has_its_mirror](#) ()
- [f360_determine_reflected_obj_Determine_Reflected_Obj__check_whether_track_that_is_not_a_mirror_track_has_its_mirror](#) ()
- [f360_determine_reflected_obj_Determine_Reflected_Obj__check_whether_mirror_track_has_its_mirror_prob_set_to_one](#) ()
- [f360_determine_reflected_obj_Determine_Reflected_Obj__check_whether_mirror_track_has_its_mirror_prob_set_to_one_ho](#) ()
- [f360_determine_reflected_obj_Determine_Reflected_Obj__check_whether_valid_track_in_front_and_on_guardrail_has_its_m](#) ()
- [f360_determine_reflected_obj_Determine_Reflected_Obj__check_whether_valid_track_behind_guardrail_has_its_mirror_prob](#) ()
- [f360_determine_reflected_obj_check_calibrations](#) ()

2.54.1 Detailed Description

2.54.2 Function Documentation

2.54.2.1 f360_determine_reflected_obj_check_calibrations()

`f360_determine_reflected_obj_check_calibrations` ()

Purpose:\n

Check calibrations correctness

Requirements CPR-4874

Precondition:\n

Same as setup (initialized calibrations)

Result:\n

Calibrations set as expected

2.54.2.2 f360_determine_reflected_obj_Determine_Reflected_Obj__check_whether_mirror_track_has_its_mirror_prob_set

```
f360_determine_reflected_obj_Determine_Reflected_Obj__check_whether_mirror_track_has_its_↵  
mirror_prob_set_to_one ( )
```

Purpose:\n Purpose of this test is to verify whether when track is classified as mirror track and host speed is below threshold it has its mirror prob set

Requirements CPR-4874 CPR-4875

Precondition:\n

Set parameters of selected track to be qualified as mirror track

Action:\n

Call tested function

Result:\n

Check whether selected track had its mirror prob set to 1

2.54.2.3 f360_determine_reflected_obj_Determine_Reflected_Obj__check_whether_mirror_track_has_its_mirror_prob_set

```
f360_determine_reflected_obj_Determine_Reflected_Obj__check_whether_mirror_track_has_its_↵  
mirror_prob_set_to_one_host_speed_threshold ( )
```

Purpose:\n Purpose of this test is to verify whether when track is classified as mirror track and host speed is above threshold it has its mirror prob set

Requirements CPR-4874 CPR-4875

Precondition:\n

Set parameters of selected track to be qualified as mirror track

Action:\n

Call tested function

Result:\n

Check whether selected track had its mirror prob set to 1

2.54.2.4 f360_determine_reflected_obj_Determine_Reflected_Obj__check_whether_track_behind_guardrail_has_its_mirror_prob_set_to_one ()

```
f360_determine_reflected_obj_Determine_Reflected_Obj__check_whether_track_behind_guardrail__↔  
has_its_mirror_prob_set_to_one ( )
```

Purpose:\n Purpose of this test is to verify whether when track is on and behind guardrail and flag indicating that it should be hidden has its mirror prob

Requirements CPR-4874 CPR-4875

Precondition:\n

Set parameters of selected track to be on and behind guardrail so it is qualified as mirror track

Action:\n

Call tested function

Result:\n

Check whether selected track had its mirror prob set to 1

2.54.2.5 f360_determine_reflected_obj_Determine_Reflected_Obj__check_whether_track_that_is_not_a_mirror_track_has_its_mirror_prob_lowered_by_calib_value ()

```
f360_determine_reflected_obj_Determine_Reflected_Obj__check_whether_track_that_is_not_a__↔  
mirror_track_has_its_mirror_prob_lowered_by_calib_value ( )
```

Purpose:\n Purpose of this test is to verify whether when track that is not mirror has its mirror prob lowered by calib value

Requirements CPR-4874 CPR-4875

Precondition:\n

Set parameters of selected track

Action:\n

Call tested function

Result:\n

Check whether selected track had its mirror prob reduced to 0.9

2.54.2.6 f360_determine_reflected_obj_Determine_Reflected_Obj__check_whether_track_that_is_not_a_mirror_track_has

```
f360_determine_reflected_obj_Determine_Reflected_Obj__check_whether_track_that_is_not_a_↵  
mirror_track_has_its_mirror_prob_reset ( )
```

Purpose:\n Purpose of this test is to verify whether when track that is not mirror and host speed is below threshold has its mirror prob reset

Requirements CPR-4874 CPR-4875

Precondition:\n

Set parameters of selected track

Action:\n

Call tested function

Result:\n

Check whether selected track had its mirror prob set to 0

2.54.2.7 f360_determine_reflected_obj_Determine_Reflected_Obj__check_whether_track_that_is_not_a_mirror_track_has

```
f360_determine_reflected_obj_Determine_Reflected_Obj__check_whether_track_that_is_not_a_↵  
mirror_track_has_its_mirror_prob_reset_host_speed_threshold ( )
```

Purpose:\n Purpose of this test is to verify whether when track that is not mirror, and host speed is above threshold has its mirror prob reset

Requirements CPR-4874 CPR-4875

Precondition:\n

Set parameters of selected track

Action:\n

Call tested function

Result:\n

Check whether selected track had its mirror prob set to calibration value

2.54.2.8 f360_determine_reflected_obj_Determine_Reflected_Obj__check_whether_valid_track_behind_guardrail_has_its_

```
f360_determine_reflected_obj_Determine_Reflected_Obj__check_whether_valid_track_behind_↵
guardrail_has_its_mirror_prob_reset ( )
```

Purpose: Purpose of this test is to verify whether when track is not on guardrail, is behind it but is not a mirror track it has its mirror_prob reset

Requirements CPR-4874 CPR-4875

Precondition:

Set parameters of analysed track to not be treated as mirror reflection behind guardrail Set parameters of track to make it not being qualified as mirror track

Action:

Call tested function

Result:

Check whether selected track had its mirror prob set to 0

2.54.2.9 f360_determine_reflected_obj_Determine_Reflected_Obj__check_whether_valid_track_in_front_and_on_guardrail_

```
f360_determine_reflected_obj_Determine_Reflected_Obj__check_whether_valid_track_in_front_and_↵
_on_guardrail_has_its_mirror_prob_reset ( )
```

sep[0].p2 = 0.0F;

Purpose: Purpose of this test is to verify whether when track is on guardrail, not behind it but is not a mirror track it has its mirror_prob reset

Requirements CPR-4874 CPR-4875

Precondition:

Set parameters of selected track to be located in front and on guardrail Set parameters of selected track to not be qualified as mirror track

Action:

describe actions

Result:

Check whether selected track had its mirror prob set to 0

2.55 F360_determine_reflected_obj_with_turned_hide_tracks_outside_↔ guardrail

Functions

- test group [f360_determine_reflected_obj_with_turned_hide_tracks_outside_guardrail](#) ()
Test group of f360_determine_reflected_obj_with_turned_hide_tracks_outside_guardrail It checks output by focusing on an object and a guardrail relation.
- [f360_determine_reflected_obj_with_turned_hide_tracks_outside_guardrail_prob_set_to_zero_because_track_is_on_but_not_l](#)
()
- [f360_determine_reflected_obj_with_turned_hide_tracks_outside_guardrail_prob_set_to_one_because_track_is_not_on_but_b](#)
()
- [f360_determine_reflected_obj_with_turned_hide_tracks_outside_guardrail_prob_set_to_zero_because_track_is_on_and_behi](#)
()
- [f360_determine_reflected_obj_with_turned_hide_tracks_outside_guardrail_prob_set_to_one_because_track_is_not_on_and](#)
()

2.55.1 Detailed Description

2.55.2 Function Documentation

2.55.2.1 f360_determine_reflected_obj_with_turned_hide_tracks_outside_guardrail_prob_set_to_one_because_track_is_n

```
f360_determine_reflected_obj_with_turned_hide_tracks_outside_guardrail_prob_set_to_one_↔  
because_track_is_not_on_and_not_behind_guardrail_and_low_init_prob ( )
```

Purpose:\n Checks if mirror probability is set to 1 when object is not on guradrail and not behind guardrail and object's current mirror probability is low

Requirements CPR-4874 CPR-4875

Precondition:\n

Object's mirror probability set to low value Object position far away from a guardrail and on the left side (inside) of the guardrail

Action:\n

Call tested function

Result:\n

Mirror probability set to 0

2.55.2.2 f360_determine_reflected_obj_with_turned_hide_tracks_outside_guardrail_prob_set_to_one_because_track_is_n

```
f360_determine_reflected_obj_with_turned_hide_tracks_outside_guardrail_prob_set_to_one_↵  
because_track_is_not_on_but_behind_guardrail ( )
```

Purpose: Checks if mirror probability is set to 1 when object is not on guardrail but it is behind guardrail

Requirements CPR-4874 CPR-4875

Precondition:

Object outside of a guardrail (i.e. its lateral position is much higher than guardrail lateral position).

Action:

Call tested function

Result:

Mirror probability set to 1

2.55.2.3 f360_determine_reflected_obj_with_turned_hide_tracks_outside_guardrail_prob_set_to_zero_because_track_is_o

```
f360_determine_reflected_obj_with_turned_hide_tracks_outside_guardrail_prob_set_to_zero_↵  
because_track_is_on_and_behind_guardrail_and_low_initial_prob ( )
```

Purpose: Checks if mirror probability is set to 0 when object is on guardrail and it is behind guardrail and object's current mirror probability is low

Requirements CPR-4874 CPR-4875

Precondition:

Object's mirror probability set to low value Object position close to a guardrail and on the right side (outside) of the guardrail Host speed set to 10

Action:

Call tested function

Result:

Mirror probability set to 1

2.55.2.4 f360_determine_reflected_obj_with_turned_hide_tracks_outside_guardrail_prob_set_to_zero_because_track_is_o

```
f360_determine_reflected_obj_with_turned_hide_tracks_outside_guardrail_prob_set_to_zero_↔  
because_track_is_on_but_not_behind_guardrail_and_low_initial_prob ( )
```

Purpose: Checks if mirror probability is set to 0 when object is on guardrail but not behind guardrail and its current mirror probability is low

Requirements CPR-4874 CPR-4875

Precondition:

Set object's mirror probability to low value Object position close to a guardrail and on the left side of the guardrail

Action:

Call tested function

Result:

Mirror probability set to 0

2.56 F360_update_exist_prob

Functions

- test group [f360_update_exist_prob_qualtest](#) ()
Test group of Update_Existence_Probability function. Tests verify wheter existence probability is updated only for selected tracks and whether it is updated properly.
- [f360_update_exist_prob_qualtest_Update_Existence_Probability__check_whether_minimum_EP_value_is_equal_to_min_thre](#) ()
- [f360_update_exist_prob_qualtest_Update_Existence_Probability__check_whether_maximum_EP_value_is_1](#) ()
- [f360_update_exist_prob_qualtest_Update_Existence_Probability__check_whether_EP_depend_onobject_state_variance](#) ()

2.56.1 Detailed Description

2.56.2 Function Documentation

2.56.2.1 f360_update_exist_prob_qualtest()

```
test group f360_update_exist_prob_qualtest ( )
```

Test group of Update_Existence_Probability function. Tests verify wheter existence probability is updated only for selected tracks and whether it is updated properly.

Setup:\n

Set up common variables Set up sensor parameters Set object vcs longitudinal position

2.56.2.2 f360_update_exist_prob_qualtest_Update_Existence_Probability__check_whether_EP_depend_onobject_state_v

```
f360_update_exist_prob_qualtest_Update_Existence_Probability__check_whether_EP_depend_onobject↔  
_state_variance ( )
```

Purpose:\n Purpose of this test is to verify whether existence probability depend on object's state variance.

Requirements FTCP-12167, FTCP-12387

Step 1:\n

Calculate EP for object track

Precondition:\n

Setting needed data

Action:\n

Call function

Step 2:\n

Calculate EP for object track from step(1) after changing state covariance

Precondition:\n

Setting needed data

Action:\n

Call tested function

Result:\n

Existence probability should change after changing state covariance between two steps.

2.56.2.3 f360_update_exist_prob_qualtest_Update_Existence_Probability__check_whether_maximum_EP_value_is_1()

```
f360_update_exist_prob_qualtest_Update_Existence_Probability__check_whether_maximum_EP_value←  
_is_1 ( )
```

Purpose:\n Purpose of this test is to verify whether maximum existence probability is equal 1.

Requirements FTCP-12236, FTCP-12387

Precondition:\n

Setting needed data

Action:\n

Call tested function

Result:\n

exist_prob should not exceed 1.0

2.56.2.4 f360_update_exist_prob_qualtest_Update_Existence_Probability__check_whether_minimum_EP_value_is_equal_

```
f360_update_exist_prob_qualtest_Update_Existence_Probability__check_whether_minimum_EP_value←  
_is_equal_to_min_threshold ( )
```

Purpose:\n Purpose of this test is to verify whether minimum existence probability is equal to minimum threshold.

Requirements FTCP-12236, FTCP-12387

Precondition:\n

Define needed data

Action:\n

Call tested function

Result:\n

Expected EP should be equald to k_exist_prob_min_allowed_exist_prob defined in calibrations.

2.57 F360_conf_track_confidence_level

Functions

- test group `f360_conf_track_confidence_level_qualtest` ()

This test group contains tests for `Conf_Track_Confidence_Level()` function regarding requirements from SRD document.

- `f360_conf_track_confidence_level_qualtest_Conf_Track_Confidence_Level__if_confidence_is_calculated_for_updated_obj_tra`
()
- `f360_conf_track_confidence_level_qualtest_Conf_Track_Confidence_Level__is_confidence_level_is_equal_or_higher_than_0`
()
- `f360_conf_track_confidence_level_qualtest_Conf_Track_Confidence_Level__is_confidence_level_is_equal_or_lower_than_1`
()
- `f360_conf_track_confidence_level_qualtest_Conf_Track_Confidence_Level__is_confidence_level_is_equal_or_higher_than_0`
()
- `f360_conf_track_confidence_level_qualtest_Conf_Track_Confidence_Level__is_confidence_level_is_equal_or_lower_than_1`
()
- `f360_conf_track_confidence_level_qualtest_Conf_Track_Confidence_Level__is_confidence_overlapping_object_is_reduced_by`
()
- `f360_conf_track_confidence_level_qualtest_Conf_Track_Confidence_Level__is_confidence_level_depend_on_object_status`
()
- `f360_conf_track_confidence_level_qualtest_Conf_Track_Confidence_Level__is_confidence_level_depend_on_number_of_det`
()
- `f360_conf_track_confidence_level_qualtest_Conf_Track_Confidence_Level__is_confidence_level_depend_on_time_since_the`
()

2.57.1 Detailed Description

2.57.2 Function Documentation

2.57.2.1 `f360_conf_track_confidence_level_qualtest()`

```
test group f360_conf_track_confidence_level_qualtest ( )
```

This test group contains tests for `Conf_Track_Confidence_Level()` function regarding requirements from SRD document.

Setup:\n

Creating required data structures and initializing calibrations.

2.57.2.2 f360_conf_track_confidence_level_qualtest_Conf_Track_Confidence_Level__if_confidence_is_calculated_for_updated_obj_tracks ()

```
f360_conf_track_confidence_level_qualtest_Conf_Track_Confidence_Level__if_confidence_is_calculated_for_updated_obj_tracks ( )
```

Purpose:\n Purpose of this test is to verify whether the tracker calculates confidence level for updated object tracks.

Requirements FTCP-12155

Precondition:\n

Setting needed data

Action:\n

Call function

Result:\n

Initial confidence level different from confidence after run Conf_Track_Confidence_Level().

2.57.2.3 f360_conf_track_confidence_level_qualtest_Conf_Track_Confidence_Level__is_confidence_level_depend_on_number_of_detections_updating_object ()

```
f360_conf_track_confidence_level_qualtest_Conf_Track_Confidence_Level__is_confidence_level_depend_on_number_of_detections_updating_object ( )
```

Purpose:\n Purpose of this test is to verify whether the confidence level depend on number of detections updating the object track.

Requirements FTCP-12149

Precondition:\n

Setting needed data

Action:\n

Call function

Result:\n

Confidence level should depend on number of detections associated to object.

2.57.2.4 f360_conf_track_confidence_level_qualtest_Conf_Track_Confidence_Level__is_confidence_level_depend_on_ob

```
f360_conf_track_confidence_level_qualtest_Conf_Track_Confidence_Level__is_confidence_level↔  
depend_on_object_status ( )
```

Purpose:\n Purpose of this test is to verify whether the confidence level depend on object track status and number of associated detections.

Requirements FTCP-12149

Precondition:\n

Setting needed data

Action:\n

Call function

Result:\n

Confidence level should be calculated different way for objects with different statuses.

2.57.2.5 f360_conf_track_confidence_level_qualtest_Conf_Track_Confidence_Level__is_confidence_level_depend_on_tim

```
f360_conf_track_confidence_level_qualtest_Conf_Track_Confidence_Level__is_confidence_level↔  
depend_on_time_since_the_object_was_updated ( )
```

Purpose:\n Purpose of this test is to verify whether the confidence level depend on time since the object track was updated.

Requirements FTCP-12149

Precondition:\n

Setting needed data

Action:\n

Call function

Result:\n

Because of different time_since_track_updated for tested objects the confidenceLevel should differ.

2.57.2.6 f360_conf_track_confidence_level_qualtest_Conf_Track_Confidence_Level__is_confidence_level_is_equal_or_higher_than_0_for_COASTED ()

f360_conf_track_confidence_level_qualtest_Conf_Track_Confidence_Level__is_confidence_level_is_equal_or_higher_than_0_for_COASTED ()

Purpose:\n Purpose of this test is to verify whether the confidence level is in range 0 to 1 for object with status F360_OBJECT_STATUS_COASTED, when confidence initial value is 0.0

Requirements FTCP-12232

Precondition:\n

Setting needed data

Action:\n

Call function

Result:\n

Confidence level should be ≥ 0 and ≤ 1 .

2.57.2.7 f360_conf_track_confidence_level_qualtest_Conf_Track_Confidence_Level__is_confidence_level_is_equal_or_higher_than_0_for_UPDATED ()

f360_conf_track_confidence_level_qualtest_Conf_Track_Confidence_Level__is_confidence_level_is_equal_or_higher_than_0_for_UPDATED ()

Purpose:\n Purpose of this test is to verify whether the confidence level is in range 0 to 1 for object with status F360_OBJECT_STATUS_UPDATED, when confidence initial value is 0.0

Requirements FTCP-12232

Precondition:\n

Setting needed data

Action:\n

Call function

Result:\n

Confidence level ≥ 0 and ≤ 1 .

2.57.2.8 f360_conf_track_confidence_level_qualtest_Conf_Track_Confidence_Level__is_confidence_level_is_equal_or_lo

```
f360_conf_track_confidence_level_qualtest_Conf_Track_Confidence_Level__is_confidence_level_↔  
is_equal_or_lower_than_1_for_COASTED ( )
```

Purpose:\n Purpose of this test is to verify whether the confidence level is in range 0 to 1 for object with status F360_OBJECT_STATUS_COASTED,
when confidence initial value is 1.0

Requirements FTCP-12232

Precondition:\n

Setting needed data

Action:\n

Call function

Result:\n

Confidence level should be ≥ 0 and ≤ 1 .

2.57.2.9 f360_conf_track_confidence_level_qualtest_Conf_Track_Confidence_Level__is_confidence_level_is_equal_or_lo

```
f360_conf_track_confidence_level_qualtest_Conf_Track_Confidence_Level__is_confidence_level_↔  
is_equal_or_lower_than_1_for_UPDATED ( )
```

Purpose:\n Purpose of this test is to verify whether the confidence level is in range 0 to 1 for object with status F360_OBJECT_STATUS_UPDATED,
when confidence initial value is 1.0

Requirements FTCP-12232

Precondition:\n

Setting needed data

Action:\n

Call function

Result:\n

Confidence level should be ≥ 0 and ≤ 1 .

2.57.2.10 f360_conf_track_confidence_level_qualtest_Conf_Track_Confidence_Level__is_confidence_overlapping_object_i

```
f360_conf_track_confidence_level_qualtest_Conf_Track_Confidence_Level__is_confidence_overlapping↵_object_is_reduced_by_factor ( )
```

Purpose:\n Purpose of this test is to verify whether the confidence level for overlapping object is reduced by multiplying it by calibration factor.

Requirements FTCP-12230

Precondition:\n

Setting needed data

Action:\n

Call function

Result:\n

Confidence level should be equal the calibration factor for overlapping objects. Initial confidence level is 1.0 so after the multiplying it by the reduction factor for overlapping objects(which is lower than 1.0) they should be equal the reduction factor and lower than initial confidence level.

2.58 F360_input_diagnostics_qualtest

Functions

- test group `f360_input_diagnostics_qualtest ()`
Set up initial values of the signals to be checked by the module. Keep in mind that class Execute methods needs to be run at least twice for the module to start detecting errors. In the first cycle no information about the previous cycle is known, so signals can't be verified for consistency.
- `f360_input_diagnostics_qualtest_No_Increase_Faults_Should_Be_Returned_When_Core_Signal_Received_Twice_Without_T`
`()`
- `f360_input_diagnostics_qualtest_Fault_Should_Be_Returned_When_Core_Info_Cnt_Loops_Lower_Than_Before`
`()`
- `f360_input_diagnostics_qualtest_Fault_Should_Be_Returned_When_Core_Info_Time_us_Lower_Than_Before`
`()`
- `f360_input_diagnostics_qualtest_Fault_Should_Be_Returned_When_Core_Info_Elapsed_Time_Below_Lower_Limit`
`()`
- `f360_input_diagnostics_qualtest_Fault_Should_Be_Returned_When_Core_Info_Elapsed_Above_Upper_Limit`
`()`
- `f360_input_diagnostics_qualtest_Fault_Should_Be_Returned_When_Host_Info_Vehicle_index_Lower_Than_Before`
`()`
- `f360_input_diagnostics_qualtest_True_Should_Be_Returned_When_Host_QF_Is_Accurate ()`
- `f360_input_diagnostics_qualtest_True_Should_Be_Returned_When_Host_QF_Is_NOT_Accurate ()`
- `f360_input_diagnostics_qualtest_True_Should_Be_Returned_When_Big_Curvature_Rear_Or_VCS_Sideslip`
`()`
- `f360_input_diagnostics_qualtest_No_Increase_Faults_Should_Be_Returned_When_Sensors_Signal_Received_Twice_Without`
`()`
- `f360_input_diagnostics_qualtest_No_Increase_Fault_Should_Be_Returned_When_Look_Index_For_Single_Sensor_Lower_T`
`()`
- `f360_input_diagnostics_qualtest_Signal_Fault_Should_Be_Returned_When_Difference_Between_Sensor_And_Tracker_Time`
`()`
- `f360_input_diagnostics_qualtest_Signal_Fault_Should_Be_Returned_When_Difference_Between_Sensor_And_Tracker_Time`
`()`
- `f360_input_diagnostics_qualtest_Signal_Fault_Should_Not_Be_Set_When_No_Sensor_Data_In_Received`
`()`
- `f360_input_diagnostics_qualtest_No_Fault_Reported_For_Sensor_When_Calibration_Not_Valid ()`
- `f360_input_diagnostics_qualtest_No_Fault_Reported_For_Sensor_When_look_index_Overflow ()`
- `f360_input_diagnostics_qualtest_Signal_Faults_Should_Be_Reported_When_Look_Index_Equal_To_Zero_Twice_In_A_Row`
`()`
- `f360_input_diagnostics_qualtest_Signal_Faults_Should_Be_Reported_When_Sensor_Polarity_Is_Equal_To_Zero`
`()`
- `f360_input_diagnostics_qualtest_Signal_Faults_Should_Be_Reported_When_Sensor_Polarity_Is_Greater_Than_Expected_P`
`()`
- `f360_input_diagnostics_qualtest_Signal_Faults_Should_Be_Reported_When_Sensor_Polarity_Is_Greater_Than_Expected_N`
`()`
- `f360_input_diagnostics_qualtest_Polarity_Fault_Should_Not_Be_Reported_When_Sensor_Polarity_Is_Valid_Positive`
`()`
- `f360_input_diagnostics_qualtest_Polarity_Fault_Should_Not_Be_Reported_When_Sensor_Polarity_Is_Valid_Negative`
`()`
- `f360_input_diagnostics_qualtest_Signal_Faults_Should_Be_Reported_When_Sensor_Mounting_Position_Is_Too_High_Positi`
`()`
- `f360_input_diagnostics_qualtest_Signal_Faults_Should_Be_Reported_When_Sensor_Mounting_Position_Is_Too_High_Nega`
`()`
- `f360_input_diagnostics_qualtest_Signal_Faults_Should_Not_Be_Reported_When_Sensor_Mounting_Position_Equal_To_Lim`
`()`
- `f360_input_diagnostics_qualtest_Signal_Faults_Should_Not_Be_Reported_When_Sensor_Mounting_Position_Equal_To_Lim`
`()`

- [f360_input_diagnostics_qualtest_Signal_Faults_Should_Not_Be_Reported_When_Sensor_Mounting_Position_Is_Within_Limits\(\)](#)
- [f360_input_diagnostics_qualtest_Signal_Faults_Should_Not_Be_Reported_When_Sensor_Mounting_Position_Is_Within_Limits\(\)](#)
- [f360_input_diagnostics_qualtest_Signal_Faults_Should_Be_Reported_When_Sensor_Azimuth_Boresight_Angle_Is_Greater_Than_Threshold\(\)](#)
- [f360_input_diagnostics_qualtest_Signal_Faults_Should_Be_Reported_When_Sensor_Azimuth_Boresight_Angle_Is_Less_Than_Threshold\(\)](#)
- [f360_input_diagnostics_qualtest_Signal_Faults_Should_Be_Reported_When_Sensor_Azimuth_Boresight_Angle_Is_Within_Limits\(\)](#)
- [f360_input_diagnostics_qualtest_Signal_Faults_Should_Be_Reported_When_Sensor_Azimuth_Boresight_Angle_Is_Within_Limits\(\)](#)

2.58.1 Detailed Description

2.58.2 Function Documentation

2.58.2.1 f360_input_diagnostics_qualtest_Fault_Should_Be_Returned_When_Core_Info_Cnt_Loops_Lower_Than_Before()

```
f360_input_diagnostics_qualtest_Fault_Should_Be_Returned_When_Core_Info_Cnt_Loops_Lower_Than_Before(
    _Before )
```

Purpose:

Execute function should return appropriate fault status when the tracker_index (core_info.cnt_loops) is lower than in the last cycle

Requirements FTCP-8837

Precondition:

Execute input diagnostics class for the first time to fill previous signal states Decrement core_info.cnt_loops signal value

Action:

Execute check

Result:

Signal fault should be reported

2.58.2.2 f360_input_diagnostics_qualtest_Fault_Should_Be_Returned_When_Core_Info_Elapsed_Above_Upper_Limit()

```
f360_input_diagnostics_qualtest_Fault_Should_Be_Returned_When_Core_Info_Elapsed_Above_Upper_↵  
Limit ( )
```

Purpose:

Execute function should return appropriate fault status when core_info.elapsed_time is above upper limit

Requirements FTCP-8604

Precondition:

Execute input diagnostics class for the first time to fill previous signal states Modify core_info.elapsed_time to above limit

Action:

Modify core_info.elapsed_time to above limit and Execute check

Result:

Signal fault should be reported

2.58.2.3 f360_input_diagnostics_qualtest_Fault_Should_Be_Returned_When_Core_Info_Elapsed_Time_Below_Lower_Lim

```
f360_input_diagnostics_qualtest_Fault_Should_Be_Returned_When_Core_Info_Elapsed_Time_Below_↵  
Lower_Limit ( )
```

Purpose:

Execute function should return appropriate fault status when core_info.elapsed_time is below lower limit

Requirements FTCP-8605

Precondition:

Execute input diagnostics class for the first time to fill previous signal states Modify core_info.elapsed_time to below limit

Action:

Execute check

Result:

Signal fault should be reported

2.58.2.4 f360_input_diagnostics_qualtest_Fault_Should_Be_Returned_When_Core_Info_Time_us_Lower_Than_Before()

f360_input_diagnostics_qualtest_Fault_Should_Be_Returned_When_Core_Info_Time_us_Lower_Than_Before ()

Purpose:

Execute function should return appropriate fault status when core_info.time_us is lower than in the last cycle

Requirements FTCP-8603

Precondition:

Execute input diagnostics class for the first time to fill previous signal states Decrease core_info.time_us

Action:

Execute signal check

Result:

Signal fault should be reported

2.58.2.5 f360_input_diagnostics_qualtest_Fault_Should_Be_Returned_When_Host_Info_Vehicle_index_Lower_Than_Before()

f360_input_diagnostics_qualtest_Fault_Should_Be_Returned_When_Host_Info_Vehicle_index_Lower_Than_Before ()

Purpose:

Execute function should return appropriate fault status when the vehicle_index is lower than in the last cycle

Requirements FTCP-11448

Precondition:

Execute input diagnostics class for the first time to fill previous signal states Decrement host.vehicle_index signal value

Action:

Execute check

Result:

Signal fault should be reported

2.58.2.6 f360_input_diagnostics_qualtest_No_Fault_Reported_For_Sensor_When_Calibration_Not_Valid()

```
f360_input_diagnostics_qualtest_No_Fault_Reported_For_Sensor_When_Calibration_Not_Valid ( )
```

Purpose:\n

Execute function must not return timestamp fault status when sensor calibration is not valid

Requirements FTCP-8592

Precondition:\n

Execute input diagnostics class for the first time to fill previous signal states Fill timestamp_us for one faulty sensor with value below specification

Action:\n

Execute signal check

Result:\n

Signal fault should NOT be reported

2.58.2.7 f360_input_diagnostics_qualtest_No_Fault_Reported_For_Sensor_When_look_index_Overflow()

```
f360_input_diagnostics_qualtest_No_Fault_Reported_For_Sensor_When_look_index_Overflow ( )
```

Purpose:\n

Execute function must not return sensor look_index_no_increase fault status during look_index rollover (overflow)

Requirements FTCP-8591

Precondition:\n

Execute input diagnostics class for the first time to fill previous signal states Simulate look index rollover (overflow)

Action:\n

Execute signal check

Result:\n

Signal fault should NOT be reported

2.58.2.8 f360_input_diagnostics_qualtest_No_Increase_Fault_Should_Be_Returned_When_Look_Index_For_Single_Sensor_Lower_Than_Before ()

f360_input_diagnostics_qualtest_No_Increase_Fault_Should_Be_Returned_When_Look_Index_For_Single_Sensor_Lower_Than_Before ()

Purpose:

Execute function should return appropriate fault status when sensors.look_index is the lower than in the last cycle

Requirements FTCP-8591

Precondition:

Execute input diagnostics class for the first time to fill previous signal states Increment sensors[i].look_index for all sensors apart from one

Action:

Execute signal check

Result:

Signal fault should be reported

2.58.2.9 f360_input_diagnostics_qualtest_No_Increase_Faults_Should_Be_Returned_When_Core_Signal_Received_Twice_Without_Change ()

f360_input_diagnostics_qualtest_No_Increase_Faults_Should_Be_Returned_When_Core_Signal_Received_Twice_Without_Change ()

Purpose:

Execute function should return appropriate faults status when core_info received is the same as in the last cycle

Requirements FTCP-8603, FTCP-8837, FTCP-11448,

Precondition:

Execute input diagnostics class for the first time to fill previous signal states

Action:

Execute signals check without incrementing core_info signal with new iteration

Result:

Signal fault should be reported

2.58.2.10 f360_input_diagnostics_qualtest_No_Increase_Faults_Should_Be_Returned_When_Sensors_Signal_Received_T

f360_input_diagnostics_qualtest_No_Increase_Faults_Should_Be_Returned_When_Sensors_Signal_↔
Received_Twice_Without_Change ()

Purpose:\n

Execute function should return appropriate faults status when sensors signal received is the same as in the last cycle

Requirements FTCP-8591

Precondition:\n

Execute input diagnostics class for the first time to fill previous signal states Increment only core_info and host signals and not sensors signals

Action:\n

Execute signals check without incrementing sensors signal with new iteration

Result:\n

Signal faults should be reported

2.58.2.11 f360_input_diagnostics_qualtest_Polarity_Fault_Should_Not_Be_Reported_When_Sensor_Polarity_Is_Valid_Neg

f360_input_diagnostics_qualtest_Polarity_Fault_Should_Not_Be_Reported_When_Sensor_Polarity_↔
Is_Valid_Negative ()

Purpose:\n

Execute function should return sensor polarity_is_invalid fault status when sensor polarity absolute value is valid negative

Requirements FTCP-8594

Precondition:\n

Set polarity to valid negative value and execute input diagnostics class for the first time to fill previous signal states

Action:\n

Execute signal check again with valid negative polarity

Result:\n

Signal fault should NOT be reported

2.58.2.12 f360_input_diagnostics_qualtest_Polarity_Fault_Should_Not_Be_Reported_When_Sensor_Polarity_Is_Valid_Pos

f360_input_diagnostics_qualtest_Polarity_Fault_Should_Not_Be_Reported_When_Sensor_Polarity_↔
Is_Valid_Positive ()

Purpose:\n

Execute function should return sensor_polarity_is_invalid fault status when sensor_polarity_absolute value is valid positive

Requirements FTCP-8594

Precondition:\n

Set polarity to valid value and execute input_diagnostics class for the first time to fill previous signal states

Action:\n

Execute signal check again with valid polarity value

Result:\n

Signal fault should NOT be reported

2.58.2.13 f360_input_diagnostics_qualtest_Signal_Fault_Should_Be_Returned_When_Difference_Between_Sensor_And_T

f360_input_diagnostics_qualtest_Signal_Fault_Should_Be_Returned_When_Difference_Between_↔
Sensor_And_Tracker_Timestamp_Too_High ()

Purpose:\n

Execute function should return appropriate fault status when difference between sensors timestamp and tracker timestamp is higher than expected

Requirements FTCP-8592

Precondition:\n

Execute input_diagnostics class for the first time to fill previous signal states Fill timestamp_us for one faulty sensor with value above specification

Action:\n

Execute signal check

Result:\n

Signal fault should be reported

2.58.2.14 f360_input_diagnostics_qualtest_Signal_Fault_Should_Be_Returned_When_Difference_Between_Sensor_And_Tracker_Timestamp_Too_Low

f360_input_diagnostics_qualtest_Signal_Fault_Should_Be_Returned_When_Difference_Between_Sensor_And_Tracker_Timestamp_Too_Low ()

Purpose:

Execute function should return appropriate fault status when difference between sensors timestamp and tracker timestamp is lower than expected

Requirements FTCP-8592

Precondition:

Execute input diagnostics class for the first time to fill previous signal states Fill timestamp_us for one faulty sensor with value below specification

Action:

Execute signal check

Result:

Signal fault should be reported

2.58.2.15 f360_input_diagnostics_qualtest_Signal_Fault_Should_Not_Be_Set_When_No_Sensor_Data_In_Received()

f360_input_diagnostics_qualtest_Signal_Fault_Should_Not_Be_Set_When_No_Sensor_Data_In_Received ()

Purpose:

Execute function should return appropriate fault status when f_data_on_stack_is_empty is true and sensors data is zero

Requirements .NA

Precondition:

Execute input diagnostics class for the first time to fill previous signal states Then set the sensor 1 data to zeros.

Action:

Execute signal check

Result:

Signal fault should not be reported because when no input data is received the timestamp difference check shall not be executed

2.58.2.16 f360_input_diagnostics_qualtest_Signal_Faults_Should_Be_Reported_When_Look_Index_Equal_To_Zero Twice

f360_input_diagnostics_qualtest_Signal_Faults_Should_Be_Reported_When_Look_Index_Equal_To_Zero_Twice_In_A_Row ()

Purpose:

Execute function should return sensor look_index_no_increase fault status when sensor look_index equal to zero two times in a row

Requirements FTCP-8591**Precondition:**

Execute input diagnostics class for the first time to fill previous signal states

Action:

Execute signal check again with look_index equal to zero

Result:

Signal fault should be reported

2.58.2.17 f360_input_diagnostics_qualtest_Signal_Faults_Should_Be_Reported_When_Sensor_Azimuth_Boresight_Angle

f360_input_diagnostics_qualtest_Signal_Faults_Should_Be_Reported_When_Sensor_Azimuth_Boresight_Angle_Is_Greater_Than_Upper_Limit ()

Purpose:

Execute function should return sensor boresight_angle_is_invalid fault status as true when sensor azimuth boresight angle is greater than PI

Requirements FTCP-10137**Precondition:**

Set mounting position within limit negative and execute input diagnostics class for the first time to fill previous signal states

Action:

Execute signal check again with azimuth boresight angle greater than max allowed value.

Result:

Signal fault should be reported

2.58.2.18 f360_input_diagnostics_qualtest_Signal_Faults_Should_Be_Reported_When_Sensor_Azimuth_Boresight_Angle

```
f360_input_diagnostics_qualtest_Signal_Faults_Should_Be_Reported_When_Sensor_Azimuth_Boresight↔  
_Angle_Is_Less_Than_Lower_Limit ( )
```

Purpose:

Execute function should return sensor boresight_angle_is_invalid fault status as true when sensor azimuth boresight angle is less than $(-1.0) \cdot \pi$

Requirements FTCP-10137

Precondition:

Set mounting position within limit negative and execute input diagnostics class for the first time to fill previous signal states

Action:

Execute signal check again with azimuth boresight angle less than minimum allowed value.

Result:

Signal fault should be reported

2.58.2.19 f360_input_diagnostics_qualtest_Signal_Faults_Should_Be_Reported_When_Sensor_Azimuth_Boresight_Angle

```
f360_input_diagnostics_qualtest_Signal_Faults_Should_Be_Reported_When_Sensor_Azimuth_Boresight↔  
_Angle_Is_Within_Lower_Limit ( )
```

Purpose:

Execute function should return sensor boresight_angle_is_invalid fault status as false when sensor azimuth boresight angle is within the lower limit

Requirements FTCP-10137

Precondition:

Set mounting position within limit negative and execute input diagnostics class for the first time to fill previous signal states

Action:

Execute signal check again with azimuth boresight angle within minimum allowed value.

Result:

Signal fault should NOT be reported

2.58.2.20 f360_input_diagnostics_qualtest_Signal_Faults_Should_Be_Reported_When_Sensor_Azimuth_Boresight_Angle

```
f360_input_diagnostics_qualtest_Signal_Faults_Should_Be_Reported_When_Sensor_Azimuth_Boresight↔  
_Angle_Is_Within_Upper_Limit ( )
```

Purpose:\n

Execute function should return sensor boresight_angle_is_invalid fault status as false when sensor azimuth boresight angle is within the upper limit

Requirements FTCP-10137

Precondition:\n

Set mounting position within limit negative and execute input diagnostics class for the first time to fill previous signal states

Action:\n

Execute signal check again with azimuth boresight angle within max allowed value.

Result:\n

Signal fault should NOT be reported

2.58.2.21 f360_input_diagnostics_qualtest_Signal_Faults_Should_Be_Reported_When_Sensor_Mounting_Position_Is_Too

```
f360_input_diagnostics_qualtest_Signal_Faults_Should_Be_Reported_When_Sensor_Mounting_Position↔  
_Is_Too_High_Negative ( )
```

Purpose:\n

Execute function should return sensor mounting_pos_is_invalid fault status when sensor mounting position is greater than expected negative

Requirements FTCP-8596

Precondition:\n

Set mounting position to above the limit negative and execute input diagnostics class for the first time to fill previous signal states

Action:\n

Execute signal check again with mounting position set above limit negative

Result:\n

Signal fault should be reported

2.58.2.22 f360_input_diagnostics_qualtest_Signal_Faults_Should_Be_Reported_When_Sensor_Mounting_Position_Is_Too_High_Positive ()

```
f360_input_diagnostics_qualtest_Signal_Faults_Should_Be_Reported_When_Sensor_Mounting_Position_Is_Too_High_Positive ( )
```

Purpose:

Execute function should return sensor mounting_pos_is_invalid fault status when sensor mounting position is greater than expected positive

Requirements FTCP-8596

Precondition:

Set mounting position to above the limit positive and execute input diagnostics class for the first time to fill previous signal states

Action:

Execute signal check again with mounting position set above limit positive

Result:

Signal fault should be reported

2.58.2.23 f360_input_diagnostics_qualtest_Signal_Faults_Should_Be_Reported_When_Sensor_Polarity_Is_Equal_To_Zero ()

```
f360_input_diagnostics_qualtest_Signal_Faults_Should_Be_Reported_When_Sensor_Polarity_Is_Equal_To_Zero ( )
```

Purpose:

Execute function should return sensor polarity_is_invalid fault status when sensor polarity absolute value is equal to zero

Requirements FTCP-8594

Precondition:

Set polarity at zero and execute input diagnostics class for the first time to fill previous signal states

Action:

Execute signal check again with polarity equal to zero

Result:

Signal fault should be reported

2.58.2.24 f360_input_diagnostics_qualtest_Signal_Faults_Should_Be_Reported_When_Sensor_Polarity_Is_Greater_Than_Expected_Negative

```
f360_input_diagnostics_qualtest_Signal_Faults_Should_Be_Reported_When_Sensor_Polarity_Is_Greater_Than_Expected_Negative ( )
```

Purpose:

Execute function should return sensor polarity_is_invalid fault status when sensor polarity absolute value is greater than expected negative

Requirements FTCP-8594**Precondition:**

Set polarity greater than expected negative and execute input diagnostics class for the first time to fill previous signal states

Action:

Execute signal check again with polarity absolute value greater than expected

Result:

Signal fault should be reported

2.58.2.25 f360_input_diagnostics_qualtest_Signal_Faults_Should_Be_Reported_When_Sensor_Polarity_Is_Greater_Than_Expected_Positive

```
f360_input_diagnostics_qualtest_Signal_Faults_Should_Be_Reported_When_Sensor_Polarity_Is_Greater_Than_Expected_Positive ( )
```

Purpose:

Execute function should return sensor polarity_is_invalid fault status when sensor polarity absolute value is greater than expected positive

Requirements FTCP-8594**Precondition:**

Set polarity value greater than expected and execute input diagnostics class for the first time to fill previous signal states

Action:

Execute signal check again with polarity greater than expected

Result:

Signal fault should be reported

2.58.2.26 f360_input_diagnostics_qualtest_Signal_Faults_Should_Not_Be_Reported_When_Sensor_Mounting_Position_E

```
f360_input_diagnostics_qualtest_Signal_Faults_Should_Not_Be_Reported_When_Sensor_Mounting_↵  
Position_Equal_To_Limit_Negative ( )
```

Purpose:\n

Execute function should NOT return sensor mounting_pos_is_invalid fault status when sensor mounting position is equal to limit negative

Requirements FTCP-8596

Precondition:\n

Set mounting position equal to the limit negative and execute input diagnostics class for the first time to fill previous signal states

Action:\n

Execute signal check again with mounting position equal to the limit negative

Result:\n

Signal fault should NOT be reported

2.58.2.27 f360_input_diagnostics_qualtest_Signal_Faults_Should_Not_Be_Reported_When_Sensor_Mounting_Position_E

```
f360_input_diagnostics_qualtest_Signal_Faults_Should_Not_Be_Reported_When_Sensor_Mounting_↵  
Position_Equal_To_Limit_Positive ( )
```

Purpose:\n

Execute function should NOT return sensor mounting_pos_is_invalid fault status when sensor mounting position is equal to calibrated limit value positive

Requirements FTCP-8596

Precondition:\n

Set mounting position equal to the limit positive and execute input diagnostics class for the first time to fill previous signal states

Action:\n

Execute signal check again with mounting position equal to limit positive

Result:\n

Signal fault should NOT be reported

2.58.2.28 f360_input_diagnostics_qualtest_Signal_Faults_Should_Not_Be_Reported_When_Sensor_Mounting_Position_Is

f360_input_diagnostics_qualtest_Signal_Faults_Should_Not_Be_Reported_When_Sensor_Mounting_↔
Position_Is_Within_Limit_Negative ()

Purpose:\n

Execute function should NOT return sensor mounting_pos_is_invalid fault status when sensor mounting position is within limit negative

Requirements FTCP-8596

Precondition:\n

Set mounting position within limit negative and execute input diagnostics class for the first time to fill previous signal states

Action:\n

Execute signal check again with mounting position within limit negative

Result:\n

Signal fault should NOT be reported

2.58.2.29 f360_input_diagnostics_qualtest_Signal_Faults_Should_Not_Be_Reported_When_Sensor_Mounting_Position_Is

f360_input_diagnostics_qualtest_Signal_Faults_Should_Not_Be_Reported_When_Sensor_Mounting_↔
Position_Is_Within_Limit_Positive ()

Purpose:\n

Execute function should NOT return sensor mounting_pos_is_invalid fault status when sensor mounting position is within limit positive

Requirements FTCP-8596

Precondition:\n

Set mounting position within limit positive and execute input diagnostics class for the first time to fill previous signal states

Action:\n

Execute signal check again with mounting position within limit positive

Result:\n

Signal fault should NOT be reported

2.58.2.30 f360_input_diagnostics_qualtest_True_Should_Be_Returned_When_Big_Curvature_Rear_Or_VCS_Sideslip()

```
f360_input_diagnostics_qualtest_True_Should_Be_Returned_When_Big_Curvature_Rear_Or_VCS_↵  
Sideslip ( )
```

Purpose:

Execute function should return appropriate fault status for different curvature_rear and vcs_sideslip values

Requirements FTCP-10857**Precondition:**

Execute input diagnostics class for the first time to fill previous signal states Set curvature_rear to big negative value

Action:

Execute check

Result:

Signal fault should be reported

Precondition:

Reset curvature_rear to 0

Action:

Execute check

Result:

Signal fault should NOT be reported

Precondition:

Set vcs_sideslip to big negative value

Action:

Execute check

Result:

Signal fault should be reported

Precondition:\n

Set vcs_sideslip to big positive value

Action:\n

Execute check

Result:\n

Signal fault should be reported

Precondition:\n

Reset vcs_sideslip to 0

Action:\n

Execute check

Result:\n

Signal fault should NOT be reported

Precondition:\n

Set curvature_rear to big negative value Set vcs_sideslip to big positive value

Action:\n

Execute check

Result:\n

Signal fault should be reported

Precondition:\n

Set curvature_rear to big positive value Set vcs_sideslip to big negative value

Action:\n

Execute check

Result:\n

Signal fault should be reported

Precondition:\n

Set curvature_rear to big positive value Set vcs_sideslip to big positive value

Action:\n

Execute check

Result:\n

Signal fault should be reported

Precondition:\n

Set curvature_rear to big negative value Set vcs_sideslip to big negative value

Action:\n

Execute check

Result:\n

Signal fault should be reported

2.58.2.31 f360_input_diagnostics_qualtest_True_Should_Be_Returned_When_Host_QF_Is_Accurate()

f360_input_diagnostics_qualtest_True_Should_Be_Returned_When_Host_QF_Is_Accurate ()

Purpose:\n

Execute function should return appropriate fault status when the host qualifiers are QF_ACCURATE

Requirements FTCP-8602, FTCP-10857, FTCP-13077, FTCP-13076

Precondition:\n

Execute input diagnostics class for the first time to fill previous signal states Set host Quality Factors to F360↔
_QF_ACCURATE

Action:\n

Execute check

Result:\n

Signal fault should NOT be reported

2.58.2.32 f360_input_diagnostics_qualtest_True_Should_Be_Returned_When_Host_QF_Is_NOT_Accurate()

```
f360_input_diagnostics_qualtest_True_Should_Be_Returned_When_Host_QF_Is_NOT_Accurate ( )
```

Purpose:

Execute function should return appropriate fault status when the host qualifiers are not QF_ACCURATE

Requirements FTCP-8602, FTCP-10857, FTCP-13077, FTCP-13076

Precondition:

Execute input diagnostics class for the first time to fill previous signal states i = 0: Set host Quality Factors to F360_QF_UNDEFINED i = 1: Set host Quality Factors to F360_QF_TEMP_UNDEFINED i = 2: Set host Quality Factors to F360_QF_INACCURATE

Action:

Execute check

Result:

Signal fault should be reported

Chapter 3

File Documentation

3.1 Fusion360/cluster_grouping/unittest/f360_cluster_grouping_↵ qualtest_unittest.cpp File Reference

Tests

- test group [f360_cluster_grouping_qualtest](#) ()
Test group for f360_cluster_grouping_qualtest.
- [f360_cluster_grouping_qualtest_Test_Cluster_Grouping_Check_If_Range_Rates_Are_Dealiased_Correctly](#) ()
- [f360_cluster_grouping_qualtest_Test_Cluster_Grouping_Check_If_Dealiased_Flag_Is_Set_Correctly_1](#) ()
- [f360_cluster_grouping_qualtest_Test_Cluster_Grouping_Check_If_Dealiased_Flag_Is_Set_Correctly_2](#) ()

3.1.1 Detailed Description

File with set of qualification tests (which are also unit test) for cluster_grouping function

3.2 Fusion360/cluster_grouping/unittest/f360_merge_two_clusters_↵ qualtest_unittest.cpp File Reference

Tests

- test group [f360_merge_two_clusters](#) ()
Test group for Merge_Two_Clusters function.
- [f360_merge_two_clusters_Test_Merge_Two_Clusters_Check_If_All_4_Dets_Are_Dealiased](#) ()
- [f360_merge_two_clusters_Test_Merge_Two_Clusters_Check_If_Dets_4_Are_Dealiased_1_Not_Dealiased](#) ()
- [f360_merge_two_clusters_Test_Merge_Two_Clusters_Check_Range_Rate_Is_Dealiased_Correctly_1](#) ()

3.2.1 Detailed Description

File with set of qualification tests (which are also unit test) for Merge_Two_Clusters function

3.3 Fusion360/clustering/unittest/f360_initialize_clusters_qualtest_↵ unittest.cpp File Reference

Tests

- test group [initialize_qualtest_clusters \(\)](#)
Test group for initialize_clusters function.
- [initialize_qualtest_clusters_Test_Initialize_Clusters_One_Detection_To_Cluster \(\)](#)
- [initialize_qualtest_clusters_Test_Initialize_Clusters_Many_Detection_To_Cluster \(\)](#)

3.3.1 Detailed Description

File with set of qualification tests (which are also unit test) for initialize_clusters function

3.4 Fusion360/common/unittest/f360_allocate_id_for_initialized_object_↵ _qualtest_unittest.cpp File Reference

Tests

- test group [f360_allocate_id_for_initialized_object \(\)](#)
- [f360_allocate_id_for_initialized_object_no_active_objects \(\)](#)
- [f360_allocate_id_for_initialized_object_single_active_object \(\)](#)
- [f360_allocate_id_for_initialized_object_objects_saturation \(\)](#)

3.4.1 Detailed Description

File with set of qualification tests (which are also unit test) for Allocate_Id_For_Initialized_Object function.

3.5 Fusion360/common/unittest/f360_find_detection_inliers_qualtest_unittest.cpp File Reference

Tests

- test group [f360_find_detection_inliers_single_object \(\)](#)
Testing of a function that downselects associated detections that have a predicted range rate similar to compensated range rate.
- [f360_find_detection_inliers_single_object_Find_Detection_Inliers_Close_Target_Detection \(\)](#)
- test group [f360_find_detection_inliers_find_valid_detections \(\)](#)
Testing of a function that finds detections that are valid candidates to be downselected to track.
- [f360_find_detection_inliers_find_valid_detections_Select_Detections_To_Mark_As_Inliers_Object_Wheel_Spin_Detection \(\)](#)
- [f360_find_detection_inliers_find_valid_detections_Select_Detections_To_Mark_As_Inliers_Pair_Wheel_Spin_Detection \(\)](#)
- [f360_find_detection_inliers_find_valid_detections_Select_Detections_To_Mark_As_Inliers_Nearby_Wheel_Spin_Detection \(\)](#)

3.5.1 Detailed Description

This file contains unit tests for testing requirements connected to f360_find_detection_inliers.cpp file

3.6 Fusion360/detection_to_track_association/unittest/f360_detection_to_track_association_qualtest_unittest.cpp File Reference

Tests

- test group [f360_Test_Determine_Association_Hypothesis_qualtest \(\)](#)
Set up an object and place detections around it with various properties and check that expected detections are associated to expected parameters. More details can be found in DFD-175.
- [f360_Test_Determine_Association_Hypothesis_qualtest_Test_Determine_Association_Hypothesis_A \(\)](#)
- [f360_Test_Determine_Association_Hypothesis_qualtest_Test_Determine_Association_Hypothesis_B \(\)](#)
- test group [f360_Test_Is_Det_Allowed_To_Associate_qualtest \(\)](#)
Included tests related to calculation of detection is allowed to associate based flags and properties.
- [f360_Test_Is_Det_Allowed_To_Associate_qualtest_Test_Is_Det_Allowed_To_Associate_Det_Not_Ok_To_Use \(\)](#)
- [f360_Test_Is_Det_Allowed_To_Associate_qualtest_Test_Is_Det_Allowed_To_Associate_Det_On_SEP_Obj_Moving_CTCA \(\)](#)
- [f360_Test_Is_Det_Allowed_To_Associate_qualtest_Test_Is_Det_Allowed_To_Associate_Mirror_Obj_Ambiguous_Detection \(\)](#)
- [f360_Test_Is_Det_Allowed_To_Associate_qualtest_Test_Is_Det_Allowed_To_Associate_Detection_Behind_SEP_Obj_Moving_C \(\)](#)
- [f360_Test_Is_Det_Allowed_To_Associate_qualtest_Test_Is_Det_Allowed_To_Associate_Object_Behind_SEP_Obj_Moving_C \(\)](#)
- [f360_Test_Is_Det_Allowed_To_Associate_qualtest_Test_Is_Det_Allowed_To_Associate_Object_Behind_SEP_Obj_Moving_C \(\)](#)
- [f360_Test_Determine_Association_Hypothesis_qualtest_Test_Determine_Association_Hypothesis_C \(\)](#)
- [f360_Test_Determine_Association_Hypothesis_qualtest_Test_Determine_Association_Hypothesis_D \(\)](#)

3.6.1 Detailed Description

This file contains unit tests to test the requirements connected to f360_detection_to_track_association.cpp file

3.7 Fusion360/detection_to_track_association/unittest/f360_mark_detections_wheel_spin_from_objects_qualtest_unittest.cpp File Reference

Tests

- test group [f360_detect_wheelspin_object_qualtest](#) ()
Group of tests checking if detections are correctly marked as object wheel spins.
- [f360_detect_wheelspin_object_qualtest_f360_detect_wheelspin_object_qualtest__correctly_marked_as_object_wheel_spin](#) ()
- [f360_detect_wheelspin_object_qualtest_f360_detect_wheelspin_object_qualtest__correctly_not_marked_as_object_wheel_spin](#) ()

3.7.1 Detailed Description

Qualification tests for FTCP-8840 requirement. Tests check if detections are marked correctly as wheel spin object type i.e. detections which are likely a result of reflections from a spinning wheel.

3.8 Fusion360/detection_to_track_association/unittest/f360_nearby_wheel_spins_qualtest_unittest.cpp File Reference

Tests

- test group [f360_detect_wheelspin_nearby_qualtest](#) ()
Group of tests checking if detections are correctly marked as nearby wheel spins.
- [f360_detect_wheelspin_nearby_qualtest_f360_detect_wheelspin_nearby_qualtest__correctly_marked_as_nearby_wheel_spin](#) ()
- [f360_detect_wheelspin_nearby_qualtest_f360_detect_wheelspin_nearby_qualtest__correctly_not_marked_as_wheel_spin](#) ()

3.8.1 Detailed Description

Qualification tests for FTCP-8840 requirement. Tests check if detection which is located close enough to another detection already marked as wheel spin is marked correctly as wheel spin nearby type

3.9 Fusion360/initialize_tracks/unittest/f360_fill_init_obj_track_props_qualtest_unittest.cpp File Reference

Tests

- test group [f360_fill_init_obj_track_props_qualtest \(\)](#)
Test group for Fill_Init_Obj_Track_Props function.
- [f360_fill_init_obj_track_props_qualtest_TestCalcInitObjTrackMovingFlagMoving \(\)](#)
- [f360_fill_init_obj_track_props_qualtest_TestCalcInitObjTrackMovingFlagMovingValidForLiberalTracking \(\)](#)
- [f360_fill_init_obj_track_props_qualtest_TestCalcInitObjTrackMovingFlagNotMovingHighFracAz \(\)](#)
- [f360_fill_init_obj_track_props_qualtest_TestCalcInitObjTrackMovingFlagNotMovingLowSpeed \(\)](#)
- [f360_fill_init_obj_track_props_qualtest_TestCalcInitObjTrackMovingFlagNotMovingLowSpeedAndHighFracAz \(\)](#)
- [f360_fill_init_obj_track_props_qualtest_TestCalcInitObjVCSHeadingMovingObj \(\)](#)
- [f360_fill_init_obj_track_props_qualtest_TestCalcInitObjVCSHeadingNotMovingObj \(\)](#)
- [f360_fill_init_obj_track_props_qualtest_TestFillInitObjTrackPropsNominal \(\)](#)
- test group [f360_init_obj_track_covariances \(\)](#)
Test group for Init_Obj_Track_Covariances function.
- [f360_init_obj_track_covariances_TestCCANominalCase \(\)](#)
- test group [f360_fill_init_obj_track_props_determine_initial_size_qualtest \(\)](#)
Test group for Determine_Initial_Size function.
- [f360_fill_init_obj_track_props_determine_initial_size_qualtest_Test_Initial_Size_Stationary_Object \(\)](#)
- [f360_fill_init_obj_track_props_determine_initial_size_qualtest_Test_Initial_Size_Slow_Moving_Object \(\)](#)
- [f360_fill_init_obj_track_props_determine_initial_size_qualtest_Test_Initial_Size_Fast_Moving_Object \(\)](#)

3.9.1 Detailed Description

File with set of qualification tests (which are also unit test) for Fill_Init_Obj_Track_Props function.

3.10 Fusion360/initialize_tracks/unittest/f360_initialize_tracks_qualtest_unittest.cpp File Reference

Tests

- test group [f360_initialize_tracked_objects \(\)](#)
Test group for Initialize_Tracks function.
- [f360_initialize_tracked_objects_Lowest_Cluster_Priority_Initialization_Try_With_Obj_Saturation \(\)](#)
- [f360_initialize_tracked_objects_Highest_Cluster_Priority_Initialization_Try_With_Obj_Saturation \(\)](#)
- [f360_initialize_tracked_objects_Single_Successfull_Object_Initialization \(\)](#)
- [f360_initialize_tracked_objects_Single_Successfull_Object_Initialization_With_Init_Scheme_Check \(\)](#)
- [f360_initialize_tracked_objects_Detections_Assignment_For_Single_Successfull_Object_Initialization \(\)](#)
- [f360_initialize_tracked_objects_Detections_Assignment_For_Single_Unsuccessfull_Object_Initialization \(\)](#)
- [f360_initialize_tracked_objects_Single_Unsuccessfull_Object_Initialization \(\)](#)
- [f360_initialize_tracked_objects_Two_Successfull_And_One_Unsuccessfull_Objects_Initialization \(\)](#)
- [f360_initialize_tracked_objects_Highest_Cluster_Priority_Reinitialization_With_Obj_Saturation \(\)](#)
- [f360_initialize_tracked_objects_Check_If_Objects_Are_Correctly_Sorted_By_Priority \(\)](#)

- [f360_initialize_tracked_objects_Check_If_Objects_Are_Correctly_Sorted_By_Longitudinal_Position \(\)](#)
- [f360_initialize_tracked_objects_Single_Non_Occluded_Cluster__Successful_Initilization \(\)](#)
- [f360_initialize_tracked_objects_Single_Occluded_Cluster__Unsuccessful_Initialization \(\)](#)
- [f360_initialize_tracked_objects_Occluded_And_Non_Occluded_Cluster__Only_One_Successfull_Initialization \(\)](#)
- [f360_initialize_tracked_objects_Occluded_Cluster_With_Detections_Valid_For_Liberal_Initilization__Successfull_Initialization \(\)](#)

3.10.1 Detailed Description

File with set of qualification tests (which are also unit test) for initialize_tracks function

3.11 Fusion360/initialize_tracks/unittest/f360_nees_cfmi_try_to_init_↵ qualtest_unittest.cpp File Reference

Tests

- test group [f360_try_to_init_obj_track_qualtest \(\)](#)
Test group for Try_To_Init_Obj_Track.
- [f360_try_to_init_obj_track_qualtest_HostOverlapCheckPosLongOverlappedInRearAxleDistPosLatNOTOverlappedtest \(\)](#)
- [f360_try_to_init_obj_track_qualtest_HostOverlapCheckPosLongOverlappedInRearAxleDistPosLatOverlappedtest \(\)](#)
- [f360_try_to_init_obj_track_qualtest_HostOverlapCheckPosLongNOTOverlapedPosLatNOTOverlappedtest \(\)](#)
- [f360_try_to_init_obj_track_qualtest_HostOverlapCheckPosLongNOTOverlapedPosLatOverlappedtest \(\)](#)
- [f360_try_to_init_obj_track_qualtest_LiberalInitializationTest \(\)](#)

3.11.1 Detailed Description

File with set of qualification tests (which are also unit test) for Try_To_Init_Obj_Track function.

3.12 Fusion360/inputs_preprocessing/unittest/f360_inputs_↵ preprocessing_qualtest_unittest.cpp File Reference

Tests

- test group [inputs_preprocessing_qualtest \(\)](#)
Add brief description of test group.
- [inputs_preprocessing_qualtest_Internal_Preprocessing_Input_Interface_Requirements \(\)](#)

3.12.1 Detailed Description

This file contains sw qualifications tests for f360_internal_preprocessing

3.12.2 Tests

3.12.2.1 inputs_preprocessing_qualtest_Internal_Preprocessing_Input_Interface_Requirements()

```
inputs_preprocessing_qualtest_Internal_Preprocessing_Input_Interface_Requirements ( )
```

Purpose:\n Test input interface requirements.

Requirements FTCP-8256, FTCP-7799

Precondition:\n

Initialize structures of the correct type as specified in requirements.

Action:\n

Call function to test that inputs, as specified in requirements, are accepted.

Result:\n

Test should compile and run. No functionality is supposed to be tested in this test.

3.13 Fusion360/internal_preprocessing/qualification_testing/qualtest/guardrail_detection_qualtest.cpp File Reference

Tests

- test group [guardrail_detection](#) ()
- [guardrail_detection_capability](#) ()
- [guardrail_detection_no_guardrail_sanity_check](#) ()

3.13.1 Detailed Description

This file contains basic unit tests for guard rail detection.

3.14 Fusion360/internal_preprocessing/unittest/f360_detect_wheelspin_pairs_qualtest_unittest.cpp File Reference

Tests

- test group [f360_detect_wheelspin_pairs_qualtest_unittest \(\)](#)
Test group including all input to Detect_Wheel_Spin_Pairs and a helper file to add detections used for testing. It also includes some constants located in Detect_Wheel_Spin_Pairs that could be used to quickly adapt tests if the internal constants change.
- [f360_detect_wheelspin_pairs_qualtest_unittest_doppler_spread_limits \(\)](#)
- [f360_detect_wheelspin_pairs_qualtest_unittest_azimuth_spread_limits \(\)](#)
- [f360_detect_wheelspin_pairs_qualtest_unittest_long_spread_limits \(\)](#)
- [f360_detect_wheelspin_pairs_qualtest_unittest_lat_spread_limits \(\)](#)
- [f360_detect_wheelspin_pairs_qualtest_unittest_long_limits \(\)](#)
- [f360_detect_wheelspin_pairs_qualtest_unittest_lat_limits \(\)](#)
- [f360_detect_wheelspin_pairs_qualtest_unittest_remarking \(\)](#)
- [f360_detect_wheelspin_pairs_qualtest_unittest_multiple_candidates \(\)](#)
- [f360_detect_wheelspin_pairs_qualtest_unittest_proximity_detections \(\)](#)
- [f360_detect_wheelspin_pairs_qualtest_unittest_max_wheelspin_pairs \(\)](#)

3.14.1 Detailed Description

Test file including combined qualification and unit tests for wheelspin detection.

3.15 Fusion360/internal_preprocessing/unittest/f360_determine_dets_on_edge_of_fov_qualtest_unittest.cpp File Reference

Tests

- test group [f360_determine_dets_on_edge_of_fov \(\)](#)
Test group that sets up two sensors with located in origo. One aligned with the x-axis and one aligned with the y-axis. Their MR FOV is 90 degree so they overlap with 45 degrees while their LR FOV is 45 degrees so they align with their respective borders. The test group also contains helper functions to add detections to the respective sensors.
- [f360_determine_dets_on_edge_of_fov_mr_detections \(\)](#)
- [f360_determine_dets_on_edge_of_fov_lr_detections \(\)](#)
- [f360_determine_dets_on_edge_of_fov_range_limits \(\)](#)

3.15.1 Detailed Description

Tests the function that determine if a detections is outside the sensor FOV but inside another sensors FOV.

3.16 Fusion360/internal_preprocessing/unittest/f360_handle_host_vehicle_clutter_qualtest_unittest.cpp File Reference

Tests

- test group [f360_handle_host_vehicle_clutter_qualtest](#) ()
Test group of HVC detections qualification tests. Tests verify whether requirements are fulfilled.
- [f360_handle_host_vehicle_clutter_qualtest_handle_host_vehicle_clutter__dets_marked_as_not_okay_to_use](#) ()
- [f360_handle_host_vehicle_clutter_qualtest_handle_host_vehicle_clutter__dets_marked_as_okay_to_use](#) ()
- [f360_handle_host_vehicle_clutter_qualtest_handle_host_vehicle_clutter__hvc_and_behind_guardrail_dets_marked_as_not_o](#) ()
- [f360_handle_host_vehicle_clutter_qualtest_handle_host_vehicle_clutter__hvc_and_on_guardrail_dets_marked_as_not_okay](#) ()

3.16.1 Detailed Description

This file contains qualification tests for content of f360_handle_host_vehicle_clutter.cpp file

3.17 Fusion360/internal_preprocessing/unittest/f360_identify_and_flag_internal_reflections_qualtest_unittest.cpp File Reference

Tests

- test group [f360_identify_and_flag_internal_reflections__scenario_based_tests](#) ()
This test group tests the functionality of the main function inside f360_identify_and_flag_internal_reflections.cpp through a set of artificial scenarios by only modifying input rather than intermittent variables and/or structures.
- [f360_identify_and_flag_internal_reflections__scenario_based_tests_flag_recurring_detections](#) ()
- [f360_identify_and_flag_internal_reflections__scenario_based_tests_reset_buffer_entries_with_low_occurance](#) ()

3.17.1 Detailed Description

This file contains qualification tests for content of f360_identify_and_flag_internal_reflections.cpp file

3.17.2 Tests

3.17.2.1 f360_identify_and_flag_internal_reflections__scenario_based_tests()

```
test_group f360_identify_and_flag_internal_reflections__scenario_based_tests ( )
```

This test group tests the functionality of the main function inside `f360_identify_and_flag_internal_reflections.cpp` through a set of artificial scenarios by only modifying input rather than intermittent variables and/or structures.

Setup:\n

Initialize and reset all variables used in function call. In `TEST_SETUP()`, the following is done;

- Set all relevant sensors for all sensors to some reasonable value
- Set the OTG speed of all sensors to be slightly higher than `internal_reflections_min_host_vel`
- Reset all entries in all buffer slots for all sensors to 0.
- Reset all entries in `raw_det_list` to 0
- Reset `f_ok_to_use` to true for for all dets in `dets_props` Setup helper function `Copy_Relevant_Detection_Properties()` that copies relevant detection properties from a single detection to another detection.

3.17.2.2 f360_identify_and_flag_internal_reflections__scenario_based_tests_flag_recurring_detections()

```
f360_identify_and_flag_internal_reflections__scenario_based_tests_flag_recurring_detections ( )
```

Purpose:\n

Verify that an empty buffer fills up and starts flagging new detections as expected for an identical detection that does not always reoccur, but often enough to eventually be flagged as an internal reflection.

Requirements FTCP-13462

Precondition:\n

- Set all relevant sensors for all sensors to some reasonable value
- Set the OTG speed of all sensors to be slightly higher than `internal_reflections.min_host_vel`
- Reset all entries in all buffer slots for all sensors to 0.
- Reset all entries in `raw_det_list` to 0
- Reset `f_ok_to_use` to true for for all dets in `dets_props`
- Set `sensors[0].internal_reflections.age_threshold` to 200
- Setup 1 detection that is relevant to buffer
- Set all other detections in `raw_det_list` to be irrelevant for buffer (i.e. by having properties such as `rscs > internal_reflections.rscs_max`).
- Feed the relevant detection into `Identify_And_Remove_Internal_Reflections()` not always but often enough to get an `occurence_rate` higher than `internal_reflections_occurrence_threshold`.
- Change the detection slot index for the relevant detection between iterations
- Call the function in this manner 200 times such that the buffer slot containing the relevant detection has an age and `occurence_count` such that it will be flagged as internal reflection.
- Setup the relevant detection in an arbitrary detection slot and store its slot index

Action:\n

Call function `Identify_And_Flag_Internal_Reflections()`.

Result:\n

Verify that the relevant detection was flagged as not ok to use Verify that all other detections is ok to use

3.17.2.3 f360_identify_and_flag_internal_reflections__scenario_based_tests_reset_buffer_entries_with_low_occurance()

```
f360_identify_and_flag_internal_reflections__scenario_based_tests_reset_buffer_entries_with_↵  
low_occurance ( )
```

Purpose:

Verify that a buffer resets for entries with a low occurrence rate by feeding the function a detection that fills a buffer slot enough times such that it will flag similar detections as internal reflections. Then call the function without any new detections such that the buffer entry's occurrence rate will drop and reset the buffer entry. Then ultimately feed the same detection again and verify that it does not get flagged.

Requirements FTCP-13462**Precondition:**

- Set all relevant sensors for all sensors to some reasonable value
- Set the OTG speed of all sensors to be slightly higher than internal_reflections.min_host_vel
- Reset all entries in all buffer slots for all sensors to 0.
- Reset all entries in raw_det_list to 0
- Reset f_ok_to_use to true for for all dets in dets_props
- Set sensors[0].internal_reflections.age_threshold to 200
- Setup 1 detection that is relevant to buffer
- Set all other detections in raw_det_list to be irrelevant for buffer (i.e. by having properties such as rcs > internal_reflections.rcs_max).
- Feed the relevant detection into Identify_And_Remove_Internal_Reflections() not always but often enough to get an occurrence_rate higher than internal_reflections.occurrence_threshold.
- Change the detection slot index for the relevant detection between iterations
- Call the function in this manner 201 times such that the buffer slot containing the relevant detection has an age and occurrence_count such that it will be flagged as internal reflection.
- Reset dets_props[0].f_ok_to_use to true as it was flagged during the 201'th call.
- Remove the detection that was relevant such that all detections are irrelevant for function.
- Call function 33 times to increase buffer age without increasing occurrence count such that the buffer slot's occurrence rate drops below internal_reflections_occurrence_lowerlimit.
- Setup the relevant detection again in raw_det_list on an arbitrary detection slot

Action:

Call function again with the same identical, relevant detection that was used during test setup.

Result:

Verify that, as the buffer has reset, the detection does is still flagged as ok to use.

3.18 Fusion360/internal_preprocessing/unittest/f360_mark_host_water_spray_detections_qualtest_unittest.cpp File Reference

Tests

- test group [f360_mark_water_spray_detections \(\)](#)
This test group contains tests for Mark_Water_Spray_Detections function.
- [f360_mark_water_spray_detections_default_test \(\)](#)
- [f360_mark_water_spray_detections_detections_outside_vcs_zone \(\)](#)
- [f360_mark_water_spray_detections_low_host_speed \(\)](#)
- [f360_mark_water_spray_detections_f_ok_to_use_false \(\)](#)
- [f360_mark_water_spray_detections_det_high_rcs \(\)](#)
- [f360_mark_water_spray_detections_det_range_rate \(\)](#)

3.18.1 Detailed Description

This file contains test for functions in f360_mark_water_spray_detections.

3.19 Fusion360/measurement_update_tracks/unittest/f360_measurement_update_tracks_qualtest_unittest.cpp File Reference

Tests

- test group [f360_msmt_update_obj_trks_cca_non_moveable_qualtest \(\)](#)
Setup general values on object states and detection properties that is needed to perform a measurement update.
- [f360_msmt_update_obj_trks_cca_non_moveable_qualtest_Msmt_Update_Obj_Trks_CCA_Non_Moveable \(\)](#)
- test group [f360_msmt_update_obj_trks_cca_moveable_qualtest \(\)](#)
Testing that a call to Msmt_Update_Obj_Trks_CCA() with one eligible detection is updating objects states.
- [f360_msmt_update_obj_trks_cca_moveable_qualtest_Msmt_Update_Obj_Trks_CCA_Moveable_Check_Update_One_Det_Obj \(\)](#)
- [f360_msmt_update_obj_trks_cca_moveable_qualtest_Msmt_Update_Obj_Trks_CCA_Moveable_Check_Update_One_Det_Obj \(\)](#)
- [f360_msmt_update_obj_trks_cca_moveable_qualtest_Msmt_Update_Obj_Trks_CCA_Moveable_Check_Head_Point_Not_Updated \(\)](#)
- test group [f360_msmt_update_obj_trks_ctca_qualtest \(\)](#)
Setup general values on object states and detection properties that is needed to perform a measurement update.
- [f360_msmt_update_obj_trks_ctca_qualtest_Msmt_Update_Obj_Trks_CTCA_Check_Update_One_Det \(\)](#)
- test group [f360_msmt_update_obj_trks_ctca_not_moving_qualtest \(\)](#)
Test group for tests related to freezing of heading and curvature for CTCA tracks.
- [f360_msmt_update_obj_trks_ctca_not_moving_qualtest_Msmt_Update_ObjTrks_CTCA__curvature_head_is_not_updated_if \(\)](#)
- test group [f360_msmt_update_object_timestamp_qualtest \(\)](#)
Create needed default variables needed for update of object measurement timestamp.
- [f360_msmt_update_object_timestamp_qualtest_Msmt_Update_Object_Timestamp_One_Det \(\)](#)

3.19.1 Detailed Description

This file contains unit tests for testing requirements connected to f360_measurement_update_tracks.cpp file

3.20 Fusion360/measurement_update_tracks/unittest/f360_msmt_↔ update_support_functions_common_qualtest_unittest.cpp File Reference

Tests

- test group [f360_msmt_update_supporting_functions_Select_Dets_For_RR_Update_qualtest](#) ()
Purpose is to very that function Select_Dets_For_RR_Update works as intended. Initialize an object and assign some detections. Assign detection properties to cover all possible branches inside the function.
- [f360_msmt_update_supporting_functions_Select_Dets_For_RR_Update_qualtest_Test_Select_Dets_For_RR_Update](#) ()

3.20.1 Detailed Description

This file contains unit tests for testing requirements connected to f360_msmt_update_support_functions_↔
common.cpp

3.21 Fusion360/post_update_track_adjustments/unittest/f360_post_↔ update_track_adjustments_qualtest_unittest.cpp File Reference

Tests

- test group [f360_update_object_track_properties_Calc_Obj_Size](#) ()
Group of tests to check that object length and width are updated under correct circumstances.
- [f360_update_object_track_properties_Calc_Obj_Size_Obj_Size_Is_Updated](#) ()
- [f360_update_object_track_properties_Calc_Obj_Size_Obj_Size_Is_Not_Updated_Obj_Coasting](#) ()
- [f360_update_object_track_properties_Calc_Obj_Size_Obj_Size_Is_Not_Updated_Obj_One_Assoc_Det](#) ()
- test group [f360_object_track_management_qualtest](#) ()
Includes tests that will test the behavior of functions implemented in f360_object_track_management.cpp.
- [f360_object_track_management_qualtest_Obj_Trk_Status_Book_Keeping_NEW](#) ()
- [f360_object_track_management_qualtest_Obj_Trk_Status_Book_Keeping_UPDATED](#) ()
- test group [f360_adjust_filtr_type_dependet_params_qualtest](#) ()
- [f360_adjust_filtr_type_dependet_params_qualtest_Adjust_Filtr_Type_Dependent_Params__switch_to_CTCA](#) ()
- test group [f360_cancel_new_updated_trk_overlapping_confirmed_trks_qualtest](#) ()
Tests that an objects is terminated correctly when overlapping with another.
- [f360_cancel_new_updated_trk_overlapping_confirmed_trks_qualtest_Cancel_New_Updated_Trk_Overlapping_Confirmed_Trk](#) ()
- [f360_cancel_new_updated_trk_overlapping_confirmed_trks_qualtest_Cancel_New_Updated_Trk_Overlapping_Confirmed_Trk](#) ()
- test group [f360_update_object_track_properties_calc_obj_height](#) ()
Group of tests to check that objects over the ground height is updated under correct circumstances.
- [f360_update_object_track_properties_calc_obj_height_obj_height_is_updated_when_valid](#) ()
- [f360_update_object_track_properties_calc_obj_height_obj_height_is_not_calculated_when_stationary](#) ()
- [f360_update_object_track_properties_calc_obj_height_obj_height_decays_when_no_dets](#) ()
- [f360_update_object_track_properties_calc_obj_height_obj_height_decay_below_zero](#) ()

3.21.1 Detailed Description

This file contains unit tests for testing the requirements connected to f360_post_update_track_adjustments module

3.22 Fusion360/pre_association_track_management/unittest/f360_pre_↵ association_track_management_qualtest_unittest.cpp File Reference

Tests

- test group [f360_object_based_water_spray_detector_qualtest \(\)](#)
Test group contains variables and test cases related to flagging of detections as water spray from objects. Three objects and five detections are defined in the test group, please see DFD-390 for a detailed description of the setup.
- [f360_object_based_water_spray_detector_qualtest_Detect_Water_Spray_From_Objects \(\)](#)
- test group [f360_object_based_angle_jump_detector_Detect_Angle_Jumps_From_Objects_qualtest \(\)](#)
Test group for testing Detect_Angle_Jumps_From_Objects()
- [f360_object_based_angle_jump_detector_Detect_Angle_Jumps_From_Objects_qualtest_flagging_correctness_of_f_ok_to_us
\(\)](#)
- test group [f360_find_tracks_on_and_behind_guardrail_qualtest \(\)](#)
Test group of Find_Tracks_On_And_Behind_Guardrail.
- [f360_find_tracks_on_and_behind_guardrail_qualtest_Flag_Objects_On_SEP_ObjectOnGuardrailLeftSide \(\)](#)
- [f360_find_tracks_on_and_behind_guardrail_qualtest_Flag_Objects_On_SEP_ObjectOnGuardrailRightSide
\(\)](#)
- test group [detecting_multibounces_from_object \(\)](#)
Test requirements w.r.t. multibounce detections.
- [detecting_multibounces_from_object_det_marked_as_multibounce \(\)](#)
- [detecting_multibounces_from_object_det_NOT_marked_as_multibounce \(\)](#)
- test group [object_wrt_to_sep_tests \(\)](#)
Test requirements w.r.t. object relation to a SEP.
- [object_wrt_to_sep_tests_obejct_marked_as_on_a_sep \(\)](#)
- [object_wrt_to_sep_tests_obejct_NOT_marked_as_on_sep \(\)](#)

3.22.1 Detailed Description

This file contains unit tests for testing the requirements connected to f360_pre_association_track_management.cpp file

3.23 Fusion360/sensor_postprocessing/unittest/f360_sensor_↵ postprocessing_qualtest_unittest.cpp File Reference

Tests

- test group [f360_Update_Detection_History_qualtest \(\)](#)
This test suit aims to test Update_Detection_History function.
- [f360_Update_Detection_History_qualtest_Test_Update_Detection_History_update_one_detection \(\)](#)
- [f360_Update_Detection_History_qualtest_Test_Update_Detection_History_old_dets_greater_than_zero \(\)](#)
- [f360_Update_Detection_History_qualtest_Test_Update_Detection_History_n_occupied_higher_than_max \(\)](#)
- [f360_Update_Detection_History_qualtest_Test_Update_Detection_History_Find_Next_Hist_Det_Idx_false \(\)](#)

3.23.1 Detailed Description

- File with set of qualification tests (which are also unit test) for Sensor_Postprocessing module functions.

3.24 Fusion360/time_update_tracks/unittest/f360_time_update_tracks_qualtest_unittest.cpp File Reference

Tests

- test group [f360_kill_coasted_tracks_qualtest](#) ()
This group provides tests ensuring specified behavior for termination of coasting objects.
- [f360_kill_coasted_tracks_qualtest_Do_not_terminate_updated_tracks](#) ()
- [f360_kill_coasted_tracks_qualtest_Termination_of_coasting_confirmed_objects](#) ()
- [f360_kill_coasted_tracks_qualtest_Termination_of_coasting_mirror_objects_variation_of_stage_time](#) ()
- [f360_kill_coasted_tracks_qualtest_Termination_of_coasting_mirror_objects_variation_of_mirror_probability](#) ()
- [f360_kill_coasted_tracks_qualtest_Non_Movable_Coasting_Obj_Outside_Front_Only_FOV_Coasting_Time](#) ()
- [f360_kill_coasted_tracks_qualtest_Non_Movable_Coasting_Obj_Outside_Front_Only_FOV_Pos](#) ()
- [f360_kill_coasted_tracks_qualtest_Non_Movable_Coasting_Obj_Outside_Front_Only_FOV_Config](#) ()
- [f360_kill_coasted_tracks_qualtest_Non_Movable_Coasting_Obj_Outside_Front_Only_FOV_Nonmoveable](#) ()

3.24.1 Detailed Description

This file contains qualification tests for content of f360_time_update_tracks.cpp file

3.25 Fusion360/track_classification/unittest/f360_object_occlusion_classification_qualtest_unittest.cpp File Reference

Tests

- test group [f360_object_occlusion_classification](#) ()
Test group dedicated to verify the functionality of the function `Object_Occlusion_Classification()`. It is done by setting up a scenario with a single forward looking sensor and 2 objects. 1 object is intended to be tested and while the other object is placed between the test object and the sensor, occluding the test object. This scenario is varied slightly to achieve different levels of occlusion.
- [f360_object_occlusion_classification_Verify_Single_Object_Status_Occluded](#) ()
- [f360_object_occlusion_classification_Verify_Single_Object_Status_On_Edge](#) ()
- [f360_object_occlusion_classification_Verify_Single_Object_Status_Visible](#) ()
- [f360_object_occlusion_classification_Verify_Single_Object_Status_Undefined_Not_Moving](#) ()
- [f360_object_occlusion_classification_Verify_All_Active_Objects_Are_Considered](#) ()

3.25.1 Detailed Description

This file contains qualification tests for content of the `f360_object_occlusion_classification.cpp` file

3.26 Fusion360/track_classification/unittest/f360_object_underdrivability_classification_qualtest_unittest.cpp File Reference

Tests

- test group `f360_object_underdrivability_classification_qt` ()
This test group will test the intended functionality of `Object_Underdrivability_Classification()`. Tests are designed to check.
- `f360_object_underdrivability_classification_qt_test_valid_moving_objs_height_above_threshold` ()
- `f360_object_underdrivability_classification_qt_test_valid_moving_objs_height_below_threshold` ()
- `f360_object_underdrivability_classification_qt_test_valid_moving_objs_behind_host` ()
- `f360_object_underdrivability_classification_qt_test_objs_invalid` ()

3.26.1 Detailed Description

This file contains unit tests for content of `f360_object_underdrivability_classification_qt.cpp` file

3.27 Fusion360/track_classification/unittest/f360_track_classification_qualtest_unittest.cpp File Reference

Tests

- test group `f360_assign_underdrivability_status_to_tracks_ocg_QT` ()
Test group of `Assign_Underdrivability_Status_To_Tracks`. Tests verify whether tracks are properly checked whether they can be marked as underdrivable, whether they do not have their underdrivability status modified when are not out of zones FOV and whether they are properly assigned new underdrivability status.
- `f360_assign_underdrivability_status_to_tracks_ocg_QT_Assign_Underdrivability_Status_To_Tracks__Object_Not_In_FOV` ()
- `f360_assign_underdrivability_status_to_tracks_ocg_QT_Assign_Underdrivability_Status_To_Tracks__Current_Status_Can_Pa` ()
- `f360_assign_underdrivability_status_to_tracks_ocg_QT_Assign_Underdrivability_Status_To_Tracks__Current_Status_Can_Pa` ()
- `f360_assign_underdrivability_status_to_tracks_ocg_QT_Assign_Underdrivability_Status_To_Tracks__Not_To_Consider_After` ()
- `f360_assign_underdrivability_status_to_tracks_ocg_QT_Assign_Underdrivability_Status_To_Tracks__Current_Status_Can_No` ()
- `f360_assign_underdrivability_status_to_tracks_ocg_QT_Assign_Underdrivability_Status_To_Tracks__Current_Status_Can_No` ()
- test group `object_motion_status_should_be_updated_qualtest` ()
Test group of `Object_Motion_Status_Should_Be_Updated` function. Tests verify whether function properly verifies whether object motion status should be updated.

- `object_motion_status_should_be_updated_qualtest_Object_Motion_Status_Should_Be_Updated__NEW_UPDATED`
()
- `object_motion_status_should_be_updated_qualtest_Object_Motion_Status_Should_Be_Updated__UPDATED`
()
- `object_motion_status_should_be_updated_qualtest_Object_Motion_Status_Should_Be_Updated__COASTED__Time_Below`
()
- `object_motion_status_should_be_updated_qualtest_Object_Motion_Status_Should_Be_Updated__COASTED__Time_Above`
()
- `object_motion_status_should_be_updated_qualtest_Object_Motion_Status_Should_Be_Updated__NEW_COASTED__Time`
()
- test group `update_object_properties_qualtest` ()
Test group of Update_Object_Properties() function. Tests verify whether object motion properties are properly updated.
- `update_object_properties_qualtest_update_object_properties__check_if_changes_moveable_flag` ()
- `update_object_properties_qualtest_update_object_properties__check_if_moveable_flag_is_not_reset` ()
- test group `check_if_object_is_suspected_stationary_qualtest` ()
Test group of Check_If_Object_Is_Suspected_Stationary() function. Test verify whether stationary suspected flag is properly set for both CTCA and CCA tracks.
- `check_if_object_is_suspected_stationary_qualtest_check_if_object_is_suspected_stationary__ctca_marked_as_stationary_if`
()
- `check_if_object_is_suspected_stationary_qualtest_check_if_object_is_suspected_stationary__ctca_marked_as_not_stationary`
()
- `check_if_object_is_suspected_stationary_qualtest_check_if_object_is_suspected_stationary__ctca_marked_as_not_stationary`
()
- `check_if_object_is_suspected_stationary_qualtest_check_if_object_is_suspected_stationary__ctca_marked_as_not_stationary`
()
- `check_if_object_is_suspected_stationary_qualtest_check_if_object_is_suspected_stationary__ctca_marked_as_not_stationary`
()
- `check_if_object_is_suspected_stationary_qualtest_check_if_object_is_suspected_stationary__cca_marked_as_stationary_if`
()
- `check_if_object_is_suspected_stationary_qualtest_check_if_object_is_suspected_stationary__cca_marked_as_stationary_if`
()
- `check_if_object_is_suspected_stationary_qualtest_check_if_object_is_suspected_stationary__cca_marked_as_not_stationary`
()
- `check_if_object_is_suspected_stationary_qualtest_check_if_object_is_suspected_stationary__cca_marked_as_not_stationary`
()
- `check_if_object_is_suspected_stationary_qualtest_check_if_object_is_suspected_stationary__cca_marked_as_not_stationary`
()
- `check_if_object_is_suspected_stationary_qualtest_check_if_object_is_suspected_stationary__cca_marked_as_not_stationary`
()
- test group `calcProbability_qualtest` ()
Test group of calcProbability function.
- `calcProbability_qualtest_calcProbability__check_if_sum_of_calculated_probabilities_is_equal_1` ()
- test group `set_object_class_qualtest` ()
Test group of setObjectClass function. Tests verify whether object class is properly assigned.
- `set_object_class_qualtest_set_object_class__check_if_prioritizes_pedestrian` ()
- `set_object_class_qualtest_set_object_class__check_if_prioritizes_motorcycles` ()
- `set_object_class_qualtest_set_object_class__check_if_prioritizes_cars` ()
- `set_object_class_qualtest_set_object_class__check_if_prioritizes_trucks` ()
- `set_object_class_qualtest_set_object_class__check_if_classifies_as_undetermined` ()
- `set_object_class_qualtest_set_object_class__check_if_min_prob_value_to_classify_is_0_65` ()
- test group `filteringAndNormalizationOfProbabilities_qualtest` ()
Test group of filteringAndNormalizationOfProbabilities function. Tests verify whether probability being undetermined is used to lower probabilities of object being specific class.

- [filteringAndNormalizationOfProbabilities_qualtest_filteringAndNormalizationOfProbabilities__check_if_undeterm_prob_value_i](#)
()
- test group [calcUndetProb_qualtest](#) ()
*Purpose of this test group is to verify whether probability unknown is set to max value when object has status: F360_↔
_OBJECT_STATUS_NEW or F360_OBJECT_STATUS_INVALID or F360_OBJECT_STATUS_NEW_UPDATED or
F360_OBJECT_STATUS_NEW_COASTED.*
- [calcUndetProb_qualtest_calcUndetProb__check_if_prob_unknown_is_ste_to_max_value_when_object_has_status_F360_OE](#)
()
- [calcUndetProb_qualtest_calcUndetProb__check_if_prob_unknown_is_ste_to_max_value_when_object_has_status_F360_OE](#)
()
- [calcUndetProb_qualtest_calcUndetProb__check_if_prob_unknown_is_ste_to_max_value_when_object_has_status_F360_OE](#)
()
- [calcUndetProb_qualtest_calcUndetProb__check_if_prob_unknown_is_ste_to_max_value_when_object_has_status_F360_OE](#)
()
- test group [f360_run_obj_class](#) ()
*Test group for run_obj_class. Purpose is to test that all object classes are possible object classifications, correspond-
ing to the requirement FTCP-11450. The classes are.*
- [f360_run_obj_class_f360_run_obj_class_bicycle](#) ()
- [f360_run_obj_class_f360_run_obj_class_pedestrian](#) ()
- [f360_run_obj_class_f360_run_obj_class_motorcycle](#) ()
- [f360_run_obj_class_f360_run_obj_class_car](#) ()
- [f360_run_obj_class_f360_run_obj_class_truck](#) ()
- [f360_run_obj_class_f360_run_obj_class_undetermined](#) ()

3.27.1 Detailed Description

This file contains qualification tests for Track Classification module.

3.28 Fusion360/track_downselection/unittest/f360_track_↔ downselection_qualtest_unittest.cpp File Reference

Tests

- test group [f360_track_downselection_qualtest](#) ()
This test group is used for creatie data structure needed in further tests.
- [f360_track_downselection_qualtest_f360_core_track_downselect__check_if_does_reduce_tracks_basing_on_priority](#)
()
- [f360_track_downselection_qualtest_f360_core_track_downselect__check_if_updated_tracks_are_prioritized_over_new_tracks](#)
()
- [f360_track_downselection_qualtest_f360_core_track_downselect__check_if_tracks_with_reduced_status_set_as_invalid_are](#)
()
- [f360_track_downselection_qualtest_f360_core_track_downselect__check_if_tracks_with_higher_confidence_level_are_prioriti](#)
()
- [f360_track_downselection_qualtest_f360_core_track_downselect__check_if_valid_trk_closer_to_host_is_prioritized](#)
()
- [f360_track_downselection_qualtest_f360_core_track_downselect__check_if_tracks_with_lower_ttc_are_prioritized](#)
()
- [f360_track_downselection_qualtest_f360_core_track_downselect__check_if_tracks_with_low_amount_of_average_dets_are_c](#)
()

- [f360_track_downselection_qualtest_f360_core_track_downselect__check_if_tracks_are_downselected_as_expected_based_on](#)
()
- [f360_track_downselection_qualtest_f360_core_track_downselect__check_if_resets_low_priority_target_reduced_id](#)
()
- [f360_track_downselection_qualtest_f360_core_track_downselect__check_if_assigns_reduced_id_to_higher_priority_targets_and](#)
()
- [f360_track_downselection_qualtest_f360_core_track_downselect__check_if_lowers_confidence_of_tracks_that_changed_their](#)
()
- [f360_track_downselection_qualtest_f360_core_track_downselect__check_if_lowers_confidence_and_priority_of_mirror_detected](#)
()
- [f360_track_downselection_qualtest_f360_core_track_downselect__check_if_lowers_confidence_of_tracks_behind_guardrail](#)
()
- test group [f360_track_downselection_calc_track_priority_qualtest](#) ()
This test group tests priority assignment strategy for non_moveable objects. It tests the following cases: If host speed should impact importance (for downselection), such that the objects in the direction of host motion, should have higher importance. For forward driving host, objects in front have higher importance. On the flip side, for reverse driving host, objects in the back of host have higher importance The importance of the non_movable objects is decided based on their vcs longitudinal position and curvi lateral position (based on host trajectory) Objects longitudinally further away from the host are given lower importance for downselection, as compared to longitudinally closer objects Objects laterally further away from the host trajectory are given lower importance for downselection, as compared to laterally closer objects.
- [f360_track_downselection_calc_track_priority_qualtest_Host_Is_Stationary](#) ()
- [f360_track_downselection_calc_track_priority_qualtest_Host_Is_Moving](#) ()
- [f360_track_downselection_calc_track_priority_qualtest_Host_Is_Reversing](#) ()
- [f360_track_downselection_calc_track_priority_qualtest_Impact_of_Host_Speeds_On_Object_Priorities](#) ()
- [f360_track_downselection_calc_track_priority_qualtest_Host_Curvature_Is_Not_Close_To_Zero](#) ()

3.28.1 Detailed Description

This file contains qualification tests for Track Downselection module.

3.29 Fusion360/track_grouping/unittest/f360_split_objects_in_orth_↔ direction_qualtest_unittest.cpp File Reference

Tests

- test group [f360_split_objects_in_orth_direction_qualtest](#) ()
*This purpose of this test group is for testing the functionality of the main function in f360_split_objects_in_orth_↔
direction.cpp file.*
- [f360_split_objects_in_orth_direction_qualtest_Verify_Split_Logic](#) ()

3.29.1 Detailed Description

This file contains qualification tests for f360_split_objects_in_orth_direction.cpp

3.30 Fusion360/track_grouping/unittest/f360_track_grouping_qualtest_↔ unittest.cpp File Reference

Tests

- test group [f360_track_grouping \(\)](#)
This purpose of this test group is for testing the functionality of the Track_Grouping() function.
- [f360_track_grouping_Test_Similar_Objects_Were_Merged \(\)](#)
- [f360_track_grouping_Test_Merged_Obj_Max_Size_Is_Not_exceed \(\)](#)
- [f360_track_grouping_Test_Remain_Dets_Assoc \(\)](#)
- [f360_track_grouping_Test_Move_All_Dets_Assoc \(\)](#)
- [f360_track_grouping_Test_Move_Dets_Assoc \(\)](#)
- [f360_track_grouping_Check_Consistency_In_Obj_Id \(\)](#)
- [f360_track_grouping_Check_Updates_In_Long_Pos_Sort \(\)](#)
- [f360_track_grouping_Check_Updates_In_Priority_Sort \(\)](#)

3.30.1 Detailed Description

This file contains qualification tests (which are also unit tests) for f360_track_grouping.cpp

3.31 Fusion360/track_validity/unittest/f360_detect_multipath_qualtest_↔ unittest.cpp File Reference

Tests

- test group [f360_detect_multipath__Detect_And_Mark_Multipath_Objects_Qual \(\)](#)
Test group of Detect_And_Mark_Multipath_Objects from f360_detect_multipath module.
- [f360_detect_multipath__Detect_And_Mark_Multipath_Objects_Qual_default_scenario_where_host_speed_is_too_high \(\)](#)
- [f360_detect_multipath__Detect_And_Mark_Multipath_Objects_Qual_default_scenario_where_host_speed_is_below_threshold \(\)](#)
- [f360_detect_multipath__Detect_And_Mark_Multipath_Objects_Qual_scenario_where__source_objects_is_stationary \(\)](#)
- [f360_detect_multipath__Detect_And_Mark_Multipath_Objects_Qual_scenario_where__source_objects_is_inactive \(\)](#)
- [f360_detect_multipath__Detect_And_Mark_Multipath_Objects_Qual_scenario_where__source_objects_is_beyond_sensor_fov \(\)](#)
- [f360_detect_multipath__Detect_And_Mark_Multipath_Objects_Qual_scenario_where__source_objects_use_CCV_filter \(\)](#)
- [f360_detect_multipath__Detect_And_Mark_Multipath_Objects_Qual_scenario_where__reflector_objects_does_not_exist \(\)](#)
- [f360_detect_multipath__Detect_And_Mark_Multipath_Objects_Qual_scenario_where__source_objects_does_not_exist \(\)](#)

3.31.1 Detailed Description

This file contains qualification tests for content of f360_detect_multipath.cpp

3.32 Fusion360/track_validity/unittest/f360_determine_reflected_obj_qualtest_unittest.cpp File Reference

Tests

- test group `f360_determine_reflected_obj` ()
Test group of f360_determine_reflected_obj.
- `f360_determine_reflected_obj_Determine_Reflected_Obj__check_whether_track_behind_guardrail_has_its_mirror_prob_set_to_one` ()
- `f360_determine_reflected_obj_Determine_Reflected_Obj__check_whether_track_that_is_not_a_mirror_track_has_its_mirror_prob_set_to_one` ()
- `f360_determine_reflected_obj_Determine_Reflected_Obj__check_whether_track_that_is_not_a_mirror_track_has_its_mirror_prob_set_to_zero` ()
- `f360_determine_reflected_obj_Determine_Reflected_Obj__check_whether_track_that_is_not_a_mirror_track_has_its_mirror_prob_set_to_zero` ()
- `f360_determine_reflected_obj_Determine_Reflected_Obj__check_whether_mirror_track_has_its_mirror_prob_set_to_one` ()
- `f360_determine_reflected_obj_Determine_Reflected_Obj__check_whether_mirror_track_has_its_mirror_prob_set_to_one_how_far` ()
- `f360_determine_reflected_obj_Determine_Reflected_Obj__check_whether_valid_track_in_front_and_on_guardrail_has_its_mirror_prob_set_to_one` ()
- `f360_determine_reflected_obj_Determine_Reflected_Obj__check_whether_valid_track_behind_guardrail_has_its_mirror_prob_set_to_one` ()
- `f360_determine_reflected_obj_check_calibrations` ()
- test group `f360_determine_reflected_obj_with_turned_hide_tracks_outside_guardrail` ()
Test group of f360_determine_reflected_obj_with_turned_hide_tracks_outside_guardrail It checks output by focusing on an object and a guardrail relation.
- `f360_determine_reflected_obj_with_turned_hide_tracks_outside_guardrail_prob_set_to_zero_because_track_is_on_but_not_on_guardrail` ()
- `f360_determine_reflected_obj_with_turned_hide_tracks_outside_guardrail_prob_set_to_one_because_track_is_not_on_but_behind_guardrail` ()
- `f360_determine_reflected_obj_with_turned_hide_tracks_outside_guardrail_prob_set_to_zero_because_track_is_on_and_behind_guardrail` ()
- `f360_determine_reflected_obj_with_turned_hide_tracks_outside_guardrail_prob_set_to_one_because_track_is_not_on_and_behind_guardrail` ()

3.32.1 Detailed Description

This file contains qualification tests for content of `f360_determine_reflected_obj.cpp`

3.33 Fusion360/track_validity/unittest/f360_track_validity_qualtest_unittest.cpp File Reference

Tests

- test group `f360_update_exist_prob_qualtest` ()

Test group of Update_Existence_Probability function. Tests verify wheter existence probability is updated only for selected tracks and whether it is updated properly.

- [f360_update_exist_prob_qualtest_Update_Existence_Probability__check_whether_minimum_EP_value_is_equal_to_min_thr](#)
()
- [f360_update_exist_prob_qualtest_Update_Existence_Probability__check_whether_maximum_EP_value_is_1](#)
()
- [f360_update_exist_prob_qualtest_Update_Existence_Probability__check_whether_EP_depend_onobject_state_variance](#)
()
- test group [f360_conf_track_confidence_level_qualtest](#) ()

This test group contains tests for Conf_Track_Confidence_Level() function regarding requirements from SRD document.

- [f360_conf_track_confidence_level_qualtest_Conf_Track_Confidence_Level__if_confidence_is_calculated_for_updated_obj_tr](#)
()
- [f360_conf_track_confidence_level_qualtest_Conf_Track_Confidence_Level__is_confidence_level_is_equal_or_higher_than_0](#)
()
- [f360_conf_track_confidence_level_qualtest_Conf_Track_Confidence_Level__is_confidence_level_is_equal_or_lower_than_1](#)
()
- [f360_conf_track_confidence_level_qualtest_Conf_Track_Confidence_Level__is_confidence_level_is_equal_or_higher_than_0](#)
()
- [f360_conf_track_confidence_level_qualtest_Conf_Track_Confidence_Level__is_confidence_level_is_equal_or_lower_than_1](#)
()
- [f360_conf_track_confidence_level_qualtest_Conf_Track_Confidence_Level__is_confidence_overlapping_object_is_reduced_by](#)
()
- [f360_conf_track_confidence_level_qualtest_Conf_Track_Confidence_Level__is_confidence_level_depend_on_object_status](#)
()
- [f360_conf_track_confidence_level_qualtest_Conf_Track_Confidence_Level__is_confidence_level_depend_on_number_of_det](#)
()
- [f360_conf_track_confidence_level_qualtest_Conf_Track_Confidence_Level__is_confidence_level_depend_on_time_since_the](#)
()

3.33.1 Detailed Description

This file contains qualification tests for Track Validity module.

3.34 StateManager/unittest/f360_input_diagnostics_qualtest_↔ unittest.cpp File Reference

Tests

- uint64_t **ms2us** (uint64_t in_ms)
- test group [f360_input_diagnostics_qualtest](#) ()
Set up initial values of the signals to be checked by the module. Keep in mind that class Execute methods needs to be run at least twice for the module to start detecting errors. In the first cycle no information about the previous cycle is known, so signals can't be verified for consistency.
- [f360_input_diagnostics_qualtest_No_Increase_Faults_Should_Be_Returned_When_Core_Signal_Received_Twice_Without_C](#)
()
- [f360_input_diagnostics_qualtest_Fault_Should_Be_Returned_When_Core_Info_Cnt_Loops_Lower_Than_Before](#)
()
- [f360_input_diagnostics_qualtest_Fault_Should_Be_Returned_When_Core_Info_Time_us_Lower_Than_Before](#)
()

- [f360_input_diagnostics_qualtest_Fault_Should_Be_Returned_When_Core_Info_Elapsed_Time_Below_Lower_Limit\(\)](#)
- [f360_input_diagnostics_qualtest_Fault_Should_Be_Returned_When_Core_Info_Elapsed_Above_Upper_Limit\(\)](#)
- [f360_input_diagnostics_qualtest_Fault_Should_Be_Returned_When_Host_Info_Vehicle_index_Lower_Than_Before\(\)](#)
- [f360_input_diagnostics_qualtest_True_Should_Be_Returned_When_Host_QF_Is_Accurate\(\)](#)
- [f360_input_diagnostics_qualtest_True_Should_Be_Returned_When_Host_QF_Is_NOT_Accurate\(\)](#)
- [f360_input_diagnostics_qualtest_True_Should_Be_Returned_When_Big_Curvature_Rear_Or_VCS_Sideslip\(\)](#)
- [f360_input_diagnostics_qualtest_No_Increase_Faults_Should_Be_Returned_When_Sensors_Signal_Received_Twice_Without\(\)](#)
- [f360_input_diagnostics_qualtest_No_Increase_Fault_Should_Be_Returned_When_Look_Index_For_Single_Sensor_Lower_Than\(\)](#)
- [f360_input_diagnostics_qualtest_Signal_Fault_Should_Be_Returned_When_Difference_Between_Sensor_And_Tracker_Time\(\)](#)
- [f360_input_diagnostics_qualtest_Signal_Fault_Should_Be_Returned_When_Difference_Between_Sensor_And_Tracker_Time\(\)](#)
- [f360_input_diagnostics_qualtest_Signal_Fault_Should_Not_Be_Set_When_No_Sensor_Data_In_Received\(\)](#)
- [f360_input_diagnostics_qualtest_No_Fault_Reported_For_Sensor_When_Calibration_Not_Valid\(\)](#)
- [f360_input_diagnostics_qualtest_No_Fault_Reported_For_Sensor_When_look_index_Overflow\(\)](#)
- [f360_input_diagnostics_qualtest_Signal_Faults_Should_Be_Reported_When_Look_Index_Equal_To_Zero_Twice_In_A_Row\(\)](#)
- [f360_input_diagnostics_qualtest_Signal_Faults_Should_Be_Reported_When_Sensor_Polarity_Is_Equal_To_Zero\(\)](#)
- [f360_input_diagnostics_qualtest_Signal_Faults_Should_Be_Reported_When_Sensor_Polarity_Is_Greater_Than_Expected_Positive\(\)](#)
- [f360_input_diagnostics_qualtest_Signal_Faults_Should_Be_Reported_When_Sensor_Polarity_Is_Greater_Than_Expected_Negative\(\)](#)
- [f360_input_diagnostics_qualtest_Polarity_Fault_Should_Not_Be_Reported_When_Sensor_Polarity_Is_Valid_Positive\(\)](#)
- [f360_input_diagnostics_qualtest_Polarity_Fault_Should_Not_Be_Reported_When_Sensor_Polarity_Is_Valid_Negative\(\)](#)
- [f360_input_diagnostics_qualtest_Signal_Faults_Should_Be_Reported_When_Sensor_Mounting_Position_Is_Too_High_Positive\(\)](#)
- [f360_input_diagnostics_qualtest_Signal_Faults_Should_Be_Reported_When_Sensor_Mounting_Position_Is_Too_High_Negative\(\)](#)
- [f360_input_diagnostics_qualtest_Signal_Faults_Should_Not_Be_Reported_When_Sensor_Mounting_Position_Equal_To_Limit_Positive\(\)](#)
- [f360_input_diagnostics_qualtest_Signal_Faults_Should_Not_Be_Reported_When_Sensor_Mounting_Position_Equal_To_Limit_Negative\(\)](#)
- [f360_input_diagnostics_qualtest_Signal_Faults_Should_Not_Be_Reported_When_Sensor_Mounting_Position_Is_Within_Limit_Positive\(\)](#)
- [f360_input_diagnostics_qualtest_Signal_Faults_Should_Not_Be_Reported_When_Sensor_Mounting_Position_Is_Within_Limit_Negative\(\)](#)
- [f360_input_diagnostics_qualtest_Signal_Faults_Should_Be_Reported_When_Sensor_Azimuth_Boresight_Angle_Is_Greater_Than_Limit_Positive\(\)](#)
- [f360_input_diagnostics_qualtest_Signal_Faults_Should_Be_Reported_When_Sensor_Azimuth_Boresight_Angle_Is_Less_Than_Limit_Positive\(\)](#)
- [f360_input_diagnostics_qualtest_Signal_Faults_Should_Be_Reported_When_Sensor_Azimuth_Boresight_Angle_Is_Within_Limit_Positive\(\)](#)
- [f360_input_diagnostics_qualtest_Signal_Faults_Should_Be_Reported_When_Sensor_Azimuth_Boresight_Angle_Is_Within_Limit_Negative\(\)](#)

3.34.1 Detailed Description

File with set of qualification tests (which are also unit test) for F360 input diagnostics module

Index

CalcProbability, 140

139

calcProbability_qualtest_calcProbability__check_if_stationary_qualtest	140	Check if object is suspected stationary, 13	

calcProbability_qualtest calcProbability check if sum of calculated probabilities is equal stationary_qualtest check if object is stationary

CalcProbability, 140

Check_if_object_is_suspected_stationary, 133

CalcUndetProb, 150

check_if_object_is_suspected_stationary_qualtest_check_if_object_is_su

calcUndetProb_qualtest, 150

Check_if_object_is_suspected_stationary, 134

Function	Description
calcUndetProb_qualtest_calcUndetProb_check_if_prob_unknown_is_stationary	Check if object is suspected stationary, 134
calcUndetProb_qualtest_calcUndetProb_check_if_prob_unknown_is_stationary_qualtest_check_if_prob_unknown_is_stationary	Check if object is suspected stationary, 134

<code>calcUnderProb_qualtest_calcUnderProb__check_if_prob_unknown_with_state_is_prime_value_stationary_with_no_test_has_status_F80_0B</code>	<code>check_if_object_is_suspected_stationary_qualtest_check_if_object_is_stationary</code>
151	Check if object is suspected stationary, 135

`calcUndetProb_qualtest_calcUndetProb__check_if_prob_unknown_with_status_of_obj` `check_if object is suspected stationary, qualtest__check_if object is suspected stationary` `Check if object is suspected stationary`

`calcUndetProb_qualtest_calcUndetProb__check_if_prob_unknown_if_is_stationary_when_object_has_status_F300_OBU`

calcUndetProb qualtest

```
check_if_object_is_suspected_stationary_qualtest_check_if_object_is_su
```

CalcUndetProb, 150

Check_if_object_is_suspected_stationary, 137

calcUndetProb_qualtest, calcUndetProb__check_if_prob_unknown_is_safe_to_max_value_when_object_has_status_F380_OBJECT, [150](#)
 CalcUndetProb, [150](#)
 Check_if_object_is_suspected_stationary, [137](#)

UndetProb qualtest c

down-is set to max value when object has s

CalcUndetProb, 151	Check_if_object_is_suspected_stationary, 138
calcUndetProb	check if object is suspected stationary
qualtest	check if object is suspected stationary
calcUndetProb	check if prob. unknown is equal to max value when object has status "band 0B REC"

CalcUndetProb. 151

Check_if_object_is_suspected_stationary, 138

```

    calcUndetProb_qualtest_calcUndetProb__check_if_prob_unknown Is a float value when object has status_F360_OBJECT
    CalcUndetProb 152 Check if object is suspected stationary, 139

```

ck if object is suscep

check if object is suspected stationary, 100

Detecting_multibounces_from_object, 101

133

```
detecting_multibounces_from_object, 101
detecting_multibounces_from_object, det
```

```

133 check_if_object_is_suspected_stationary_qualtest_check_if_object_is_suspected_stationary__cca_marked_as_not_stationary_
101 detecting_multibounces_from_object_det_marked_as_multibounce_

```

```

133         detecting_multibounces_from_object_det NOT marked as multibound
check_if_object_is_suspected_stationary_qualtest_check_if_object_is_suspected_stationary__cca_marked_as_not_stationary_
134         detecting_multibounces_from_object_det NOT marked as multibound

```

```

130         detecting_multibounces_from_object
check_if_object_is_suspected_stationary_qualtest_check_if_object_is_suspected_stationary_cca_marked_as_not_stationary
134     """

```

check_if_object_is_suspected_stationary_qualtest_check_if_object_is_suspected_stationary_qualtest - Detecting multibounces from object, cca marked as not stationary

```

133         detecting_multibounces_from_object_det_NOI_marked_as_multibounce
check_if_object_is_suspected_stationary_qualtest_check_if_object_is_suspected_stationary_qualtest_cca_marked_as_stationary_if_d
125         Detecting_multibounces_from_object, for

```

```
check_if_object_is_suspected_stationary_qualtest_check_if_object_is_suspected_stationary_qualmarked_as_stationary_if_m
```

```

136         r360_adjust_fit_type_dependet_params_qualtest,
check_if_object_is_suspected_stationary_qualtest_check_if_object_is_suspected_stationary_ctca_marked_as_not_stationary
137     )

```

137	t360_adjust_fittr_type_dependet_params_qualtest_Adjust_Fittr_Type_
check_if_object_is_suspected_stationary_qualtest_check_if_object_is_suspected_stationary__ctca_marked_as_not_stationary	
187	

```

137         f360_adjust_filt_type_dependent_params_qualtest
check_if_object_is_suspected_stationary_qualtest_check_f360adjust_filt_type_dependent_params_qualtest_as_not_stationary
138

```

[illegible]

```
138 F360_adjust_filtr_type_dependet_params_qualtest,
check_if_object_is_suspected_stationary_qualtest_check_if_object_is_suspected_stationary__ctca_marked_as_stationary_if_c
```

F360_allocate_id_for_initialized_object, [24](#) [92](#)
 f360_allocate_id_for_initialized_object, [24](#) [f360_cancel_new_updated_trk_overlapping_confirmed_trks_qualtest](#)
 f360_allocate_id_for_initialized_object_no_active_objects, [24](#) [93](#)
 f360_cancel_new_updated_trk_overlapping_confirmed_trks_qualtest
 f360_allocate_id_for_initialized_object_objects_saturation, [24](#) [F360_cancel_new_updated_trk_overlapping_confirmed_trks_qualtest](#)
 f360_cancel_new_updated_trk_overlapping_confirmed_trks_qualtest
 f360_allocate_id_for_initialized_object_single_active_object, [25](#) [F360_cancel_new_updated_trk_overlapping_confirmed_trks_qualtest](#)
 F360_cancel_new_updated_trk_overlapping_confirmed_trks_qualtest
 f360_allocate_id_for_initialized_object [92](#)
 F360_allocate_id_for_initialized_object, [24](#) [f360_cancel_new_updated_trk_overlapping_confirmed_trks_qualtest](#)
 f360_allocate_id_for_initialized_object_no_active_objects [F360_cancel_new_updated_trk_overlapping_confirmed_trks_qualtest](#)
 F360_allocate_id_for_initialized_object, [24](#) [93](#)
 f360_allocate_id_for_initialized_object_objects_saturation [F360_cluster_grouping_qualtest, 17](#)
 F360_allocate_id_for_initialized_object, [24](#) [f360_cluster_grouping_qualtest, 17](#)
 f360_allocate_id_for_initialized_object_single_active_object [f360_cluster_grouping_qualtest_Test_Cluster_Grouping_Check_If_D](#)
 F360_allocate_id_for_initialized_object, [25](#) [17](#)
 F360_assign_underdrivability_status_to_tracks_ocg_QT, [f360_cluster_grouping_qualtest_Test_Cluster_Grouping_Check_If_D](#)
 [123](#) [18](#)
 f360_assign_underdrivability_status_to_tracks_ocg_QT, [f360_cluster_grouping_qualtest_Test_Cluster_Grouping_Check_If_R](#)
 [123](#) [18](#)
 f360_assign_underdrivability_status_to_tracks_ocg_QT_Assign_Underdrivability_Status_To_Tracks_Current_Status_Can_Not_Pas
 [123](#) [F360_cluster_grouping_qualtest, 17](#)
 f360_assign_underdrivability_status_to_tracks_ocg_QT_Assign_Underdrivability_Status_To_Tracks_Grouping_Status_Can_Decl
 [124](#) [F360_cluster_grouping_qualtest, 17](#)
 f360_assign_underdrivability_status_to_tracks_ocg_QT_Assign_Underdrivability_Status_To_Tracks_Grouping_Status_Can_Decl
 [124](#) [F360_cluster_grouping_qualtest, 18](#)
 f360_assign_underdrivability_status_to_tracks_ocg_QT_Assign_Underdrivability_Status_To_Tracks_Grouping_Status_Can_Ra
 [125](#) [F360_cluster_grouping_qualtest, 18](#)
 f360_assign_underdrivability_status_to_tracks_ocg_QT_Assign_Underdrivability_Status_To_Tracks_Grouping_Status_Can_Ra
 [125](#) [F360_conf_track_confidence_level_qualtest, 186](#)
 f360_assign_underdrivability_status_to_tracks_ocg_QT_Assign_Underdrivability_Status_Level_Tracks_Object_Track_In_Confide
 [126](#) [186](#)
 f360_assign_underdrivability_status_to_tracks_ocg_QT [f360_conf_track_confidence_level_qualtest_Conf_Track_Confidence](#)
 F360_assign_underdrivability_status_to_tracks_ocg_QT, [187](#)
 [123](#) [f360_conf_track_confidence_level_qualtest_Conf_Track_Confidence](#)
 f360_assign_underdrivability_status_to_tracks_ocg_QT_Assign_Underdrivability_Status_To_Tracks_Current_Status_Can_Not_Pas
 F360_assign_underdrivability_status_to_tracks_ocg_QT, [f360_conf_track_confidence_level_qualtest_Conf_Track_Confidence](#)
 [123](#) [188](#)
 f360_assign_underdrivability_status_to_tracks_ocg_QT_Assign_Underdrivability_Status_To_Tracks_Current_Status_Can_Not_Pas
 F360_assign_underdrivability_status_to_tracks_ocg_QT, [188](#)
 [124](#) [f360_conf_track_confidence_level_qualtest_Conf_Track_Confidence](#)
 f360_assign_underdrivability_status_to_tracks_ocg_QT_Assign_Underdrivability_Status_To_Tracks_Current_Status_Can_Pass_Us
 F360_assign_underdrivability_status_to_tracks_ocg_QT, [f360_conf_track_confidence_level_qualtest_Conf_Track_Confidence](#)
 [124](#) [189](#)
 f360_assign_underdrivability_status_to_tracks_ocg_QT_Assign_Underdrivability_Status_To_Tracks_Current_Status_Can_Pass_Us
 F360_assign_underdrivability_status_to_tracks_ocg_QT, [190](#)
 [125](#) [f360_conf_track_confidence_level_qualtest_Conf_Track_Confidence](#)
 f360_assign_underdrivability_status_to_tracks_ocg_QT_Assign_Underdrivability_Status_To_Tracks_Not_To_Consider_After_Status
 F360_assign_underdrivability_status_to_tracks_ocg_QT, [f360_conf_track_confidence_level_qualtest](#)
 [125](#) [F360_conf_track_confidence_level, 186](#)
 f360_assign_underdrivability_status_to_tracks_ocg_QT_Assign_Underdrivability_Status_Level_Tracks_Object_Track_In_Confide
 F360_assign_underdrivability_status_to_tracks_ocg_QT, [F360_conf_track_confidence_level, 186](#)
 [126](#) [f360_conf_track_confidence_level_qualtest_Conf_Track_Confidence_Level](#)
 F360_cancel_new_updated_trk_overlapping_confirmed_trks_qualtest [F360_conf_track_confidence_level, 187](#)
 [92](#) [f360_conf_track_confidence_level_qualtest_Conf_Track_Confidence_Level](#)
 f360_cancel_new_updated_trk_overlapping_confirmed_trks_qualtest [F360_conf_track_confidence_level, 187](#)
 [92](#) [f360_conf_track_confidence_level_qualtest_Conf_Track_Confidence_Level](#)
 f360_cancel_new_updated_trk_overlapping_confirmed_trks_qualtest [F360_conf_track_confidence_level, 188](#)
 [92](#) [F360_cancel_new_updated_trk_overlapping_confirmed_trks_qualtest](#)

f360_conf_track_confidence_level_qualtest_Conf_Track_Confidence	F360_detect_wheelspin_nearby_qualtest_38
F360_conf_track_confidence_level, 188	f360_detect_wheelspin_nearby_qualtest_f360_detect_wheelspin_ne
f360_conf_track_confidence_level_qualtest_Conf_Track_Confidence	38level_is_confidence_level_is_equal_or_higher_than_0_for_U
F360_conf_track_confidence_level, 189	f360_detect_wheelspin_nearby_qualtest_f360_detect_wheelspin_ne
f360_conf_track_confidence_level_qualtest_Conf_Track_Confidence	38level_is_confidence_level_is_equal_or_lower_than_1_for_CO
F360_conf_track_confidence_level, 189	f360_detect_wheelspin_nearby_qualtest
f360_conf_track_confidence_level_qualtest_Conf_Track_Confid	F360_detect_wheelspin_nearby_qualtest_38or_lower_than_1_for_UP
F360_conf_track_confidence_level, 190	f360_detect_wheelspin_nearby_qualtest_f360_detect_wheelspin_nearby_
f360_conf_track_confidence_level_qualtest_Conf_Track_Confid	F360_detect_wheelspin_nearby_qualtest_38object_is_reduced_by_factor
F360_conf_track_confidence_level, 190	f360_detect_wheelspin_nearby_qualtest_f360_detect_wheelspin_nearby_
F360_detect_multipath__Detect_And_Mark_Multipath_Objects	F360_detect_wheelspin_nearby_qualtest, 38
170	F360_detect_wheelspin_object_qualtest, 36
f360_detect_multipath__Detect_And_Mark_Multipath_Obj	F360_detect_wheelspin_object_qualtest, 36
170	f360_detect_wheelspin_object_qualtest_f360_detect_wheelspin_obje
f360_detect_multipath__Detect_And_Mark_Multipath_Objects	36Qual default_scenario_where_host_speed_is_below_threshold
170	f360_detect_wheelspin_object_qualtest_f360_detect_wheelspin_obje
f360_detect_multipath__Detect_And_Mark_Multipath_Objects	36Qual default_scenario_where_host_speed_is_too_high,
171	f360_detect_wheelspin_object_qualtest
f360_detect_multipath__Detect_And_Mark_Multipath_Obj	F360_detect_wheelspin_object_qualtest_36its_does_not_exist,
171	f360_detect_wheelspin_object_qualtest_f360_detect_wheelspin_obje q
f360_detect_multipath__Detect_And_Mark_Multipath_Obj	F360_detect_wheelspin_object_qualtest_36its_does_not_exist,
172	f360_detect_wheelspin_object_qualtest_f360_detect_wheelspin_obje q
f360_detect_multipath__Detect_And_Mark_Multipath_Obj	F360_detect_wheelspin_object_qualtest_36is_beyond_sensor_fov,
172	F360_detect_wheelspin_pairs, 62
f360_detect_multipath__Detect_And_Mark_Multipath_Obj	F360_detect_wheelspin_pairs_qualtest_unittest_azimuth_spread_limi
173	f360_detect_wheelspin_pairs_qualtest_unittest_doppler_spread_limi
f360_detect_multipath__Detect_And_Mark_Multipath_Objects	62Qual scenario_where_source_objects_is_stationary,
173	f360_detect_wheelspin_pairs_qualtest_unittest_doppler_spread_limi
f360_detect_multipath__Detect_And_Mark_Multipath_Objects	63Qual scenario_where_source_objects_use_CCV_filter,
174	f360_detect_wheelspin_pairs_qualtest_unittest_lat_limits,
f360_detect_multipath__Detect_And_Mark_Multipath_Objects	Qual63
F360_detect_multipath__Detect_And_Mark_Multipath_Obj	F360_detect_wheelspin_pairs_qualtest_unittest_lat_spread_limits,
170	64
f360_detect_multipath__Detect_And_Mark_Multipath_Objects	F360_detect_wheelspin_pairs_qualtest_unittest_long_spread_limi
F360_detect_multipath__Detect_And_Mark_Multipath_Objects	63Qual,
170	f360_detect_wheelspin_pairs_qualtest_unittest_long_spread_limits,
f360_detect_multipath__Detect_And_Mark_Multipath_Objects	Qual65default_scenario_where_host_speed_is_too_high
F360_detect_multipath__Detect_And_Mark_Multipath_Obj	F360_detect_wheelspin_pairs_qualtest_unittest_max_wheelspin_pai
171	65
f360_detect_multipath__Detect_And_Mark_Multipath_Objects	F360_detect_wheelspin_pairs_qualtest_unittest_multiple_candidates,
F360_detect_multipath__Detect_And_Mark_Multipath_Objects	60Qual,
171	f360_detect_wheelspin_pairs_qualtest_unittest_proximity_detections
f360_detect_multipath__Detect_And_Mark_Multipath_Objects	Qual66scenario_where_source_objects_does_not_exist
F360_detect_multipath__Detect_And_Mark_Multipath_Obj	F360_detect_wheelspin_pairs_qualtest_unittest_remarking,
172	67
f360_detect_multipath__Detect_And_Mark_Multipath_Obj	F360_detect_wheelspin_pairs_qualtest_unittest_is_beyond_sensor_fov
F360_detect_multipath__Detect_And_Mark_Multipath_Obj	F360_detect_wheelspin_pairs, 62
172	f360_detect_wheelspin_pairs_qualtest_unittest_azimuth_spread_limits
f360_detect_multipath__Detect_And_Mark_Multipath_Objects	F360_detect_wheelspin_pairs_qualtest_unittest_doppler_spread_limits
F360_detect_multipath__Detect_And_Mark_Multipath_Objects	Qual63F360_detect_wheelspin_pairs, 63
173	F360_detect_wheelspin_pairs, 63
f360_detect_multipath__Detect_And_Mark_Multipath_Obj	F360_detect_wheelspin_pairs_qualtest_unittest_lat_limits
F360_detect_multipath__Detect_And_Mark_Multipath_Obj	F360_detect_wheelspin_pairs, 63
173	f360_detect_wheelspin_pairs_qualtest_unittest_lat_spread_limits
f360_detect_multipath__Detect_And_Mark_Multipath_Objects	F360_detect_wheelspin_pairs_qualtest_unittest_long_limits
F360_detect_multipath__Detect_And_Mark_Multipath_Objects	Qual64F360_detect_wheelspin_pairs, 64
174	F360_detect_wheelspin_pairs, 64
F360 detect wheelspin nearby qualtest, 38	f360 detect wheelspin pairs qualtest unittest long spread limits

f360_fill_init_obj_track_props_qualtest_TestCalclnitObjTrackMovingFlagNotMovingLowSpeedAndHighFracAz,	71
42	f360_handle_host_vehicle_clutter_qualtest_handle_host_vehicle_clutter
f360_fill_init_obj_track_props_qualtest_TestCalclnitObjVCSHeadingMovingObj,	71
42	f360_handle_host_vehicle_clutter_qualtest_handle_host_vehicle_clutter
f360_fill_init_obj_track_props_qualtest_TestCalclnitObjVCSHeadingNotMovingObj,	72
43	f360_handle_host_vehicle_clutter_qualtest_handle_host_vehicle_clutter
f360_fill_init_obj_track_props_qualtest_TestFillInitObjTrackPropsNominal,	38
43	f360_handle_host_vehicle_clutter_qualtest
f360_fill_init_obj_track_props_qualtest	F360_handle_host_vehicle_clutter_qualtest, 71
F360_fill_init_obj_track_props_qualtest, 40	f360_handle_host_vehicle_clutter_qualtest_handle_host_vehicle_clutter
f360_fill_init_obj_track_props_qualtest_TestCalclnitObjTrackMovingFlagMovingHostVehicleClutterQualtest,	71
F360_fill_init_obj_track_props_qualtest, 40	f360_handle_host_vehicle_clutter_qualtest_handle_host_vehicle_clutter
f360_fill_init_obj_track_props_qualtest_TestCalclnitObjTrackMovingFlagMovingValidForClericalTracking,	71
F360_fill_init_obj_track_props_qualtest, 40	f360_handle_host_vehicle_clutter_qualtest_handle_host_vehicle_clutter
f360_fill_init_obj_track_props_qualtest_TestCalclnitObjTrackMovingFlagMovingHostHighFractAz,	72
F360_fill_init_obj_track_props_qualtest, 41	f360_handle_host_vehicle_clutter_qualtest_handle_host_vehicle_clutter
f360_fill_init_obj_track_props_qualtest_TestCalclnitObjTrackMovingFlagMovingLowSpeed,	72
F360_fill_init_obj_track_props_qualtest, 41	f360_identify_and_flag_internal_reflections_scenario_based_tests
f360_fill_init_obj_track_props_qualtest_TestCalclnitObjTrackMovingFlagMovingSpotAndHighFracAz,	unittest.cpp,
F360_fill_init_obj_track_props_qualtest, 42	221
f360_fill_init_obj_track_props_qualtest_TestCalclnitObjVCSHeadingMovingObj,	f360_identify_and_flag_internal_reflections_scenario_based_tests_flag
F360_fill_init_obj_track_props_qualtest, 42	f360_identify_and_flag_internal_reflections_qualtest_unittest.cpp,
f360_fill_init_obj_track_props_qualtest_TestCalclnitObjVCSHeadingNotMovingObj	221
F360_fill_init_obj_track_props_qualtest, 43	f360_identify_and_flag_internal_reflections_scenario_based_tests_reset
f360_fill_init_obj_track_props_qualtest_TestFillInitObjTrackPropsNominal,	f360_identify_and_flag_internal_reflections_qualtest_unittest.cpp,
F360_fill_init_obj_track_props_qualtest, 43	222
F360_find_detection_inliers_find_valid_detections, 27	f360_identify_and_flag_internal_reflections_qualtest_unittest.cpp
f360_find_detection_inliers_find_valid_detections_Select_Detections_To_Mark_As_Inliers_Object_Wheel_Spin_Detection,	221
27	f360_identify_and_flag_internal_reflections_scenario_based_tests
f360_find_detection_inliers_find_valid_detections_Select_Detections_To_Mark_As_Inliers_Object_Wheel_Spin_Detection,	222
27	f360_identify_and_flag_internal_reflections_scenario_based_tests
f360_find_detection_inliers_find_valid_detections_Select_Detections_To_Mark_As_Inliers_Object_Wheel_Spin_Detection,	222
28	f360_initialize_tracked_objects
f360_find_detection_inliers_find_valid_detections_Select_Detections_To_Mark_As_Inliers_Object_Wheel_Spin_Detection	45
F360_find_detection_inliers_find_valid_detections,	f360_init_obj_track_covariances, 45
27	f360_init_obj_track_covariances_TestCCANominalCase,
f360_find_detection_inliers_find_valid_detections_Select_Detections_To_Mark_As_Inliers_Object_Wheel_Spin_Detection	45
F360_find_detection_inliers_find_valid_detections,	f360_init_obj_track_covariances
27	F360_init_obj_track_covariances, 45
f360_find_detection_inliers_find_valid_detections_Select_Detections_To_Mark_As_Inliers_Object_Wheel_Spin_Detection	45
F360_find_detection_inliers_find_valid_detections,	F360_init_obj_track_covariances, 45
28	f360_initialize_tracked_objects
F360_find_detection_inliers_single_object, 26	F360_initialize_tracks, 48
F360_find_tracks_on_and_behind_guardrail_qualtest,	f360_initialize_tracked_objects_Check_If_Objects_Are_Correctly_Sorted
99	F360_initialize_tracks, 48
f360_find_tracks_on_and_behind_guardrail_qualtest_F360_initialize_tracked_objects_Check_If_Objects_Are_Correctly_Sorted	49
99	F360_initialize_tracks, 49
f360_find_tracks_on_and_behind_guardrail_qualtest_F360_initialize_tracked_objects_Check_If_Objects_Are_Correctly_Sorted	49
99	F360_initialize_tracks, 49
f360_find_tracks_on_and_behind_guardrail_qualtest_Flag_Objects_Are_Separate_Or_Assigned_For_Single_Unsu	50
F360_find_tracks_on_and_behind_guardrail_qualtest,	F360_initialize_tracks, 50
99	f360_initialize_tracked_objects_Highest_Cluster_Priority_Initialization_Try
f360_find_tracks_on_and_behind_guardrail_qualtest_Flag_Objects_Are_Separate_Or_Assigned_For_Single_Unsu	50
F360_find_tracks_on_and_behind_guardrail_qualtestf360_initialize_tracked_objects_Highest_Cluster_Priority_Reinitialization_V	51
99	F360_initialize_tracks, 51
F360_handle_host_vehicle_clutter_qualtest, 71	f360_initialize_tracked_objects_Lowest_Cluster_Priority_Initialization_Try
f360_handle_host_vehicle_clutter_qualtest, 71	F360_initialize_tracks, 51
f360 handle host vehicle clutter qualtest handle host vehicle clutter qualtest handle host vehicle clutter qualtest	f360_initialize_tracked_objects_Lowest_Cluster_Priority_Initialization_Try

f360_input_diagnostics_qualtest_Fault_Should_Be_Returned_When_Host_Info_Vehicle_Index_Lower_Than_Before	F360_input_diagnostics_qualtest, 194	f360_input_diagnostics_qualtest_Fault_Should_Be_Returned_When_Host_Info_Vehicle_Index_Lower_Than_Before	F360_input_diagnostics_qualtest, 210
f360_input_diagnostics_qualtest_Fault_Should_Be_Returned_When_Host_Info_Vehicle_Index_Lower_Than_Before	F360_input_diagnostics_qualtest, 194	f360_inputs_preprocessing_qualtest_Internal_Preprocessing_Input_Interface	
f360_input_diagnostics_qualtest_Fault_Should_Be_Returned_When_Host_Info_Vehicle_Index_Lower_Than_Before	F360_input_diagnostics_qualtest, 195	f360_kill_coasted_tracks_qualtest	108
f360_input_diagnostics_qualtest_No_Fault_Reported_For_Sensor_When_Calibration_Invalid	F360_input_diagnostics_qualtest, 195	f360_kill_coasted_tracks_qualtest_Do_not_terminate_updated_tracks	
f360_input_diagnostics_qualtest_No_Fault_Reported_For_Sensor_When_Calibration_Invalid	F360_input_diagnostics_qualtest, 196	f360_kill_look_index_Overflow	
f360_input_diagnostics_qualtest_No_Increase_Fault_Should_Be_Returned_When_Look_Index_For_Single_Sensor_Lower_Than_Before	F360_input_diagnostics_qualtest, 196	f360_kill_coasted_tracks_qualtest_Non_Movable_Coasting_Obj_Outside_Vcs_Zone	
f360_input_diagnostics_qualtest_No_Increase_Faults_Should_Be_Returned_When_Core_Signal_Received_Twice_Without_Change	F360_input_diagnostics_qualtest, 197	f360_kill_coasted_tracks_qualtest_Non_Movable_Coasting_Obj_Outside_Vcs_Zone	
f360_input_diagnostics_qualtest_No_Increase_Faults_Should_Be_Returned_When_Sensors_Signal_Received_Twice_Without_Change	F360_input_diagnostics_qualtest, 197	f360_kill_coasted_tracks_qualtest_Non_Movable_Coasting_Obj_Outside_Vcs_Zone	
f360_input_diagnostics_qualtest_Polarity_Fault_Should_Not_Be_Reported_When_Sensor_Polarity_Is_Valid_Negative	F360_input_diagnostics_qualtest, 198	f360_kill_coasted_tracks_qualtest_Termination_of_coasting_confirmed_object	
f360_input_diagnostics_qualtest_Polarity_Fault_Should_Not_Be_Reported_When_Sensor_Polarity_Is_Valid_Positive	F360_input_diagnostics_qualtest, 198	f360_kill_coasted_tracks_qualtest_Termination_of_coasting_mirror_object	
f360_input_diagnostics_qualtest_Signal_Fault_Should_Be_Returned_When_Difference_Between_Sensor_And_Tracker_Timestamp_Is_Greater_Than_Limit	F360_input_diagnostics_qualtest, 199	f360_kill_coasted_tracks_qualtest_Termination_of_coasting_mirror_object	
f360_input_diagnostics_qualtest_Signal_Fault_Should_Be_Returned_When_Difference_Between_Sensor_And_Tracker_Timestamp_Is_Greater_Than_Limit	F360_input_diagnostics_qualtest, 199	f360_kill_coasted_tracks_qualtest	
f360_input_diagnostics_qualtest_Signal_Fault_Should_Not_Be_Reported_When_Calibration_Invalid	F360_input_diagnostics_qualtest, 200	f360_kill_coasted_tracks_qualtest_Do_not_terminate_updated_tracks	
f360_input_diagnostics_qualtest_Signal_Faults_Should_Be_Reported_When_Look_Index_Falls_To_Zero_Twice_In_A_Row	F360_input_diagnostics_qualtest, 200	f360_kill_coasted_tracks_qualtest_Non_Movable_Coasting_Obj_Outside_Vcs_Zone	
f360_input_diagnostics_qualtest_Signal_Faults_Should_Be_Reported_When_Sensor_Sight_Angle_Is_Greater_Than_Limit	F360_input_diagnostics_qualtest, 201	f360_kill_coasted_tracks_qualtest_Non_Movable_Coasting_Obj_Outside_Vcs_Zone	
f360_input_diagnostics_qualtest_Signal_Faults_Should_Be_Reported_When_Sensor_Sight_Angle_Is_Less_Than_Limit	F360_input_diagnostics_qualtest, 201	f360_kill_coasted_tracks_qualtest_Non_Movable_Coasting_Obj_Outside_Vcs_Zone	
f360_input_diagnostics_qualtest_Signal_Faults_Should_Be_Reported_When_Sensor_Sight_Angle_Is_Within_Lower_Limit	F360_input_diagnostics_qualtest, 202	f360_kill_coasted_tracks_qualtest_Non_Movable_Coasting_Obj_Outside_Vcs_Zone	
f360_input_diagnostics_qualtest_Signal_Faults_Should_Be_Reported_When_Sensor_Sight_Angle_Is_Within_Upper_Limit	F360_input_diagnostics_qualtest, 202	f360_kill_coasted_tracks_qualtest_Termination_of_coasting_confirmed_object	
f360_input_diagnostics_qualtest_Signal_Faults_Should_Be_Reported_When_Sensor_Mounting_Position_Is_Too_High_Negative	F360_input_diagnostics_qualtest, 203	f360_kill_coasted_tracks_qualtest_Termination_of_coasting_mirror_object	
f360_input_diagnostics_qualtest_Signal_Faults_Should_Be_Reported_When_Sensor_Mounting_Position_Is_Too_High_Positive	F360_input_diagnostics_qualtest, 203	f360_kill_coasted_tracks_qualtest_Termination_of_coasting_mirror_object	
f360_input_diagnostics_qualtest_Signal_Faults_Should_Be_Reported_When_Sensor_Mounting_Position_Is_Within_Limit_Negative	F360_input_diagnostics_qualtest, 204	F360_mark_water_spray_detections	74
f360_input_diagnostics_qualtest_Signal_Faults_Should_Be_Reported_When_Sensor_Mounting_Position_Is_Within_Limit_Positive	F360_input_diagnostics_qualtest, 204	f360_mark_water_spray_detections_default_test	
f360_input_diagnostics_qualtest_Signal_Faults_Should_Be_Reported_When_Sensor_Polarity_Is_Greater_Than_Expected_Positive	F360_input_diagnostics_qualtest, 205	f360_mark_water_spray_detections_det_high_rcs	
f360_input_diagnostics_qualtest_Signal_Faults_Should_Not_Be_Reported_When_Sensor_Mounting_Position_Equal_To_Limit_Negative	F360_input_diagnostics_qualtest, 205	f360_mark_water_spray_detections_det_range_rate	
f360_input_diagnostics_qualtest_Signal_Faults_Should_Not_Be_Reported_When_Sensor_Mounting_Position_Equal_To_Limit_Positive	F360_input_diagnostics_qualtest, 206	f360_mark_water_spray_detections_detections_outside_vcs_zone	
f360_input_diagnostics_qualtest_Signal_Faults_Should_Not_Be_Reported_When_Sensor_Mounting_Position_Is_Within_Limit_Negative	F360_input_diagnostics_qualtest, 206	f360_mark_water_spray_detections_f_ok_to_use_false	
f360_input_diagnostics_qualtest_Signal_Faults_Should_Not_Be_Reported_When_Sensor_Mounting_Position_Is_Within_Limit_Positive	F360_input_diagnostics_qualtest, 207	f360_mark_water_spray_detections_low_host_speed	
f360_input_diagnostics_qualtest_True_Should_Be_Returned_When_Big_Curvature_Rear_Or_VCS_Sideslip	F360_input_diagnostics_qualtest, 207	f360_mark_water_spray_detections	
f360_input_diagnostics_qualtest_True_Should_Be_Returned_When_Host_Info_Vehicle_Index_Lower_Than_Before	F360_input_diagnostics_qualtest, 210	f360 mark water spray detections default test	

- F360_object_based_water_spray_detector_qualtest, 97
 f360_object_based_water_spray_detector_qualtest, 97
 f360_object_based_water_spray_detector_qualtest_DetectWaterSprayFromObjects, 97
 f360_object_based_water_spray_detector_qualtest 97
 F360_object_based_water_spray_detector_qualtest, 97
 f360_object_based_water_spray_detector_qualtest_DetectWaterSprayFromObjects, 97
 F360_object_based_water_spray_detector_qualtest, 97
 F360_object_occlusion_classification, 115
 f360_object_occlusion_classification, 115
 f360_object_occlusion_classification_Verify_All_Active_Objects_Are_Considered, 115
 f360_object_occlusion_classification_Verify_Single_Object_Status_Accepted, 116
 f360_object_occlusion_classification_Verify_Single_Object_Status_Accepted_Edge, 117
 f360_object_occlusion_classification_Verify_Single_Object_Status_Accepted_Not_Moving, 117
 f360_object_occlusion_classification_Verify_Single_Object_Status_Accepted_Visible, 118
 f360_object_occlusion_classification
 F360_object_occlusion_classification, 115
 f360_object_occlusion_classification_Verify_All_Active_Objects_Are_Considered, 115
 F360_object_occlusion_classification, 115
 f360_object_occlusion_classification_Verify_Single_Object_Status_Accepted, 116
 F360_object_occlusion_classification, 116
 f360_object_occlusion_classification_Verify_Single_Object_Status_Accepted_Edge, 117
 F360_object_occlusion_classification, 117
 f360_object_occlusion_classification_Verify_Single_Object_Status_Accepted_Not_Moving, 117
 F360_object_occlusion_classification, 117
 f360_object_occlusion_classification_Verify_Single_Object_Status_Accepted_Visible, 118
 F360_object_occlusion_classification, 118
 F360_object_track_management_qualtest, 88
 f360_object_track_management_qualtest, 88
 f360_object_track_management_qualtest_Obj_Trk_Status_Book_Keeping_NEW, 88
 f360_object_track_management_qualtest_Obj_Trk_Status_Book_Keeping_UPDATED, 89
 f360_object_track_management_qualtest
 F360_object_track_management_qualtest, 88
 f360_object_track_management_qualtest_Obj_Trk_Status_Book_Keeping_NEW, 88
 F360_object_track_management_qualtest, 88
 f360_object_track_management_qualtest_Obj_Trk_Status_Book_Keeping_UPDATED, 89
 F360_object_track_management_qualtest, 89
 F360_object_underdrivability_classification_qt, 120
 f360_object_underdrivability_classification_qt, 120
 f360_object_underdrivability_classification_qt_test_objs_invalid, 120
 f360_object_underdrivability_classification_qt_test_valid_moving_objs_behind_host, 121
 f360_object_underdrivability_classification_qt_test_valid_moving_objs_in_orth_direction, 121
 f360_object_underdrivability_classification_qt_test_valid_moving_objs_in_orth_direction_qualtest, 122
 f360_object_underdrivability_classification_qt 120
 F360_object_underdrivability_classification_qt, 120
 f360_object_underdrivability_classification_qt_test_objs_invalid, 120
 f360_object_underdrivability_classification_qt_test_valid_moving_objs_behind_host, 121
 f360_object_underdrivability_classification_qt_test_valid_moving_objs_in_orth_direction, 121
 f360_object_underdrivability_classification_qt_test_valid_moving_objs_in_orth_direction_qualtest, 122
 F360_object_underdrivability_classification_qt 31
 F360_object_underdrivability_classification_qt, 31
 f360_object_underdrivability_classification_qt_test_objs_invalid, 31
 f360_object_underdrivability_classification_qt_test_valid_moving_objs_behind_host, 31
 f360_object_underdrivability_classification_qt_test_valid_moving_objs_in_orth_direction, 31
 f360_object_underdrivability_classification_qt_test_valid_moving_objs_in_orth_direction_qualtest, 31
 F360_Test_Determine_Association_Hypothesis_qualtest, 31
 F360_Test_Is_Det_Allowed_To_Associate_qualtest, 31

File	Line	Function	Line	Function
F360_track_grouping,	168		F360_update_object_track_properties_obj_height,	
f360_track_grouping_Test_Similar_Objects_Were_Merged			94	
F360_track_grouping,	168		f360_update_object_track_properties_calc_obj_height_obj_height_decays	
F360_try_to_init_obj_track_qualtest,	57		F360_update_object_track_properties_obj_height,	
f360_try_to_init_obj_track_qualtest,	57		94	
f360_try_to_init_obj_track_qualtest_HostOverlapCheckPosLongNOTOverlappedPosLatOverlappedtest,			f360_update_object_track_properties_obj_height,	
57			95	
f360_try_to_init_obj_track_qualtest_HostOverlapCheckPosLongNOTOverlappedPosLatOverlappedtest,			f360_update_object_track_properties_obj_height,	
57			95	
f360_try_to_init_obj_track_qualtest_HostOverlapCheckPosLongOverlappedInHearAxleDistPosLatNOTOverlappedtest,			F360_update_object_track_properties_Calc_Obj_Size,	
58			86	
f360_try_to_init_obj_track_qualtest_HostOverlapCheckPosLongOverlappedInHearAxleDistPosLatOverlappedtest,			F360_update_object_track_properties_Obj_Size,	
58			86	
f360_try_to_init_obj_track_qualtest_LiberalInitializationTest,			F360_update_object_track_properties_Calc_Obj_Size_Obj_Size_Is_Not_U	
59			86	
f360_try_to_init_obj_track_qualtest			F360_update_object_track_properties_Obj_Size,	
F360_try_to_init_obj_track_qualtest,	57		86	
f360_try_to_init_obj_track_qualtest_HostOverlapCheckPosLongNOTOverlappedPosLatNOTOverlappedtest,			f360_update_object_track_properties_Calc_Obj_Size_Obj_Size_Is_Not_U	
F360_try_to_init_obj_track_qualtest,	57		F360_update_object_track_properties_Obj_Size,	
f360_try_to_init_obj_track_qualtest_HostOverlapCheckPosLongNOTOverlappedPosLatOverlappedtest			86	
F360_try_to_init_obj_track_qualtest,	57		f360_update_object_track_properties_Calc_Obj_Size_Obj_Size_Is_Updated	
f360_try_to_init_obj_track_qualtest_HostOverlapCheckPosLongOverlappedInHearAxleDistPosLatNOTOverlappedtest			87	
F360_try_to_init_obj_track_qualtest,	58		F360_update_object_track_properties_obj_height	94
f360_try_to_init_obj_track_qualtest_HostOverlapCheckPosLongOverlappedInHearAxleDistPosLatOverlappedtest			f360_update_object_track_properties_calc_obj_height,	
F360_try_to_init_obj_track_qualtest,	58		94	
f360_try_to_init_obj_track_qualtest_LiberalInitializationTest			f360_update_object_track_properties_calc_obj_height_obj_height_d	
F360_try_to_init_obj_track_qualtest,	59		94	
f360_Update_Detection_History_qualtest			f360_update_object_track_properties_calc_obj_height_obj_height_d	
Sensor_Postprocessing,	105		94	
f360_Update_Detection_History_qualtest_Test_Update_Detection_History_Find_Next_Hist_Det_Idx_false			f360_update_object_track_properties_calc_obj_height_obj_height_is	
Sensor_Postprocessing,	105		95	
f360_Update_Detection_History_qualtest_Test_Update_Detection_History_n_occupied_higher_than_max			f360_update_object_track_properties_calc_obj_height_obj_height_is	
Sensor_Postprocessing,	105		95	
f360_Update_Detection_History_qualtest_Test_Update_Detection_History_old_dets_greater_than_zero			F360_update_object_track_properties_Obj_Size,	86
Sensor_Postprocessing,	106		f360_update_object_track_properties_Calc_Obj_Size,	
f360_Update_Detection_History_qualtest_Test_Update_Detection_History_update_one_detection			86	
Sensor_Postprocessing,	106		f360_update_object_track_properties_Calc_Obj_Size_Obj_Size_Is_M	
F360_update_exist_prob,	183		86	
f360_update_exist_prob_qualtest,	183		f360_update_object_track_properties_Calc_Obj_Size_Obj_Size_Is_M	
f360_update_exist_prob_qualtest_Update_Existence_Probability_check_whether_EP_depend_onobject_state_variance,			86	
183			f360_update_object_track_properties_Calc_Obj_Size_Obj_Size_Is_U	
f360_update_exist_prob_qualtest_Update_Existence_Probability_check_whether_maximum_EP_value_is_1,			87	
184			FilteringAndNormalizationOfProbabilities,	145
f360_update_exist_prob_qualtest_Update_Existence_Probability_check_whether_minimum_EP_value_is_equal_to_min_thres			145	
184			f360_run_obj_class_f360_run_obj_class_bicycle,	
f360_update_exist_prob_qualtest			146	
F360_update_exist_prob,	183		f360_run_obj_class_f360_run_obj_class_car,	146
f360_update_exist_prob_qualtest_Update_Existence_Probability_check_whether_EP_depend_onobject_state_variance			f360_run_obj_class_f360_run_obj_class_motorcycle,	
F360_update_exist_prob,	183		146	
f360_update_exist_prob_qualtest_Update_Existence_Probability_check_whether_maximum_EP_value_is_1			f360_run_obj_class_f360_run_obj_class_pedestrian,	
F360_update_exist_prob,	184		147	
f360_update_exist_prob_qualtest_Update_Existence_Probability_check_whether_minimum_EP_value_is_equal_to_min_threshold			147	
F360_update_exist_prob,	184		f360_run_obj_class_f360_run_obj_class_undetermined,	
f360_update_object_track_properties_calc_obj_height			148	
F360_update_object_track_properties_obj_height,			filteringAndNormalizationOfProbabilities_qualtest_filteringAndNormal	
94				
f360 update object track properties calc obj height obj height decays below zero				

filteringAndNormalizationOfProbabilities_qualtest_filteringAndNormalizationOfProbabilities, 148
 Fusion360/cluster_grouping/unittest/f360_cluster_grouping_qualtest_unittest.cpp, 213
 Fusion360/cluster_grouping/unittest/f360_merge_two_clusters_qualtest_unittest.cpp, 213
 Fusion360/cluster_grouping/unittest/f360_track_validity/unittest/f360_detect_multipath_qualtest_unittest.cpp, 213
 Fusion360/clustering/unittest/f360_initialize_clusters_qualtest_unittest.cpp, 214
 Fusion360/common/unittest/f360_allocate_id_for_initialized_objects_qualtest_unittest.cpp, 214
 Fusion360/common/unittest/f360_find_detection_inliers_qualtest_unittest.cpp, 215
 Fusion360/detection_to_track_association/unittest/f360_detection_to_track_association_qualtest_unittest.cpp, 215
 Fusion360/detection_to_track_association/unittest/f360_mark_detections_wheel_spin_from_objects_qualtest_unittest.cpp, 216
 Fusion360/detection_to_track_association/unittest/f360_nearby_wheel_spins_qualtest_unittest.cpp, 216
 Fusion360/initialize_tracks/unittest/f360_fill_init_obj_track_props_qualtest_unittest.cpp, 217
 Fusion360/initialize_tracks/unittest/f360_initialize_tracks_qualtest_unittest.cpp, 217
 Fusion360/initialize_tracks/unittest/f360_nees_cfmi_try_to_init_qualtest_unittest.cpp, 218
 Fusion360/inputs_preprocessing/unittest/f360_inputs_preprocessing_qualtest_unittest.cpp, 218
 Fusion360/internal_preprocessing/qualification_testing/qualtest/guardrail_detection_qualtest.cpp, 219
 Fusion360/internal_preprocessing/unittest/f360_detect_wheelspin_pairs_qualtest_unittest.cpp, 220
 Fusion360/internal_preprocessing/unittest/f360_determine_dets_on_edge_of_fov_qualtest_unittest.cpp, 220
 Fusion360/internal_preprocessing/unittest/f360_handle_host_vehicle_clutter_qualtest_unittest.cpp, 221
 Fusion360/internal_preprocessing/unittest/f360_identify_and_flag_internal_reflections_qualtest_unittest.cpp, 221
 Fusion360/internal_preprocessing/unittest/f360_mark_host_water_spray_detections_qualtest_unittest.cpp, 224
 Fusion360/measurement_update_tracks/unittest/f360_measurement_update_tracks_qualtest_unittest.cpp, 224
 Fusion360/measurement_update_tracks/unittest/f360_msmt_update_qualtest_unittest.cpp, 225
 Fusion360/post_update_track_adjustments/unittest/f360_post_update_track_adjustments_qualtest_unittest.cpp, 225
 Fusion360/pre_association_track_management/unittest/f360_pre_association_track_management_qualtest_unittest.cpp, 226
 Fusion360/sensor_postprocessing/unittest/f360_sensor_postprocessing_qualtest_unittest.cpp, 226
 Fusion360/time_update_tracks/unittest/f360_time_update_tracks_qualtest_unittest.cpp, 227
 Fusion360/track_classification/unittest/f360_object_occlusion_classification_qualtest_unittest.cpp, 227
 Fusion360/track_classification/unittest/f360_object_underdrivable_classification_qualtest_unittest.cpp, 228
 Fusion360/track_classification/unittest/f360_track_classification_qualtest_unittest.cpp, 228
 Fusion360/track_downselection/unittest/f360_track_downselection_qualtest_unittest.cpp, 230

Object_motion_status_should_be_updated, 128	f360_track_downselection_qualtest_f360_core_track_downselect_c
object_motion_status_should_be_updated_qualtest_Object_Motion_Status_Should_Be_Updated_NEW_UPDATED	514
Object_motion_status_should_be_updated, 129	f360_track_downselection_qualtest_f360_core_track_downselect_c
object_motion_status_should_be_updated_qualtest_Object_Motion_Status_Should_Be_Updated_UPDATED	514
Object_motion_status_should_be_updated, 129	f360_track_downselection_qualtest_f360_core_track_downselect_c
Object_wrt_to_sep_tests, 103	155
object_wrt_to_sep_tests, 103	f360_track_downselection_qualtest_f360_core_track_downselect_c
object_wrt_to_sep_tests_obejct_marked_as_on_a_sep, 103	155
object_wrt_to_sep_tests_obejct_NOT_marked_as_on_sep, 103	f360_track_downselection_qualtest_f360_core_track_downselect_c
object_wrt_to_sep_tests	156
Object_wrt_to_sep_tests, 103	f360_track_downselection_qualtest_f360_core_track_downselect_c
object_wrt_to_sep_tests_obejct_marked_as_on_a_sep	157
Object_wrt_to_sep_tests, 103	f360_track_downselection_qualtest_f360_core_track_downselect_c
object_wrt_to_sep_tests_obejct_NOT_marked_as_on_sep	157
Object_wrt_to_sep_tests, 103	f360_track_downselection_qualtest_f360_core_track_downselect_c
Sensor_Postprocessing, 105	158
f360_Update_Detection_History_qualtest, 105	158
f360_Update_Detection_History_qualtest_Test_Update_Detection_History_End_Next_Hist_Det_Box_false	159
f360_Update_Detection_History_qualtest_Test_Update_Detection_History_SeenCopiesHigherThanMax	159
f360_Update_Detection_History_qualtest_Test_Update_Detection_History_OldDataGreaterthanZero	161
f360_Update_Detection_History_qualtest_Test_Update_Detection_History_update_one_detection	161
Set_object_class, 141	f360_track_downselection_calc_track_priority_qualtest_Host_Is_Mov
set_object_class_qualtest_set_object_class__check_if_classifies_as_undetermined,	161
set_object_class_qualtest_set_object_class__check_if_min_prob_value_to_classify_is_0_65,	162
set_object_class_qualtest_set_object_class__check_if_prioritizes_cars,	162
set_object_class_qualtest_set_object_class__check_if_prioritizes_motorcycles,	163
set_object_class_qualtest_set_object_class__check_UpdateObjectProperties,	131
set_object_class_qualtest_set_object_class__check_if_prioritizes_objects,	131
set_object_class_qualtest_set_object_class__check_if_classifies_update_object_properties_qualtest_update_object_properties__check	131
Set_object_class, 141	131
set_object_class_qualtest_set_object_class__check_if_min_prob_value_to_classify_is_0_65	131
Set_object_class, 141	Update_object_properties, 131
set_object_class_qualtest_set_object_class__check_if_prioritizes_objects	Update_object_properties, 131
Set_object_class, 142	Update_object_properties, 131
set_object_class_qualtest_set_object_class__check_if_prioritizes_pedestrian	Update_object_properties, 131
Set_object_class, 143	Update_object_properties, 131
set_object_class_qualtest_set_object_class__check_if_prioritizes_trucks	
Set_object_class, 143	
StateManager/unittest/f360_input_diagnostics_qualtest_unittest.cpp,	
234	
Track_Downselection, 153	
f360_track_downselection_qualtest_f360_core_track_downselect__check_if_assigns_reduced_id_to_higher_priority_targets_and	153